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Informatics Engineering-Software and
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Syrian System for Marketing and handicrafts

A senior 2 project report

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SUPERVISION CERTIFICATION

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entitled

[Syrian System for Marketing and handicrafts]

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Abstract

In the face of rapid changes and high competition in the world of business management, e-commerce platforms have become one of the most prominent means of online marketing and sales. Providing a secure and user-friendly digital environment for artisans and handmade business owners to showcase and promote their products is a vital necessity to ensure the success of buying and selling operations.

This report presents the project "Syrian System for Marketing and Handicrafts" as an important step towards enabling handicraft owners to reach their customers electronically and providing an effective marketing channel that facilitates interaction between business owners and those interested in purchasing their products.

The system aims to give artisans, productive families in Syria, and home-based entrepreneurs the opportunity to showcase their skills and products directly and securely on a reliable and easy-to-use technological platform. It also seeks to support the local economy and enhance self-sufficiency by providing an electronic business environment that contributes to the growth of small businesses.

In this context, the project aims to stimulate local economic development by promoting trade exchange among different segments of society and creating new opportunities for commercial growth, and marketing through effective promotional methods.

This report highlights the core functions of the system, the adopted technological framework, the challenges facing the project, in addition to the fact that this system represents a strategic step towards a more prosperous future for the handicraft sector in Syria, ensuring that the needs of all market players are met in a smooth and effective manner.

ملخص

في ظل التغيرات السريعة والتنافسية العالية في عالم إدارة الأعمال، أصبحت منصات التجارة الإلكترونية من أبرز وسائل التسويق والبيع عبر الإنترنت. يُعد توفير بيئة رقمية آمنة وسهلة الاستخدام للحرفيين وأصحاب الأعمال اليدوية لعرض منتجاتهم وترويجها ضرورة حيوية لضمان نجاح عمليات البيع والشراء.

يعرض هذا التقرير مشروع "النظام السوري للتسويق للأعمال اليدوية" كخطوة مهمة نحو تمكين أصحاب الأعمال اليدوية من الوصول إلى عملائهم إلكترونياً، وتوفير قناة تسويقية فعالة تسهل التفاعل بين أصحاب العمل والراغبين بشراء منتجاتهم.

يهدف النظام إلى منح الحرفيين والأسر المنتجة في سوريا وأصحاب المشاريع المنزلية فرصة عرض مهاراتهم ومنتجاتهم بشكل مباشر وآمن على منصة تكنولوجية موثوقة وسهلة الاستخدام. كما يسعى إلى دعم الاقتصاد المحلي وتعزيز الاكتفاء الذاتي من خلال توفير بيئة تجارية إلكترونية تساهم في نمو الأعمال الصغيرة.

في هذا السياق، يرمي المشروع إلى تحفيز التنمية الاقتصادية المحلية عبر تعزيز التبادل التجاري بين شرائح المجتمع المختلفة، وخلق فرص جديدة للنمو التجاري، والتسويق بطرق ترويجية فعّالة.

يسلّط هذا التقرير الضوء على الوظائف الأساسية للنظام، والإطار التكنولوجي المعتمد، والتحديات التي تواجه المشروع، بالإضافة إلى أنّ هذا النظام يمثل خطوة استراتيجية نحو مستقبل أكثر ازدهاراً لقطاع الأعمال اليدوية في سوريا، مع ضمان تلبية احتياجات جميع الفاعلين في السوق بشكل سلس وذي جدوى.

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Glossary Table

Term	Definition
Software Requirements Specification (SRS)	A formal document that precisely describes the functional and non-functional requirements of the system and serves as a primary reference throughout the analysis, design, implementation, and testing phases.
Minimum Viable Product (MVP)	An initial version of the system that contains only the core functionalities and is used to validate the idea and reduce development risks before expanding the system features.
Functional Requirement	A requirement that defines a function or behavior the system must perform in response to a user interaction or a specific event.
Non-Functional Requirement	A requirement that defines the quality attributes of the system, such as performance, security, scalability, and usability, without specifying a direct function.
System Scope	The functional and technical boundaries that the system is committed to, including the supported features and the defined development phases of the project.
System Architecture	The overall structure of the system that illustrates the organization of its main components and the way they interact to achieve the system's functionalities.
CRUD Operations	The fundamental operations used for data management within the system, including Create, Read, Update, and Delete.
Entity Relationship Diagram (ERD)	A diagram used to represent entities and the relationships between them within a relational database.
Use Case	A formal description of a specific interaction between the system and an actor in order to achieve a particular function.
Activity Diagram	A diagram that illustrates the sequence of activities and workflows within a specific process in the system.
Sequence Diagram	A diagram that shows the chronological interaction between system components during the execution of a specific use case.
Requirement Traceability Matrix (RTM)	A matrix used to trace requirements from the analysis phase through design and implementation to testing.

Chapter1

Introduction

1. Introduction

1.1 Objective

The Syrian System for Marketing & Handicrafts aims to create a simple and secure electronic business environment in Syria to facilitate communication between sellers, buyers, and bazaar organizers.

1.2 Project Scope

The Syrian System for Marketing and Handicrafts was designed to help artisans, home-based producers, and owners of small handmade businesses showcase and sell their products online in Syria. It provides them with a marketing channel that supports self-sufficiency and the local economy. The project focuses on facilitating interaction between artisans and people interested in purchasing or supporting their products. The project scope can be divided into phases:

1.2.1 Phase One:

Minimum Viable Product (MVP) – focusing on the core functionalities:

- **Account Management:**

Creating seller, buyer, and admin accounts, as well as adding and editing account information.

- **Shop Management:**

- **Seller:**

Adding shop information (basic information about the seller and their business project), editing and deleting shop information, and displaying the seller's products on the shop page.

- **Admin:**

Viewing all shops available in the system, creating a shop for a seller, and deleting a shop from the system.

- **Shopping Basket Management:**
 - **Buyer:**

Adding products to the basket, removing products from it, and viewing basket contents before placing a purchase request.
- **Order and Purchase Management:**
 - **Seller:**

Tracking purchase orders (accepting/rejecting them).
 - **Buyer:**

Requesting the purchase of products available in the basket, choosing a communication method via phone number or in-system messages, and tracking order status (approved/rejected/pending review).
- **Product Management:**
 - **Seller:**

Managing products by adding, editing, and deleting them, as well as managing product categories (food products, embroidery, decorations, handmade soap, accessories, etc.).
 - **Buyer:**

Viewing products and their detailed information.
- **Product Search:**
 - **Buyer:**

Searching for products by category.
- **Comment Management:**
 - **Seller & Buyer:**

Viewing comments on products.
 - **Buyer:**

Adding comments on products after purchase.

1.2.2 Phase Two:

- **Delivery Options:**

- **Buyer:**

- Choosing the delivery method during purchase (local delivery / hand delivery / shipping).

- **Order Status Tracking Management:**

- **Seller:**

- After approving the order, the seller can update its status according to the following sequence:

- Delivered, Out for Delivery, Prepared, In Preparation.

- **Buyer:**

- Viewing the order status at any time through the order page.

- **Seller Reply System for Comments:**

- **Seller:**

- Replying to buyer comments, where the reply appears below the buyer's comment on the product page.

- **Advertisement Management:**

- **Seller:**

- Adding promotional advertisements for their shop and products.

- **Buyer:**

- Viewing advertisements in a dedicated section for buyers.

- **Temporary Offers Management:**

- **Seller:**

- Adding temporary offers for products with promotional deals or discounts, as well as editing and deleting offers.

- **Buyer:**

- Viewing offers and discounts.

- **Product Filtering by Price:**

- **Buyer:**

- Allowing buyers to sort products within the same category from the lowest price to the highest.

- **Wishlist Management:**

- **Buyer:**

- Adding products to the wishlist, viewing the list, or removing products from it later.

- **Reporting System:**

- **Seller:**

- Viewing simple reports including:

- Number of received orders.
 - Number of displayed products.
 - Number of visitors to their shop.

- **Direct Messaging System Between Buyer and Seller:**

- Sending and receiving messages within the system (chat) to facilitate communication regarding orders or products.

- **Smart Recommendation Feature (AI Feature):**

- A simple system that suggests suitable products as gifts for buyers based on:

- Budget, gender, age, and relationship.
 - The selected occasion (birthday, friend's graduation party, visiting a patient, etc.).

- The feature relies on the Groq API for gift recommendations.

1.2.3 Phase Three:

- **Bazaar Management:**

- **Bazaar Organizer:**

- Adding bazaar information and reservable tables, as well as editing and deleting this information.

- **Seller:**

- Viewing bazaars and the details of each bazaar.

- **Bazaar Table Reservation Management:**

- **Bazaar Organizer:**

- Viewing bazaar table reservations and accepting or rejecting reservation requests.

- **Seller:**

- Reserving a table in a bazaar and canceling the reservation before the bazaar begins.

- Tracking reservation status (pending / approved / rejected).

- **Notification System:**

Displaying notifications for each user as follows:

- **Bazaar Organizer:**

- 1. Receiving a new reservation request from a seller.
 - 2. Reservation cancellation by the seller before the bazaar date.

- **Seller:**

- 1. Receiving a new purchase request from a buyer.
 - 2. Receiving a new comment on one of their products.
 - 3. Approval of a bazaar table reservation request.
 - 4. Rejection of a bazaar table reservation request.

- **Buyer:**

- 1. Approval of the basket purchase request by the seller.

2. Rejection of the basket purchase request by the seller.
 3. Updating order status such as In Preparation / Prepared / Out for Delivery / Delivered.
 4. Receiving a reply to one of their comments.
- **Note:** The Minimum Viable Product (MVP) is an initial version of the product that contains only the core functionalities. It is used to test the idea and gather user feedback before expanding the development process. The MVP aims to reduce development risks and ensure that the product meets actual market needs.

1.3 Technical Scope of the Project:

- The defined technical scope includes all the necessary elements for building an effective and secure electronic marketing system that supports artisans, home-based producers, and small home business owners.
- **1.3.1 Frontend:**
- The interfaces were designed and developed using HTML, CSS, and Bootstrap, which contributed to creating responsive and user-friendly interfaces with consistent styling across all system pages.
- **1.3.2 Backend:**
- The backend was developed using the Django framework with the Python programming language. Django is responsible for handling business logic, managing login and authorization processes, processing requests, and connecting the system components together.
- **1.3.3 Database:**
- A relational MySQL database was used to manage all system data such as users, orders, and other information. In addition, Django's built-in ORM technology was utilized to interact efficiently with the database, ensuring data integrity and simplifying storage and retrieval operations.

Chapter2

Project management

1. Introduction

In this chapter, we will explore the project management phase in depth, which is considered a crucial aspect of ensuring the project's success. We will examine the Project Charter, Project Plan, Statement of Work (SOW), Work Breakdown Structure (WBS), stakeholder analysis, and risk management strategies in order to guide and control the project effectively from initiation to completion.

2. Project Charter

The Project Charter is a formal document that serves as an official authorization to initiate the project. It also acts as a reference point throughout the project lifecycle, providing a clear understanding of the project's purpose and establishing a foundation for decision-making and project management.

Item	Description
Project Title	Syrian System for Marketing and Handicrafts
Project Start Date	February 13, 2026
Project End Date	May 5, 2026
Project Supervisors	Dr. Akram Massouh and Engineer Anas Abdul Aziz
Project Methodology	Agile Scrum (adapted for individual development)

▪ **Project Objectives:**

- Enabling artisans, small business owners, and home-based producers in Syria to create their own online shops to easily showcase their products without requiring technical expertise.
- Providing a dashboard for sellers to manage their shops, add, edit, and delete products, display products in an organized manner with details such as prices and images, and track incoming orders in order to make appropriate decisions.
- Enabling buyers to browse products and shops, add products to the shopping basket, and complete purchase requests easily and securely.
- Providing buyers with the option to choose the delivery method (local delivery / hand delivery / shipping) when placing an order.
- Enabling buyers to track the status of purchase requests through different stages (pending review, approved, rejected).
- Enabling buyers to track the order status after approval through its different stages (in preparation, prepared, out for delivery, delivered).
- Enabling the admin to manage the system through an Admin Dashboard that includes shop management, creating shops for new sellers, and deleting shops when necessary.
- Supporting a comment system that allows buyers to add their opinions about products after purchase, enhancing transparency and contributing to building trust between sellers and buyers.
- Providing a reply system for sellers to respond to buyer comments in order to improve interaction and strengthen mutual trust.
- Supporting search and filtering features to enable buyers to easily find products by category or price.
- Enabling buyers to manage their wishlist and save products they wish to return to later.

- Providing simple reports for sellers regarding their shop performance (number of orders, number of products, and number of visitors).
- Enabling sending and receiving messages between sellers and buyers through an internal messaging system to facilitate direct communication regarding products and orders.
- Supporting an advertisement system and Flash Offers feature to enhance promotion and increase sales.
- Providing an AI-based smart feature that suggests suitable gift products based on the budget and selected occasion.
- Providing a dashboard for bazaar organizers to manage bazaars, add, edit, and delete bazaar information, display bazaars with details such as table reservation prices and locations, and track incoming table reservation requests.
- Enabling sellers to browse bazaars and their details in order to reserve a table or cancel the reservation before the bazaar starts, as well as track reservation requests (pending review / approved / rejected).
- Supporting a notification system for the bazaar organizer when a new reservation request is received from a seller or when a reservation is canceled before the bazaar date.
- Supporting a notification system for sellers when a new purchase request is received from a buyer, when a new comment is added to one of their products, when a bazaar table reservation request is approved, and when a bazaar table reservation request is rejected.
- Supporting a notification system for buyers when a basket purchase request is approved or rejected by the seller, when the order status is updated such as in preparation / prepared / out for delivery / delivered, and when a reply is added to their comment on a product.

▪ **Project Approach:**

The project was developed incrementally in phases according to priority as follows:

1. Account Management:

Developing a login and registration system for users (buyers, sellers, bazaar organizers, and admins), with customized permissions for each type. It was implemented within the first Sprint to ensure secure access and account control.

2. Shop Management:

Enabling the seller to create, edit, and delete their shop, and link it to their account. The tasks focused on user interfaces and data integration with accounts. It also enabled the admin to manage shops by creating shops for sellers or deleting them from the system.

3. Product Management:

Allowing products to be added to shops, edited, and deleted, with support for images and prices. This was implemented in a separate Sprint to ensure the completion of the seller's core operations.

4. Product Category Management:

Developing a product categorization system through manageable categories controlled by the seller, making searching and organization within the system easier.

5. Basket Management:

Enabling the buyer to add products to the basket, update quantities, or remove products before confirming the order. The focus was on user experience and ease of use.

6. Order Management:

Creating purchase requests for the basket, with the ability for the seller to approve or reject the order. The tasks also included selecting the delivery method for the order.

7. Filtering and Search System:*

Improving the product display interface by supporting category-based search and price filtering (from lowest to highest) to facilitate finding the required products.

8. Comment Management: *

Allowing buyers to comment on products after completing an order, and allowing sellers to reply to buyer comments with replies displayed below each comment. User authorization was verified to ensure credibility, and each comment was linked to the relevant order and product.

9. Order Tracking Management: *

Developing a mechanism for tracking order status after seller approval, where the seller can update the status through the stages: (Delivered → Out for Delivery → Prepared → In Preparation), while the buyer can view the status at any time.

10. Advertisement Management: *

Providing sellers with the ability to add promotional advertisements about their products or shops, with a dedicated section for buyers to view these advertisements.

11. Flash Offers Management: *

Developing a system that allows sellers to add temporary offers or discounts on specific products, enabling buyers to easily browse them through a dedicated interface.

12.Wishlist Management: *

Developing a system that allows buyers to add products to a wishlist for later viewing or removal.

13.Reporting System: *

Adding an interface for sellers that displays simplified reports about their shop performance, including the number of received orders, displayed products, and shop visitors.

14.Direct Messaging System: *

Developing an internal chat system between sellers and buyers to facilitate direct communication regarding products and orders in an organized and secure manner.

15.Smart Recommendation Feature (AI Feature): *

Adding an intelligent system based on the Groq API, using product prices and descriptions to suggest suitable gift products for buyers according to the selected occasion (such as birthdays, graduations, visiting a patient, etc.), gender, age, and relationship.

16.Bazaar Management: **

Allowing bazaar organizers to add, edit, and delete bazaars, including bazaar location, table reservation price, and bazaar images, with a dedicated section for sellers to browse these bazaars and their details.

17.Bazaar Table Reservation Management: **

Creating table reservation requests in bazaars by sellers and tracking reservation status, with the ability for the bazaar organizer to approve or reject the request.

18.Notification System: **

Providing a notification system where each user (buyer, seller, and

bazaar organizer) receives notifications related only to their activities within the system in order to enhance user interaction.

Note: Features marked with (*) were implemented in Graduation Project 1, while features marked with (**) were implemented in Graduation Project 2.

▪ **Stakeholders:**

Stakeholder	Role	Responsibility
Dr. Akram Massoh	Academic Supervisor	Guides the project, reviews progress, and provides academic feedback.
Eng. Anas Abdulaziz	Project Manager & Technical Supervisor	Provides technical guidance, reviews implementation, and assists in solving technical issues.
Nour Nasan	Software Engineer	- Develop backend functionalities, implement authentication, handle database integrations. - Design and implement user interfaces, connect frontend with backend. - Frontend development, ensure a seamless user experience.

3. (SOW- Statement of Work)

The Statement of Work (SOW) is a comprehensive document that defines the project scope, including specific tasks, deliverables, timeline, and responsibilities. It provides a clear understanding of what must be accomplished, the project objectives, and the success criteria.

3.1 Project Description:

The Syrian System for Marketing and Handicrafts project aims to facilitate online marketing and sales operations, as well as organize reservable bazaars. The system includes creating shops for sellers and product reviews to increase

trust between sellers and buyers, along with an interactive interface to enhance the shopping experience.

3.2 Project Scope:

The Syrian System for Marketing and Handicrafts was designed to create a simple and secure e-commerce platform aimed at enabling artisans, home-based producers, and small home business owners to showcase and sell their products. The system focuses on providing a marketing channel that supports self-sufficiency and strengthens the local economy, while ensuring a smooth user experience for sellers, buyers, and bazaar organizers.

3.3 Project Objectives:

- Developing an electronic environment that enhances the growth and expansion of artisans' businesses and local trade.
- Providing an easy-to-use user interface for displaying products and services.
- Enhancing digital marketing for artisans through effective promotional tools.

3.4 Deliverables:

- Project Plan.
- SRS (Software Requirements Specification) document.
- Final project report.
- Backend and frontend components — the Syrian System for Marketing and Handicrafts itself.

3.5 Project Requirements:

Technologies and Tools:

- Programming Languages: Python, HTML, CSS.
- Frameworks: Django (backend), Bootstrap (frontend).
- Database: MySQL.

3.6 Assumptions:

- Availability of regular communication with supervisors for guidance and feedback.
- Access to the required resources and references necessary for completing the project.
- Ability to manage time independently in order to accomplish tasks.
- Availability of the required tools and software for project implementation.

3.7 Project Resources:

Human Resources:

- Dr. Akram Massouh: Academic Supervisor.
- Engineer Anas Abdul Aziz: Project Manager.
- Nour Al-Huda Nasan: SE-Developer (Backend & Frontend).

3.8 Schedule:

Project Title	Syrian System for Marketing and Handicrafts
Project Start Date	February 13, 2026
First Seminar	March 14, 2026
Second Seminar	April 25, 2026
Technical Committee	May 10, 2026
Project End Date	May 20, 2026

4. Project plan_Gantt Chart :

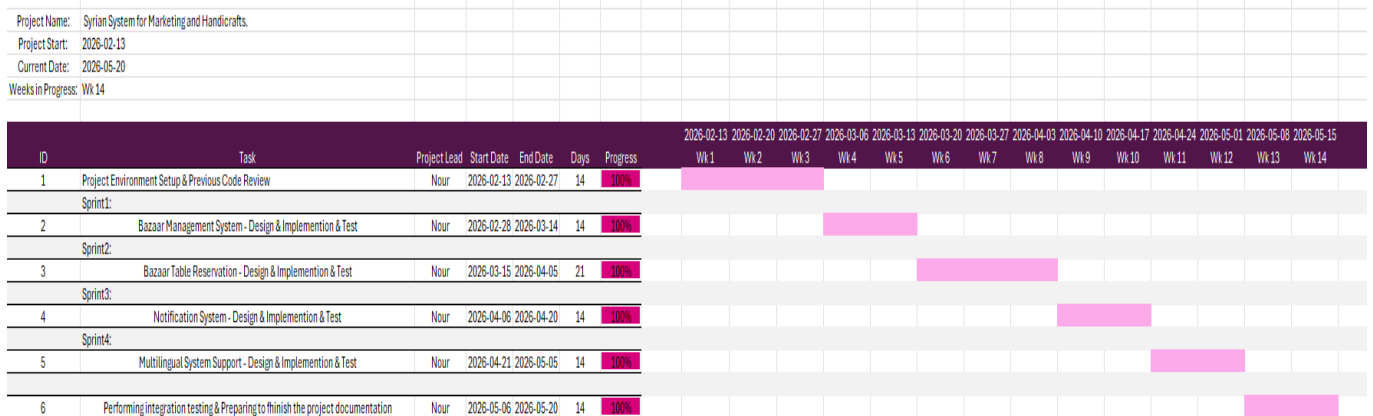


Figure 1: Gantt Chart

5. Risk management

Risk Title	Risk Description	Raised Date	Tracking Frequency	State	Impact	Mitigation Plan
Schedule Delay	Failure to adhere to the project timeline due to workload pressure and handling multiple tasks individually.	2026-02-28	Weekly	Active	High	Dividing the project into sprints, using tools such as Gantt Charts, and setting realistic weekly goals.
Difficulty Learning New Technologies	Possibility of facing difficulties while learning or integrating new libraries in Django or AI components.	2025-10-20	Weekly	Active	Medium	Allocating extra time for self-learning and research, and testing new technologies in small projects before integrating them into the main system.
Interface Integration Issues	Conflicts between the Frontend (Bootstrap) and Backend (Django) interfaces during development.	2026-03-15	Daily	Active	High	Performing integration testing frequently and relying on detailed API documentation.
Workload Pressure and Fatigue	Implementing all project parts individually may lead to fatigue or reduced productivity.	2025-11-05	Weekly	Active	High	Organizing time and prioritizing tasks effectively.
Testing Limitations	Limited ability to conduct extensive testing due to the absence of specialized testing users.	2025-11-15	Weekly	Active	Medium	Conducting accurate manual testing using sample data.
Complexity of the	The smart recommendation feature may	2025-12-20	Weekly	Potential	Medium	Starting the AI feature development

Risk Title	Risk Description	Raised Date	Tracking Frequency	State	Impact	Mitigation Plan
AI Feature	require additional logic or larger-scale data processing than expected.					early and initially relying on simple filtering algorithms before improving it later.
Delay in Preparing Final Documentation	Possibility of delaying the preparation of the final report and presentation due to the accumulation of technical tasks.	2026-05-05	Weekly	Potential	Medium	Updating documentation continuously after each Sprint instead of postponing it until the end of the project.
Data Loss or Technical Failures	Possibility of losing files or the database due to system failures or unexpected errors.	2026-02-28	Weekly	Active	High	Performing regular backups of the code and database, and using GitHub to securely maintain project versions.

Table 1: Risk Management

6. WBS (Work Breakdown Structure)

A diagram known as the Work Breakdown Structure (WBS). The WBS diagram is used in project management to divide large work into smaller sections that are easier to manage and track more effectively.

Usually, a WBS diagram is represented in a hierarchical form that shows the sequential breakdown of activities and tasks within the project. The large project (parent project) appears at the top, followed by levels of sub-elements branching from it. These are further divided into smaller and more detailed work elements until each task is clearly defined.

The WBS diagram provides a comprehensive view of the project structure and organization, and it helps define the scope of work clearly and assign specific responsibilities for each project element. It is also used to facilitate planning, resource organization, cost estimation, and monitoring work progress.



Work Breakdown Structure (WBS)

Syrian System for Marketing and Handicrafts

Project Lead / Resource: Nour Nasan

Baseline: 2026-02-13 → 2026-05-20

WBS ID	Task / Deliverable	Start Date	End Date	Duration (Days)	Resource
1.0	Project Environment Setup & Previous Code Review	2026-02-13	2026-02-27	14	Nour Nasan
1.1	Requirements Analysis & Planning	2026-02-13	2026-02-15	3	Nour Nasan

1.2	Development Environment Setup	2026-02-16	2026-02-18	3	Nour Nasan
1.3	Database Configuration & Migration	2026-02-19	2026-02-21	3	Nour Nasan
1.4	Codebase Review & Refactoring	2026-02-22	2026-02-25	4	Nour Nasan
1.5	Initial Testing & Bug Fixing	2026-02-26	2026-02-27	2	Nour Nasan
2.0	Bazaar Management System - Design & Implementation & Test	2026-02-28	2026-03-14	14	Nour Nasan
2.1	Requirements & UI/UX Design	2026-02-28	2026-03-02	3	Nour Nasan
2.2	Database Schema & API Design	2026-03-03	2026-03-05	3	Nour Nasan
2.3	Backend Development (Add/Edit/Delete Bazaar)	2026-03-06	2026-03-09	4	Nour Nasan
2.4	Frontend Development (Bazaar Display for Sellers)	2026-03-10	2026-03-12	3	Nour Nasan
2.5	Testing & Review	2026-03-13	2026-03-14	2	Nour Nasan
3.0	Bazaar Table Reservation - Design & Implementation & Test	2026-03-15	2026-04-05	21	Nour Nasan
3.1	Reservation Requirements & Workflow Analysis	2026-03-15	2026-03-18	4	Nour Nasan
3.2	Database Models & API for Reservations	2026-03-19	2026-03-22	4	Nour Nasan
3.3	Backend Development (Create Reservation Request)	2026-03-23	2026-03-27	5	Nour Nasan

3.4	Frontend Development (Accept/Reject for Organizers)	2026-03-28	2026-04-01	5	Nour Nasan
3.5	Testing & Review	2026-04-02	2026-04-05	4	Nour Nasan
4.0	Notification System - Design & Implementation & Test	2026-04-06	2026-04-20	14	Nour Nasan
4.1	Notification Requirements Analysis (Per User Type)	2026-04-06	2026-04-08	3	Nour Nasan
4.2	Notification System Design & Architecture	2026-04-09	2026-04-11	3	Nour Nasan
4.3	Backend Development (Notification Service)	2026-04-12	2026-04-15	4	Nour Nasan
4.4	Frontend Development (Notification Display)	2026-04-16	2026-04-18	3	Nour Nasan
4.5	Testing & Review	2026-04-19	2026-04-20	2	Nour Nasan
5.0	Multilingual System Support - Design & Implementation & Test	2026-04-21	2026-05-05	14	Nour Nasan
5.1	Multilingual Requirements Analysis & Planning	2026-04-21	2026-04-23	3	Nour Nasan
5.2	Language File Preparation & Integration Strategy	2026-04-24	2026-04-26	3	Nour Nasan
5.3	Frontend & Backend Localization Implementation	2026-04-27	2026-04-30	4	Nour Nasan
5.4	Testing & Review of Multilingual Features	2026-05-01	2026-05-05	5	Nour Nasan

6.0	Performing integration testing & preparing to finish the project documentation	2026-05-06	2026-05-20	14	Nour Nasan
6.1	Comprehensive Integration Testing	2026-05-06	2026-05-10	5	Nour Nasan
6.2	Bug Fixing & Performance Optimization	2026-05-11	2026-05-15	5	Nour Nasan
6.3	Final Project Documentation & Presentation Preparation	2026-05-16	2026-05-20	5	Nour Nasan

Figure 2: Work Breakdown Structure (WBS)

7. Quality Management

Quality management was emphasized in the Syrian System for Marketing Handicrafts project to ensure the development of a stable, reliable, and user-friendly system. This was achieved through the application of a set of practices and procedures aimed at improving software quality, reducing programming errors, and ensuring the system's compliance with the specified requirements.

Code Review:

The source code was reviewed continuously throughout the development process to ensure code organization, maintainability, and reduction of programming errors, in addition to verifying compatibility with the adopted system architecture using Django.

Functional Testing:

Functional tests were conducted on all parts of the system after completing the development of each feature, such as product management, orders, messaging, bazaars, and the notification system. This was done to verify the correctness of the expected results and detect programming errors at an early stage.

Authorization and Security Validation:

A permission system based on user roles within the system (buyer, seller, admin, and bazaar organizer) was implemented to ensure that each user could only access the pages and functions assigned to them.

In addition, data validation was applied within forms and application interfaces to prevent the entry of incorrect or incompatible data into the system.

User Interface Improvement:

Efforts were made to improve the user experience and design interfaces that are clear and easy to use, while maintaining consistency in colors and navigation across system pages.

Test Management and Requirements Traceability:

Initial and execution test cases were used along with the Requirement Traceability Matrix (RTM) to link functional requirements with testing and software implementation processes, which helped verify that all required system functionalities were successfully implemented.

Quality Assurance Tools and Techniques:

Tool / Technique	Purpose
Django Validation	Validating input data
Manual Testing	Manually testing software functionalities
Git & GitHub	Tracking code modifications and version management
RTM	Linking requirements with testing and implementation
Role-Based Access	Verifying user permissions

These procedures contributed to improving system quality, reducing programming errors, and ensuring the project's compliance with the predefined functional and non-functional requirements.

8. Software Configuration Management (SCM)

Software Configuration Management (SCM) was adopted in the Syrian System for Marketing Handicrafts project with the aim of organizing the project development process, tracking code modifications, and ensuring system stability throughout the different development phases. Git was used as the version control system, in addition to GitHub for hosting the project and managing updates and modifications in an organized manner.

Version Control Tools:

Git was used as a version control system to track all code modifications applied to the project, allowing previous versions to be restored when needed and preventing the loss of important data or modifications.

GitHub was also used to host the project and manage source code files, in addition to organizing the development process and monitoring different updates.

Branching Strategy:

A simple branching strategy was adopted to suit the nature of the individual project. The main branch (main) was primarily used to maintain the stable version of the project, while additional branches were created when developing new features or testing specific modifications before merging them into the final version.

Branch	Purpose
main	Contains the stable and official version of the project.
feature branches	Used for developing and testing new features before merging them into the main version.

Change Management and Backups:

GitHub was used to maintain continuous backup copies of the project, which contributed to protecting the source files from loss, in addition to facilitating change management and tracking the project's progress throughout the different development stages.

Commits were also used continuously to document code modifications and add a brief description for each change implemented in the system.

Benefits of Software Configuration Management within the Project:

- Organizing the project development process and tracking code modifications.
- Maintaining the stability of the main system version.
- Facilitating the restoration of previous versions when needed.
- Providing continuous backup copies of the project.
- Reducing the possibility of file loss or errors during development.

9. Summary

Effective project management is considered the foundation for developing successful software systems. It ensures that projects are completed within scope and schedule while taking resources into consideration. Strong project management also enhances the delivery of high-quality software systems that provide innovative solutions, achieve objectives, and meet stakeholder expectations. In this way, effective management bridges the gap between ambitious ideas and tangible results.

Chapter3

Fundamental Concepts and Literature Review

1. Introduction

In this chapter, I will present a reference study aimed at analyzing systems similar to my project, the Syrian System for Marketing and Handicrafts. I will discuss a group of electronic platforms that provide services similar or partially related to the objectives of my project, with the aim of identifying the most important features and technologies used in these systems, as well as exploring their strengths and weaknesses in order to benefit from them in improving and developing my system.

2. Basic Concepts

This section includes an explanation of the fundamental concepts and terms related to my project:

- **E-Marketing:**

The use of digital methods to promote products and services through the Internet, including email, social media, and search engines. It aims to reach the target audience efficiently and at a lower cost compared to traditional marketing.

- **E-Platforms:**

Digital environments that allow users to display their products or services and interact with buyers or other users. In the context of e-commerce, the platform acts as an intermediary that facilitates buying and selling operations between sellers and consumers.

- **Supporting the Local Economy:**

Refers to enhancing production and purchasing within a specific geographical area with the aim of strengthening local communities and preserving the cultural and productive identity of the region. This is achieved by providing direct sales channels for local producers.

- **Logistics:**

The process of planning, implementing, and controlling the efficient flow and storage of goods, services, and related information from the point of origin to the point of consumption. In e-commerce, logistics includes inventory management, packaging, shipping, and delivery.

- **Handicrafts:**

Products that are mainly manufactured or shaped manually, or using simple tools, and are characterized by a unique artistic or cultural touch that reflects the artisan's skill. They represent the core products targeted by the platform.

- **Productive Families:**

Families that rely on producing simple goods or services (often handmade or home-based) as a primary or additional source of income. This group represents the main target audience of the platform in order to support them economically.

- **AI in E-commerce:**

The use of artificial intelligence technologies, such as machine learning, to improve user experience, provide personalized product recommendations, analyze purchasing behavior, and manage inventory efficiently. In my project, this is represented by the "Smart Recommendation" feature.

- **Marketplace Model:**

A business model that allows multiple sellers to display their products on a single platform, where the platform acts as a third party facilitating transactions between sellers and buyers in exchange for a commission or subscription fee.

3. Literature Review

The purpose of this section is to analyze and evaluate systems and methodologies related to my system. This study aims to provide a comprehensive overview of platforms operating in the field of digital marketing for handicraft products or related domains, in order to understand general trends in the digital market and benefit from previous experiences in developing my own system. The analysis includes comparisons between four systems and their main features:

- Etsy

It is a global electronic platform specialized in selling handmade, artistic, and vintage products. Etsy allows independent sellers to create online shops to display and sell their products directly to consumers, and it is considered the world's largest specialized marketplace in this field [1].

Advantages:

- Easy to use: The user interface is simple and supports smooth navigation, making the shopping experience easy for users.
- Dedicated shop for each seller: Each user is given a dedicated page to manage and sell products, enhancing independent brand identity.
- Rating and review system: Buyers can evaluate products, which increases trust and reliability within the platform community.

Disadvantages:

- High commission fees: The platform charges fees for every sale (marketplace model), which may affect the profits of small sellers.
- High competition: Due to the huge number of sellers, making it difficult to stand out without strong marketing campaigns.

- Targets a global market: Which may not directly serve local artisans in areas with specific logistical and financial challenges.

Main Features:

- Effective rating and review system: Helps build trust among users.
- Shop customization: The ability to customize the shop interface to reflect the seller's brand identity.
- Analytics and statistics tools for sellers: Help them track performance and sales.
- Support for multiple payment systems: Provides various electronic payment options.
- SouqFann
It is an Arab e-commerce platform (originating in Jordan) that aims to empower local artisans and creators to sell their products online, with a focus on strengthening the local economy and supporting home-based businesses in the Arab region [2].

Advantages:

- Local support: The platform focuses on supporting artisans in the Arab world.
- Simple interface: Easy for sellers and buyers to use, and specifically designed for the Arabic language.
- Supervision and assistance: Provides support and supervision for artisans in marketing and professionally presenting their products.

Disadvantages:

- Limited payment options: Compared to global platforms, due to restrictions on electronic payment systems in the region.

- Limited geographical reach: Mainly focuses on the Jordanian market.
- Limited promotional campaigns: Which reduces the visibility of some products compared to platforms with large marketing budgets.

Main Features:

- Dedicated shop for each seller: Artisans can manage their own shops.
- Direct support from management: Supervision and assistance are available for sellers.
- Effective local delivery: Products are delivered within a local range.
- Product display by category: Products can be browsed according to clear classifications.

- Amazon Handmade

It is an initiative launched by Amazon in 2015, and it is a store within the main Amazon platform dedicated exclusively to selling handmade products. This store is characterized by strict joining requirements to ensure that products are genuinely handmade, providing a trusted environment for artisans [3].

Advantages:

- Strong user experience: An interactive and smooth interface that benefits from Amazon's expertise in e-commerce.
- Global reach and customer base: Benefiting from Amazon's huge customer base, which ensures unprecedented reach.
- Accurate rating system: Allows users to share their experiences with products while benefiting from Amazon's trusted review system.

Disadvantages:

- High fees: It imposes a high commission (approximately 15%), in addition to professional selling plan fees.
- Indirect competition: Despite the specialization, sellers still compete for attention within Amazon's crowded environment.
- Logistical challenges: Small artisans may face challenges in meeting Amazon's complex shipping and packaging requirements.

Main Features:

- Detailed rating and review system: Enhances customer trust in the product.
- Accurate product filtering and categorization: Facilitates the search process.
- Utilizing Amazon logistics services (FBA): An option for storing and shipping products.

- BeeOrder

It is a Syrian application dedicated to ordering and delivering food and groceries, and it is considered one of the first local platforms that succeeded in building a logistics network within Syria. Although its specialization is different (food services), it represents an important reference model in logistics and order management within the local Syrian environment [4].

Advantages:

- Easy ordering: A simple interface for adding products and completing orders.

- Fast user experience: Designed to reduce the number of steps required to achieve the goal.
- Effective notification system: Keeps users updated about order status.

Disadvantages:

- Does not support artisans: The application is dedicated only to food services.
- No direct communication with sellers: The process is handled only through the system.
- Limited specialization: It is not suitable for displaying and selling diverse products that require artistic presentation.

Main Features:

- Order management and tracking: The order status and execution can be tracked.
- Shopping basket: Facilitates collecting orders before payment.
- Order status notifications: To keep users continuously informed.

Note: BeeOrder differs from our project in terms of objective (food delivery) and target audience (restaurants and consumers only), but it is an important reference in local logistics workflows and order management mechanisms.

Conclusion:

The “Syrian System for Marketing and Handicrafts” project aims to provide a digital platform that supports local artisans and enables them to easily sell their products online in Syria. Through studying similar systems, we find that Etsy and SouqFann are the closest in terms of concept and target audience,

and their focus on the artisan community and advanced seller tools should be utilized.

Meanwhile, Amazon Handmade represents an advanced professional model for managing shops and sellers within an organized global environment, but it lacks the local specificity and the appropriate linguistic and logistical support for Arab markets. This is what the proposed system seeks to overcome through its locally oriented design for Syrian artisans.

As for BeeOrder, it serves as an important reference in logistics workflows and order management mechanisms within the Syrian local environment.

Therefore, this study contributes to understanding the strengths of existing systems and benefiting from them in developing a platform that efficiently serves artisans, while focusing on competitive features such as the direct messaging system and the AI-based smart recommendation feature, in order to provide added value that is not available in the other reference systems.

The following is a summary of the distinctive services provided by each of the four mentioned e-marketing systems.

(*) indicates features that will be implemented in the future.

System Feature	Etsy	SouqFann	Amazon Handmade	BeeOrder	My System
Support for Local Artisans	✓	✓	✓	×	✓
Support for Bazaar Reservation Management	×	×	×	×	✓
Ability to Filter Products by Price	✓	✓	✓	×	✓
Support for Local Marketing Advertisements	×	✓	×	✓	✓
Smart Gift Recommendation System Based on Occasion	×	×	×	×	✓
Order Status Tracking System	✓	×	✓	✓	✓
Product Categories and Classifications	✓	✓	✓	×	✓
Ability to Display Temporary Offers	×	✓	✓	✓	✓
Multiple Languages	✓	✓	✓	×	✓*

Table 2: Comparative Analysis of Similar Systems and the Target System

Chapter4

System Analysis

1. Introduction

This chapter focuses on the detailed analysis of the Syrian System for Marketing and Handicrafts. The main objective is to analyze the needs of artisans and small business owners, as well as the environment in which they operate, which helps identify the essential requirements and core functionalities that the marketing platform should include.

The chapter explores the functionalities related to marketing products and services, in addition to providing a comprehensive analysis of use cases and workflows. This provides a complete understanding of how artisans interact with the platform and how they can achieve their business objectives through it. This analysis forms the foundation for designing and implementing the proposed marketing solutions.

2. Feasibility Study

The feasibility study aims to evaluate the possibility of implementing the Syrian System for Marketing Handicrafts project from technical, economic, and operational perspectives, in order to verify the system's ability to achieve the required objectives and efficiently meet user needs.

Technical Feasibility:

The project has high technical feasibility due to the availability of the technologies and software tools required for system development. The project was developed using the Django framework based on the Python programming language, in addition to using HTML, CSS, and Bootstrap for developing the user interfaces.

A MySQL database was also used for data management, along with Git and GitHub for version control and tracking code modifications. The system also relies on external integration with the Groq API to provide the Smart Recommendation feature.

The required development environment and software tools are fully available, making the implementation and maintenance of the project feasible and effective.

Economic Feasibility:

The project is characterized by relatively low development costs, as it relies on open-source technologies such as Django, MySQL, and Bootstrap, which reduces the need for expensive software licenses.

In addition, the system contributes to supporting artisans and small business owners by providing an electronic platform that helps them showcase their products and reach a larger number of customers, which provides economic benefits for users.

Operational Feasibility:

The system is characterized by ease of use and clear interfaces, allowing users to interact with the system without requiring advanced technical expertise.

The system also supports several user types such as buyers, sellers, bazaar organizers, and administrators.

The system helps organize buying and selling operations, communication between users, and the management of orders and bazaars effectively, which increases the possibility of adopting and using it in a real working environment.

Based on the previous study, the Syrian System for Marketing Handicrafts project is considered technically, economically, and operationally feasible.

3. Software Requirement Specification (SRS)

The Software Requirement Specification (SRS) is considered an important document in the software development process. The SRS is used to describe the functional and non-functional requirements of the software system to be developed. Software Requirement Specifications usually include elements such as:

- 1. System Description:**

Provides a comprehensive understanding of the system to be developed and used.

- 2. Functional Requirements:**

Describe the activities and operations that the system must perform.

- 3. Performance Requirements:**

Define how the system should perform under different conditions and requirements.

- 4. Interface Requirements:**

Specify how the system interacts with users and with other systems.

- 5. Security Requirements:**

Describe how data and information are protected and how system integrity is ensured.

- 6. Maintenance Requirements:**

Explain how the system will be maintained and further developed in the future.

The existence of an SRS helps achieve effective communication between the software development team and the clients or users, and ensures an accurate understanding of the system requirements. It can also be used as a reference during software development to verify that all requirements are implemented correctly. Ultimately, the SRS is considered a fundamental document that helps define, document, and organize software requirements clearly and professionally, making it easier for the development team to successfully fulfill those requirements.

3. Software Requirement Specification (SRS)

Version 0.3

Revision History:

Date	Reason for Change	Version
20/6/2025		0.1
18/10/2025	Development of the first version	0.2
13/2/2026	Development of the second version	0.3

1. Introduction

1.1 Purpose

This document defines the requirements of the “Syrian System for Marketing and Handicrafts,” which is a system designed to enable artisans and handicraft business owners to showcase their products and enhance sales and marketing opportunities for their businesses by improving user experience and increasing access to handmade products.

1.2 Project Scope

The Syrian System for Marketing and Handicrafts was designed to create a simple and secure e-commerce platform aimed at enabling artisans, productive families, and home-based handicraft business owners to display and sell their products, while ensuring a smooth user experience for sellers, buyers, and bazaar organizers.

1.2.1 High-Level Requirements

- **Accounts Management:**

This functionality allows role separation, enabling users to create their accounts (seller / buyer / bazaar organizer), log into their accounts as sellers, buyers, bazaar organizers, or administrators, edit or delete their account information, and securely log out.

- **Shops Management:**

The system enables sellers to create their own shops containing identifying information about their business projects. It also allows sellers to edit or delete their shop information, while displaying their products on their personal shop page. In addition, the system enables administrators to manage shops by creating shops for sellers or deleting them.

- **Products Management:**

According to the user type in the system:

- Seller can:
 - Add new products, edit product details, and delete products.
 - Manage product categories to classify them into specific categories (such as accessories, embroidery, decorations, etc.).
- Buyer can:
 - View product details and browse products by category.
- **Basket Management:**

The system provides buyers with the ability to add products to their basket, remove products from the basket, and view the basket before confirming the order.
- **Orders Management:**

According to the user type in the system:

 - Seller is allowed to:
 - View purchase requests and approve or reject orders.
 - Buyer can:
 - Send purchase requests for products available in the basket.
 - Choose the communication method (phone / in-system messages, etc.).
 - Choose the delivery method during purchase (local delivery / hand delivery / shipping).
 - Track order status (pending / approved / rejected).

- **Comments Management:** According to the user type in the system:
 - Buyer:
 - The system provides the buyer with the ability to write comments on products after purchase, and allows all buyers to view product comments.
 - Seller is allowed to:
 - The system provides the seller with the ability to reply to buyer comments and view comments related to their products.
- **Filter & Search Management:**

The system allows users to filter products according to specific criteria such as category or price range, in addition to sorting them in ascending or descending order based on price, which contributes to improving the browsing experience and speeding up access to the required products.
- **Order Tracking:**

The system enables the seller to update the order status after approval through specific stages (Delivered, Out for Delivery, Prepared, In Preparation), and allows the buyer to track the order status to ensure transparency and ease of follow-up.
- **Advertisements Management:**

The system allows sellers to create promotional advertisements for their products or shops, where these advertisements are displayed in a dedicated interface for buyers to increase interaction with products and enhance sales.
- **Flash Offers Management:**

The system provides a feature that enables sellers to create temporary offers and discounts on specific products, which are clearly displayed

in a dedicated buyer interface to encourage purchases during limited periods.

- **Wishlist Management:**

The system allows buyers to add favorite products to a dedicated list that they can revisit later or remove products from, making it easier to track products they wish to purchase in the future.

- **Reports System:**

The system provides sellers with an interface containing simplified reports that enable them to monitor their shop performance, such as the number of received orders, the number of displayed products, and the number of visitors to their shop, supporting better development decisions.

- **Messaging System:**

The system provides an internal communication channel between sellers and buyers to exchange messages regarding products or orders in a secure and organized manner, without the need to use external communication methods.

- **AI Feature:**

The system includes a smart tool based on analyzing product descriptions to provide personalized recommendations for buyers regarding suitable gift products according to their budget and selected occasion (such as birthdays, graduations, visiting a patient, etc.).

- **Bazaar Management:** According to the user type in the system:

- Bazaar Organizer can:

- Add new bazaars, edit bazaar details, and delete bazaars.

- Seller can:

- View bazaar details and browse available bazaars.

- **Bazaar Table Reservation Management:**

According to the user type in the system:

- **Bazaar Organizer is allowed to:**

- View reservation requests and approve or reject requests.

- **Seller can:**

- Send a reservation request for a table in a bazaar.
- Cancel the table reservation before the bazaar start date.
- Track the reservation request status (pending / approved / rejected).

- **Notification System:**

According to the user type in the system, each user receives personalized notifications.

- **Bazaar Organizer:**

- Notification upon receiving a new reservation request from a seller.
- Notification when a seller cancels a reservation before the bazaar date.

- **Seller:**

- Notification upon receiving a new purchase request from a buyer.
- Notification when a new comment is added to one of their products.
- Notification when a bazaar table reservation request is approved.

- Notification when a bazaar table reservation request is rejected.
- Buyer:
 - Notification when a basket purchase request is approved by the seller.
 - Notification when a basket purchase request is rejected by the seller.
 - Notification when the order status is updated, such as In Preparation / Prepared / Out for Delivery / Delivered.
 - Notification when a reply is added to their comment on a product.

1.2.2 Actors

- **Seller:**

One of the main actors in the system, responsible for managing their electronic shop and its content. The seller can create a personal shop containing identifying information about their business activity, with the ability to edit or delete it when needed. The seller can also add new products, update product details such as name, price, and product image, delete unavailable products, and organize them into specific categories to facilitate their display to buyers.

The seller is also responsible for managing incoming orders to the shop, where they can view, approve, or reject orders and update their status through the order tracking stages (In Preparation, Prepared, Out for Delivery, Delivered), ensuring accurate monitoring of sales operations.

In addition, the seller can view comments related to their products and reply to them, which enhances communication and credibility with buyers. The seller is also able to create promotional advertisements for products or the shop, and add temporary offers to attract buyers, in addition to viewing comprehensive reports about shop performance, including the number of orders, products, and visitors.

The system also provides the seller with the ability to communicate directly with buyers through a secure and organized internal messaging system that facilitates exchanging inquiries related to orders and products. Furthermore, the seller can browse available bazaars, view the location of each bazaar and the table reservation price in order to participate and sell products, track reservation request status, and cancel reservations before the bazaar begins if desired, providing flexibility in handling reservation requests. The seller also receives notifications regarding orders, reservations, and comments.

- **Buyer:**

The buyer is the user who browses electronic shops and searches for products suitable for their needs. The system allows the buyer to add products to the shopping basket, modify quantities or remove items from it, and review the basket before completing the purchase process. After confirming the order, the buyer can choose the appropriate communication method with the seller and select the delivery method, whether local delivery, hand delivery, or shipping. The buyer can also track the order status throughout its different stages until receiving the product.

The system also supports advanced search and filtering for buyers based on category or price, making it easier to access the required products. After receiving the product, the buyer can write comments

based on their experience with the products, in addition to viewing comments from other buyers.

The system also allows buyers to save products in a wishlist for future reference, and browse available promotional advertisements and temporary offers. Buyers also benefit from the Smart Recommendation feature, which helps them find suitable products as gifts according to their budget. In addition, buyers can use the internal messaging system to communicate directly with sellers regarding products or orders, and receive appropriate notifications related to their orders.

- **Admin:**

The administrator manages shops and can create shops for sellers or remove them from the system if they do not comply with the platform's content policies.

- **Bazaar Organizer:**

One of the main actors in the system, responsible for managing and organizing bazaars within the platform. The system allows the bazaar organizer to create new bazaars containing detailed information such as the bazaar name, location, bazaar image, table reservation price, start date and time, and end date and time, with the ability to edit this information or delete the bazaar when needed.

The bazaar organizer can also view reservation requests submitted by sellers and approve or reject them according to the number of available tables, which helps organize participation in bazaars in a structured and flexible manner. The organizer can also monitor approved and rejected reservations, manage participants in each bazaar, and receive notifications about new reservations or when a seller cancels a reservation before the bazaar start date, making it easier to efficiently manage and organize events within the system.

2. General Description

2.1 Product Perspective

The Syrian System for Marketing and Handicrafts described in this document is considered an innovative product that offers many advantages to users. The system is distinguished by providing a simple and secure electronic business environment that facilitates communication between sellers and buyers. It also contributes to supporting artisans and handicraft business owners by providing a marketing channel that enhances self-sufficiency and improves the shopping experience for users in an innovative and effective way.

2.2 Product Features

- **Account Management:**

Creating accounts, logging into accounts, and securely logging out (seller – buyer - bazar organizer– admin).

- **Shop Management:**

Creating a shop for the seller, including adding, editing, and deleting business information.

Displaying products on the shop page.

Creating or deleting a shop (admin).

- **Product Management:**

Adding, editing, and deleting product information (seller).

Classifying products into specific categories (seller).

Displaying products (buyer).

- **Basket Management:**

Adding products to the basket, removing products from it, and viewing the basket before confirming the order (buyer).

- **Orders Management:**

Viewing purchase requests and approving or rejecting orders (seller).

Sending purchase requests for products available in the basket,

choosing the communication method, and selecting the delivery method (buyer).

Tracking the purchase order status (buyer).

- **Comments Management:**

Adding comments about products after purchase (buyer).

Viewing product comments (seller – buyer).

Replying to buyer comments on the seller's products (seller).

- **Order Tracking Management:**

Enabling order status updates after approval through specific stages (seller).

Tracking the order status to identify the current order stage until delivery (buyer).

- **Filtering and Search:**

Searching for products using product categories (buyer).

Filtering by price from lowest to highest to facilitate access to the required products (buyer).

- **Advertisements Management:**

Adding, editing, and deleting promotional advertisements for products (seller).

Displaying advertisements in a dedicated interface for buyers (buyer).

- **Flash Offers Management:**

Adding, editing, and deleting temporary offers and discounts on specific products (seller).

Displaying offers to buyers in a dedicated interface to encourage purchases (buyer).

- **Wishlist Management:**

Adding products to the wishlist for later viewing or removal (buyer).

- **Reports Management:**

Displaying reports related to shop performance (seller).

- **Direct Messaging System:**

Providing an internal communication method between sellers and buyers to exchange messages regarding products or orders in a secure and organized manner.

- **Smart Recommendation Feature (AI Feature):**

Suggesting suitable gift products for buyers based on a specific budget, gender, age, and relationship, with the ability to customize recommendations according to the occasion (such as birthdays, graduation parties, visiting a patient, etc.).

- **Bazaar Management:**

Adding, editing, and deleting bazaar information such as location, bazaar image, start and end date and time, and table reservation price (bazaar organizer).

Displaying available bazaars and their details for sellers in order to participate in them (seller).

- **Bazaar Table Reservation Management:**

Sending table reservation requests to participate in bazaars and tracking request status (seller).

Approving or rejecting table reservation requests submitted by sellers (bazaar organizer).

Canceling table reservations before the bazaar start date (seller).

- **Notification System:**

Sending notifications to sellers regarding new orders, comments, and bazaar reservation status (seller).

Sending notifications to buyers regarding order status and purchase updates (buyer).

Sending notifications to bazaar organizers regarding new reservation requests or reservation cancellations (bazaar organizer).

2.3 User Classes and Characteristics

2.3.1 Seller

Responsibilities:

The seller is responsible for managing their electronic shop and supervising all related operations, including creating and updating shop information and managing products through adding, editing, and deleting them. The seller also classifies products into suitable categories to facilitate their display to buyers and processes purchase requests by reviewing, approving, or rejecting them.

In addition, the seller updates order statuses through their different stages and responds to messages exchanged with buyers as well as buyer comments in order to maintain effective communication. The seller can also browse available bazaars, send table reservation requests to participate in bazaars, track reservation status or cancel reservations before the bazaar start date, in addition to monitoring notifications related to orders, comments, and reservations.

Privileges:

The seller has full control over their shop, including managing the shop profile, managing products and their categories, tracking orders, and viewing their status. The seller also has the right to create promotional advertisements and temporary offers, access shop performance reports, and use the direct messaging system to communicate with buyers within the system.

The seller can also browse available bazaars and submit reservation requests to participate in them, while receiving notifications related to order status, reservations, and comments.

2.3.2 Buyer

Responsibilities:

The buyer browses products and shops through the platform, adds or removes products from the shopping basket, and confirms orders after reviewing them. The buyer also selects the communication method and delivery method, and tracks the order status until receiving it. After completing the purchase process, the buyer expresses their opinion by writing comments on products, in addition to following notifications related to order status and updates.

Privileges:

The buyer has the ability to browse products using the search system or the filtering and sorting system based on price or category. The buyer can also track the status of their orders at any time, save products to a wishlist, and browse advertisements and temporary offers.

In addition, the buyer can benefit from the Smart Recommendation feature (AI Feature), which recommends suitable products according to a specific budget, in addition to communicating with sellers through the internal messaging system and receiving notifications related to orders and purchase status.

2.3.3 Admin

Responsibilities:

The administrator is responsible for the overall supervision of the system through shop management.

Privileges:

The administrator has the authority to create or delete shops and supervise overall platform activity to ensure that users comply with system policies and maintain the quality of displayed content.

2.3.4 Bazaar Organizer

Responsibilities:

The bazaar organizer is responsible for managing bazaars and supervising all related operations, including creating bazaars and updating their information such as location, bazaar image, table reservation price, and start and end dates and times. The organizer also reviews table reservation requests submitted by sellers and approves or rejects them, in addition to monitoring reservations, managing bazaar participants, and following notifications related to new reservations or canceled reservations.

Privileges:

The bazaar organizer has full control over their bazaars, including adding, editing, or deleting bazaars, and managing sellers' table reservation requests. The organizer can also track reservation status and participants in each bazaar, and receive notifications related to new reservation requests or reservation cancellations within the system.

3. System Features

3.1 Functional Requirements

3.1.1 Account Management

- **Requirement 1.1:**

The system shall allow the user to create an account.

- **Requirement 1.2:**

The system shall allow the user to log in using their username and password.

- **Requirement 1.3:**

The system shall allow the user to log out.

3.1.2 Shop Management

- **Requirement 2.1:**

The system shall allow the seller to add their shop information.

- **Requirement 2.2:**

The system shall allow the seller to edit their shop information.

- **Requirement 2.3:**

The system shall allow the seller to delete their shop.

- **Requirement 2.4:**

The system shall allow the administrator to create a new shop for any seller when needed.

- **Requirement 2.5:**

The system shall allow the administrator to delete a shop from the system.

- **Requirement 2.6:**

The system shall allow the administrator to view registered shops.

- **Requirement 2.7:**

The system shall display all products belonging to a specific shop on a separate shop page.

3.1.3 Product Management

- **Requirement 3.1:**
The system shall allow the seller to add new products.
- **Requirement 3.2:**
The system shall allow the seller to edit product information.
- **Requirement 3.3:**
The system shall allow the seller to delete products.
- **Requirement 3.4:**
The system shall allow the seller to display their products on their personal shop page.
- **Requirement 3.5:**
The system shall allow the seller to classify products into specific categories.
- **Requirement 3.6:**
The system shall allow the buyer to view products.
- **Requirement 3.7:**
The system shall allow the buyer to view product details (name, price, image, category).

3.1.4 Basket Management

- **Requirement 4.1:**
The system shall allow the buyer to add products to the basket.
- **Requirement 4.2:**
The system shall allow the buyer to remove products from the basket.
- **Requirement 4.3:**
The system shall allow the buyer to view the basket contents before confirming the purchase request.

3.1.5 Orders Management

- **Requirement 5.1:**
The system shall allow the seller to view all purchase requests received by their shop.
- **Requirement 5.2:**
The system shall allow the seller to approve purchase requests.
- **Requirement 5.3:**
The system shall allow the seller to reject purchase requests.
- **Requirement 5.4:**
The system shall allow the buyer to send purchase requests for products available in the basket.
- **Requirement 5.5:**
The system shall allow the buyer to choose the communication method with the seller (phone / email).
- **Requirement 5.6:**
The system shall allow the buyer to choose the delivery method (local delivery / hand delivery / shipping).
- **Requirement 5.7:**
The system shall allow the buyer to view the order status (Pending Review – Approved – Rejected).
- **Requirement 5.8:**
The system shall allow the seller to update the order status according to the implementation stages (In Preparation – Prepared – Out for Delivery – Delivered).
- **Requirement 5.9:**
The system shall allow the buyer to view their order status at any time.
- **Requirement 5.10:**
The system shall display the buyer's previous order history.

3.1.6 Comments Management

- **Requirement 6.1:**

The system shall allow the buyer to add a comment about a product after the purchase process.

- **Requirement 6.2:**

The system shall allow both the seller and the buyer to view product comments.

- **Requirement 6.3:**

The system shall allow the seller to reply to comments related to their products.

3.1.7 Search and Browsing

- **Requirement 7.1:**

The system shall allow the buyer to search for products using product categories.

- **Requirement 7.2:**

The system shall allow filtering results by price from lowest to highest and vice versa.

3.1.8 Advertisements Management

- **Requirement 8.1:**

The system shall allow the seller to create a promotional advertisement for a product or their shop.

- **Requirement 8.2:**

The system shall allow the seller to edit advertisement details.

- **Requirement 8.3:**

The system shall allow the seller to delete unwanted advertisements.

- **Requirement 8.4:**

The system shall display all active advertisements in a dedicated section for buyers.

3.1.9 Flash Offers Management

- **Requirement 9.1:**

The system shall allow the seller to create a temporary offer for a specific product.

- **Requirement 9.2:**

The system shall allow the seller to edit offer details before or during its active period.

- **Requirement 9.3:**

The system shall allow the seller to delete a temporary offer before or after its expiration.

- **Requirement 9.4:**

The system shall allow the buyer to view all temporary offers within a dedicated interface.

- **Requirement 9.5:**

The system shall automatically deactivate temporary offers after the specified period ends.

3.1.10 Wishlist Management

- **Requirement 10.1:**

The system shall allow the buyer to add products to the wishlist.

- **Requirement 10.2:**

The system shall allow the buyer to remove products from the wishlist.

- **Requirement 10.3:**

The system shall allow the buyer to view all products added to the wishlist.

- **Requirement 10.4:**

The system shall allow the buyer to navigate directly to the product page from the wishlist.

- **Requirement 10.5:**

The system shall permanently save each user's wishlist to display it upon future logins.

3.1.11 Reports System

- **Requirement 11.1:**

The system shall display the number of received orders.

- **Requirement 11.2:**

The system shall display the number of products available in the **shop**.

- **Requirement 11.3:**

The system shall display the number of visitors to the seller's shop.

- **Requirement 11.4:**

The system shall allow the seller to view general reports about shop activity on the platform (number of visitors – number of products – number of orders).

3.1.12 Direct Messaging System

- **Requirement 12.1:**

The system shall allow the creation of a direct conversation between the buyer and the seller regarding a specific product or order.

- **Requirement 12.2:**

The system shall allow sending and receiving text messages within the platform in a secure and organized manner.

- **Requirement 12.3:**

The system shall allow users to view previous conversation history.

- **Requirement 12.4:**

The system shall support securely attaching images or links within messages.

- **Requirement 12.5:**

The system shall display notifications when a new message is received by the user.

3.1.13 AI Recommendation Feature

- **Requirement 13.1:**

The system shall allow the buyer to enter criteria such as budget, gender, age, relationship, and occasion type (birthday, graduation party, visiting a patient, Mother's Day, etc.).

- **Requirement 13.2:**

The system shall analyze the input and recommend suitable products to the buyer as gifts.

- **Requirement 13.3:**

The system shall rely on the use of the [Groq API](#) while considering the entered criteria.

- **Requirement 13.4:**

The system shall allow displaying the list of recommended products interactively.

- **Requirement 13.5:**

The system shall allow the buyer to access the product details page by clicking on the product.

3.1.14 Bazaar Management

- **Requirement 14.1:**

The system shall allow the bazaar organizer to add information for a new bazaar.

- **Requirement 14.2:**

The system shall allow the bazaar organizer to edit bazaar information.

- **Requirement 14.3:**

The system shall allow the bazaar organizer to delete an existing bazaar.

- **Requirement 14.4:**

The system shall allow the seller to browse all available bazaars.

- **Requirement 14.5:**

The system shall allow the seller to view bazaar details.

3.1.15 Bazaar Table Reservation Management

- **Requirement 15.1:**

The system shall allow the seller to request the reservation of one table within the bazaar.

- **Requirement 15.2:**

The system shall allow the seller to cancel the reservation request before the bazaar start date.

- **Requirement 15.3:**

The system shall allow the bazaar organizer to view incoming reservation requests.

- **Requirement 15.4:**

The system shall allow the bazaar organizer to approve reservation requests.

- **Requirement 15.5:**

The system shall allow the bazaar organizer to reject reservation requests.

3.1.16 Notification System

- **Requirement 16.1:**

The system shall allow the buyer to view a notification when a basket purchase request is approved by the seller.

- **Requirement 16.2:**

The system shall allow the buyer to view a notification when a basket purchase request is rejected by the seller.

- **Requirement 16.3:**

The system shall allow the buyer to view a notification when the order status is updated, such as In Preparation / Prepared / Out for Delivery / Delivered.

- **Requirement 16.4:**

The system shall allow the buyer to view a notification when a reply is added to their comment.

- **Requirement 16.5:**

The system shall allow the seller to view a notification when a new purchase request is received from a buyer.

- **Requirement 16.6:**

The system shall allow the seller to view a notification when a new comment is added to one of their products.

- **Requirement 16.7:**

The system shall allow the seller to view a notification when a bazaar table reservation request is approved.

- **Requirement 16.8:**

The system shall allow the seller to view a notification when a bazaar table reservation request is rejected.

- **Requirement 16.9:**

The system shall allow the bazaar organizer to view a notification when a new reservation request is received from a seller.

- **Requirement 16.10:**

The system shall allow the bazaar organizer to view a notification when a reservation is canceled by the seller before the bazaar date.

3.2 Non-Functional Requirements

- **Scalability:** The system shall be scalable to support an increasing number of users without performance degradation.
- **Availability:** The system shall maintain operational readiness to ensure continuous availability of shop management features.
- **Security:** Implementing basic authentication and encrypted password storage.
- **Usability:** The system shall support easy switching of the user interface language between Arabic and English.
- **Maintainability:** The code shall be modular and well documented and Future updates shall be supported.

4. System Requirements

Req-ID	Requirement Title	Category	Priority
REQ-1.1	The system must allow the user to create an account.	Account Management	1
REQ-1.2	The system must allow the user to log in using a username and password.	Account Management	1
REQ-1.3	The system must allow the user to log out of their account.	Account Management	1
REQ-2.1	The system must allow the seller to add their shop information.	Shop Management	1
REQ-2.2	The system must allow the seller to edit their shop information.	Shop Management	1
REQ-2.3	The system must allow the seller to delete their shop.	Shop Management	2
REQ-2.4	The system must allow the administrator to create a shop for any seller when needed.	Shop Management	2
REQ-2.5	The system must allow the administrator to delete a shop.	Shop Management	2
REQ-2.6	The system must allow the administrator to View registered shops.	Shop Management	2
REQ-2.7	The system allows displaying all products belonging to a specific shop on a dedicated shop page.	Shop Management	1
REQ-3.1	The system must allow the seller to add new products.	Product Management	1
REQ-3.2	The system must allow the seller to edit product information.	Product Management	1

REQ-3.3	The system must allow the seller to delete products.	Product Management	1
REQ-3.4	The system must display the seller's products on their personal shop page.	Product Management	1
REQ-3.5	The system must allow the seller to classify products into specific categories.	Product Management	1
REQ-3.6	The system must allow the buyer to view products.	Product Management	1
REQ-3.7	The system must allow the buyer to view product details.	Product Management	1
REQ-4.1	The system must allow the buyer to add products to the basket.	Basket Management	1
REQ-4.2	The system must allow the buyer to remove products from the basket.	Basket Management	1
REQ-4.3	The system must allow the buyer to view the contents of the basket before confirming the order.	Basket Management	1
REQ-5.1	The system must allow the seller to view incoming purchase orders to their shop.	Order & Purchase Management	1
REQ-5.2	The system must allow the seller to accept purchase orders.	Order & Purchase Management	1
REQ-5.3	The system must allow the seller to reject purchase orders.	Order & Purchase Management	1
REQ-5.4	The system must allow the buyer to place a purchase order for products in the cart.	Order & Purchase Management	1

REQ-5.5	The system must allow the buyer to choose a communication method.	Order & Purchase Management	2
REQ-5.6	The system must allow the buyer to choose a delivery method.	Order & Purchase Management	2
REQ-5.7	The system must allow the buyer to view their order status (Under Review / Accepted / Rejected).	Order & Purchase Management	1
REQ-5.8	The system must allow the seller to update the order status according to execution stages.	Order & Purchase Management	1
REQ-5.9	The system must allow the buyer to view their order status at any time (Being Prepared / Prepared / Out for Delivery / Delivered).	Order & Purchase Management	1
REQ-5.10	The system must allow the buyer to view their previous order history.	Order & Purchase Management	2
REQ-6.1	The system must allow the buyer to add a comment on a product after a purchase.	Comment Management	2
REQ-6.2	The system must allow the seller and buyer to view comments on products.	Comment Management	2
REQ-6.3	The system must allow the seller to reply to comments on their products.	Comment Management	2
REQ-7.1	The system must allow the buyer to search for products by category.	Search & View	1

REQ-7.2	The system must allow the buyer to filter product results by price from lowest to highest and vice versa.	Search & View	1
REQ-8.1	The system must allow the seller to create a promotional advertisement about their products or shop.	Advertisement Management	1
REQ-8.2	The system must allow the seller to edit advertisement details.	Advertisement Management	1
REQ-8.3	The system must allow the seller to delete unwanted advertisements.	Advertisement Management	1
REQ-8.4	The system must allow the buyer to view all active advertisements.	Advertisement Management	1
REQ-9.1	The system must allow the seller to create a Flash offer on a specific product.	Flash Offers Management	1
REQ-9.2	The system must allow the seller to edit Flash offer details.	Flash Offers Management	1
REQ-9.3	The system must allow the seller to delete a Flash offer.	Flash Offers Management	1
REQ-9.4	The system must allow the buyer to view all Flash offers.	Flash Offers Management	1
REQ-9.5	The system must automatically deactivate Flash offers after the specified period has ended.	Flash Offers Management	1
REQ-10.1	The system must allow the buyer to add products to the favorites list.	WishList Management	2
REQ-10.2	The system must allow the buyer to remove products from the favorites list.	WishList Management	2

REQ-10.3	The system must allow the buyer to view all products added to the favorites list.	WishList Management	2
REQ-10.4	The system must allow the buyer to navigate directly to the product page from the favorites list.	WishList Management	1
REQ-10.5	The system must permanently save the Wishlist for each user to display it upon subsequent login.	WishList Management	1
REQ-11.1	The system must display the number of received orders for the seller.	Reporting System	2
REQ-11.2	The system must display the number of products listed in the shop for the seller.	Reporting System	2
REQ-11.3	The system must display the number of shop visitors for the seller.	Reporting System	1
REQ-11.4	The system must display general reports about shop activity for the seller.	Reporting System	1
REQ-12.1	The system must allow the creation of a direct conversation between the buyer and seller.	Direct Messaging System	1
REQ-12.2	The system must allow the seller and buyer to send and receive text messages.	Direct Messaging System	1
REQ-12.3	The system must allow the seller and buyer to view previous conversation history.	Direct Messaging System	2
REQ-12.4	The system must allow the seller and buyer to attach images or links within messages.	Direct Messaging System	1
REQ-12.5	The system must display notifications for the seller and buyer when a new message arrives.	Direct Messaging System	1

REQ-13.1	The system must allow the buyer to enter criteria such as budget or occasion type.	Smart Suggestions Feature	1
REQ-13.2	The system must analyze the inputs and suggest suitable products for the buyer as gifts.	Smart Suggestions Feature	1
REQ-13.3	The system must make suggestions using Groq API taking into account the specified criteria.	Smart Suggestions Feature	1
REQ-13.4	The system must allow the buyer to view the list of suggested products.	Smart Suggestions Feature	1
REQ-13.5	The system must allow the buyer to access the product details page by clicking on it.	Smart Suggestions Feature	1
REQ-14.1	The system must allow the bazaar organizer to create a new bazaar.	Bazaar Management	1
REQ-14.2	The system must allow the bazaar organizer to edit bazaar data.	Bazaar Management	1
REQ-14.3	The system must allow the bazaar organizer to delete an existing bazaar.	Bazaar Management	1
REQ-14.4	The system must allow the seller to view all available bazaars.	Bazaar Management	1
REQ-14.5	The system must allow the seller to view bazaar details.	Bazaar Management	1
REQ-15.1	The system must allow the seller to request a table reservation within the bazaar.	Bazaar Table Reservation Management	1

REQ-15.2	The system must allow the seller to cancel a reservation request before the bazaar start date.	Bazaar Table Reservation Management	1
REQ-15.3	The system must allow the bazaar organizer to view incoming reservation requests.	Bazaar Table Reservation Management	1
REQ-15.4	The system must allow the bazaar organizer to accept reservation requests.	Bazaar Table Reservation Management	1
REQ-15.5	The system must allow the bazaar organizer to reject reservation requests.	Bazaar Table Reservation Management	1
REQ-16.1	The system must allow the buyer to view a notification when a purchase order is accepted.	Notification System	1
REQ-16.2	The system must allow the buyer to view a notification when a purchase order is rejected.	Notification System	1
REQ-16.3	The system must allow the buyer to view a notification when the order status is updated.	Notification System	1
REQ-16.4	The system must allow the buyer to view a notification when their comment is replied to.	Notification System	2
REQ-16.5	The system must allow the seller to view a notification when a new order arrives.	Notification System	1
REQ-16.6	The system must allow the seller to view a notification when a new comment is added to a product.	Notification System	2

REQ-16.7	The system must allow the seller to view a notification when a bazaar reservation is accepted.	Notification System	1
REQ-16.8	The system must allow the seller to view a notification when a bazaar reservation is rejected.	Notification System	1
REQ-16.9	The system must allow the bazaar organizer to view a notification when a new reservation request arrives.	Notification System	1
REQ-16.10	The system must allow the bazaar organizer to view a notification when a seller cancels their reservation.	Notification System	1
NFR-1	The system must be scalable to support an increasing number of users without performance degradation.	Scalability	1
NFR-2	The system must maintain operational readiness to ensure continuous availability of features.	Availability	1
NFR-3	Basic authentication must be implemented and passwords must be stored in encrypted form.	Security	1
NFR-4	The system must support switching the user interface language between Arabic and English easily.	Usability	2
NFR-5	The code must be standardized and well-documented for ease of maintenance.	Maintainability	1

Table 3: System Requirements

5. Requirements Modeling

5.1 Use Case Diagram

Use case diagrams represent the system behavior and help capture the system requirements.

High level-use case diagram

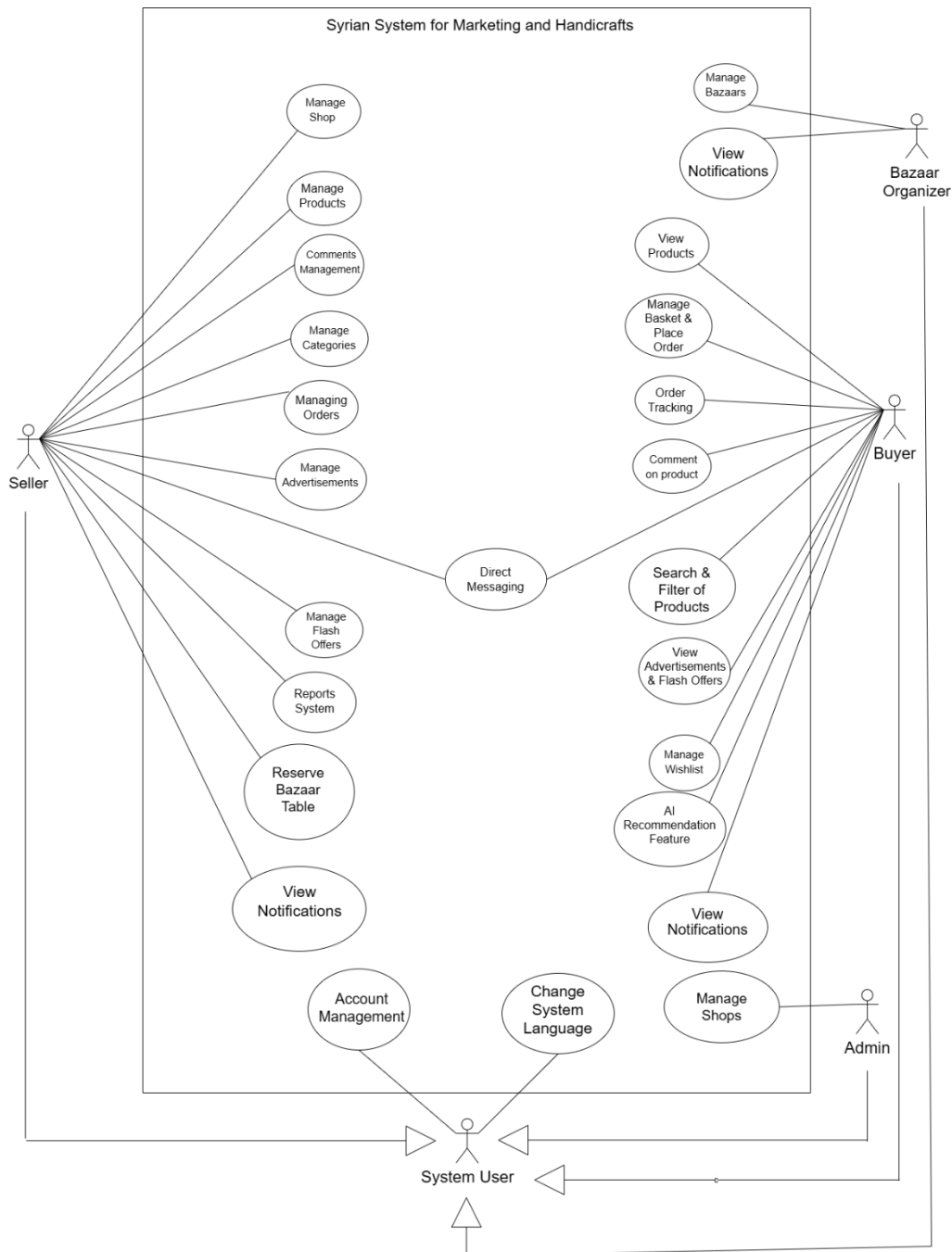


Figure 3: High level-use case diagram

Low level-use case diagram

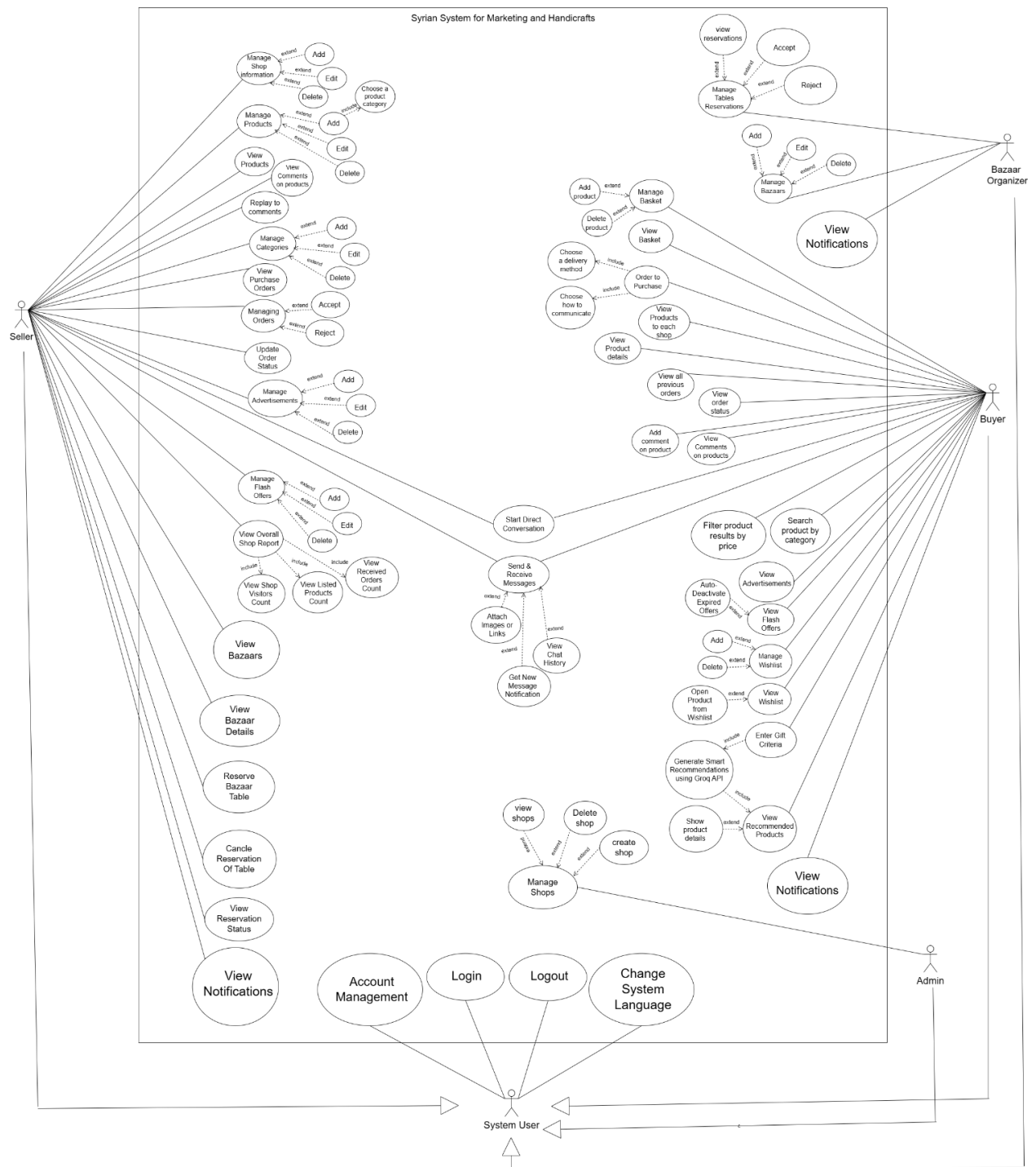


Figure 4: Low level-use case diagram

5.2 System Features – Use Case Specifications and Activity Diagrams

(System features – use case specifications – Activity diagrams):

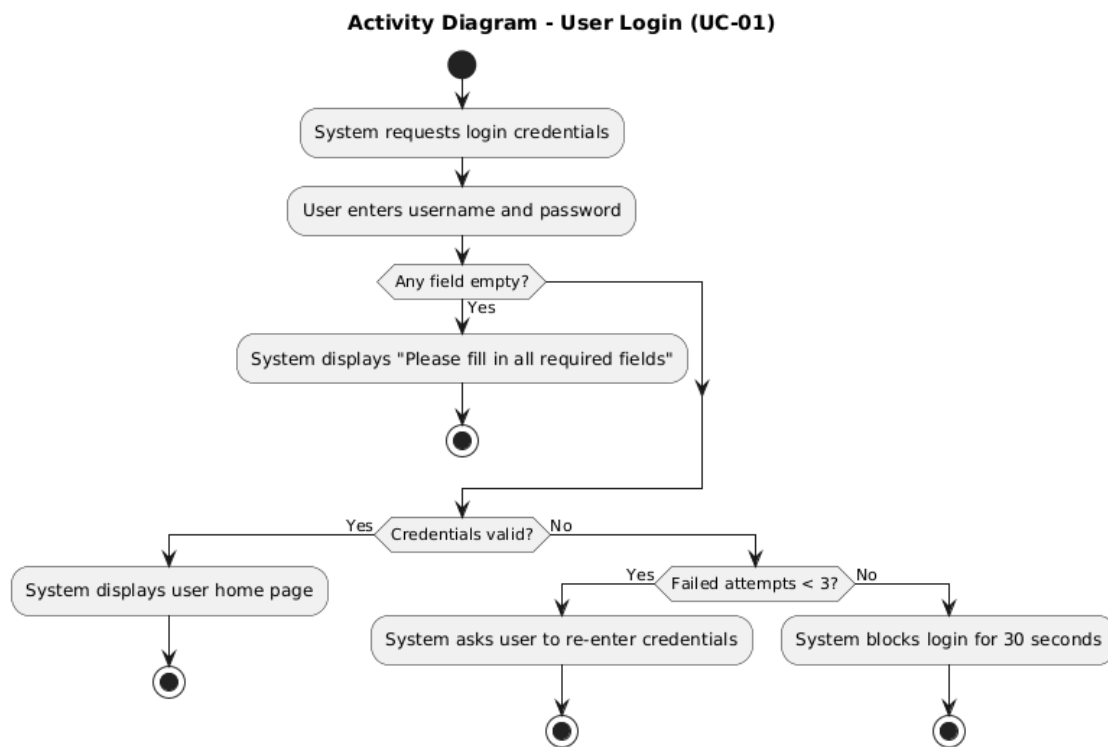
- Login (UC-01):

Table 4: Use Case Specification – Login

Use Case ID	UC-01
Use Case Name	Login
Actors	Users
Description	Log in to the user's account
Precondition	The user must have an existing account.
Main flow	1- The system requests login information. 2- The user enters their data (username – password) into the system. 3- The system verifies the accuracy of the entered data. 4- The system displays the main page for the user's account.
Alternative flow	A1: If the user's data is incorrect for the first or second attempt, the flow resumes after step 2 of the main flow. A2: If the user's data is incorrect for the third attempt, the system asks the user to try again after 30 seconds, and the use case ends. A3: If a field is left empty, the system requests that all fields be completed.
Postconditions	The user has successfully logged in to their account.

Login – Activity Diagram:

Figure 5: Login Activity Diagram



- Logout (UC-02):

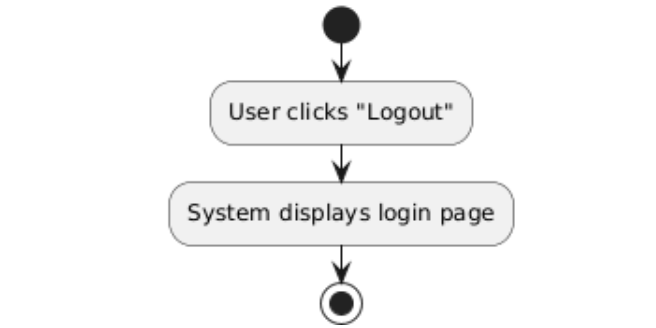
Table 5: Use Case Specification – Logout

Use Case ID	UC-02
Use Case Name	Logout
Actors	Users
Description	Log out of the user's account.
Precondition	The user must have previously logged in to their account.
Main flow	1- The user selects "Logout". 2- The system displays the login page.
Alternative flow	None
Postconditions	The user has successfully logged out of their account.

Logout – Activity Diagram:

Figure 6: Logout Activity Diagram

Activity Diagram - User Logout (UC-02)



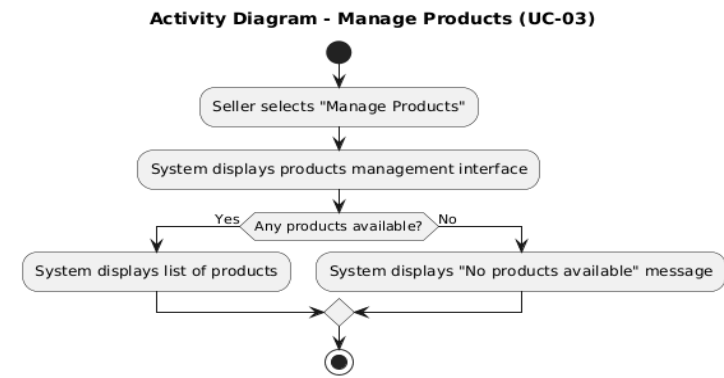
- Product Management (UC-03):

Table 6: Use Case Specification – Product Management

Use Case ID	UC-03
Use Case Name	Product Management
Actors	Seller
Description	The seller manages products.
Precondition	The seller must have previously logged in to their account.
Main flow	1- The seller selects "Product Management". 2- The system displays the products interface and management options (Add – Edit – Delete).
Alternative flow	A1: If the seller has no registered products, the system displays a message: "No products currently available."
Postconditions	The product management interface is displayed.

Product Management – Activity Diagram:

Figure 7: Product Management Activity Diagram



- Add Products (UC-04):

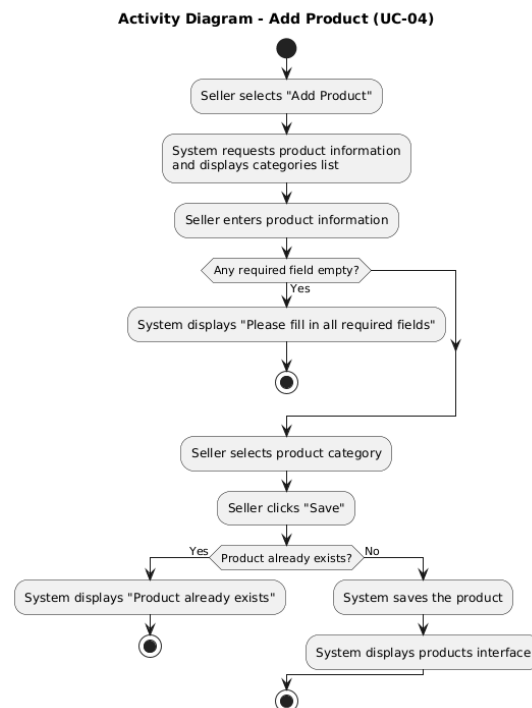
Table 7: Use Case Specification – Add Products

Use Case ID	UC-04
Use Case Name	Add Products
Actors	Seller
Description	Allows the seller to add a product to the system.
Precondition	The seller must have previously logged in to their account and selected the Product Management page.
Main flow	1- The seller selects "Add Product". 2- The system requests the required information for the product and displays a list of product categories. 3- The seller enters the product information. 4- The seller selects a category for the product. 5- The seller clicks "Save". 6- The system displays the products interface.

Alternative flow	A1: After step 5, if a previously existing product is added, the system displays a message that the product already exists, and the use case fails. A2: After step 3, if the seller does not enter mandatory fields, the system displays a message: "Please enter all required information."
Postconditions	The product has been successfully added.

Add Products – Activity Diagram:

Figure 8: Add Products Activity Diagram



- Edit Products (UC-05):

Table 8: Use Case Specification – Edit Products

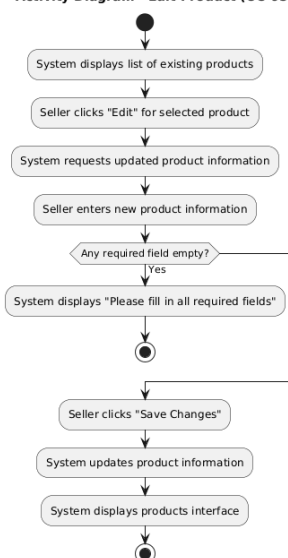
Use Case ID	UC-05
Use Case Name	Edit Products
Actors	Seller

Description	Allows the seller to edit information for existing products in the system.
Precondition	The seller must have previously logged in to their account and selected the Product Management page.
Main flow	1- The system displays the list of existing products. 2- The seller clicks "Edit" for the product to be edited. 3- The system requests the required information for editing. 4- The seller enters the new product information. 5- The seller clicks "Save Changes". 6- The system displays the products interface.
Alternative flow	A1: After step 4, if the seller does not enter mandatory fields, the system displays a message: "Please enter all required information."
Postconditions	The product data has been successfully edited.

Edit Products – Activity Diagram:

Figure 9: Edit Products Activity Diagram

Activity Diagram - Edit Product (UC-05)



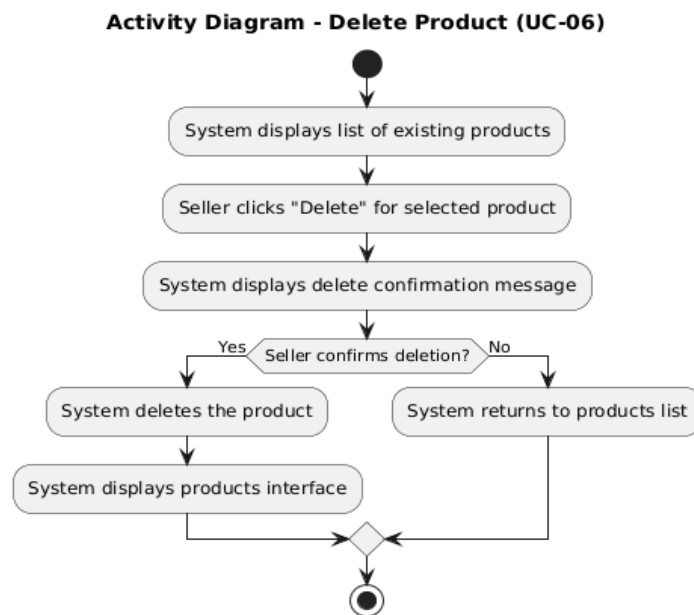
- Delete Product (UC-06):

Table 9: Use Case Specification – Delete Product

Use Case ID	UC-6
Use Case Name	Delete Product
Actors	Seller
Description	Allows the seller to delete existing products from the system.
Precondition	The seller must have previously logged in to their account and selected the Product Management page.
Main flow	1- The system displays the list of existing products. 2- The seller clicks "Delete" for the product to be deleted. 3- The system displays a deletion confirmation message. 4- The seller confirms the deletion by clicking "Yes". 5- The system displays the products interface.
Alternative flow	A1: Returning to step 1 of the main flow, the seller selects "No".
Postconditions	The product has been successfully deleted.

Delete Product – Activity Diagram:

Figure 10: Delete Product Activity Diagram



- Add Products to Basket (UC-07):

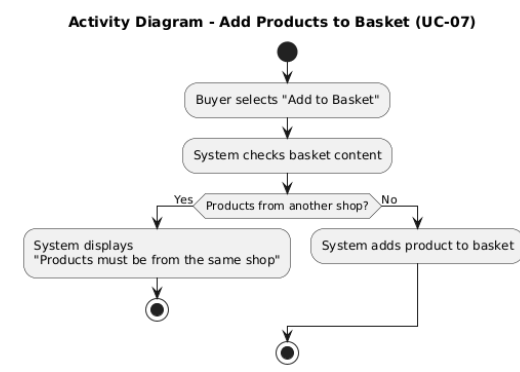
Table 10: Use Case Specification – Add Products to Basket

Use Case ID	UC-07
Use Case Name	Add Products to Basket
Actors	Buyer
Description	Allows the buyer to add products to the basket.
Precondition	The buyer must have previously logged in to their account and be browsing products.
Main flow	1- The buyer selects "Add Product to Basket". 2- The system saves the products in the basket.
Alternative flow	A1: After step 1, if the buyer attempts to add products from a different shop to the basket, the system displays a

	message: "Products in a single cart must be purchased from one shop."
Postconditions	The products have been successfully added to the basket.

Add Products to Basket – Activity Diagram:

Figure 11: Add Products to Basket Activity Diagram



- Delete Products from Basket (UC-08):

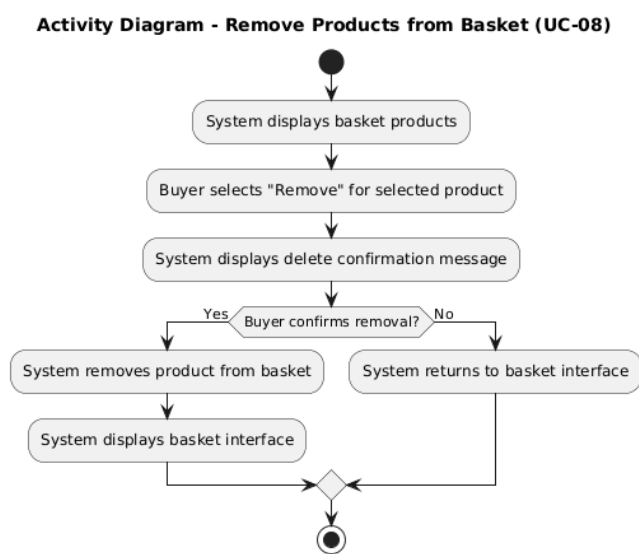
Table 11: Use Case Specification – Delete Products from Basket

Use Case ID	UC-08
Use Case Name	Delete Products from Basket
Actors	Buyer
Description	Allows the buyer to delete products from the basket.
Precondition	The buyer must have previously logged in to their account and selected basket management.
Main flow	1- The system displays the list of products in the basket. 2- The buyer clicks "Delete" for the product to be deleted. 3- The system displays a deletion confirmation message. 4- The buyer confirms the deletion by clicking "Yes".

	5- The system displays the basket management interface.
Alternative flow	A1: Returning to step 1 of the main flow, the buyer selects "No".
Postconditions	The product has been successfully deleted from the basket.

Delete Products from Basket – Activity Diagram:

Figure 12: Delete Products from Basket Activity Diagram



- Order Management (Accept Purchase Orders) (UC-09):

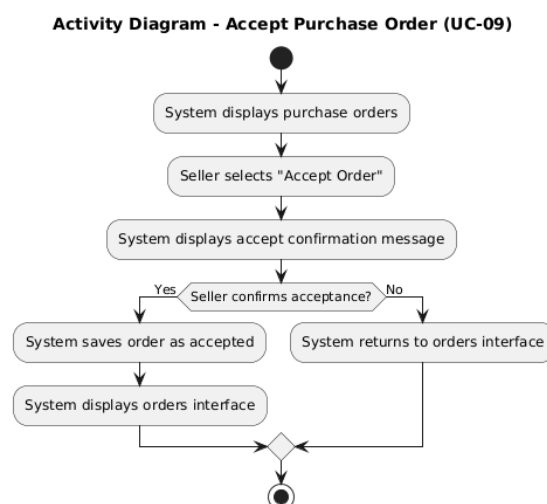
Table 12: Use Case Specification – Order Management (Accept Purchase Orders)

Use Case ID	UC-9
Use Case Name	Accept Purchase Orders
Actors	Seller
Description	Allows the seller to accept a purchase order.
Precondition	The seller must have previously logged in to their account and selected the Purchase Order Management page.
Main flow	1- The system displays the purchase orders.

	2- The seller selects "Accept Order" for any order. 3- The system displays an order acceptance confirmation message. 4- The seller confirms acceptance by clicking "Yes". 5- The system displays the orders interface.
Alternative flow	A1: Returning to step 4 of the main flow, the seller does not confirm acceptance and clicks "Back".
Postconditions	The purchase order is saved within the accepted orders.

Accept Purchase Orders – Activity Diagram:

Figure 13: Accept Purchase Orders Activity Diagram



- Order Management (Reject Purchase Orders) (UC-10):

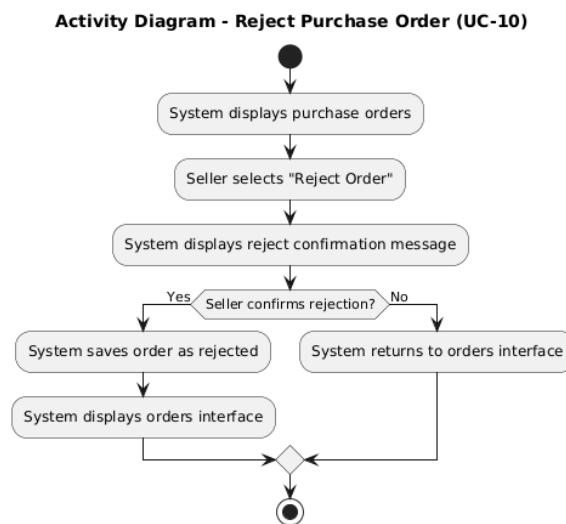
Table 13: Use Case Specification – Order Management (Reject Purchase Orders)

Use Case ID	UC-10
Use Case Name	Reject Purchase Orders
Actors	Seller
Description	Allows the seller to reject a purchase order.

Precondition	The seller must have previously logged in to their account and selected the Purchase Order Management page.
Main flow	1- The system displays the purchase orders. 2- The seller selects "Reject Order" for any order. 3- The system displays an order rejection confirmation message. 4- The seller confirms rejection by clicking "Yes". 5- The system displays the orders interface.
Alternative flow	A1: Returning to step 4 of the main flow, the seller does not confirm rejection and clicks "Back".
Postconditions	The purchase order is saved within the rejected orders.

Reject Purchase Orders – Activity Diagram:

Figure 14: Reject Purchase Orders Activity Diagram



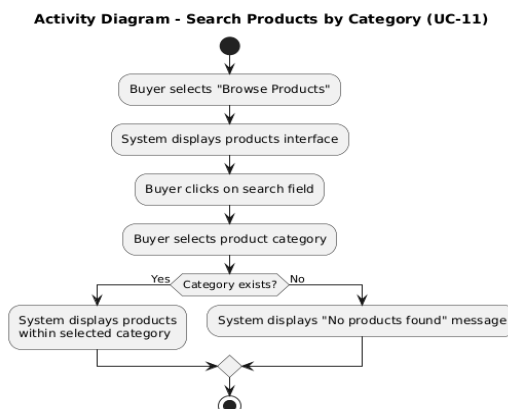
- Search for Products by Category (UC-11):

Table 14: Use Case Specification – Search for Products by Category

Use Case ID	UC-11
Use Case Name	Search for Products by Category
Actors	Buyer
Description	Allows the buyer to search for products by category.
Precondition	The buyer must have previously logged in to their account.
Main flow	1- The buyer clicks "View Products". 2- The system displays the products interface. 3- The buyer clicks on the search box. 4- The buyer selects the desired product category. 5- The system displays the products matching the selected category.
Alternative flow	A1: If the required product category does not exist in the system, the system displays a "Not Found" message.
Postconditions	Products within the selected category are displayed to the buyer after searching.

Search for Products by Category – Activity Diagram:

Figure 15: Search for Products by Category Activity Diagram



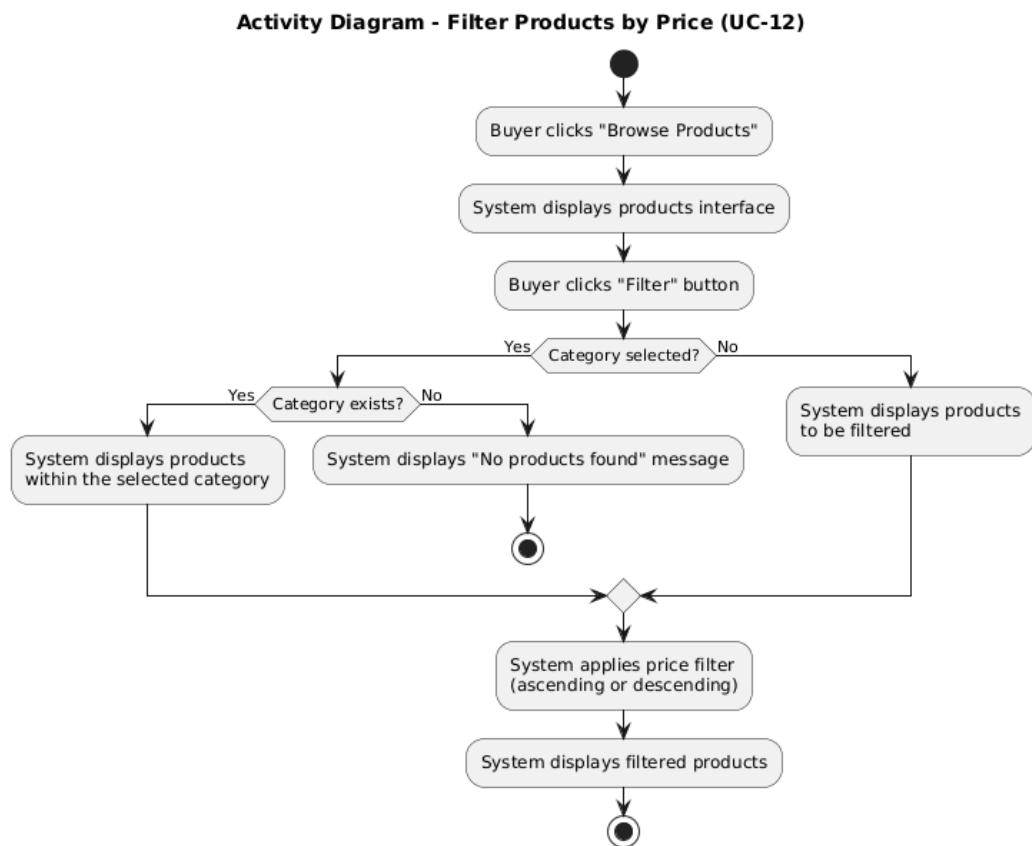
- Filter Products by Price (UC-12):

Table 15: Use Case Specification – Filter Products by Price

Use Case ID	UC-12
Use Case Name	Filter Products by Price.
Actors	Buyer
Description	Allows the buyer to filter products by price from lowest to highest, and vice versa.
Precondition	The buyer must have previously logged in to their account.
Main flow	1- The buyer clicks "View Products". 2- The system displays the products interface. 3- The buyer clicks the filter button. 4- The system displays the filtered products.
Alternative flow	A1: Returning to step 3 of the main flow, if the buyer selects a product category, the system displays the filtered products within a specific category. A2: If the product category to be filtered does not exist in the system, the system displays a "Not Found" message.
Postconditions	Products are displayed to the buyer according to the specified filter.

Filter Products by Price – Activity Diagram:

Figure 16: Filter Products by Price Activity Diagram



- Add Comments (UC-13):

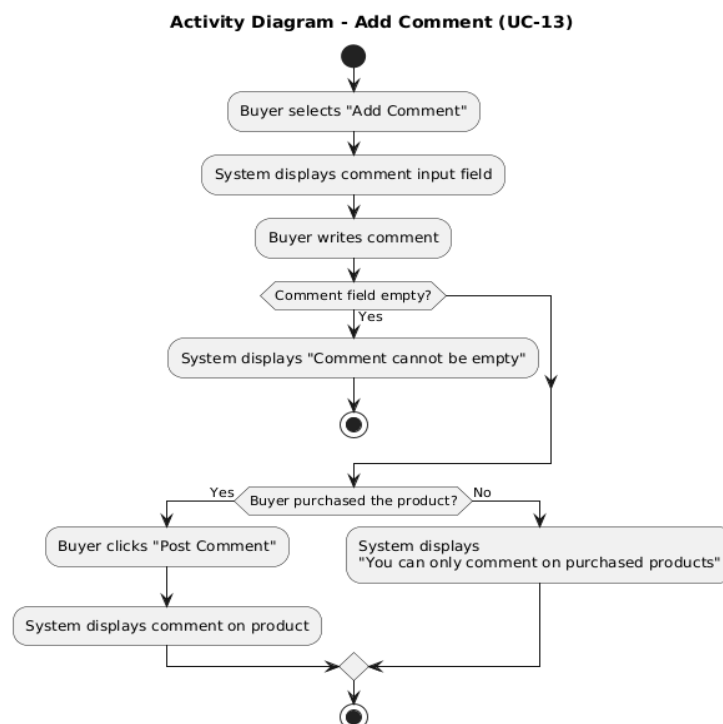
Table 16: Use Case Specification – Add Comments

Use Case ID	UC-13
Use Case Name	Add Comments
Actors	Buyer
Description	Allows the buyer to add comments on products after a purchase.
Precondition	The buyer must have previously logged in to their account, and the purchase order status must be "Delivered".
Main flow	1- The buyer selects "Add Comment".

	2- The system displays a text box for writing the comment. 3- The buyer clicks on the text box. 4- The buyer writes the comment and clicks "Post Comment". 5- The system displays the comment on the product.
Alternative flow	A1: If the user attempts to add a comment on a product they did not purchase, the system displays an alert: "You can only comment on products you have purchased." A2: If the comment field is left empty, the system displays a message: "An empty comment cannot be posted."
Postconditions	The comment has been successfully added.

Add Comments – Activity Diagram:

Figure 17: Add Comments Activity Diagram



- Reply to Comments (UC-14):

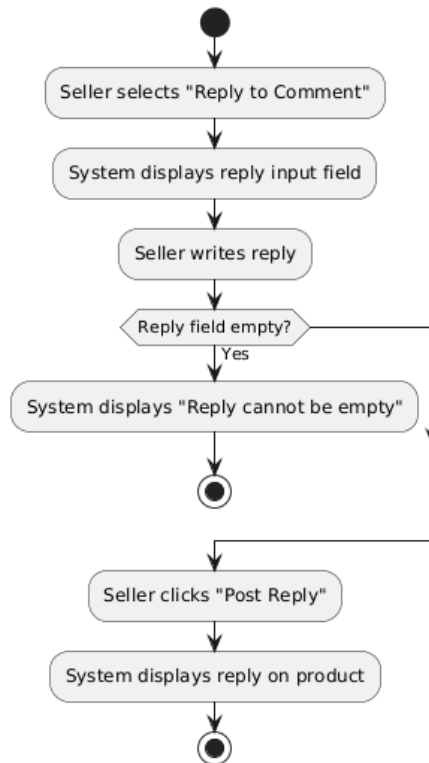
Table 17: Use Case Specification – Reply to Comments

Use Case ID	UC-14
Use Case Name	Reply to Comments
Actors	Seller
Description	Allows the seller to reply to buyer comments on products belonging to their shop.
Precondition	The seller must have previously logged in to their account.
Main flow	1- The seller selects "Reply to Comment". 2- The system displays a text box for writing the comment. 3- The seller clicks on the text box. 4- The seller writes the reply and clicks "Post Comment". 5- The system displays the comment on the product.
Alternative flow	A1: If the seller leaves the comment field empty, the system displays a message: "An empty comment cannot be posted."
Postconditions	The comment has been successfully added.

Reply to Comments – Activity Diagram:

Figure 18: Reply to Comments Activity Diagram

Activity Diagram - Reply to Comment (UC-14)



- Place a Purchase Order for Products (UC-15):

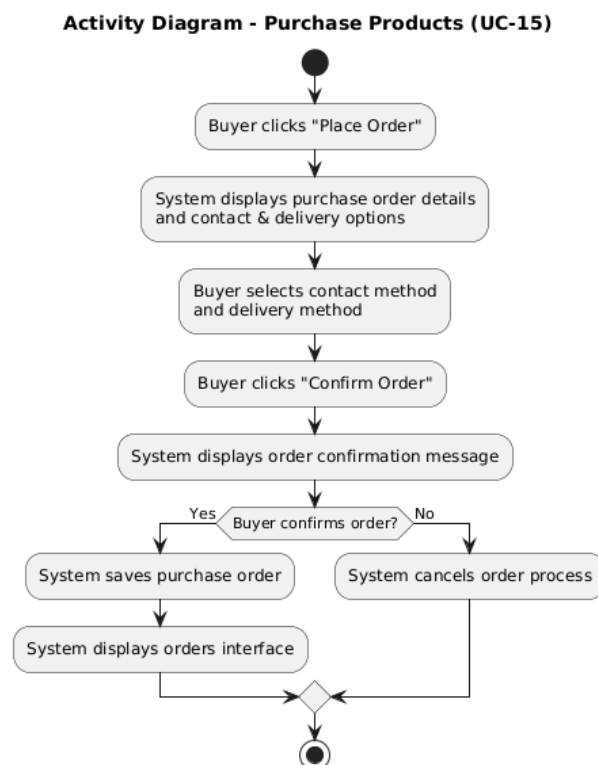
Table 18: Use Case Specification – Place a Purchase Order for Products

Use Case ID	UC-15
Use Case Name	Place a Purchase Order for Products
Actors	Buyer
Description	Allows the buyer to purchase products.
Precondition	The buyer must have previously logged in to their account and selected the shopping cart page.
Main flow	1- The buyer clicks "Place Order" for the shopping cart.

	2- The system displays the purchase order details page and shows a list of communication methods and delivery options. 3- The buyer selects a communication method and delivery option. 4- The buyer clicks "Confirm Order". 5- The system displays an order confirmation message. 6- The buyer confirms the order by clicking "Yes". 7- The system displays the orders interface.
Alternative flow	A1: Returning to step 1 of the main flow, the user does not confirm the order and clicks "No".
Postconditions	The order has been successfully saved within the orders.

Place a Purchase Order for Products – Activity Diagram:

Figure 19: Place a Purchase Order for Products Activity Diagram



- View Order Status (UC-16):

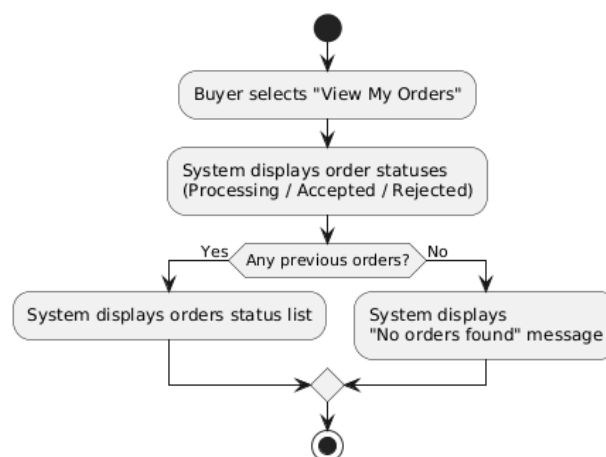
Table 19: Use Case Specification – View Order Status

Use Case ID	UC-16
Use Case Name	View Order Status
Actors	Buyer
Description	Allows the buyer to View the status of their orders.
Precondition	The buyer must have previously logged in to their account and selected the orders page.
Main flow	1- The buyer clicks "View My Orders". 2- The system displays the order status (Pending / Accepted / Rejected).
Alternative flow	A1: If the user has no previous orders, the system displays a message: "No orders yet."
Postconditions	The order status has been successfully displayed to the buyer.

View Order Status – Activity Diagram:

Figure 20: View Order Status Activity Diagram

Activity Diagram - View Order Status (UC-16)



- Manage Order Tracking After Acceptance (UC-17):

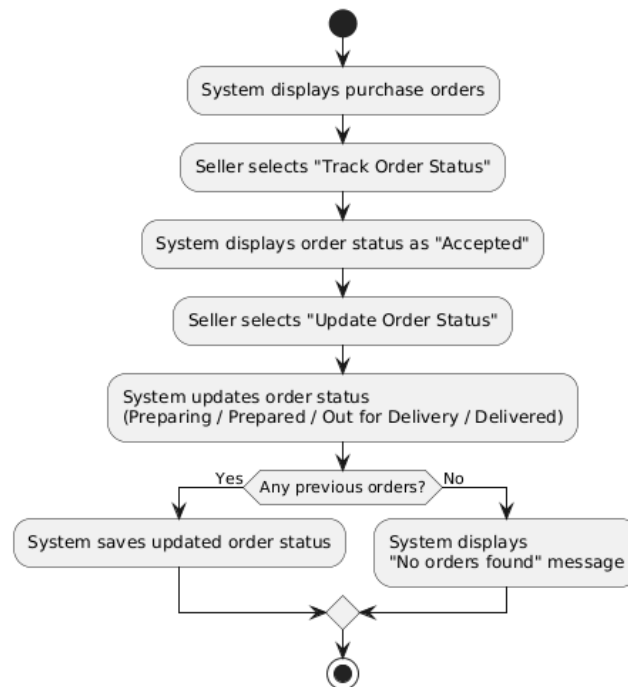
Table 20: Use Case Specification – Manage Order Tracking After Acceptance

Use Case ID	UC-17
Use Case Name	Manage Order Tracking After Acceptance
Actors	Seller
Description	Allows the seller to manage the tracking of their orders.
Precondition	The seller must have previously logged in to their account and selected the Purchase Order Management page.
Main flow	1- The system displays the purchase orders. 2- The seller selects "Track Order Status" for any order. 3- The system displays the order status as accepted. 4- The seller selects "Update Order Status". 5- The system displays the updated order status (Being Prepared / Prepared / Out for Delivery / Delivered).
Alternative flow	A1: If the user has no previous orders, the system displays a message: "No orders yet."
Postconditions	The order status has been successfully saved.

Manage Order Tracking After Acceptance – Activity Diagram:

Figure 21: Manage Order Tracking After Acceptance Activity Diagram

Activity Diagram - Manage Order Tracking After Acceptance (UC-17)



- Track Orders After Acceptance (UC-18):

Table 21: Use Case Specification – Track Orders After Acceptance

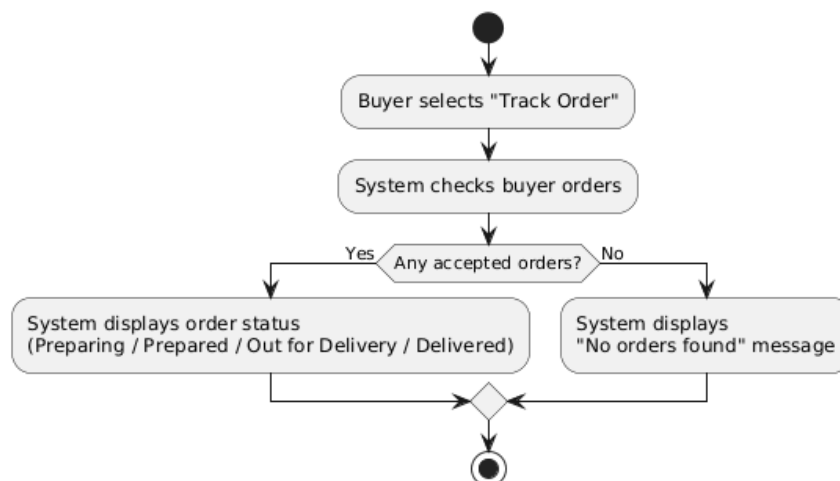
Use Case ID	UC-18
Use Case Name	Track Orders After Acceptance
Actors	Buyer
Description	Allows the buyer to View the status of their orders after acceptance.
Precondition	The buyer must have previously logged in to their account and selected the orders page.
Main flow	1- The buyer clicks "Track Order" for any of their orders.

	2- The system displays the order status (Being Prepared / Prepared / Out for Delivery / Delivered).
Alternative flow	A1: If the user has no previous orders, the system displays a message: "No orders yet."
Postconditions	The status of accepted orders has been successfully displayed to the buyer.

Track Orders After Acceptance – Activity Diagram:

Figure 22: Track Orders After Acceptance Activity Diagram

Activity Diagram - Track Orders After Acceptance (UC-18)



- Advertisement Management (UC-19):

Table 22: Use Case Specification – Advertisement Management

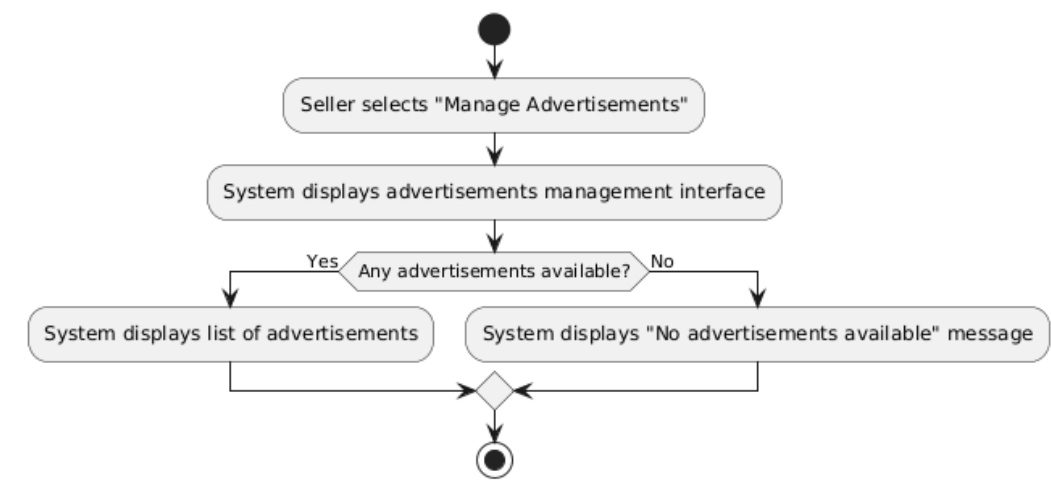
Use Case ID	UC-19
Use Case Name	Advertisement Management
Actors	Seller
Description	The seller manages advertisements.
Precondition	The seller must have previously logged in to their account.

Main flow	1- The seller selects "Advertisement Management". 2- The system displays the advertisements interface and management options (Add – Edit – Delete).
Alternative flow	A1: If the seller has no registered advertisements, the system displays a message: "No advertisements currently available."
Postconditions	The advertisement management interface is displayed.

Advertisement Management – Activity Diagram:

Figure 23: Advertisement Management Activity Diagram

Activity Diagram - Manage Advertisements (UC-19)



- Add Advertisements (UC-20):

Table 23: Use Case Specification – Add Advertisements

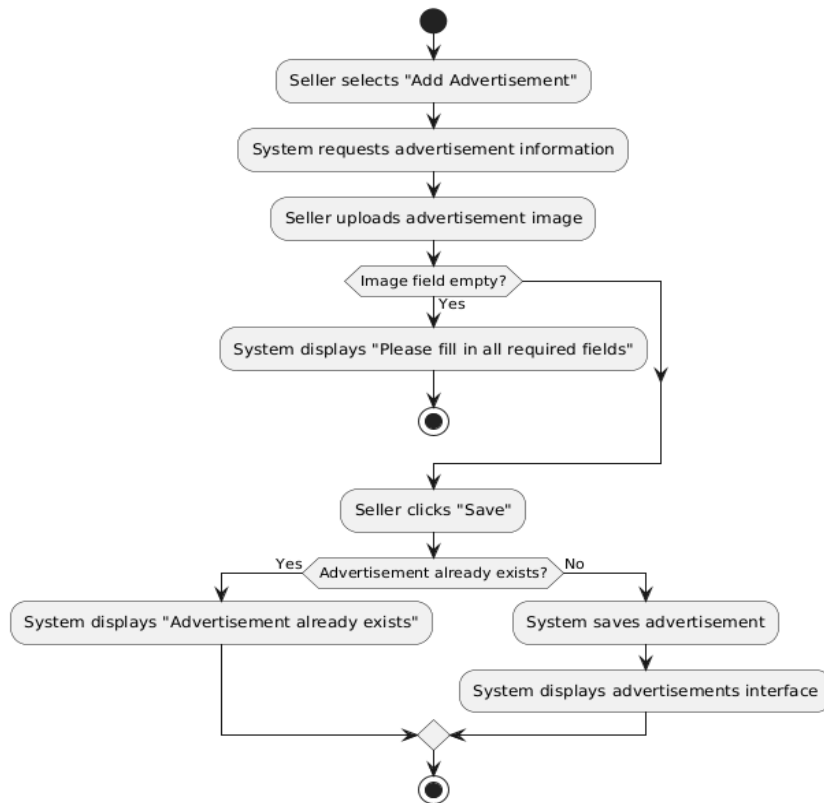
Use Case ID	UC-20
Use Case Name	Add Advertisements
Actors	Seller
Description	Allows the seller to add an advertisement to the system.

Precondition	The seller must have previously logged in to their account and selected the Advertisement Management page.
Main flow	1- The seller selects "Add Advertisement". 2- The system requests the required information for the advertisement. 3- The seller uploads the advertisement image. 4- The seller clicks "Save". 5- The system displays the products interface.
Alternative flow	A1: After step 4, if a previously existing advertisement is added, the system displays a message that the advertisement already exists, and the use case fails. A2: After step 2, if the seller does not enter the mandatory image field, the system displays a message: "Please enter all required information."
Postconditions	The advertisement has been successfully added.

Add Advertisements – Activity Diagram:

Figure 24: Add Advertisements Activity Diagram

Activity Diagram - Add Advertisement (UC-20)



- Edit Advertisements (UC-21):

Table 24: Use Case Specification – Edit Advertisements

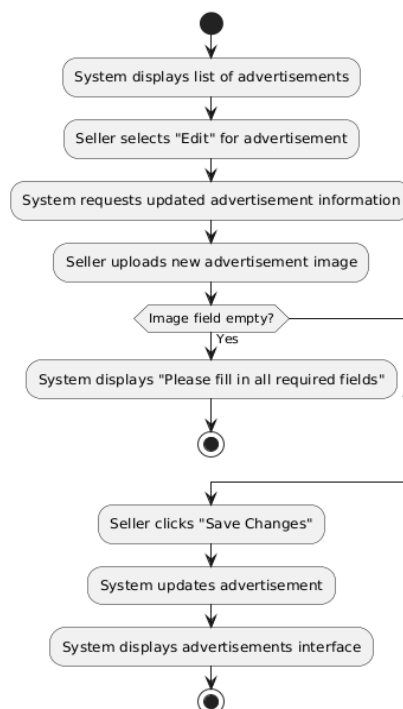
Use Case ID	UC-21
Use Case Name	Edit Advertisements
Actors	Seller
Description	Allows the seller to edit information for existing advertisements in the system.
Precondition	The seller must have previously logged in to their account and selected the Advertisement Management page.

Main flow	1- The system displays the list of existing advertisements. 2- The seller clicks "Edit" for the advertisement to be edited. 3- The system requests the required information for editing. 4- The seller uploads the new advertisement image. 5- The seller clicks "Save Changes". 6- The system displays the advertisements interface.
Alternative flow	A1: After step 4, if the seller does not enter the mandatory image field, the system displays a message: "Please enter all required information."
Postconditions	The advertisement has been successfully edited.

Edit Advertisements – Activity Diagram:

Figure 25: Edit Advertisements Activity Diagram

Activity Diagram - Edit Advertisement (UC-21)



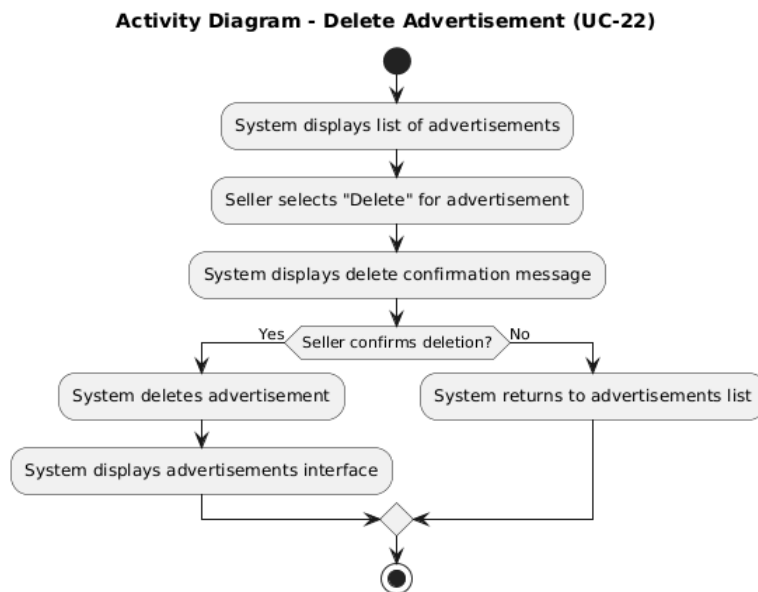
- Delete Advertisements (UC-22):

Table 25: Use Case Specification – Delete Advertisements

Use Case ID	UC-22
Use Case Name	Delete Advertisements
Actors	Seller
Description	Allows the seller to delete existing advertisements from the system.
Precondition	The seller must have previously logged in to their account and selected the Advertisement Management page.
Main flow	1- The system displays the list of existing advertisements. 2- The seller clicks "Delete" for the advertisement to be deleted. 3- The system displays a deletion confirmation message. 4- The seller confirms the deletion by clicking "Yes". 5- The system displays the advertisements interface.
Alternative flow	A1: Returning to step 1 of the main flow, the seller selects "No".
Postconditions	The advertisement has been successfully deleted.

Delete Advertisements – Activity Diagram:

Figure 26: Delete Advertisements Activity Diagram



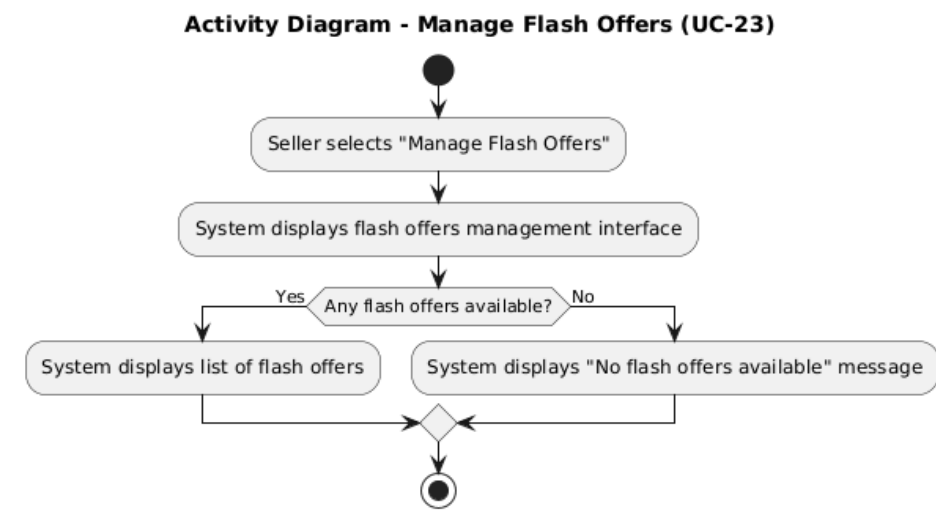
- Flash Offers Management (UC-23):

Table 26: Use Case Specification – Flash Offers Management

Use Case ID	UC-23
Use Case Name	Flash Offers Management.
Actors	Seller
Description	The seller manages Flash offers.
Precondition	The seller must have previously logged in to their account.
Main flow	1- The seller selects "Flash Offers Management". 2- The system displays the Flash offers interface and management options (Add – Edit – Delete).
Alternative flow	A1: If the seller has no registered Flash offers, the system displays a message: "No Flash offers currently available."
Postconditions	The Flash offers management interface is displayed.

Flash Offers Management – Activity Diagram:

Figure 27: Flash Offers Management Activity Diagram



- Add Flash Offers (UC-24):

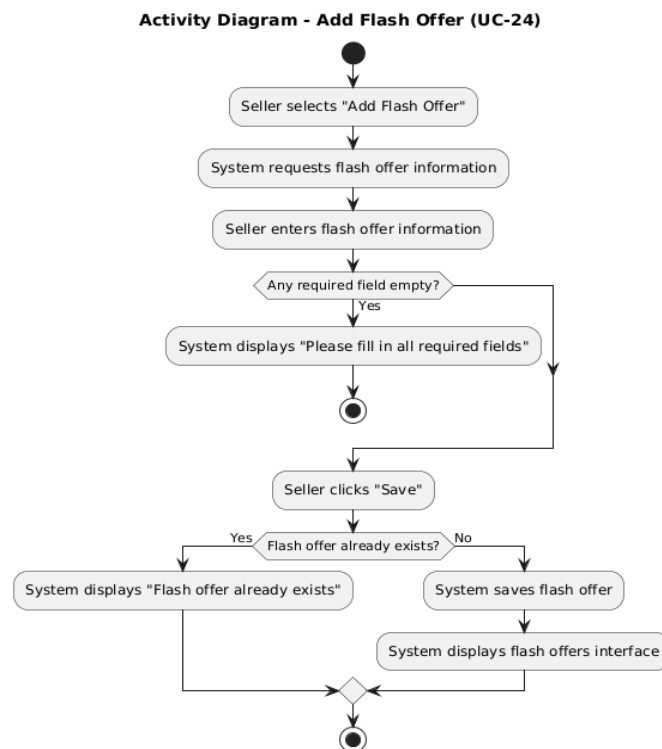
Table 27: Use Case Specification – Add Flash Offers

Use Case ID	UC-24
Use Case Name	Add Flash Offers
Actors	Seller
Description	Allows the seller to add a Flash offer to the system.
Precondition	The seller must have previously logged in to their account and selected the Flash Offers Management page.
Main flow	1- The seller selects "Add Flash Offer". 2- The system requests the required information for the Flash offer. 3- The seller enters the Flash offer information. 4- The seller clicks "Save". 5- The system displays the Flash offers interface.

Alternative flow	A1: After step 4, if a previously existing Flash offer is added, the system displays a message that the Flash offer already exists, and the use case fails. A2: After step 2, if the seller does not enter mandatory fields, the system displays a message: "Please enter all required information."
Postconditions	The Flash offer has been successfully added.

Add Flash Offers – Activity Diagram:

Figure 28: Add Flash Offers Activity Diagram



- Edit Flash Offers (UC-25):

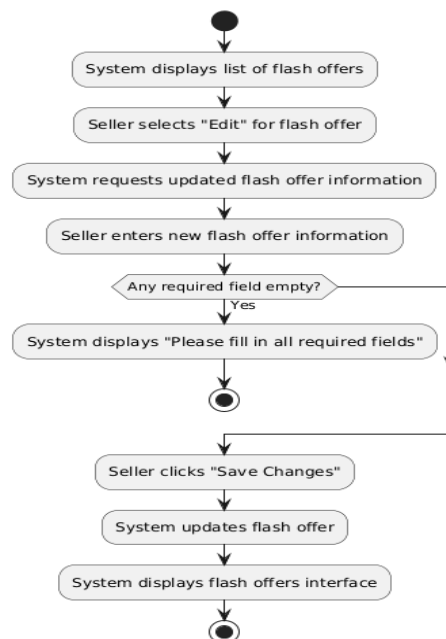
Table 28: Use Case Specification – Edit Flash Offers

Use Case ID	UC-25
Use Case Name	Edit Flash Offers
Actors	Seller
Description	Allows the seller to edit information for existing Flash offers in the system.
Precondition	The seller must have previously logged in to their account and selected the Flash Offers Management page.
Main flow	1- The system displays the list of existing Flash offers. 2- The seller clicks "Edit" for the Flash offer to be edited. 3- The system requests the required information for editing. 4- The seller enters the new information for the Flash offer. 5- The seller clicks "Save Changes". 6- The system displays the Flash offers interface.
Alternative flow	A1: After step 4, if the seller does not enter mandatory fields, the system displays a message: "Please enter all required information."
Postconditions	The Flash offer has been successfully edited.

Edit Flash Offers – Activity Diagram:

Figure 29: Edit Flash Offers Activity Diagram

Activity Diagram - Edit Flash Offer (UC-25)



- Delete Flash Offers (UC-26):

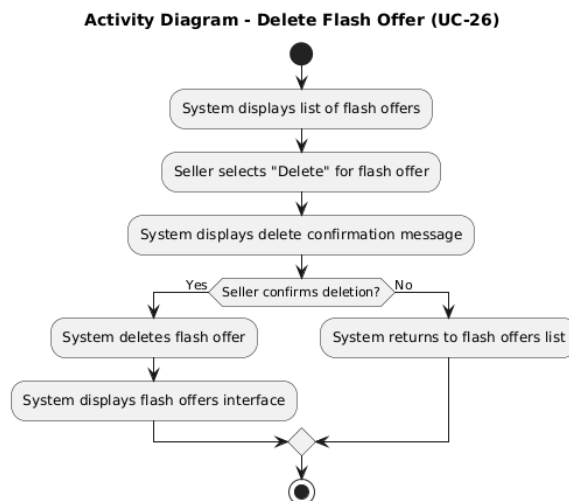
Table 29: Use Case Specification – Delete Flash Offers

Use Case ID	UC-26
Use Case Name	Delete Flash Offers
Actors	Seller
Description	Allows the seller to delete existing Flash offers from the system.
Precondition	The seller must have previously logged in to their account and selected the Flash Offers Management page.
Main flow	1- The system displays the list of existing Flash offers. 2- The seller clicks "Delete" for the Flash offer to be deleted. 3- The system displays a deletion confirmation message.

	4- The seller confirms the deletion by clicking "Yes". 5- The system displays the Flash offers interface.
Alternative flow	A1: Returning to step 1 of the main flow, the seller selects "No".
Postconditions	The Flash offer has been successfully deleted.

Delete Flash Offers – Activity Diagram:

Figure 30: Delete Flash Offer Activity Diagram



- Add Products to Wishlist (UC-27):

Table 30: Use Case Specification – Add Products to Favorites List

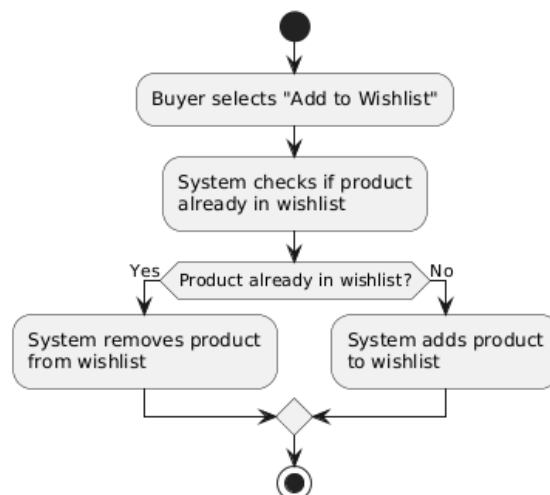
Use Case ID	UC-27
Use Case Name	Add Products to Favorites List
Actors	Buyer
Description	Allows the buyer to add products to the favorites list.
Precondition	The buyer must have previously logged in to their account and be browsing products.

Main flow	1- The buyer selects the heart icon "Add Product to Favorites List". 2- The system saves the products in the favorites list.
Alternative flow	A1: After step 1, if the product was previously added to the favorites list, it will be removed from it.
Postconditions	The products have been successfully added to the cart.

Add Products to Wishlist– Activity Diagram:

Figure 31: Add Products to Wishlist Activity Diagram

Activity Diagram - Add Products to Wishlist (UC-27)



- View Wishlist (UC-28):

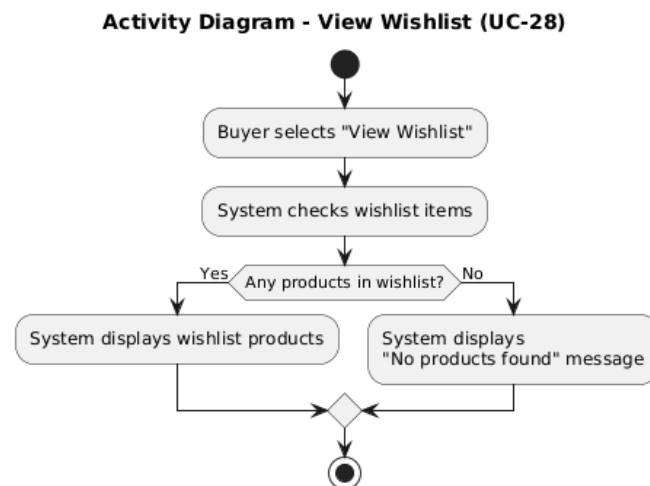
Table 31: Use Case Specification – View Wishlist

Use Case ID	UC-28
Use Case Name	View Wishlist
Actors	Buyer
Description	Allows the buyer to View their personal favorites list.
Precondition	The buyer must have previously logged in to their account.
Main flow	1- The buyer clicks "View Favorites List".

Alternative flow	2- The system displays the products in the favorites list.
	A1: If the user has no previously added products in the favorites list, the system displays a message: "No products yet."
Postconditions	The products in the Wishlist have been successfully displayed to the buyer.

View Wishlist– Activity Diagram:

Figure 32: View Wishlist Activity Diagram



- View Reports for Seller (UC-29):

Table 32: Use Case Specification – View Reports for Seller

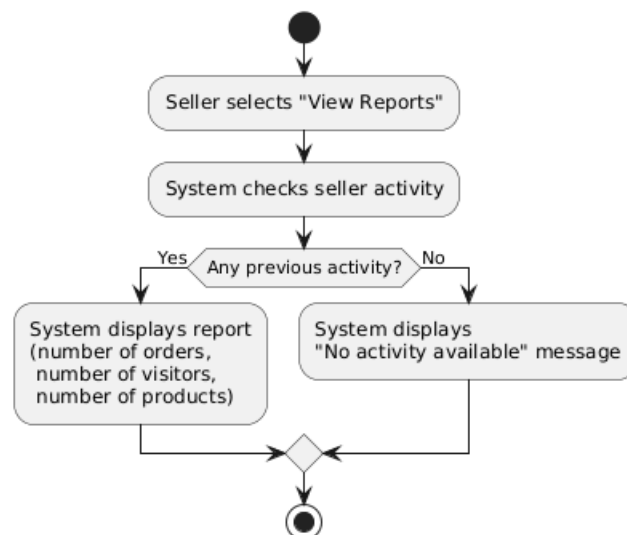
Use Case ID	UC-29
Use Case Name	View Reports for Seller
Actors	Seller
Description	Allows the seller to View reports related to their shop.
Precondition	The seller must have previously logged in to their account.
Main flow	1- The seller selects to view the reports page.

	2- The system displays a report on the number of received orders, number of shop visitors, and number of products.
Alternative flow	A1: If the user has no previous activity, the system displays a message: "No activity yet."
Postconditions	The report has been successfully displayed to the seller.

View Reports for Seller – Activity Diagram:

Figure 33: View Reports for Seller Activity Diagram

Activity Diagram - View Reports for Seller (UC-29)



- Direct Messages Between Seller and Buyer (UC-30):

Table 33: Use Case Specification – Direct Messages Between Seller and Buyer

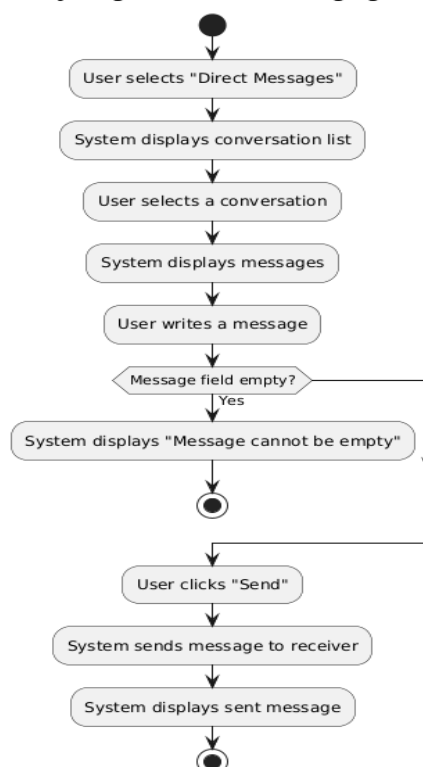
Use Case ID	UC-30
Use Case Name	Direct Messages Between Seller and Buyer
Actors	Seller, Buyer
Description	Allows the buyer and seller to exchange direct messages regarding products or orders.

Precondition	The seller or buyer must have previously logged in to their account.
Main flow	1- The user selects "Conversation". 2- The system displays the list of conversations. 3- The user selects a specific conversation. 4- The user writes the message. 5- The user clicks the send button. 6- The system sends the message to the other party.
Alternative flow	A1: If the message field is left empty, the system displays an error message and the message is not sent.
Postconditions	The message has been successfully sent and appears within the conversation.

Direct Messages Between Seller and Buyer – Activity Diagram:

Figure 34: Direct Messages Between Seller and Buyer Activity Diagram

Activity Diagram - Direct Messaging (UC-30)



- Gift Suggestions Using Artificial Intelligence (UC-31):

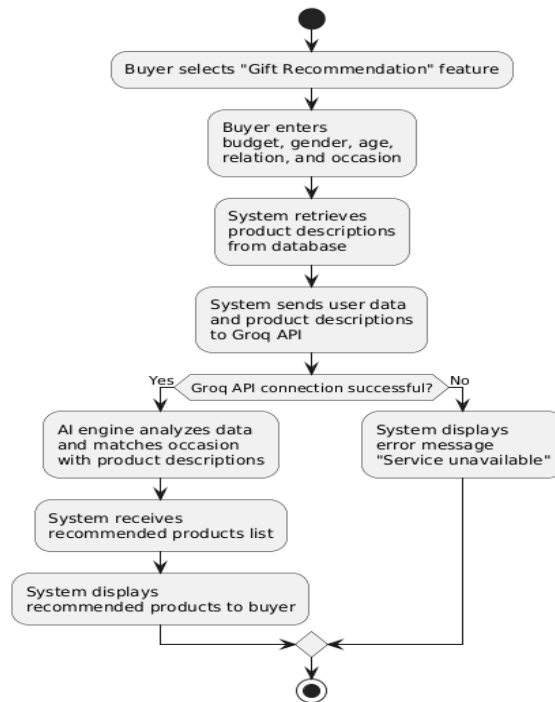
Table 34: Use Case Specification – Gift Suggestions Using Artificial Intelligence

Use Case ID	UC-31
Use Case Name	Gift Suggestions Using Artificial Intelligence
Actors	Buyer
Description	This use case allows the buyer to receive suggestions for suitable products as gifts based on budget and occasion type using artificial intelligence.
Precondition	The buyer must have previously logged in to their account, and products with text descriptions must be available in the database.
Main flow	1- The buyer selects the gift suggestion feature. 2- The buyer enters the available budget, gender, age, relationship, and occasion type. 3- The system collects suitable product descriptions from the database. 4- The system sends the data to the Groq API. 5- The AI analyzes the data and matches the occasion with product descriptions. 6- The system receives a list of suggested products. 7- The system displays the list of suggested products to the buyer.
Alternative flow	A1: If the connection with the Groq API fails, the system displays an appropriate error message to the user.
Postconditions	Suitable gift suggestions are successfully displayed to the buyer.

Gift Suggestions Using Artificial Intelligence – Activity Diagram:

Figure 35: Gift Suggestions Using Artificial Intelligence Activity Diagram

Activity Diagram - Gift Recommendation Using AI (UC-31)



- Bazaar Management (UC-32):

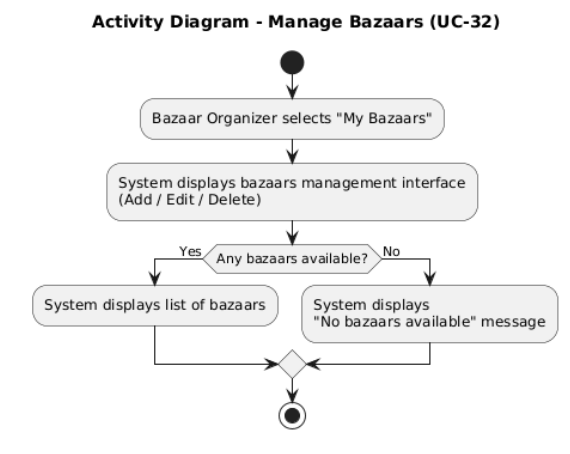
Table 35: Use Case Specification – Bazaar Management

Use Case ID	UC-32
Use Case Name	Bazaar Management.
Actors	Bazaar Organizer
Description	The bazaar organizer manages bazaars.
Precondition	The bazaar organizer must have previously logged in to their account.
Main flow	1- The bazaar organizer selects "My Bazaars". 2- The system displays the bazaars list interface and management options (Add – Edit – Delete).

Alternative flow	A1: If the organizer has no registered bazaars, the system displays a message: "No bazaars currently available."
Postconditions	The bazaar management interface is displayed.

Bazaar Management – Activity Diagram:

Figure 36: Bazaar Management Activity Diagram



- Add Bazaars (UC-33):

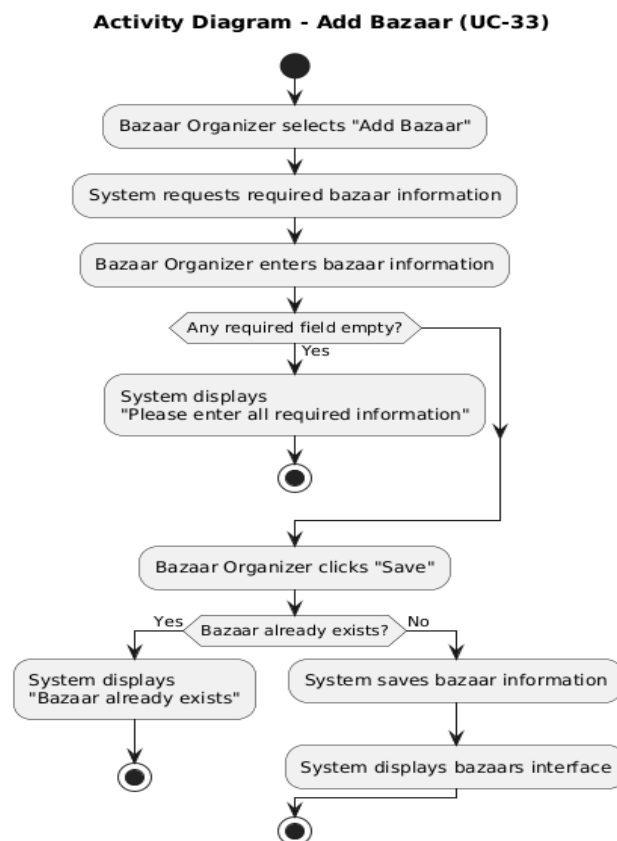
Table 36: Use Case Specification – Add Bazaars

Use Case ID	UC-33
Use Case Name	Add Bazaars
Actors	Bazaar Organizer
Description	Allows the bazaar organizer to add a bazaar to the system.
Precondition	The bazaar organizer must have previously logged in to their account and selected the "My Bazaars" page.
Main flow	1- The bazaar organizer selects "Add Bazaar". 2- The system requests the required information for the bazaar. 3- The bazaar organizer enters the bazaar information.

	4- The bazaar organizer clicks "Save". 5- The system displays the bazaars interface.
Alternative flow	A1: After step 4, if a previously existing bazaar is added, the system displays a message that the bazaar already exists, and the use case fails. A2: After step 2, if the bazaar organizer does not enter mandatory fields, the system displays a message: "Please enter all required information."
Postconditions	The bazaar has been successfully added.

Add Bazaars – Activity Diagram:

Figure 37: Add Bazaars Activity Diagram



- Edit Bazaars (UC-34):

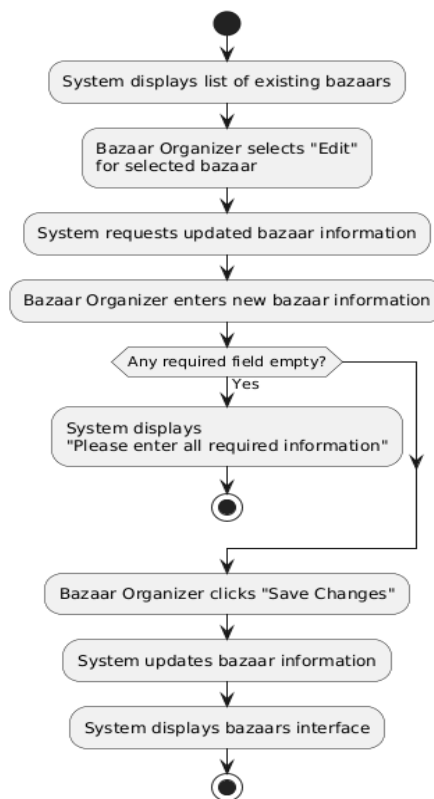
Table 37: Use Case Specification – Edit Bazaars

Use Case ID	UC-34
Use Case Name	Edit Bazaars
Actors	Bazaar Organizer
Description	Allows the bazaar organizer to edit information for existing bazaars in the system.
Precondition	The bazaar organizer must have previously logged in to their account and selected the "My Bazaars" page.
Main flow	1- The system displays the list of existing bazaars. 2- The bazaar organizer clicks "Edit" for the bazaar to be edited. 3- The system requests the required information for editing. 4- The bazaar organizer enters the new information for the bazaar. 5- The bazaar organizer clicks "Save Changes". 6- The system displays the bazaars interface.
Alternative flow	A1: After step 4, if the bazaar organizer does not enter mandatory fields, the system displays a message: "Please enter all required information."
Postconditions	The bazaar has been successfully edited.

Edit Bazaars – Activity Diagram:

Figure 38: Edit Bazaars Activity Diagram

Activity Diagram - Edit Bazaar (UC-34)



- Delete Bazaars (UC-35):

Table 38: Use Case Specification – Delete Bazaars

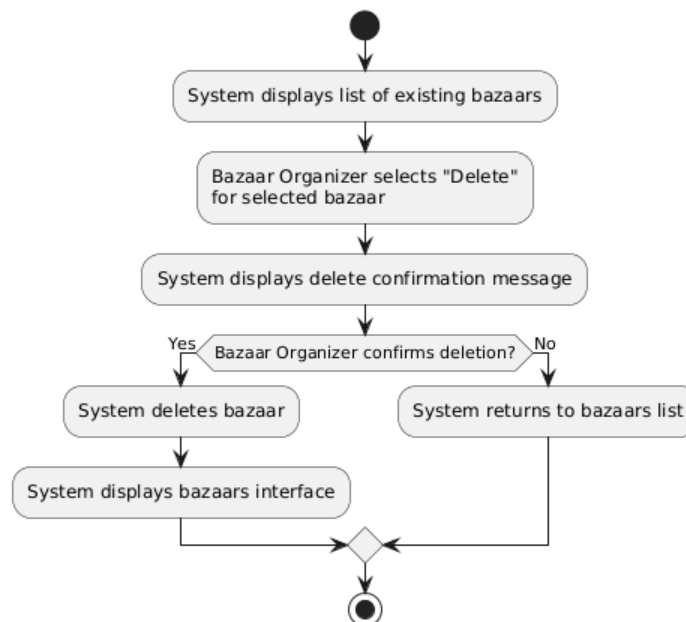
Use Case ID	UC-35
Use Case Name	Delete Bazaars
Actors	Bazaar Organizer
Description	Allows the bazaar organizer to delete existing bazaars from the system.
Precondition	The bazaar organizer must have previously logged in to their account and selected the "My Bazaars" page.
Main flow	1- The system displays the list of existing bazaars.

	2- The bazaar organizer clicks "Delete" for the bazaar to be deleted. 3- The system displays a deletion confirmation message. 4- The bazaar organizer confirms the deletion by clicking "Yes". 5- The system displays the bazaars interface after deletion.
Alternative flow	A1: Returning to step 1 of the main flow, the bazaar organizer selects "No".
Postconditions	The bazaar has been successfully deleted.

Delete Bazaars – Activity Diagram:

Figure 39: Delete Bazaars Activity Diagram

Activity Diagram - Delete Bazaar (UC-35)



- View Bazaars (UC-36):

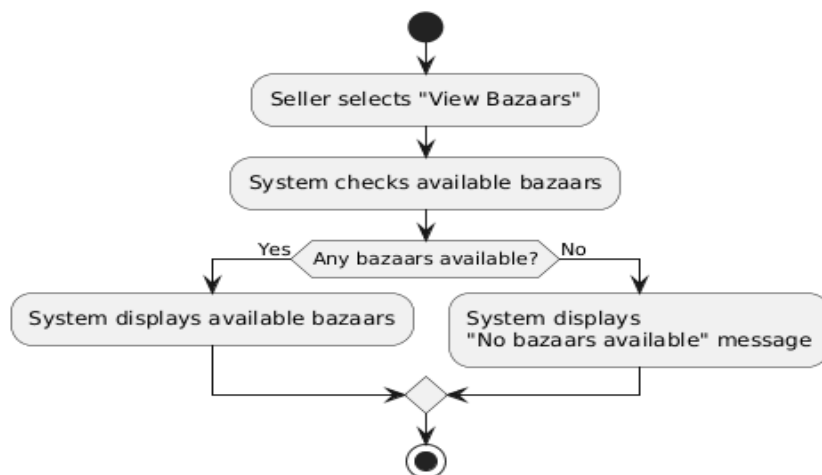
Table 39: Use Case Specification – View Bazaars

Use Case ID	UC-36
Use Case Name	View Bazaars
Actors	Seller
Description	Allows the seller to View available bazaars.
Precondition	The seller must have previously logged in to their account.
Main flow	1- The seller clicks "View Bazaars". 2- The system displays the available bazaars.
Alternative flow	A1: If no bazaars are registered, the system displays a message: "No bazaars currently available."
Postconditions	The available bazaars have been successfully displayed.

View Bazaars – Activity Diagram:

Figure 40: View Bazaars Activity Diagram

Activity Diagram - View Bazaars (UC-36)



- View Bazaar Details (UC-37):

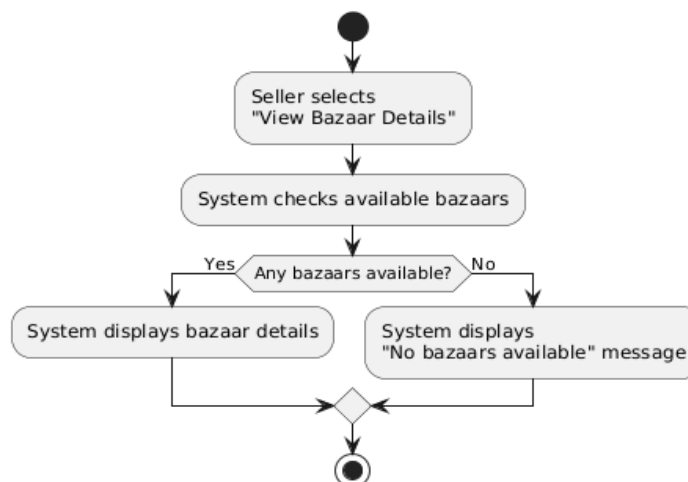
Table 40: Use Case Specification – View Bazaar Details

Use Case ID	UC-37
Use Case Name	View Bazaar Details
Actors	Seller
Description	Allows the seller to View the details of available bazaars.
Precondition	The seller must have previously logged in to their account and selected the bazaars page.
Main flow	1- The seller clicks "View Bazaar Details". 2- The system displays the bazaar details.
Alternative flow	A1: If no bazaars are registered, the system displays a message: "No bazaars currently available."
Postconditions	The bazaar details have been successfully displayed.

View Bazaar Details – Activity Diagram:

Figure 41: View Bazaar Details Activity Diagram

Activity Diagram - View Bazaar Details (UC-37)



- Request a Table Reservation at the Bazaar (UC-38):

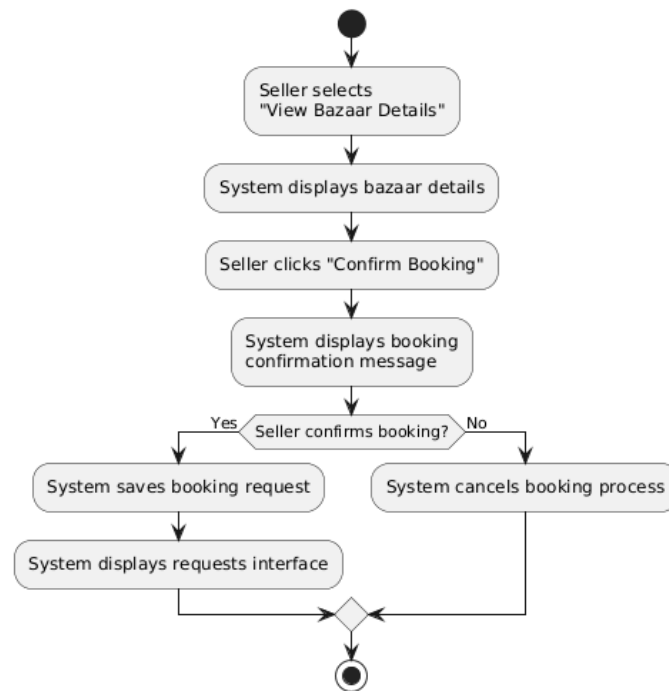
Table 41: Use Case Specification – Request a Table Reservation at the Bazaar

Use Case ID	UC-38
Use Case Name	Request a Table Reservation at the Bazaar
Actors	Seller
Description	Allows the seller to reserve a table at the bazaar.
Precondition	The seller must have previously logged in to their account and selected the bazaars page.
Main flow	1- The seller clicks "View Bazaar Details". 2- The system displays the bazaar details. 3- The seller clicks "Confirm Reservation" for the bazaar. 4- The system displays an order confirmation message. 5- The seller confirms the reservation by clicking "Yes". 6- The system displays the orders interface.
Alternative flow	A1: Returning to step 5 of the main flow, the user does not confirm the reservation and clicks "No".
Postconditions	The reservation request has been successfully saved within the orders.

Request a Table Reservation at the Bazaar – Activity Diagram:

Figure 42: Request a Table Reservation at the Bazaar Activity Diagram

Activity Diagram - Request Bazaar Table Booking (UC-38)



- Cancel a Table Reservation at the Bazaar (UC-39):

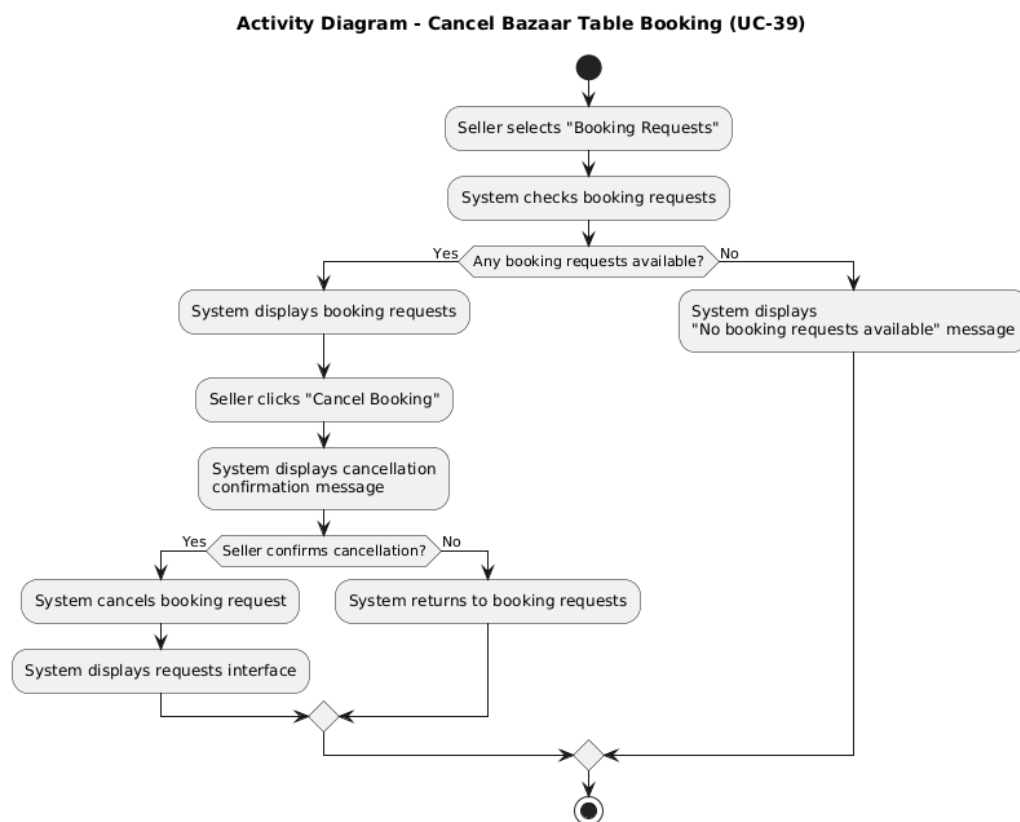
Table 42: Use Case Specification – Cancel a Table Reservation at the Bazaar

Use Case ID	UC-39
Use Case Name	Cancel a Table Reservation at the Bazaar
Actors	Seller
Description	Allows the seller to cancel a table reservation at the bazaar before the bazaar start date.
Precondition	The seller must have previously logged in to their account.
Main flow	1- The seller clicks "Reservation Requests". 2- The system displays the reservation requests. 3- The seller clicks "Cancel Reservation" for the bazaar.

Alternative flow	4- The system displays an order confirmation message. 5- The seller confirms the cancellation by clicking "Yes". 6- The system displays the orders interface.
	A1: Returning to step 5 of the main flow, the user does not confirm the cancellation and clicks "No". A2: If no bazaar reservation requests are registered, the system displays a message: "No requests currently available."
Postconditions	The reservation request has been successfully cancelled.

Cancel a Table Reservation at the Bazaar – Activity Diagram:

Figure 43: Cancel a Table Reservation at the Bazaar Activity Diagram



- View Bazaar Reservation Requests (UC-40):

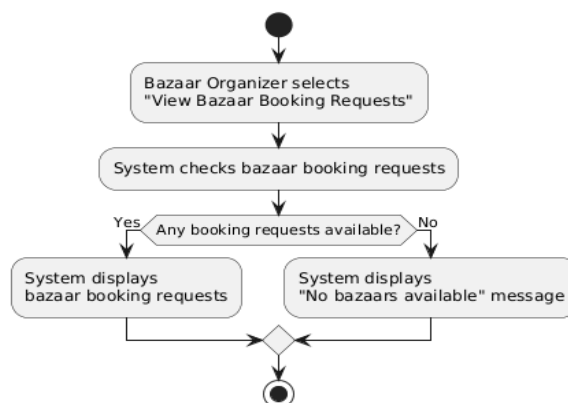
Table 43: Use Case Specification – View Bazaar Reservation Requests

Use Case ID	UC-40
Use Case Name	View Bazaar Reservation Requests
Actors	Bazaar Organizer
Description	Allows the bazaar organizer to View bazaar reservation requests.
Precondition	The bazaar organizer must have previously logged in to their account.
Main flow	1- The seller clicks "View Bazaar Reservation Requests". 2- The system displays their bazaar reservation requests.
Alternative flow	A1: If no bazaars are registered, the system displays a message: "No bazaars currently available."
Postconditions	The available bazaars have been successfully displayed.

View Bazaar Reservation Requests – Activity Diagram:

Figure 44: View Bazaar Reservation Requests Activity Diagram

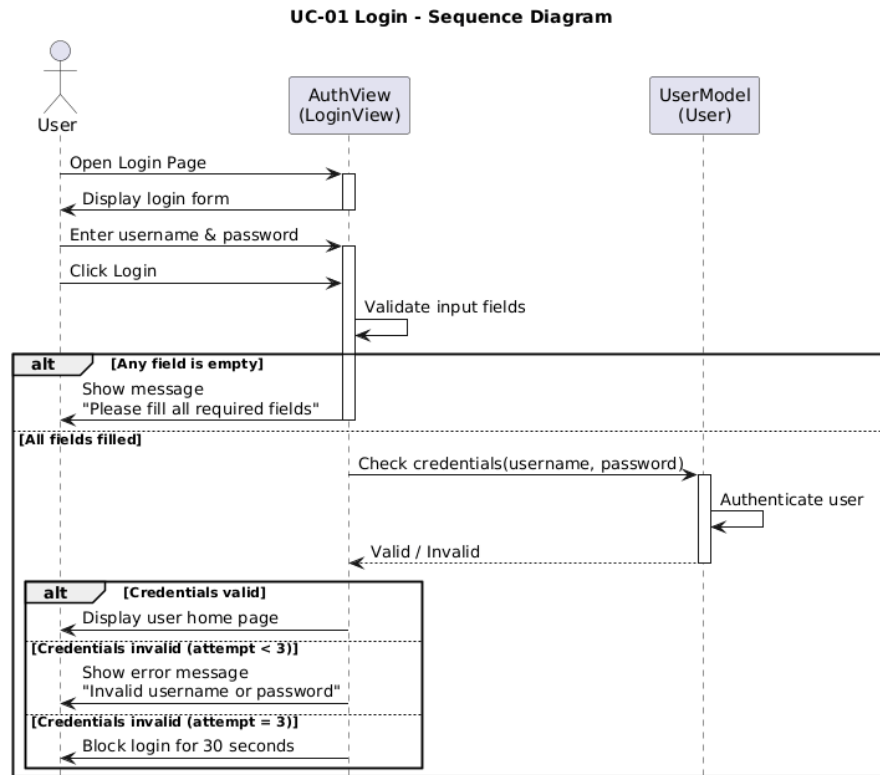
Activity Diagram - View Bazaar Booking Requests (UC-40)



5.3 Sequence diagrams

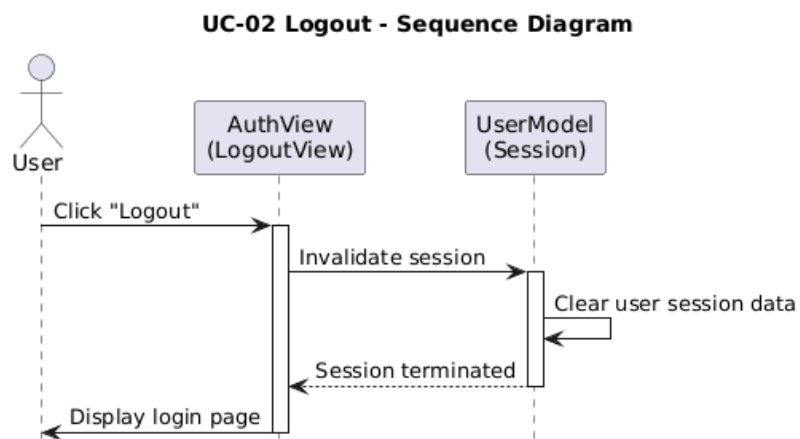
- Login (UC-01) Sequence diagram:

Figure 45 Login sequence diagram



- Logout (UC-02) Sequence diagram:

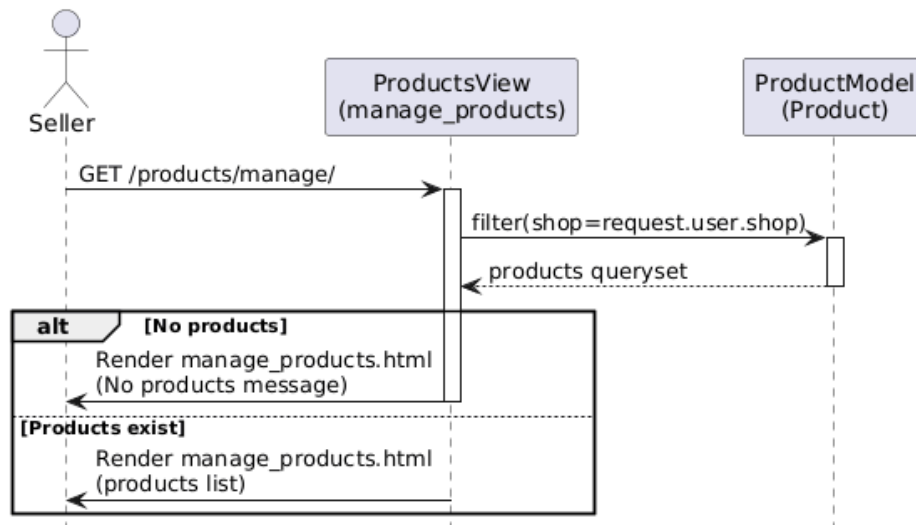
Figure 46 Logout sequence diagram



- Product Management (UC-03) Sequence diagram:

Figure 47 Product management sequence diagram

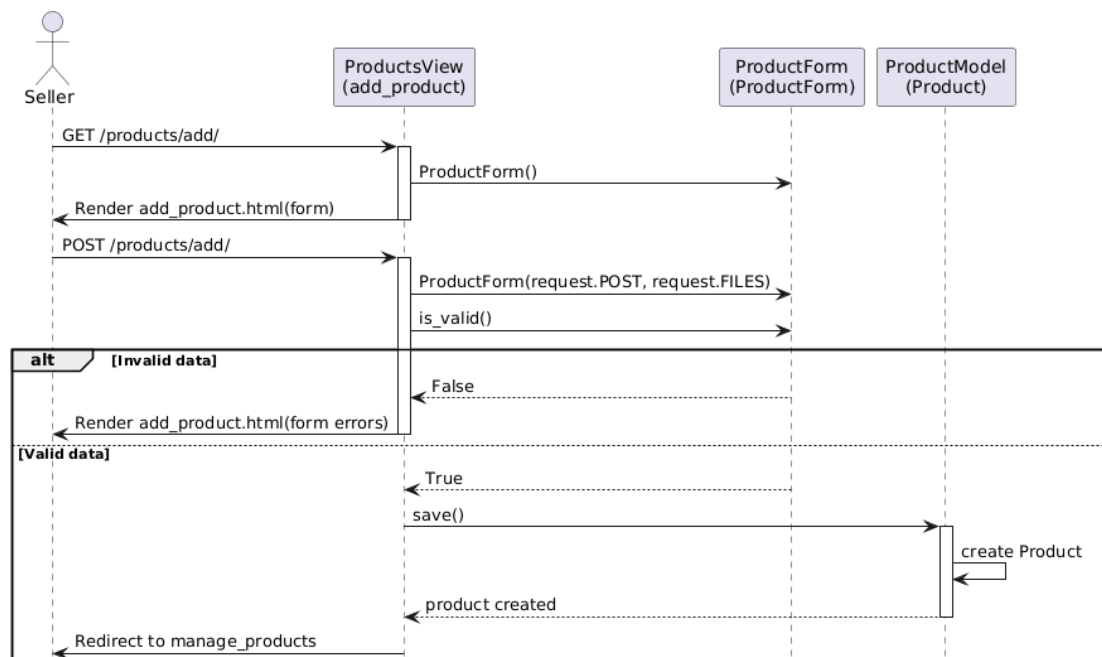
UC-03 Manage Products - Sequence Diagram



- Add Products (UC-04) Sequence diagram:

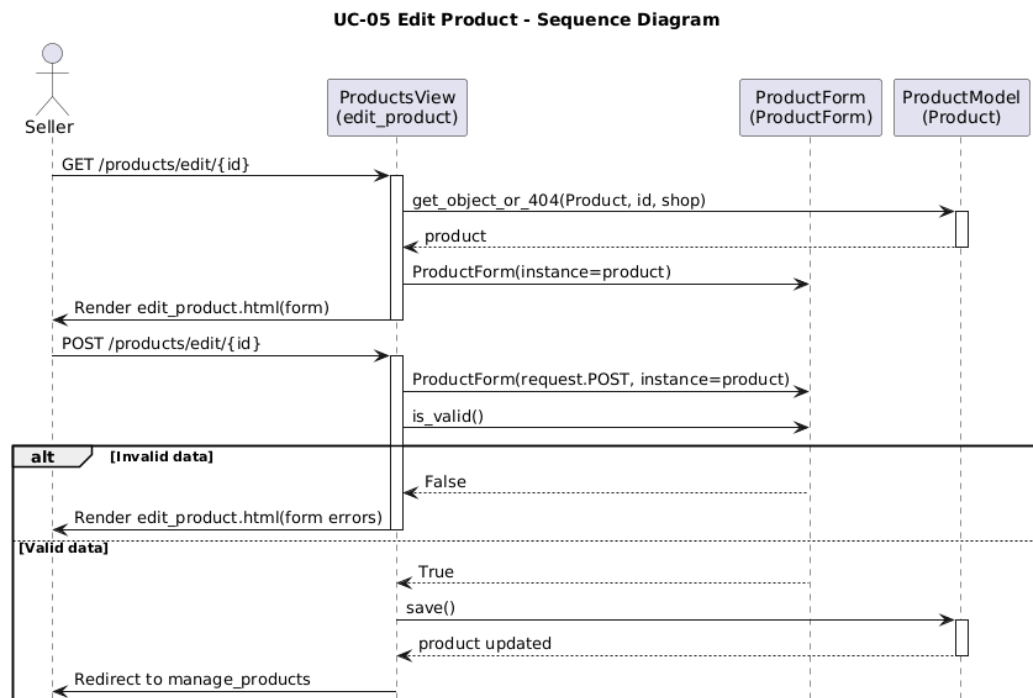
Figure 48 Add products sequence diagram

UC-04 Add Product - Sequence Diagram



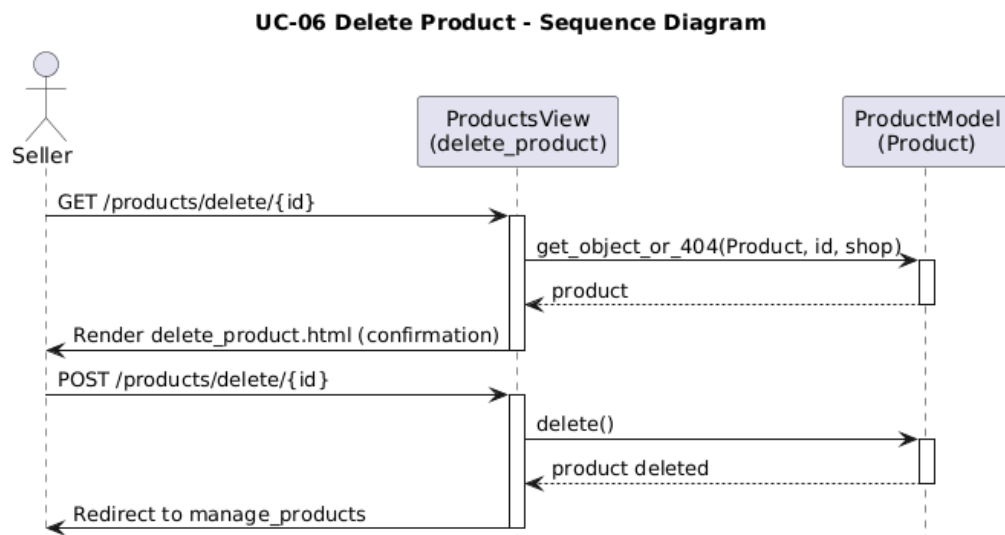
- Edit Products (UC-05) Sequence diagram:

Figure 49 Edit products sequence diagram



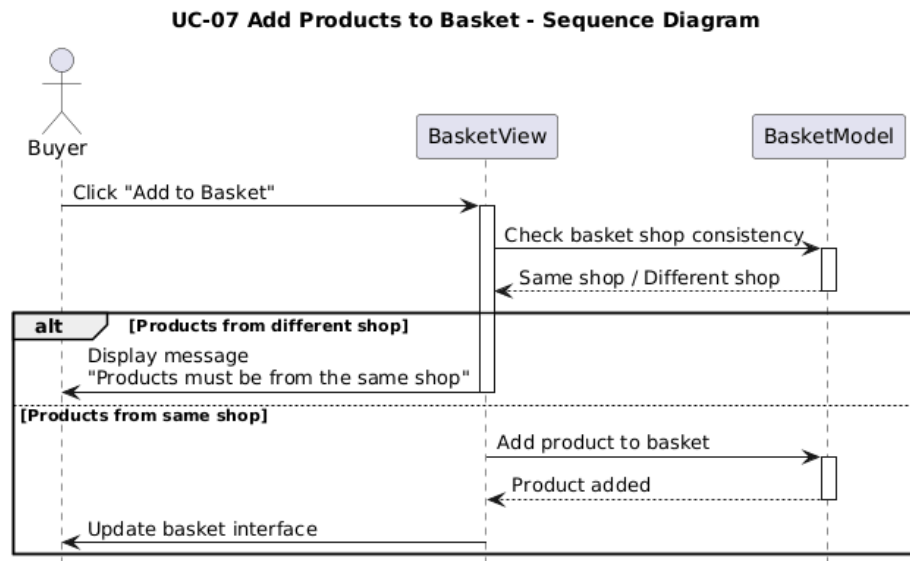
- Delete Product (UC-06) Sequence diagram:

Figure 50 Delete product sequence diagram



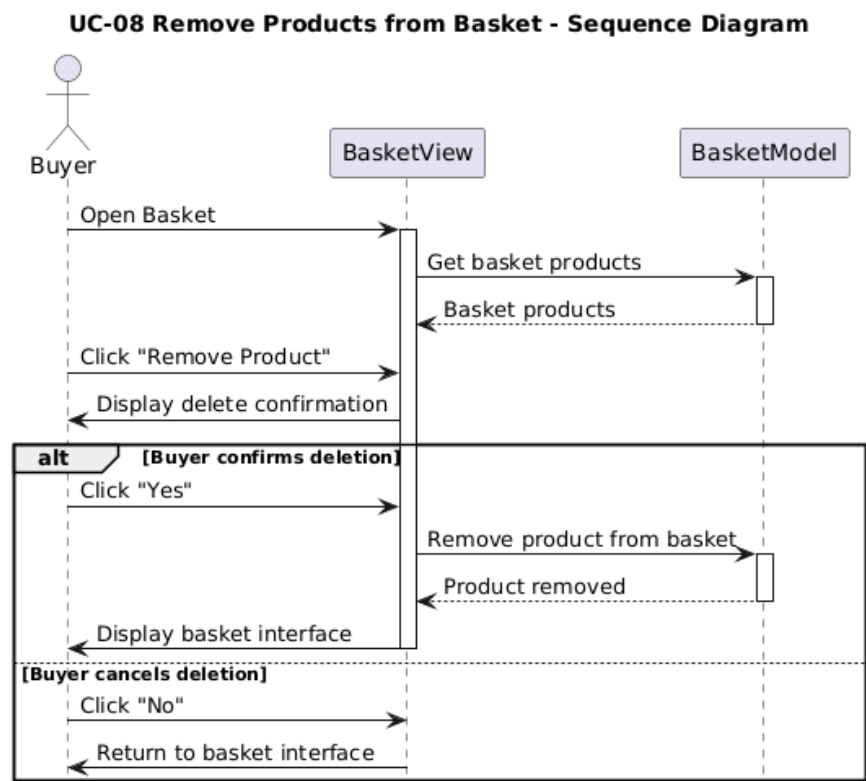
- Add Products to Basket (UC-07) Sequence diagram:

Figure 51 Add products to basket sequence diagram



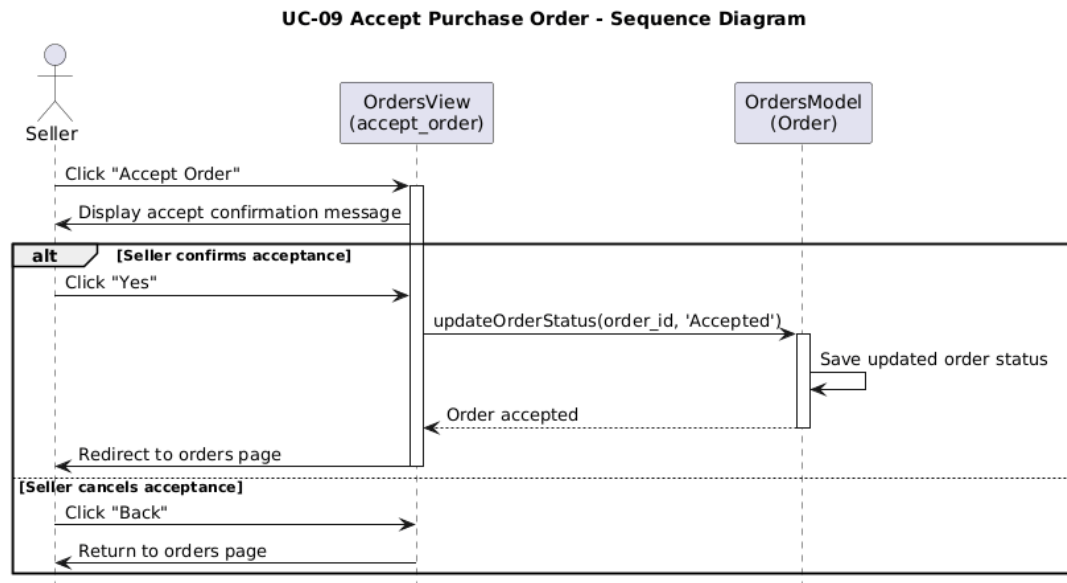
- Remove Products from Basket(UC-08) Sequence diagram :

Figure 52 Remove products from basket sequence diagram



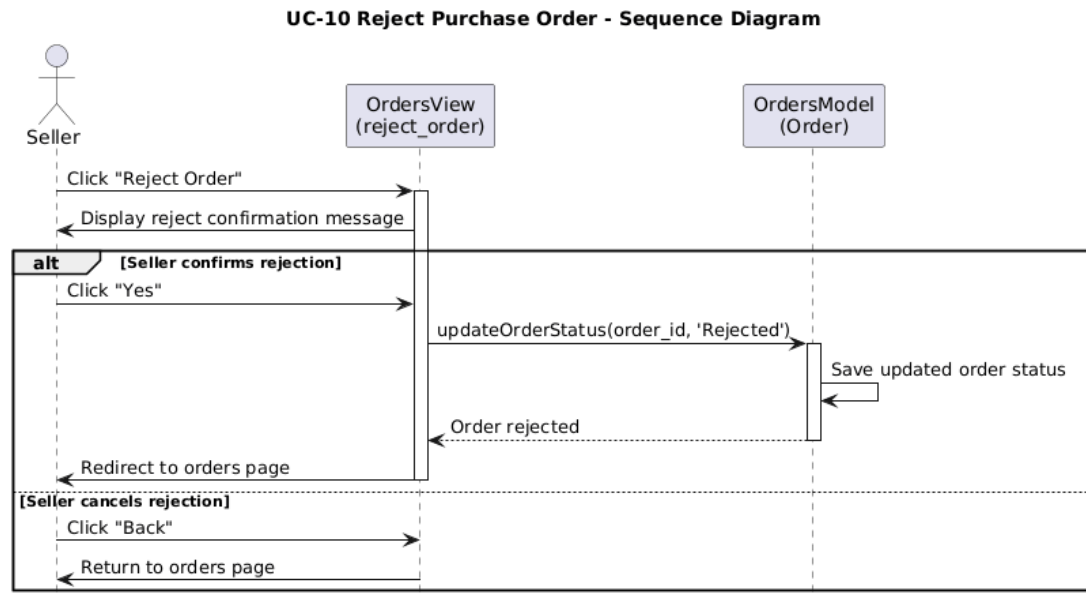
- Accept Purchase Orders (UC-09) Sequence diagram:

Figure 53 Accept purchase order sequence diagram



- Reject Purchase Orders (UC-10) Sequence diagram:

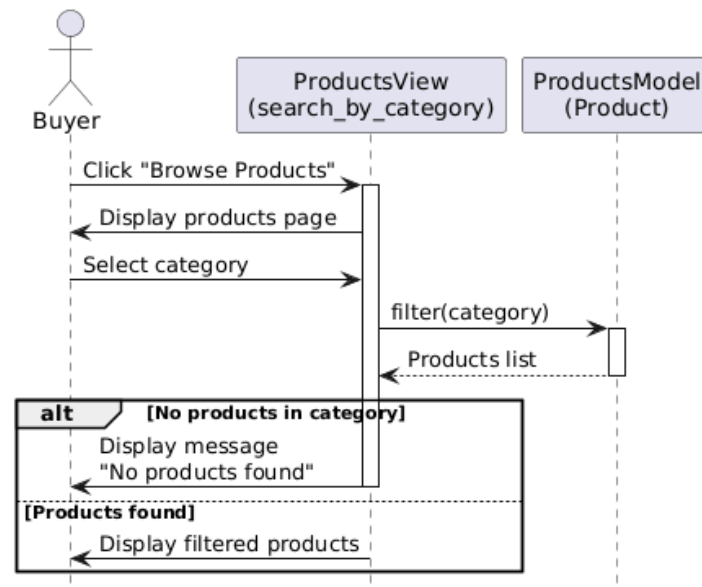
Figure 54 Reject purchase order sequence diagram



- Search Products by Category (UC-11) Sequence diagram:

Figure 55 Search products by category sequence diagram

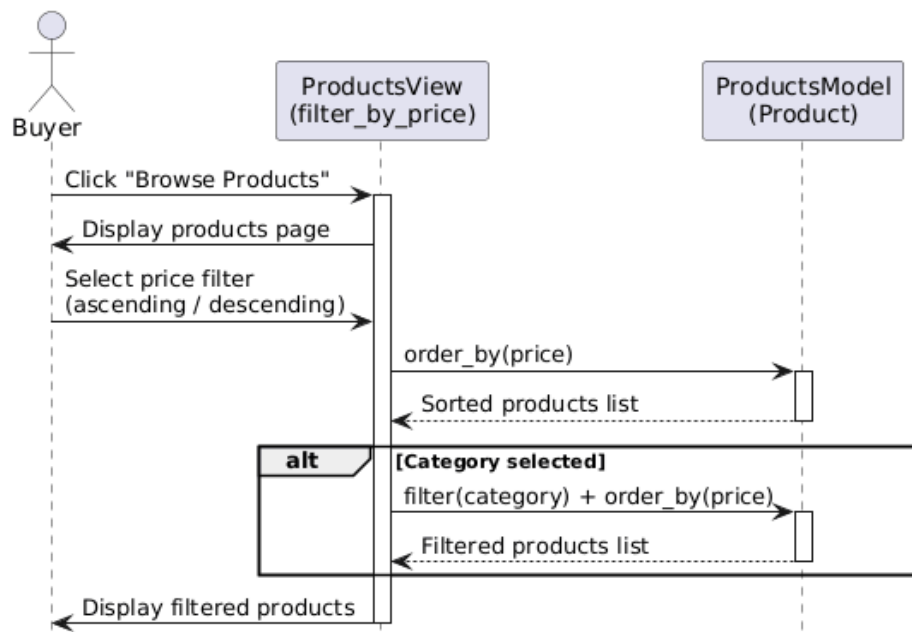
UC-11 Search Products by Category - Sequence Diagram



- Filter Products by Price (UC-12) Sequence diagram:

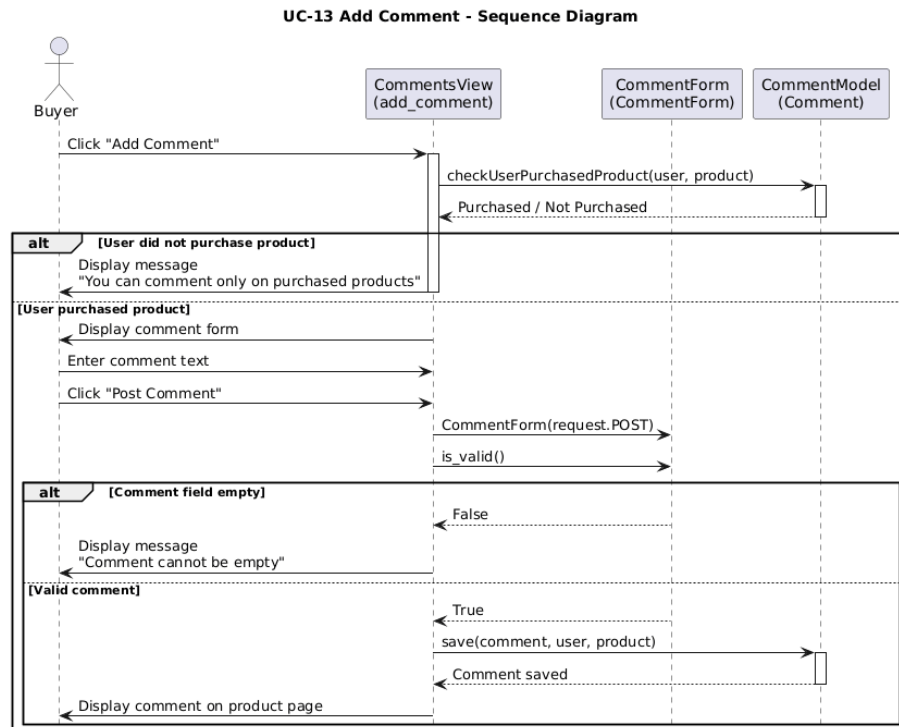
Figure 56 Filter products by price sequence diagram

UC-12 Filter Products by Price - Sequence Diagram



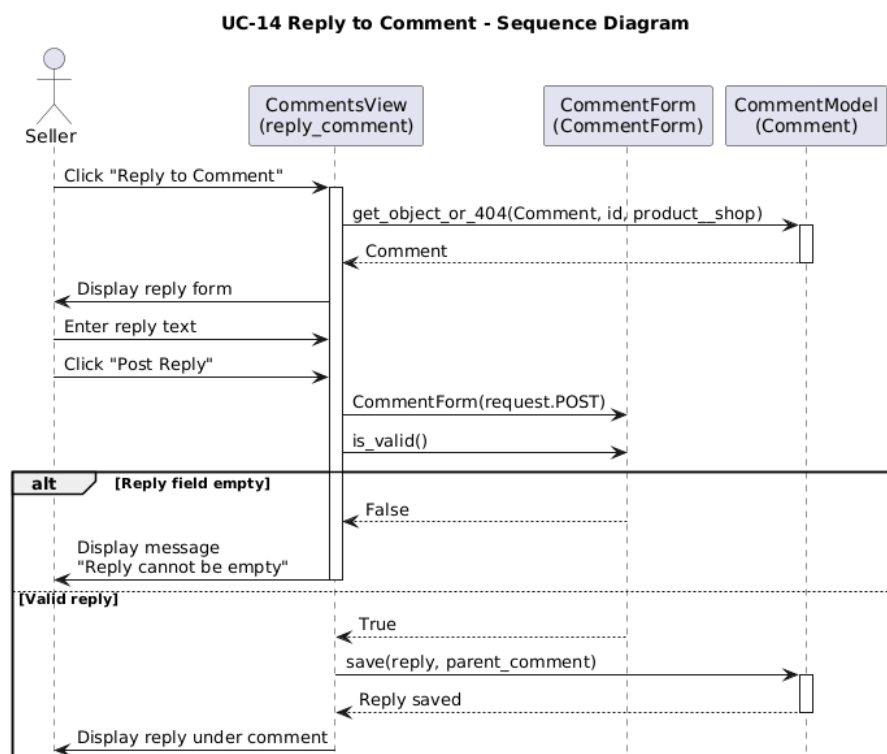
- Add Comments (UC-13) Sequence diagram:

Figure 57 Add comments sequence diagram



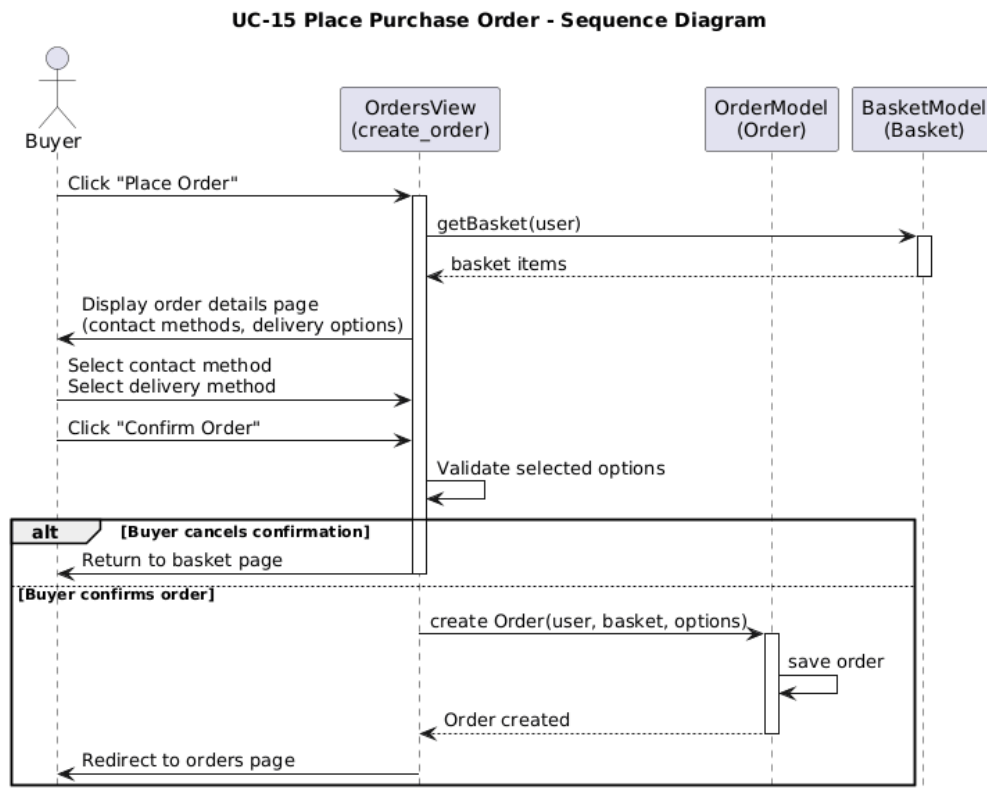
- Reply to Comments (UC-14) Sequence diagram:

Figure 58 Reply to comments sequence diagram



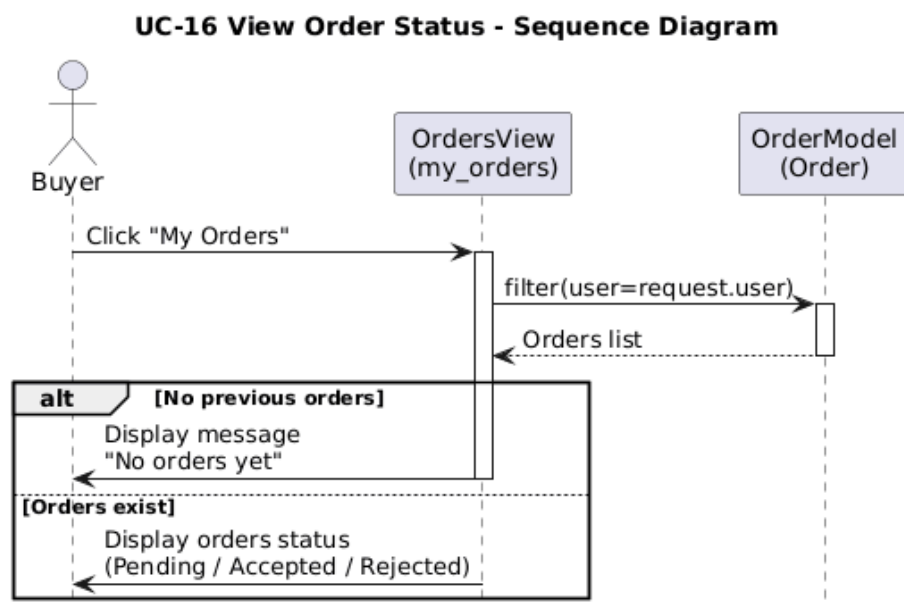
- Request Product Purchase (UC-15) Sequence diagram:

Figure 59 Request product purchase sequence diagram



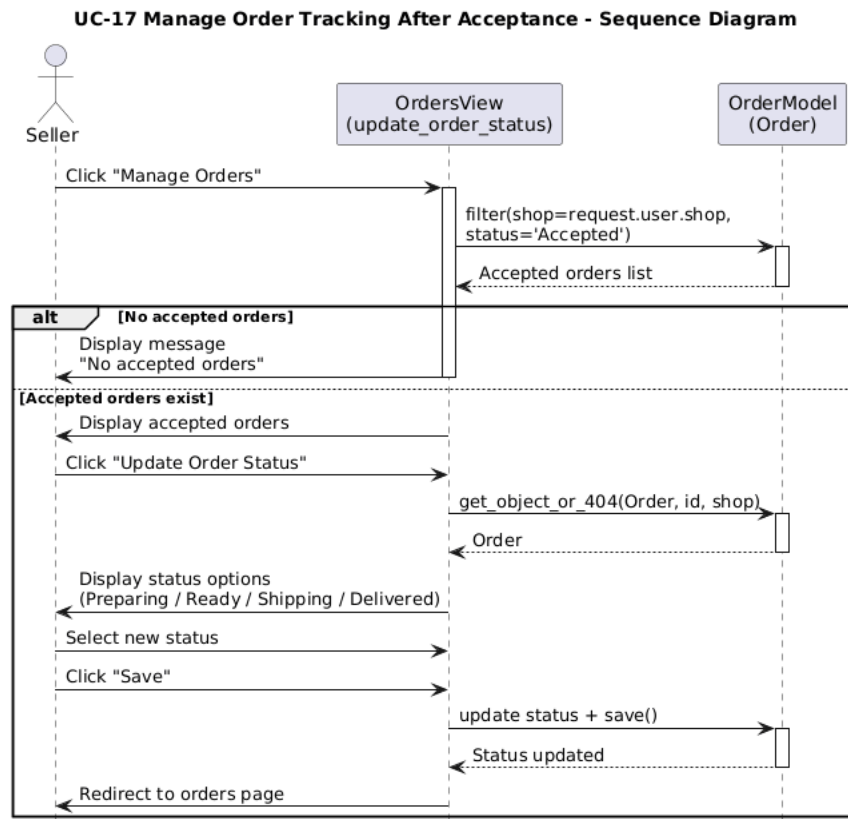
- View Order Status (UC-16) Sequence diagram:

Figure 60 View order status sequence diagram



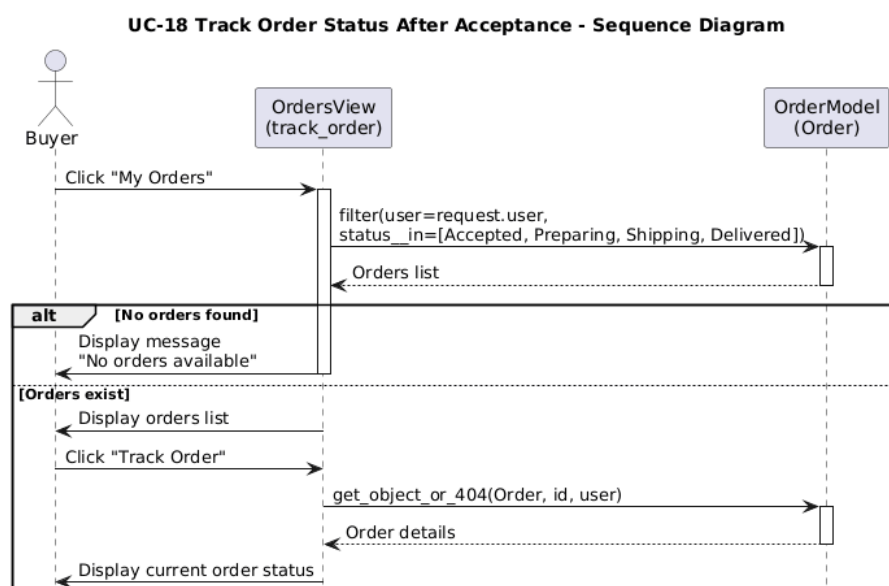
- Manage Order Tracking after Acceptance (UC-17) Sequence diagram:

Figure 61 Manage order tracking after acceptance sequence diagram



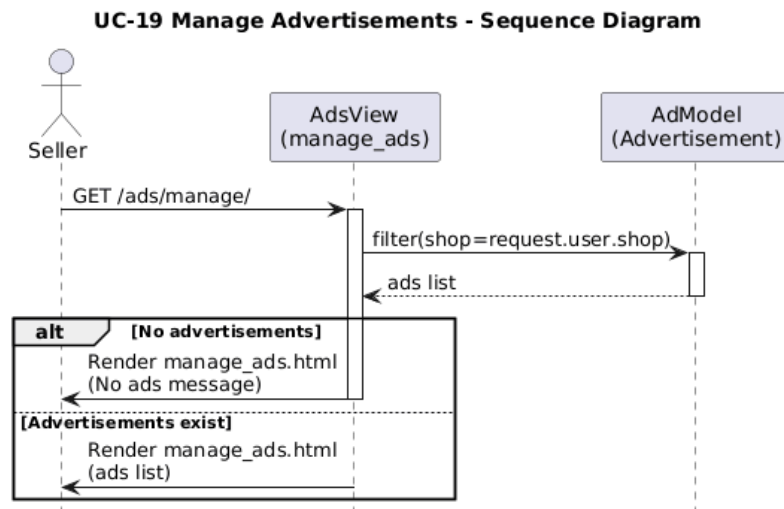
- Track Orders after Acceptance (UC-18) Sequence diagram:

Figure 62 Track orders after acceptance sequence diagram



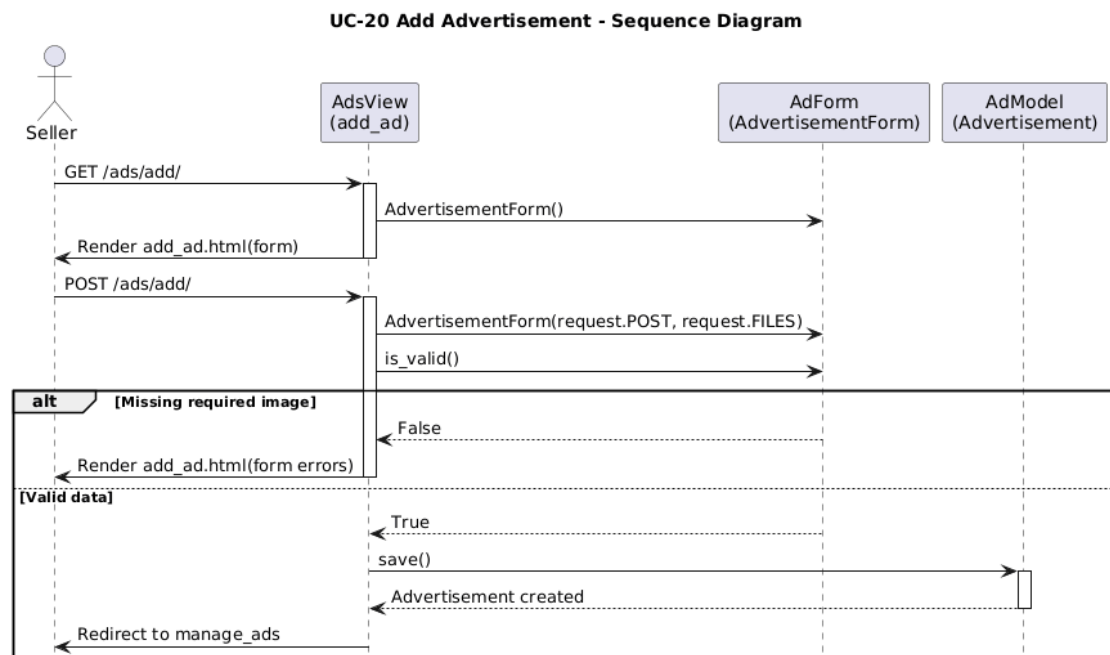
- Advertisement Management (UC-19) Sequence diagram:

Figure 63 Advertisement management sequence diagram



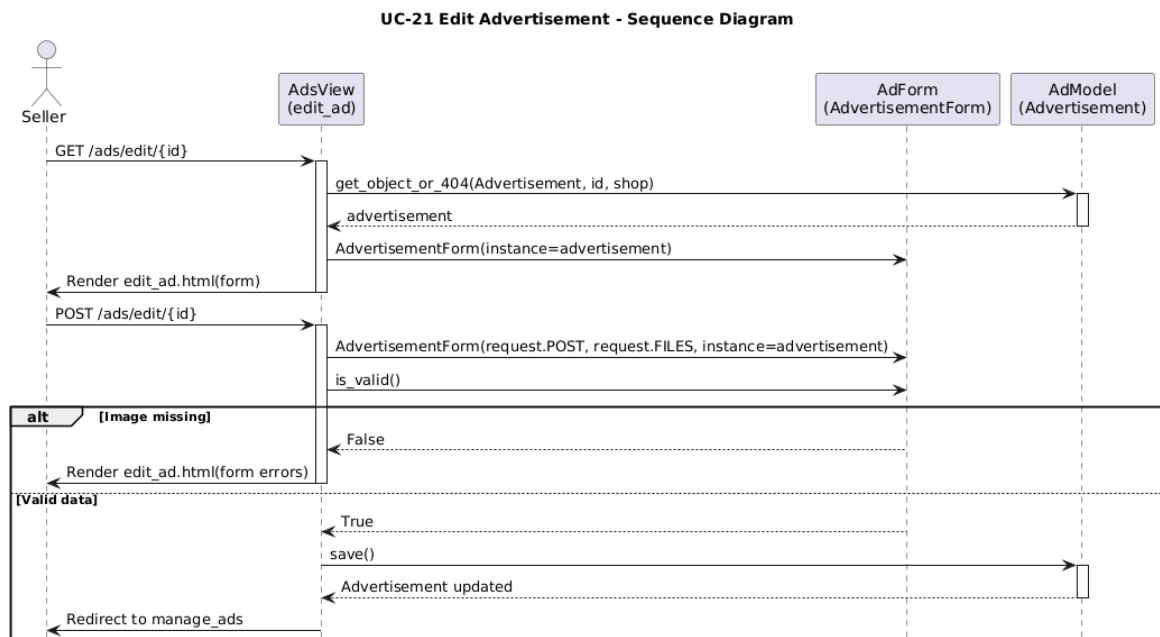
- Add Advertisements (UC-20) Sequence diagram:

Figure 64 Add advertisements sequence diagram



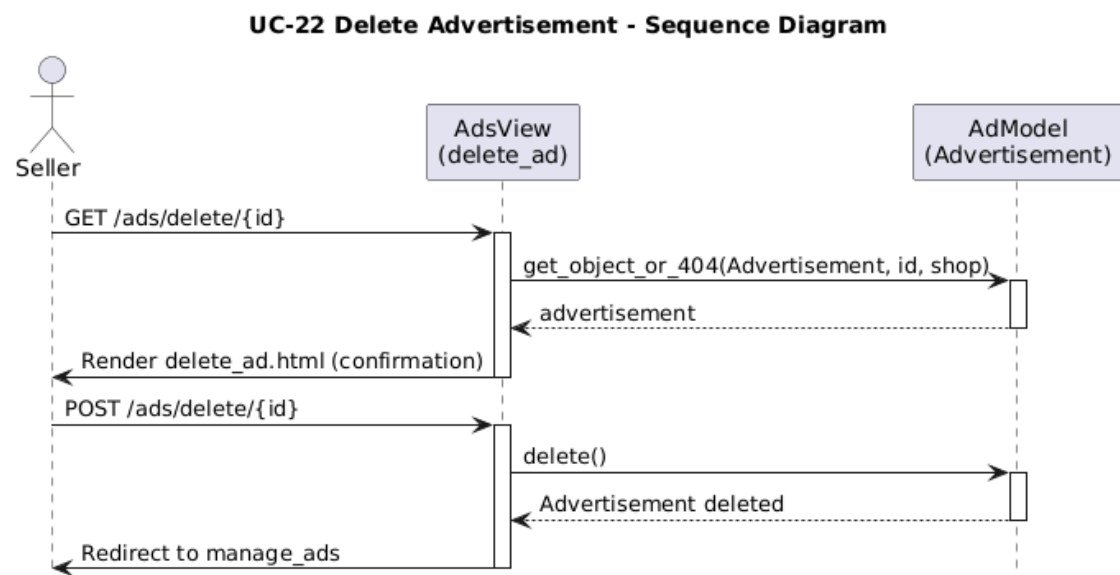
- Edit Advertisements (UC-21) Sequence diagram:

Figure 65 Edit advertisements sequence diagram



- Delete Advertisements (UC-22) Sequence diagram:

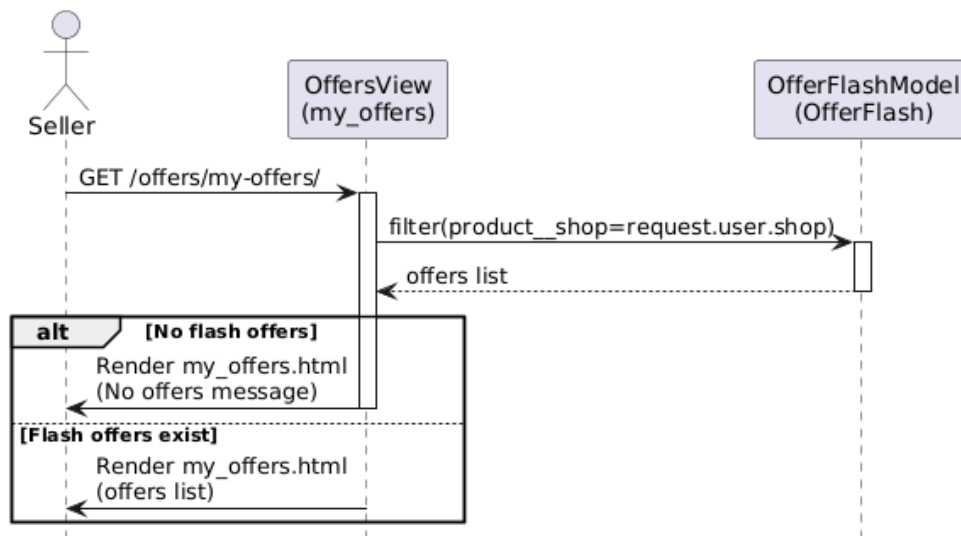
Figure 66 Delete advertisements sequence diagram



- Manage Flash Offers (UC-23) Sequence diagram :

Figure 67 Manage flash offers sequence diagram

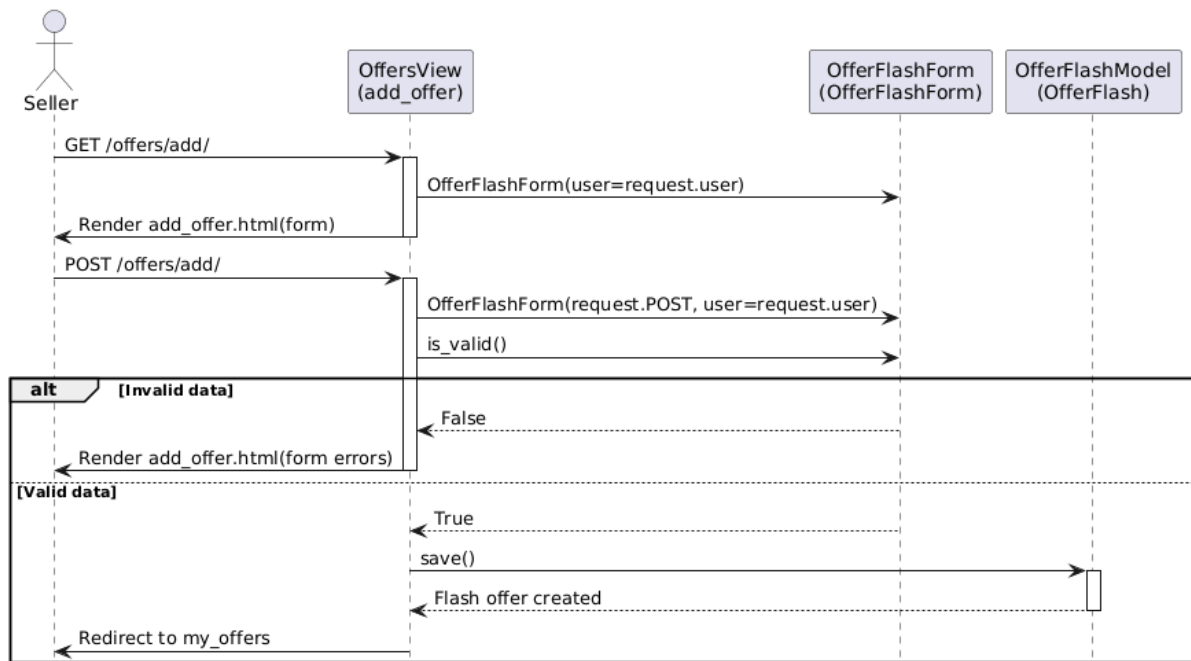
UC-23 Manage Flash Offers - Sequence Diagram



- Add Flash Offers (UC-24) Sequence diagram :

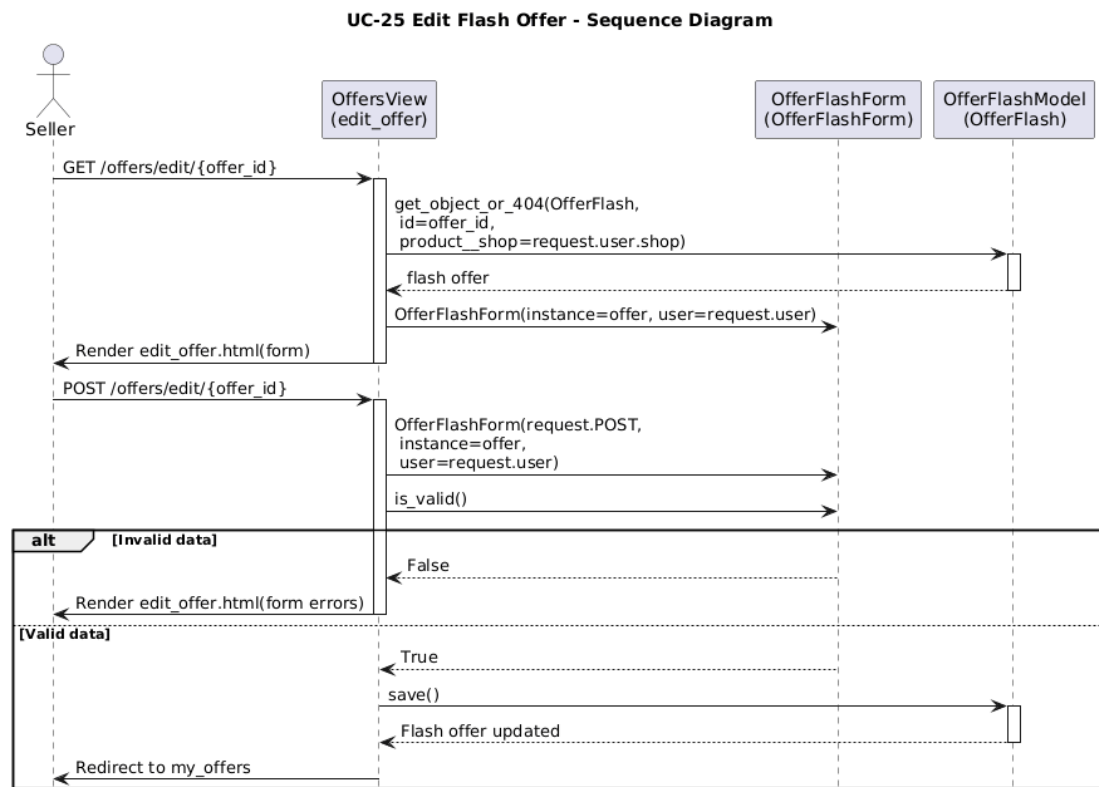
Figure 68 Add flash offers sequence diagram

UC-24 Add Flash Offer - Sequence Diagram



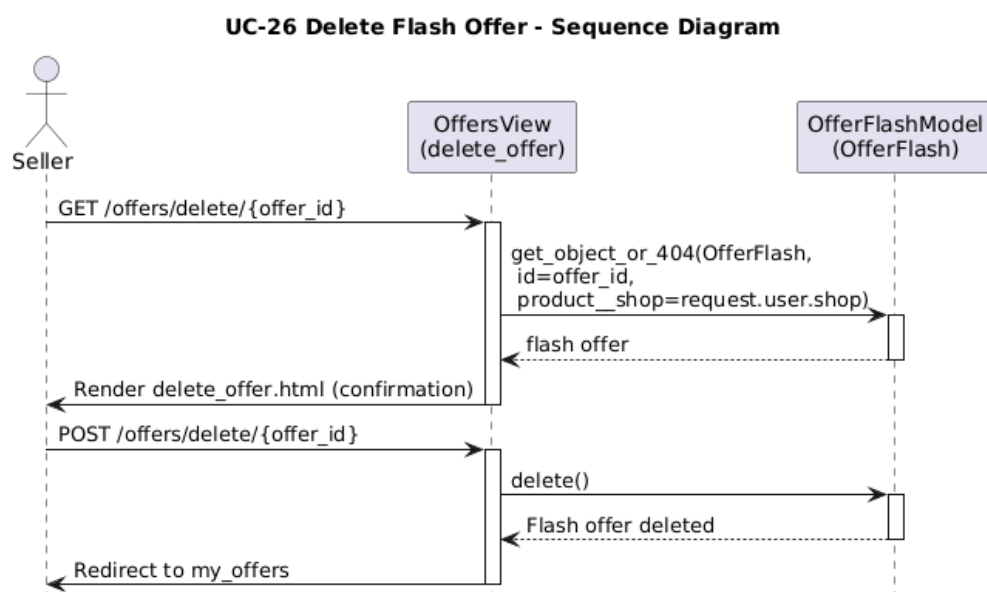
- Edit Flash Offers (UC-25) Sequence diagram:

Figure 69 Sequence diagram for modifying flash offers



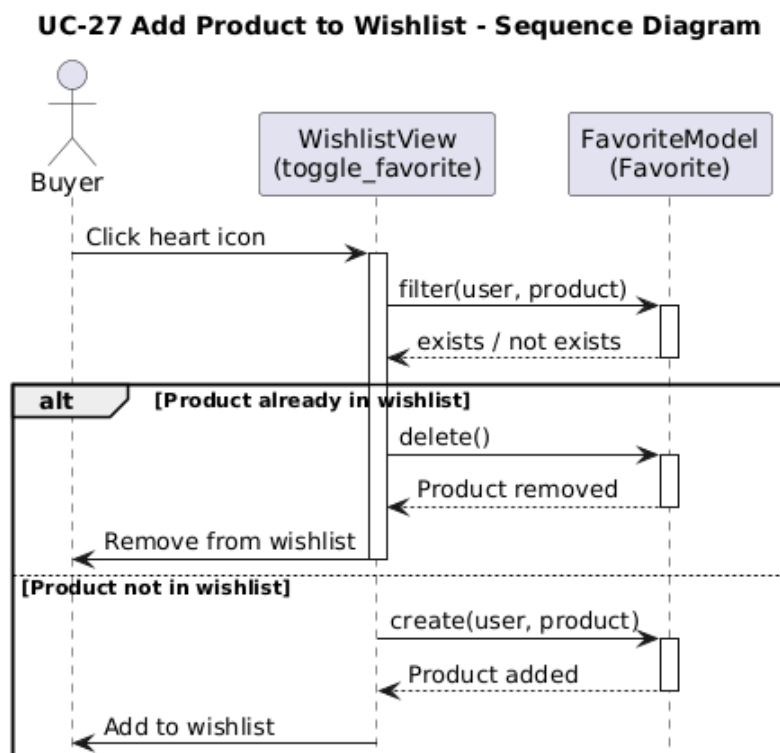
- Deleting Flash offers (UC-26) Sequence diagram:

Figure 70 Sequence diagram for deleting flash offers



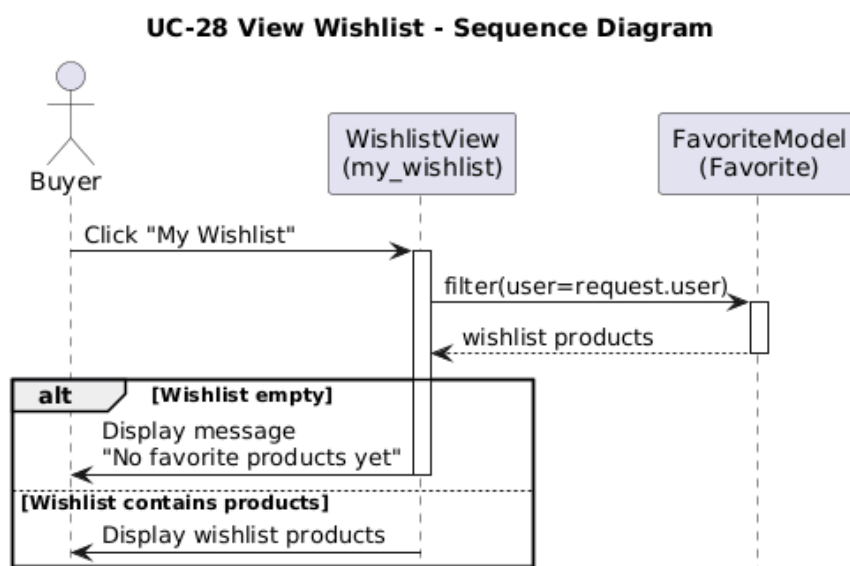
- Adding products to the wishlist (UC-27) Sequence diagram:

Figure 71 Sequence diagram for adding products to the wishlist



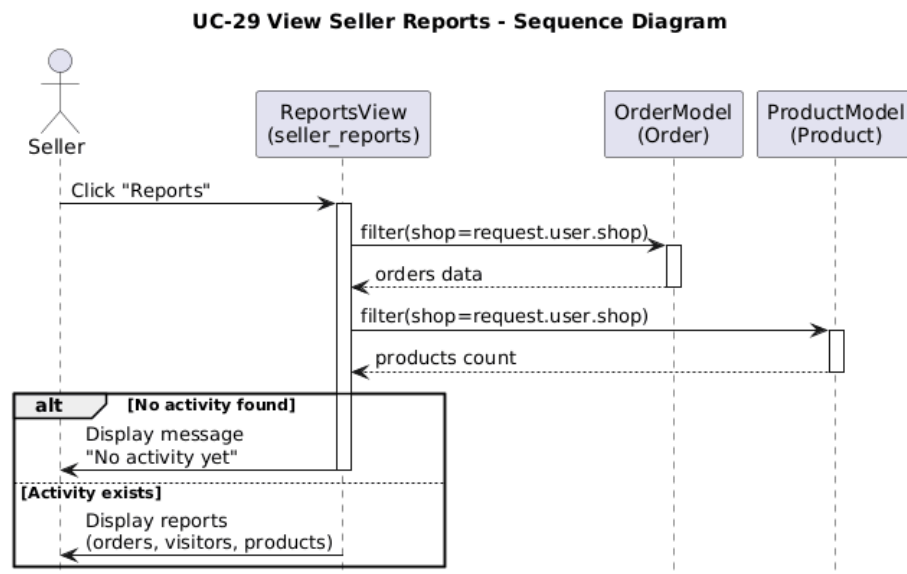
- Viewing the wishlist (UC-28) Sequence diagram:

Figure 72 Sequence diagram for viewing the wishlist



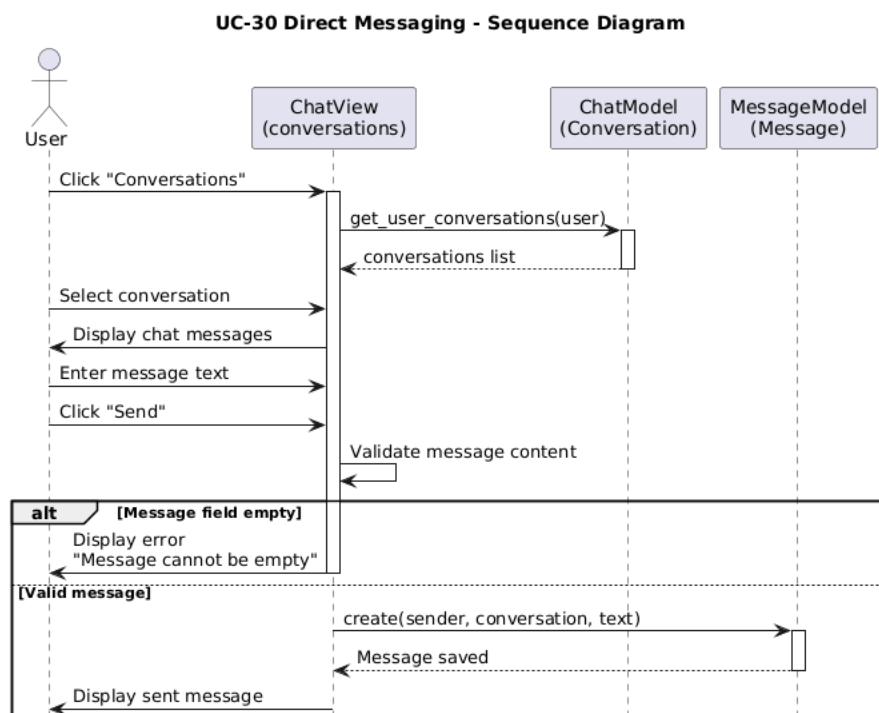
- Viewing reports for the seller (UC-29) Sequence diagram:

Figure 73 Sequence diagram for viewing reports for the seller



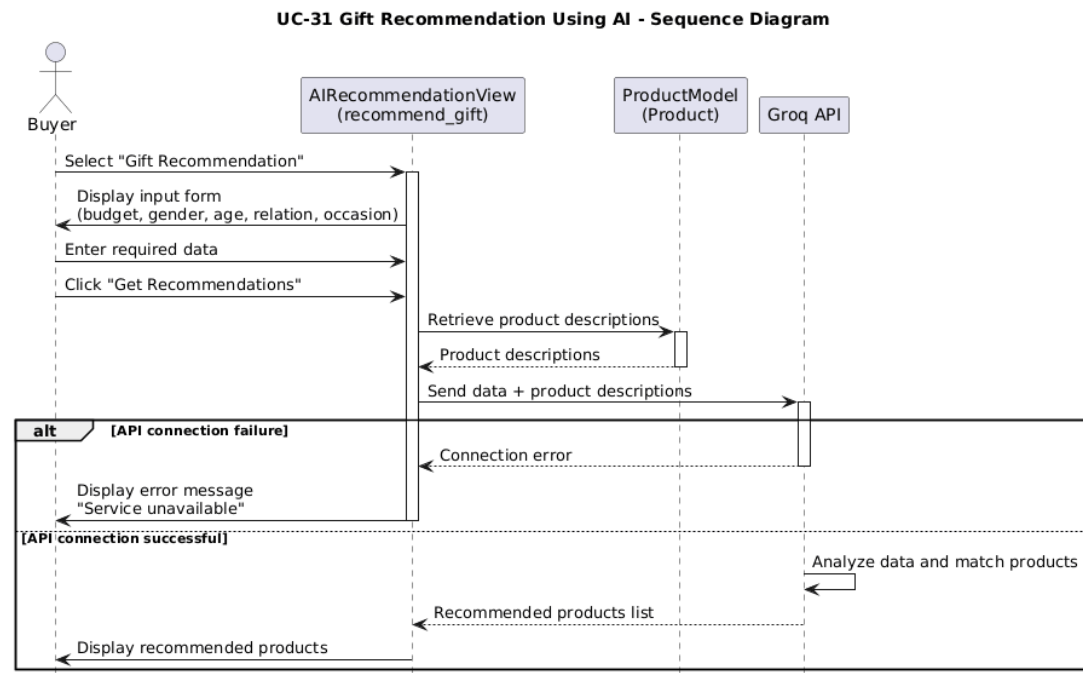
- Direct messaging between seller and buyer (UC-30) Sequence diagram:

Figure 74 Sequence diagram for direct messaging between seller and buyer



- Gift suggestion using artificial intelligence (UC-31) Sequence diagram:

Figure 75 Sequence diagram for gift suggestion using artificial intelligence



5.4 Analysis class diagram

Figure 76 High level class diagram

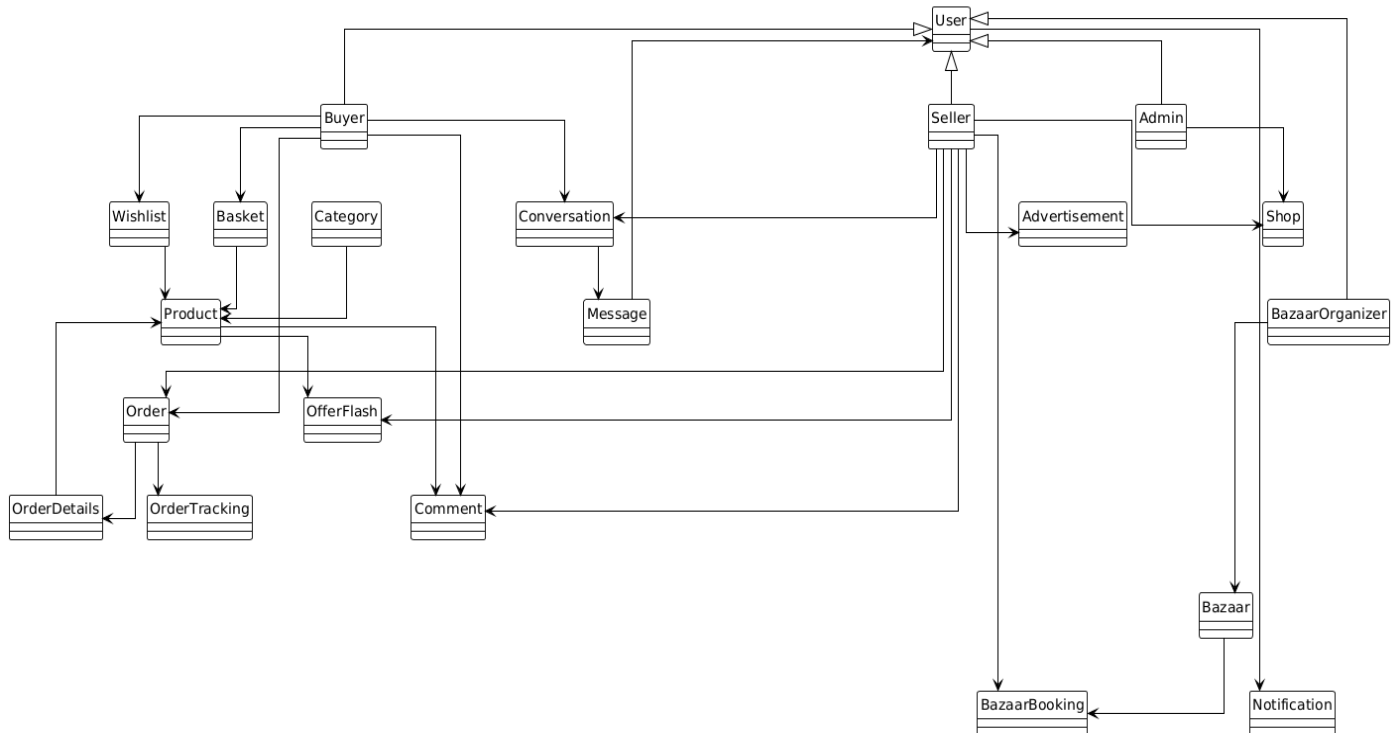
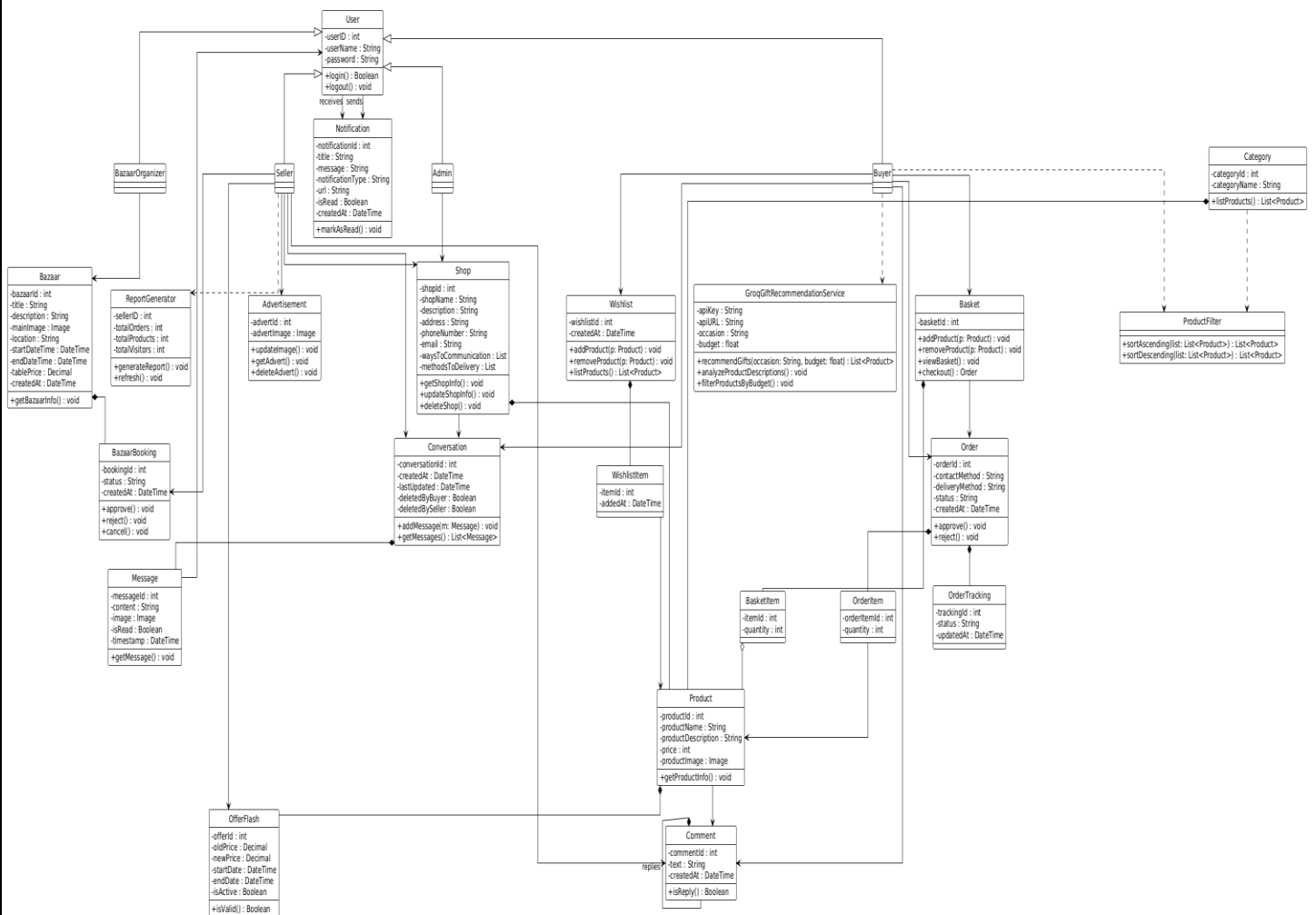
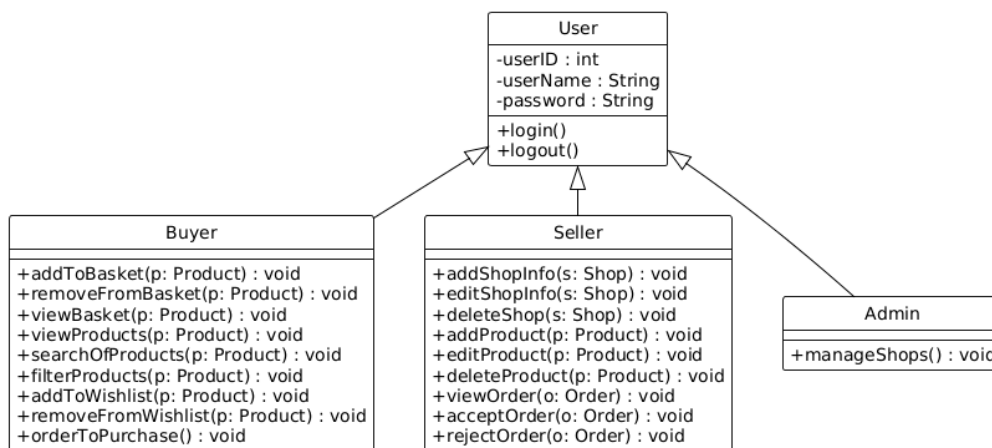


Figure 77 Low level class diagram



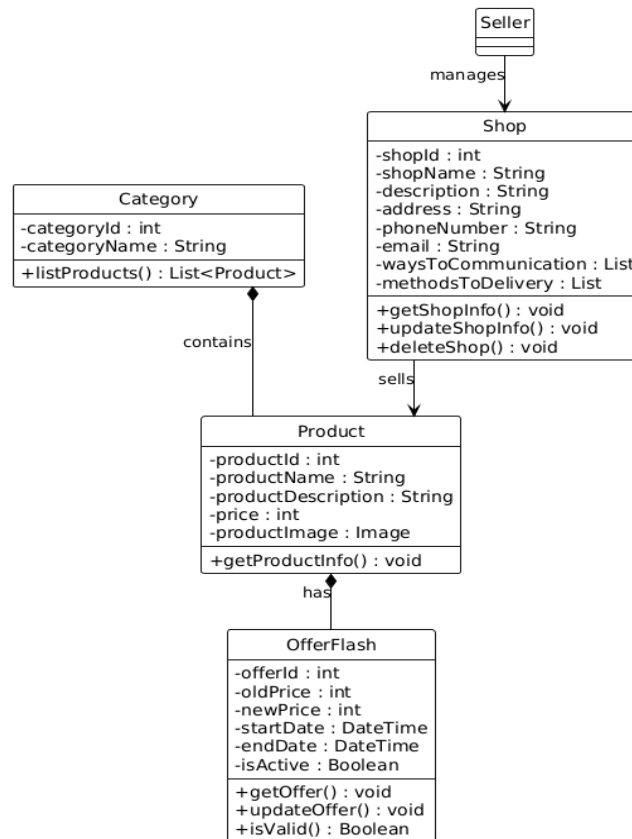
5.4.1 Users class diagram

Figure 78 Users class



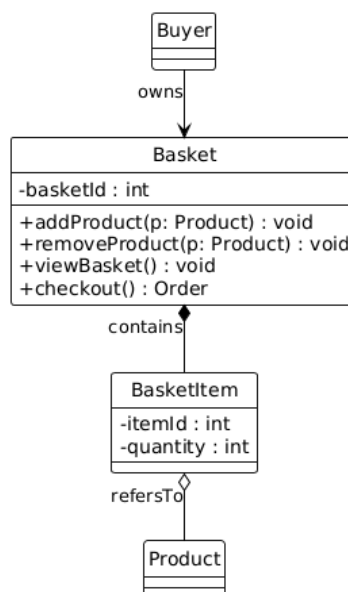
5.4.2 Shop & Products class diagram

Figure 79 Shop & Products class



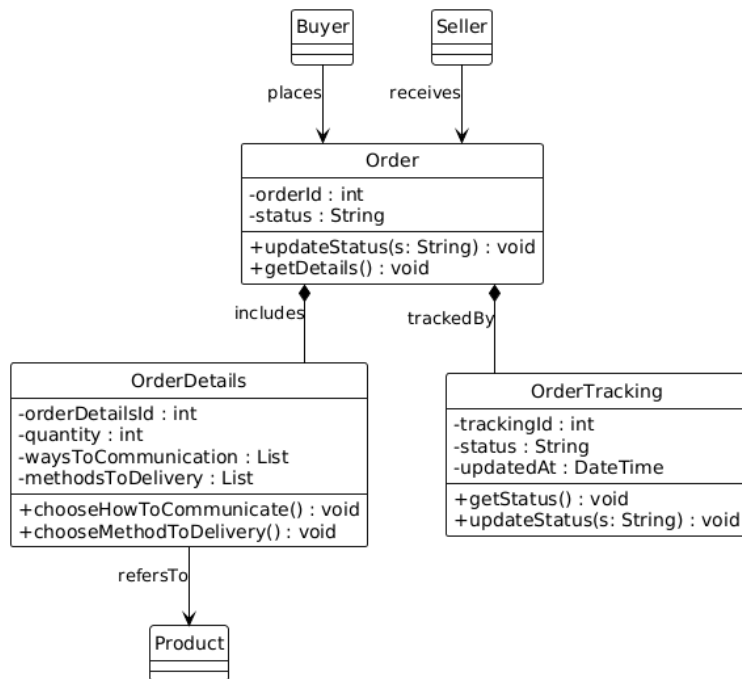
5.4.3 Basket class diagram

Figure 80 Basket class



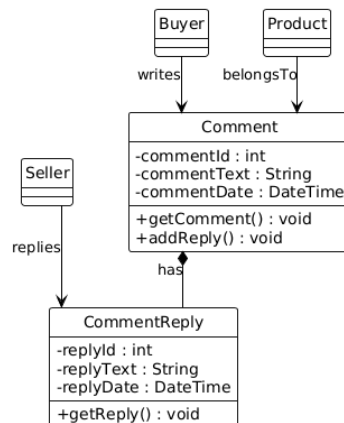
5.4.4 Orders class diagram

Figure 81 Orders class



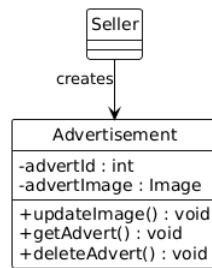
5.4.5 Comments class diagram

Figure 82 Comments class



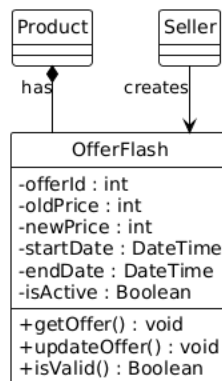
5.4.6 Advertisements class diagram

Figure 83 Advertisements class



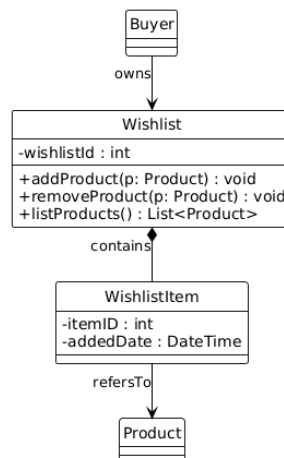
5.4.7 Offer Flash class diagram

Figure 84 Offer Flash class



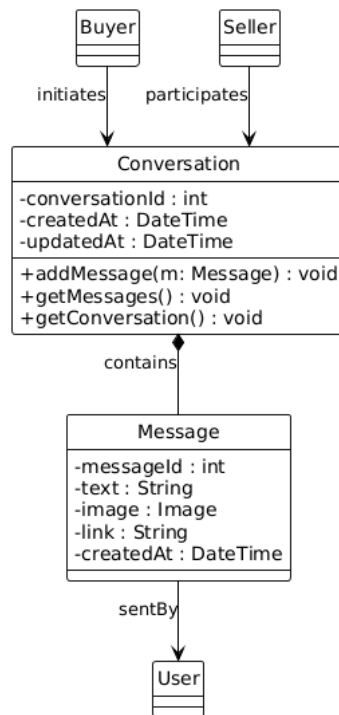
5.4.8 Wishlist class diagram

Figure 85 Wishlist class



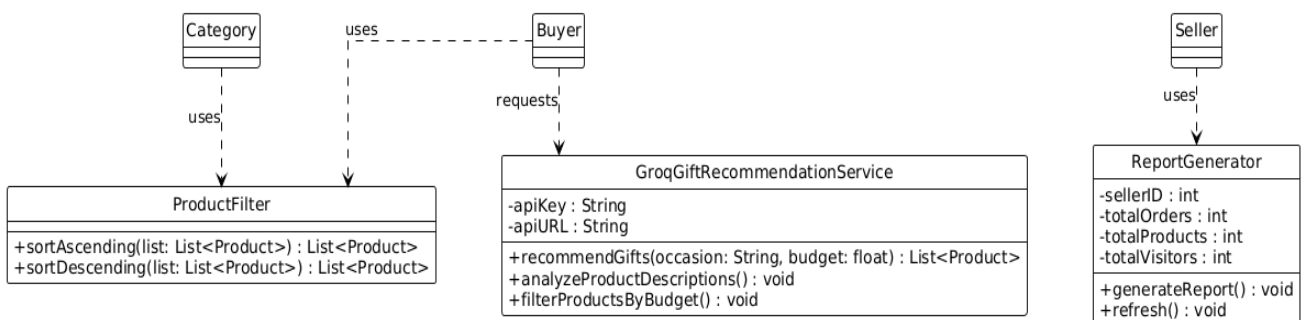
5.4.9 Messages class diagram

Figure 86 Messages class



5.4.10 Service classes diagram

Figure 87 Services class



Note on the Service Classes diagram:

This set of classes represents functional services (Service Classes) within the system; they are not independent entities possessing persistent data or ownership relationships with other objects. These services act as supportive

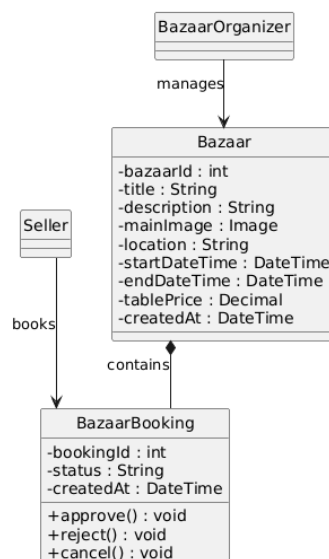
components invoked only when needed, performing specific operations
.without belonging directly to any part of the system's core data structure

This section includes the following classes:

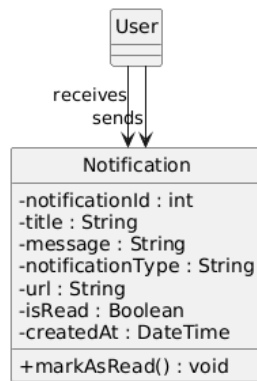
- ProductFilter is responsible for filtering and sorting products according to specific criteria.
- ReportGenerator generates reports for sellers based on performance data.
- GroqGiftRecommenderService an external service for handling intelligent suggestions based on the user's occasion and budget.

The relationships here are not ownership relationships (Composition/Aggregation), but dependency relationships. In other words, some entities like Buyer or Seller only invoke these services when needed to perform certain functions, without any permanent structural linkage to the system.

5.4.11 Bazar class diagram



5.4.12 Notification class diagram



5.5 Entity Relationship (ER) diagram

Figure 88 High level ERD

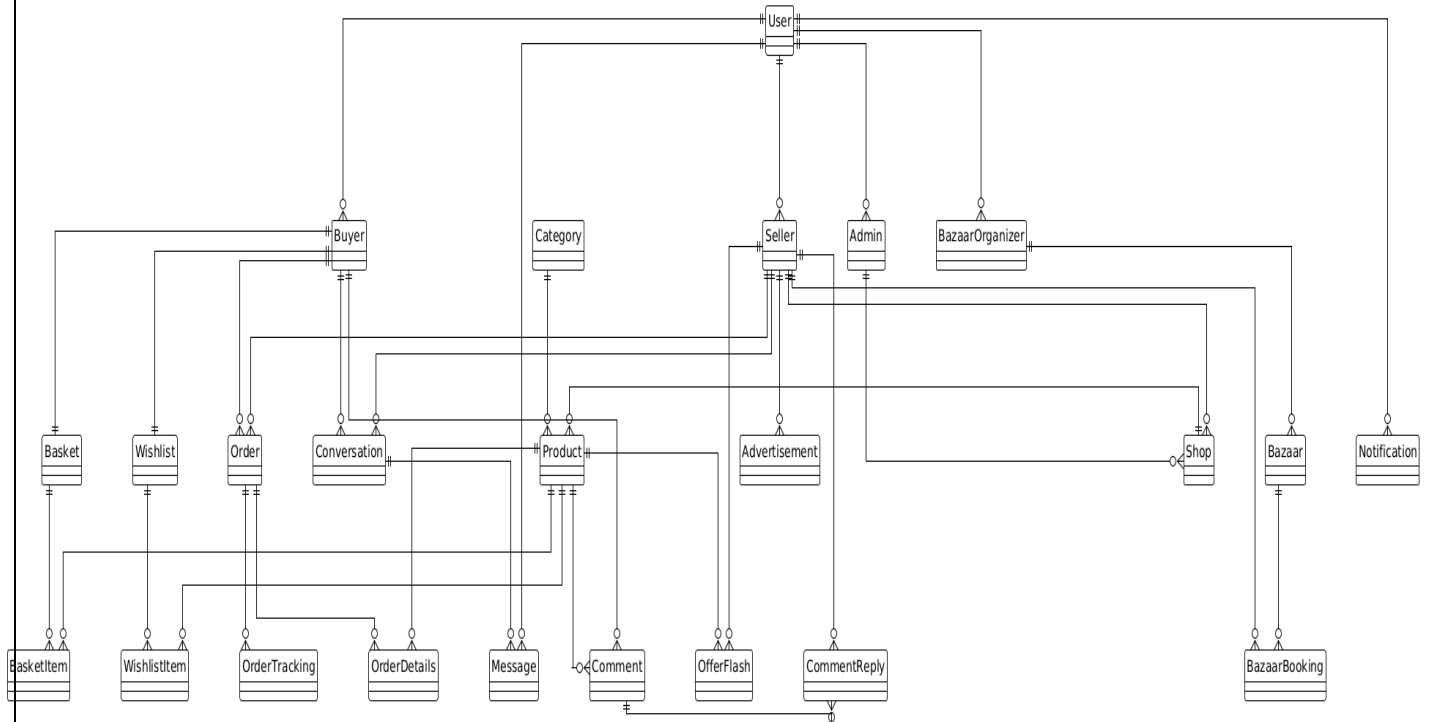
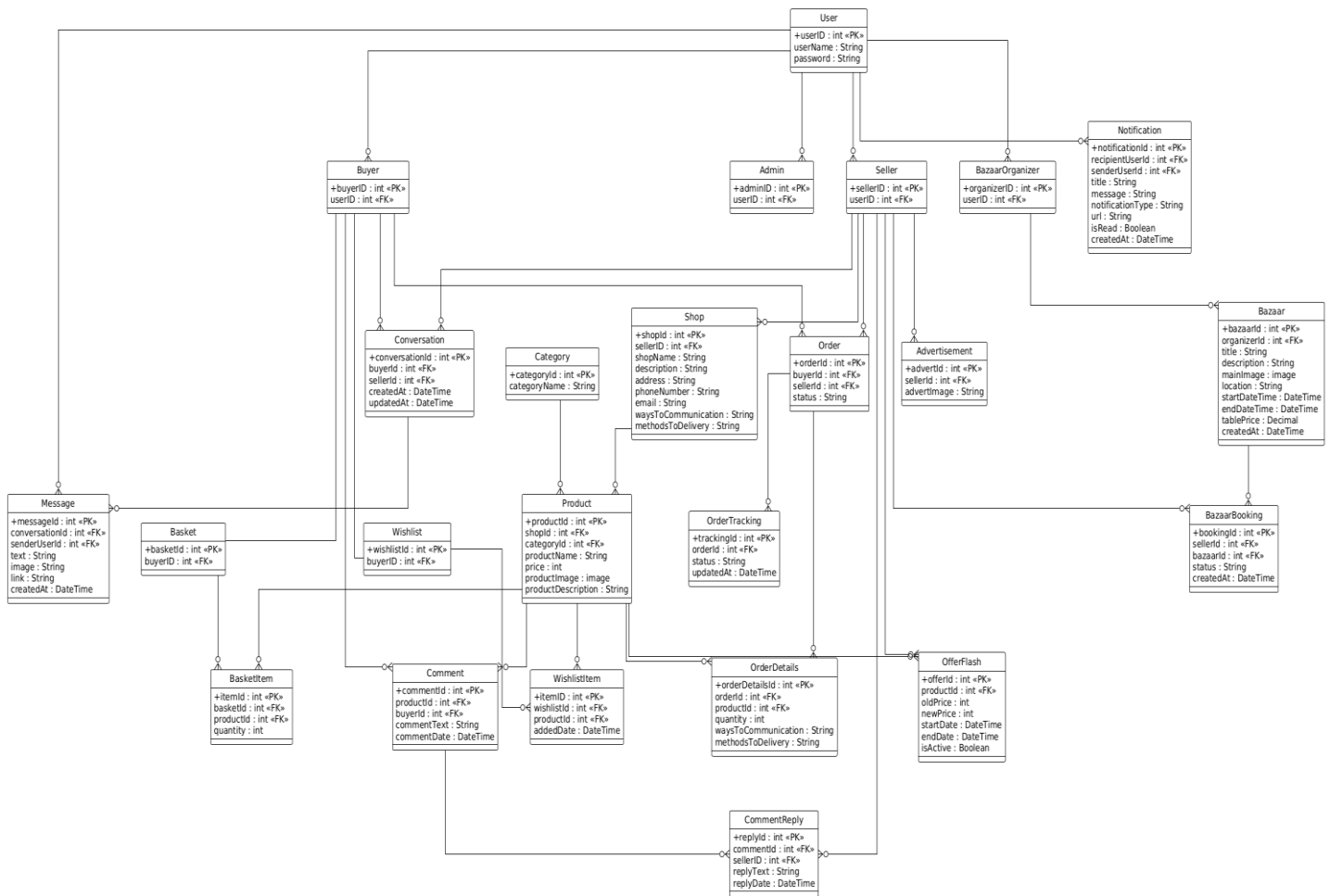


Figure 89 Low level ERD



6. User Stories

User Stories were adopted within the Scrum methodology to flexibly describe system requirements with a focus on user needs and the actual value the system provides. These stories help organize development phases and link functional requirements with sprints, testing, and software implementation.

Account Management:

As a user, I want to create a new account so that I can use the system.

As a user, I want to log into the system so that I can access my account.

As a user, I want to log out of the system so that I can maintain the security of my account.

As a user, I want to edit my profile so that I can update my information.

As a user, I want to upload a profile picture so that I can personalize my account.

As a user, I want to change my password so that I can protect my account.

As a system administrator, I want to manage user accounts so that I can monitor the system.

Shop Management:

As a seller, I want to create a new shop so that I can display my products.

As a seller, I want to edit the shop information so that I can update its data.

As a seller, I want to delete the shop so that I can remove it from the system.

As a seller, I want to view my shop information so that I can monitor it.

As a buyer, I want to browse shops so that I can shop for products.

As a system administrator, I want to view all shops so that I can monitor the system.

As a system administrator, I want to delete a non-compliant shop so that I can maintain platform quality.

Product Management:

As a seller, I want to add a new product so that I can list it for sale.

As a seller, I want to edit product data so that I can update the price and description.

As a seller, I want to delete a product so that I can remove unavailable items.

As a seller, I want to view all my products so that I can manage them.

As a buyer, I want to browse products so that I can select suitable items.

As a buyer, I want to view product details so that I can know its specifications.

As a buyer, I want to see the product image so that I can visually assess it.

As a buyer, I want to know the product price so that I can compare between products.

Category Management:

As a seller, I want to create product categories so that I can organize them.

As a seller, I want to edit categories so that I can update their names.

As a seller, I want to delete unused categories so that I can manage the shop.

As a buyer, I want to browse products by category so that I can find products easily.

As a buyer, I want to view all categories so that I can navigate within the system.

Search and Filter by Price:

As a buyer, I want to search for products so that I can find specific items.

As a buyer, I want to filter products by category so that I can reach suitable products.

As a buyer, I want to sort products by price so that I can compare them.

As a buyer, I want to search within a specific shop so that I can find particular products.

Basket Management:

As a buyer, I want to add products to the basket so that I can purchase them later.

As a buyer, I want to remove products from the basket so that I can modify my purchases.

As a buyer, I want to adjust the product quantity in the basket so that I can control the order.

As a buyer, I want to view the basket contents so that I can review the products.

As a buyer, I want to know the total basket price so that I can make a purchase decision.

As a buyer, I want to prevent adding products from different sellers to the same basket

so that the system logic is maintained.

Order Management:

As a buyer, I want to place a purchase order so that I can complete the buying process.

As a buyer, I want to choose a communication method so that I can confirm the order.

As a buyer, I want to view my past orders so that I can track them.

As a buyer, I want to view order details so that I can know the order status.

As a seller, I want to view the orders for my shop so that I can manage them.

As a seller, I want to accept orders so that I can proceed with sales.

As a seller, I want to reject orders so that I can manage the available products.

As a seller, I want to update the order status so that I can track the shipping process.

As a buyer, I want to know the current order status so that I can track the purchase.

Comments Management:

As a buyer, I want to add a comment on a product I purchased so that I can share my opinion.

As a buyer, I want to edit my comment so that I can update my review.

As a buyer, I want to delete my comment so that I can remove it.

As a seller, I want to reply to buyers' comments so that I can improve communication.

As a user, I want to view comments so that I can evaluate product quality.

As a user, I want to see seller responses so that I can understand product details.

Advertisements Management:

As a seller, I want to add a new advertisement so that I can promote my shop.

As a seller, I want to edit the advertisement so that I can update its content.

As a seller, I want to delete the advertisement so that I can remove old ads.

As a buyer, I want to view advertisements so that I can discover featured products.

Flash Offers Management:

As a seller, I want to create a flash offer on a product so that I can increase sales.

As a seller, I want to edit the flash offer so that I can update the price or date.

As a seller, I want to delete the flash offer so that I can terminate it.

As a buyer, I want to view products with offers so that I can benefit from discounts.

WishList Management:

As a buyer, I want to add a product to wishlist so that I can refer to it later.

As a buyer, I want to remove a product from wishlist so that I can manage my list.

As a buyer, I want to view the wishlist list so that I can quickly access products.

Messaging System:

As a buyer, I want to send a message to the seller so that I can inquire about products.

As a seller, I want to reply to buyers' messages so that I can communicate with them.

As a user, I want to view all conversations so that I can follow the messages.

As a user, I want to see the number of unread messages so that I can know new messages.

As a user, I want to prevent other users from accessing my conversations to maintain privacy.

Reports System:

As a seller, I want to view sales reports so that I can analyze shop performance.

As a seller, I want to know the best-selling products so that I can improve marketing.

As a seller, I want to view order statistics so that I can monitor business activity.

AI-based Gift Suggestions Feature:

As a buyer, I want to enter the occasion and budget so that I can get suitable suggestions.

As a buyer, I want to receive product suggestions using AI so that I can choose an appropriate gift.

As a buyer, I want to view the suggested products directly within the system so that I can purchase them.

Bazaar Management:

As a bazaar organizer, I want to add a new bazaar so that I can organize events.

As a bazaar organizer, I want to edit bazaar information so that I can update details.

As a bazaar organizer, I want to delete a bazaar so that I can remove unavailable events.

As a bazaar organizer, I want to view reservation requests so that I can manage them.

As a seller, I want to browse all bazaars so that I can participate in them.

As a seller, I want to view bazaar details so that I can know the location and cost.

Bazaar Table Reservation Management:

As a seller, I want to send a table reservation request so that I can participate in the bazaar.

As a seller, I want to track the reservation request status so that I can know if it is accepted or rejected.

As a seller, I want to cancel the reservation request before the bazaar date so that I can manage my participations.

As a bazaar organizer, I want to approve reservation requests so that I can organize participants.

As a bazaar organizer, I want to reject booking requests so that I can control the number of participants.

Notifications System:

As a user, I want to receive notifications related to my activities so that I stay continuously informed.

As a seller, I want to receive a notification when a new order arrives so that I can follow up.

As a seller, I want to receive a notification when a new comment is added so that I can respond to it.

As a buyer, I want to receive a notification when the order is accepted or rejected so that I can know the order status.

As a buyer, I want to receive a notification when the order status is updated so that I can track the shipment.

As a seller, I want to receive a notification when a bazaar booking is accepted or rejected so that I can follow up on the request.

As a bazaar organizer, I want to receive a notification when a new booking request is sent so that I can review it.

7. Story Mapping

The Story Mapping diagram organizes the system's functions based on the user's journey within the system, where features are grouped into main groups called Epics (user activities), then arranged and prioritized into core features (MVP), medium-priority features, and low-priority features.

Epic (User Activity)	High Priority (MVP)	Medium Priority	Low Priority
User Management	Register, Login, Logout	Edit profile, Change password	-
Shop Management	Create shop, View shops	Edit/Delete shop	-
Product Management	Add products, View products, View product details	Edit/Delete products	Product image improvements
Category Management	Add categories, View categories	Edit/Delete categories	-
Search & Filtering	Search products, Filter by category	Sort products by price	-
Basket Management	Add to basket, View basket	Remove products, Update quantity	-
Orders Management	Create orders, Approve/Reject orders	Track order status, Select contact method	Delivery method enhancements
Comments & Replies	Add comments, View comments	Reply to comments	Edit/Delete comments

Advertisement Management	Add advertisements	Edit/Delete advertisements	-
Flash Offers Management	Create flash offers	Edit/Delete flash offers	-
Wishlist Management	Add to wishlist, View wishlist	Remove from wishlist	-
Messaging System	Send messages, View conversations	Unread messages counter	Attachments support
Reports System	View sales reports	View order statistics	Advanced analytics
AI Recommendation System	Generate AI gift recommendations	Enter budget and occasion details	Advanced recommendation customization
Bazaar Management	Create bazaars, View bazaars	Edit/Delete bazaars	-
Bazaar Booking Management	Request table booking, Approve/Reject requests	Track booking status	Cancel booking requests
Notifications	Receive notifications	Order notifications, Comment notifications	Advanced notification settings
Admin Management	Manage shops	Delete inappropriate shops	-

Table 44 Story Mapping

8. Product Backlog

The following table displays the Product Backlog list prioritized for the ,Syrian Handicrafts Marketing System, including each associated User Story its related Epic, as well as the expected priority level for each feature.

ID	Epic	User Story	Priority
PB-01	User Management	As a user, I want to register so that I can access the system.	High
PB-02	User Management	As a user, I want to log in securely so that I can access my account.	High
PB-03	User Management	As a user, I want to log out so that I can protect my account.	Medium
PB-04	User Management	As a user, I want to update my profile so that I can keep my information current.	Medium
PB-05	Shop Management	As a seller, I want to create a shop so that I can display my products.	High
PB-06	Shop Management	As a seller, I want to edit and delete my shop so that I can manage my store.	Medium
PB-07	Shop Management	As a buyer, I want to view shops so that I can browse products.	Medium

PB-08	Product Management	As a seller, I want to add products so that I can sell them online.	High
PB-09	Product Management	As a seller, I want to edit and delete products so that I can manage my inventory.	High
PB-10	Product Management	As a buyer, I want to view products so that I can purchase suitable items.	High
PB-11	Product Management	As a buyer, I want to view detailed product information so that I can make informed decisions.	High
PB-12	Category Management	As a seller, I want to manage categories so that I can organize products.	Medium
PB-13	Search & Filtering	As a buyer, I want to search products so that I can quickly find products.	High
PB-14	Search & Filtering	As a buyer, I want to filter products by category and price so that I can compare products.	High
PB-15	Basket Management	As a buyer, I want to add products to the basket so that I can purchase them later.	High

PB-16	Basket Management	As a buyer, I want to remove products from the basket so that I can update my order.	Medium
PB-17	Orders Management	As a buyer, I want to create purchase orders so that I can complete purchases.	High
PB-18	Orders Management	As a seller, I want to approve or reject orders so that I can manage purchases.	High
PB-19	Orders Management	As a seller, I want to update order status so that I can track delivery progress.	High
PB-20	Orders Management	As a buyer, I want to track my orders so that I can know the order status.	High
PB-21	Comments Management	As a buyer, I want to add comments on purchased products so that I can share feedback.	Medium
PB-22	Comments Management	As a seller, I want to reply to comments so that I can communicate with customers.	Medium
PB-23	Advertisement Management	As a seller, I want to add advertisements so that I can promote products.	Medium

PB-24	Flash Offers Management	As a seller, I want to create flash offers so that I can attract buyers.	Medium
PB-25	Wishlist Management	As a buyer, I want to add products to my wishlist so that I can save them for later.	Medium
PB-26	Wishlist Management	As a buyer, I want to remove products from my wishlist so that I can manage saved items.	Low
PB-27	Messaging System	As a buyer, I want to send messages to sellers so that I can ask about products.	High
PB-28	Messaging System	As a seller, I want to reply to buyers so that I can communicate with customers.	High
PB-29	Messaging System	As a user, I want to view my conversations so that I can track messages.	Medium
PB-30	Reports System	As a seller, I want to view sales reports so that I can analyze store performance.	Medium
PB-31	AI Recommendation System	As a buyer, I want AI gift suggestions so that I can find suitable gifts.	High

PB-32	AI Recommendation System	As a buyer, I want to enter budget and occasion details so that AI can recommend products.	Medium
PB-33	Bazaar Management	As a bazaar organizer, I want to create bazaars so that I can organize events.	High
PB-34	Bazaar Management	As a bazaar organizer, I want to edit and delete bazaars so that I can manage events.	Medium
PB-35	Bazaar Management	As a seller, I want to browse bazaars so that I can participate in them.	Medium
PB-36	Bazaar Booking Management	As a seller, I want to request a table booking so that I can join bazaars.	High
PB-37	Bazaar Booking Management	As a seller, I want to track booking requests so that I can know their status.	High
PB-38	Bazaar Booking Management	As a bazaar organizer, I want to approve or reject booking requests so that I can manage participants.	High
PB-39	Bazaar Booking Management	As a seller, I want to cancel a booking request so that I can	Medium

		manage my participation.	
PB-40	Notifications	As a user, I want to receive notifications so that I can stay informed.	High
PB-41	Notifications	As a buyer, I want to receive order status notifications so that I can follow my purchases.	High
PB-42	Notifications	As a seller, I want to receive notifications about comments and orders so that I can manage my store.	High
PB-43	Notifications	As a bazaar organizer, I want to receive booking notifications so that I can manage requests.	Medium
PB-44	Admin Management	As an admin, I want to manage shops so that I can monitor the system.	Medium
PB-45	Admin Management	As an admin, I want to delete inappropriate shops so that I can maintain platform quality.	Low

Table 45 Product Backlog

9. Initial Test Cases

Table 46 Test cases for account management

Initial Test Cases – Account Management

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-01	Login with valid credentials	1- Open the website 2- Navigate to the login page 3- Enter a valid username and password 4- Click the login button	Successfully log in and be directed to the appropriate dashboard
TC-02	Login with incorrect password	1- Open the website 2- Go to the login page 3- Enter a valid username and incorrect password 4- Click login	An error message appears indicating invalid login credentials
TC-03	Login with invalid username	1- Open the website 2- Go to the login page 3- Enter a non-existent username 4- Click the login button	An error message appears indicating invalid login credentials
TC-04	Login leaving fields empty	1- Open the website 2- Go to the login page 3- Leave the fields empty 4- Click the login button	A message appears requesting all required fields to be filled
TC-05	Login with three consecutive failed attempts	1- Open the website 2- Go to the login page 3- Enter incorrect credentials three times 4- Click the login button	The user is temporarily blocked with a message to try again later

TC-06	Logout from account	1- Successfully log in 2- Click the logout button	User is logged out and redirected to the login page
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Table 47 Test cases for shop management

Initial Test Cases – Shop Management

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-07	Create a new shop by the seller	1- Open the website 2- Log in as a seller 3- Choose to create a shop 4- Enter all required information 5- Click save	The shop is successfully created and displayed on the seller's page
TC-08	Creating a shop with an empty field	1- Open the website 2- Log in as a seller 3- Select Create Shop 4- Leave one of the fields empty 5- Click Save	A message appears requesting to enter all required information
TC-09	Edit shop information by the seller	1- Log in as a seller 2- Select Manage Shop 3- Click Edit 4- Update the information 5- Click Save Changes	Shop information is updated successfully
TC-10	Edit shop information with	1- Log in as a seller 2- Select Manage Shop 3- Click Edit 4- Leave one of the fields	A message appears requesting to enter

	leaving a field empty	empty 5- Click Save	all required information
TC-11	Delete shop by the seller	1- Log in as a seller 2- Select Manage Shop 3- Click Delete Shop 4- Confirm deletion	The shop is successfully deleted from the system
TC-12	Create a new shop by the administrator	1- Log in as an administrator 2- Select Create Shop 3- Specify the seller 4- Enter shop information 5- Click Save	A new shop is created and linked to the specified seller
TC-13	Delete shop by the administrator	1- Log in as an administrator 2- Display shop list 3- Select a shop 4- Click Delete 5- Confirm deletion	The shop is successfully deleted from the system

Table 48 Test cases for product management

Initial Test Cases – Product Management

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-14	Add a new product by the seller	1- Open the website 2- Log in as a seller 3- Select Manage Products 4- Click Add Product 5- Enter all product	The product is added successfully and appears among the shop's products

		information 6- Click Save	
TC-15	Add a product with leaving a field empty	1- Log in as a seller 2- Manage Products 3- Add a new product 4- Leave one of the fields empty 5- Click Save	A message appears requesting to enter all required information
TC-16	Add an existing product	1- Log in as a seller 2- Manage Products 3- Add product 4- Enter data of an existing product 5- Click Save	A message appears indicating that the product already exists
TC-17	Edit product information by the seller	1- Log in as a seller 2- Manage Products 3- Select a product 4- Click Edit 5- Update the information 6- Click Save	Product information is updated successfully
TC-18	Edit product with leaving a field empty	1- Log in as a seller 2- Manage Products 3- Select a product 4- Click Edit 5- Leave one of the fields empty 6- Click Save	A message appears requesting to enter all required information
TC-19	Delete product by the seller	1- Log in as a seller 2- Manage Products 3- Select a product	The product is deleted successfully from the shop

		4- Click Delete 5- Confirm deletion	
TC-20	Unauthorized user attempts to edit a product	1- Log in as a buyer 2- Attempt to access product management 3- Attempt to edit a product	The user is prevented from executing the operation and access is denied

Table 49 Test cases for category management

Initial Test Cases – Category Management

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-21	Add a new category by the seller	1- Open the website 2- Log in as a seller 3- Select Manage Categories 4- Click Add Category 5- Enter category name 6- Click Save	The category is added successfully and appears in the category list
TC-22	Add category leaving the category name empty	1- Log in as Seller 2- Manage Categories 3- Click on Add Category 4- Leave the category name empty 5- Click Save	An alert message requesting to enter the category name appears
TC-23	Add an existing category	1- Log in as Seller 2- Manage Categories 3- Add a new category 4- Enter an already	A message appears indicating the category already exists

		existing category name 5- Click Save	
TC-24	Edit an existing category name	1- Log in as Seller 2- Manage Categories 3- Select a category 4- Click Edit 5- Enter the new name 6- Click Save	The category name is updated successfully
TC-25	Edit category leaving the name empty	1- Log in as Seller 2- Manage Categories 3- Select a category 4- Click Edit 5- Leave the category name empty 6- Click Save	An alert message requesting to enter the category name appears
TC-26	Delete a category by the Seller	1- Log in as Seller 2- Manage Categories 3- Select a category 4- Click Delete 5- Confirm deletion	The category is successfully deleted from the system
TC-27	Attempt to manage categories by unauthorized user	1- Log in as Buyer 2- Attempt to access Manage Categories	User is prevented from accessing Manage Categories

Table 50 Test cases for basket management

Initial Test Cases – Basket Management

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-28	Add product to basket from the same shop	1- Open website 2- Log in as Buyer 3- Browse products 4- Select a product 5- Click Add to Basket	Product is successfully added to the basket
TC-29	Add another product from the same shop to the basket	1- Log in as Buyer 2- Browse products from the same shop 3- Add a second product to the basket	The second product is successfully added to the basket
TC-30	Add product from a different shop to the basket	1- Log in as Buyer 2- Add a product from first shop 3- Attempt to add a product from a different shop	Warning message appears indicating that purchase must be from only one shop
TC-31	Delete product from basket	1- Log in as Buyer 2- Go to the basket page 3- Select delete product 4- Confirm deletion	Product is successfully deleted from the basket
TC-32	View basket contents	1- Log in as Buyer 2- Go to the basket page	All products in the basket are displayed
TC-33	Access basket without login	1- Open website 2- Attempt to access the basket page directly	Redirected to the login page

TC-34	Attempt to manage the basket by unauthorized user	1- Log in as Seller 2- Attempt to add a product to the basket	The user is prevented from performing the operation
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Table 51 Test cases for orders management

Initial Test Cases – Orders Management

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-35	Creating a Purchase Order by the Buyer	1- Log in as a buyer 2- Go to the basket 3- Click confirm order 4- Choose communication method 5- Choose delivery method 6- Confirm the order	The order is successfully created and saved with status Under Review
TC-36	Attempt to create an order with an empty basket	1- Log in as a buyer 2- Go to the basket 3- Click confirm order	A message appears indicating there are no products in the basket
TC-37	Accepting the Purchase Order by the Seller	1- Log in as a seller 2- View orders 3- Select an order 4- Click accept order 5- Confirm acceptance	The order is accepted and its status is updated to Accepted
TC-38	Rejecting the Purchase Order by the Seller	1- Log in as a seller 2- View orders 3- Select an order	The order is rejected and its status is updated to Rejected

		4- Click reject order 5- Confirm rejection	
TC-39	Viewing Order Status by the Buyer	1- Log in as a buyer 2- Go to the orders page	The order is displayed with its current status
TC-40	Update Order Status to Being Prepared	1- Log in as a seller 2- Select an accepted order 3- Click update order status 4- Choose Being Prepared	The order status is successfully updated to Being Prepared
TC-41	Update Order Status to Prepared	1- Log in as a seller 2- Select an order Being Prepared 3- Update status to Prepared	The order status is updated to Prepared
TC-42	Update Order Status to Out for Delivery	1- Log in as a seller 2- Select a prepared order 3- Update status to Out for Delivery	The order status is updated to Out for Delivery
TC-43	Update Order Status to Delivered	1- Log in as a seller 2- Select an order Out for Delivery 3- Update status to Delivered	The order status is updated to Delivered and the order is closed
TC-44	Attempt to update the status of a non-accepted order	1- Log in as a seller 2- Select a Rejected or Under Review order	The operation is prevented with an explanatory message

		3- Attempt to update its status	
TC-45	Attempt to track an order by an unauthorized user	1- Log in as a seller 2- Attempt to track an order not belonging to their shop	The user is prevented from accessing the order

Table 52 Test cases for search and filtering

Initial Test Cases – Search & Filtering (Category & Price Only)

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-46	Display Products by Selected Category	1- Open the products page 2- Select a specific category from the category list	Only products belonging to the selected category are displayed
TC-47	Displaying products within a category that contains no products	1- Open the products page 2- Select an empty category	A message appears indicating that there are no products in this category
TC-48	Filtering products by price from lowest to highest	1- Open the products page 2- Select filtering by price from lowest to highest	Products are sorted in ascending order by price
TC-49	Filtering products by price from highest to lowest	1- Open the products page 2- Select filtering by price from highest to lowest	Products are sorted in descending order by price
TC-50	Applying price filtering within a specific category	1- Open the products page 2- Select a specific category 3- Choose price filtering	Products of the selected category are displayed sorted by price

Table 53 Test cases for comments management

Initial Test Cases – Comments Management

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-51	Adding a comment after purchasing and receiving the product	1- Log in as a buyer 2- Navigate to the completed order page 3- Open the product page 4- Write a comment 5- Click send	The comment is successfully added and displayed below the product
TC-52	Attempting to add a comment without purchasing the product	1- Log in as a buyer 2- Open a product page that was not purchased 3- Write a comment 4- Click send	Adding the comment is prevented with an explanatory message
TC-53	Attempting to add an empty comment	1- Log in as a buyer 2- Open a purchased product page 3- Leave the comment field empty 4- Click send	A message appears requesting to enter comment text
TC-54	Displaying comments related to a product	1- Open a product page	All comments associated with the product are displayed
TC-55	Seller replying to a comment	1- Log in as a seller 2- Open the product page 3- Select a comment	The seller's reply is successfully added below the comment

		4- Write a reply 5- Click send	
TC-56	Attempting for the seller to reply with an empty comment	1- Log in as a seller 2- Open the product page 3- Select a comment 4- Leave the reply empty 5- Click send	A message appears requesting to enter reply text
TC-57	Attempting to add a reply by an unauthorized user	1- Log in as a buyer 2- Attempt to reply to a comment	The user is prevented from performing the operation

Table 54 Test cases for advertisements management

Initial Test Cases – Advertisements Management

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-58	Adding a new advertisement by the seller	1- Log in as a seller 2- Navigate to advertisement management 3- Click add advertisement 4- Enter advertisement data and attach an image 5- Click save	The advertisement is successfully added and displayed within the advertisements list
TC-59	Adding an advertisement without attaching an image	1- Log in as a seller 2- Manage advertisements 3- Add a new advertisement 4- Leave the image field	A message appears requesting to attach an image to the advertisement

		empty 5- Click save	
TC-60	Edit an existing advertisement by the seller	1- Log in as a seller 2- Navigate to Advertisement Management 3- Select an advertisement 4- Click Edit 5- Update the data 6- Click Save	The advertisement data is updated successfully
TC-61	Edit an advertisement leaving a mandatory field empty	1- Log in as a seller 2- Advertisement Management 3- Select an advertisement 4- Click Edit 5- Leave one of the fields empty 6- Click Save	A message appears requesting all required fields to be filled
TC-62	Delete an advertisement by the seller	1- Log in as a seller 2- Advertisement Management 3- Select an advertisement 4- Click Delete 5- Confirm deletion	The advertisement is successfully deleted from the system
TC-63	Attempt to manage advertisements by an unauthorized user	1- Log in as a buyer 2- Attempt to access Advertisement Management	The user is prevented from accessing Advertisement Management

Table 55 Test cases for flash offers management

Initial Test Cases – Flash Offers Management

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-66	Add a new flash offer by the seller	1- Log in as a seller 2- Navigate to Flash Offers Management 3- Click Add Offer 4- Select the product 5- Enter the new price and offer duration 6- Click Save	The flash offer is created successfully and activated
TC-67	Add a flash offer leaving a mandatory field empty	1- Log in as a seller 2- Flash Offers Management 3- Add a new offer 4- Leave one of the mandatory fields empty 5- Click Save	A message appears requesting to enter all required information
TC-68	Add a flash offer with an invalid price	1- Log in as a seller 2- Flash Offers Management 3- Add an offer 4- Enter a price less than zero or higher than the original price 5- Save	Offer creation is rejected with an explanatory message
TC-69	Edit an existing flash offer	1- Log in as a seller 2- Flash Offers Management 3- Select an offer	The flash offer is updated successfully

		4- Modify the price or duration 5- Save changes	
TC-70	Delete a flash offer by the seller	1- Log in as a seller 2- Flash Offers Management 3- Select an offer 4- Click Delete 5- Confirm deletion	The flash offer is successfully deleted from the system
TC-71	Automatic expiration of flash offer duration	1- An active flash offer exists 2- The specified offer duration has ended	The offer is stopped automatically and the product returns to its original price
TC-72	Attempt to manage flash offers by an unauthorized user	1- Log in as a buyer 2- Attempt to add or modify a flash offer	The user is prevented from performing the operation

Table 56 Test cases for wishlist management

Initial Test Cases – Wishlist Management

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-73	Add a product to the wishlist	1- Open the website 2- Log in as a buyer 3- Browse products 4- Select a product 5- Click the Add to Wishlist icon	The product is successfully added to the wishlist list
TC-74	Remove a product from the wishlist	1- Log in as a buyer 2- Open the wishlist list	The product is successfully

		3- Select a product 4- Click Remove from Wishlist	removed from the wishlist
TC-75	Display Favorite List	1- Log in as a buyer 2- Navigate to the WishList page	All products added to the WishList are displayed
TC-76	Display Favorite List When Empty	1- Log in as a buyer 2- Navigate to the Favorite List page without adding products	A message indicating that the Favorite List is empty appears
TC-77	Attempt to Use Favorite List Without Logging In	1- Open the website 2- Attempt to add a product to the Favorite List	Redirected to the login page
TC-78	Attempt to Manage Favorite List by Unauthorized User	1- Log in as a seller 2- Attempt to add a product to the Favorite List	The user is prevented from performing the operation

Table 57 Test cases for reports system

Initial Test Cases – Reports System

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-79	Display Orders Report by Seller	1- Log in as a seller 2- Navigate to the Reports system 3- Select the Orders Report	The Orders Report for the seller's shop is displayed

TC-80	Display Orders Report When No Orders Exist	1- Log in as a seller 2- Navigate to the Reports system 3- Select the Orders Report when no data exists	A message indicating no data is available to display appears
TC-81	Attempt to Access Reports System by Unauthorized User	1- Log in as a buyer 2- Attempt to access the Reports page	User is prevented from accessing the Reports system

Table 58 Test cases for messaging system

Initial Test Cases – Messaging System

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-82	Send Valid Text Message	1- Log in as a user 2- Open a chat 3- Write a message 4- Click send	The message is sent successfully and appears in the chat
TC-83	Attempt to Send Empty Message	1- Log in as a user 2- Open a chat 3- Leave the message field empty 4- Click send	An alert message requesting to enter the message text appears
TC-84	Display Chat History	1- Log in as a user 2- Navigate to the Messages page	All user's chats are displayed
TC-85	Attempt to Access Chat Not Belonging to User	1- Log in as a user 2- Attempt to open a chat not belonging to them	User is prevented from accessing the chat

Table 59 Test cases for AI gift recommendation feature

Initial Test Cases – AI Gift Recommendation

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-86	Get Gift Suggestion Using Artificial Intelligence	1- Log in as a buyer 2- Navigate to Gift Suggestion feature 3- Enter the required data 4- Click suggest	A list of suggested products is displayed
TC-87	Attempt to Use Feature with Incomplete Data	1- Log in as a buyer 2- Open Gift Suggestion feature 3- Leave some fields empty 4- Click suggest	A message requesting all required data to be entered appears
TC-88	Failure to connect to the AI service	1- Log in as a buyer 2- Enter valid data 3- Attempt to get a suggestion when the connection fails	A message appears indicating that the service is currently unavailable

Table 60 Test cases for bazaar management

Initial Test Cases – Bazaar Management

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-89	Create a new bazaar by the bazaar organizer	1- Log in as a bazaar organizer 2- Navigate to Bazaar Management	The bazaar is created successfully and displayed in the bazaar list

		3- Click Add Bazaar 4- Enter all bazaar details 5- Click Save	
TC-90	Create a bazaar leaving a mandatory field empty	1- Log in as a bazaar organizer 2- Manage Bazaars 3- Add a new bazaar 4- Leave a mandatory field empty 5- Click Save	A message appears requesting to enter all required information
TC-91	Edit an existing bazaar's data	1- Log in as a bazaar organizer 2- Manage Bazaars 3- Select a bazaar 4- Click Edit 5- Update the data 6- Click Save	The bazaar data is updated successfully
TC-92	Delete a bazaar by the bazaar organizer	1- Log in as a bazaar organizer 2- Manage Bazaars 3- Select a bazaar 4- Click Delete 5- Confirm deletion	The bazaar is successfully deleted from the system
TC-93	Attempt to manage bazaars by an unauthorized user	1- Log in as a buyer or seller 2- Attempt to access Bazaar Management	The user is prevented from accessing Bazaar Management

Table 61 Test cases for bazaar table booking management

Initial Test Cases – Bazaar Table Booking Management

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-94	Send a table booking request in a bazaar	1- Log in as a seller 2- Browse bazaars 3- Select a bazaar 4- Click Book Table 64- Confirm booking	The booking request is sent successfully with status Pending Review
TC-95	Attempt to book more than one table in the same bazaar	1- Log in as a seller 2- Submit first booking request 3- Attempt to submit another booking request for the same bazaar	A message appears indicating that booking more than one table in the same bazaar is not allowed
TC-96	Accept booking request by the bazaar organizer	1- Log in as a bazaar organizer 2- Review booking requests 3- Select a request 4- Click Accept	The booking request is accepted and its status is updated to Accepted
TC-97	Reject booking request by the bazaar organizer	1- Log in as a bazaar organizer 2- Review booking requests 3- Select a request 4- Click Reject	The booking request is rejected and its status is updated to Rejected
TC-98	Cancel booking request before the bazaar date	1- Log in as a seller 2- Go to booking requests 3- Select booking	The booking request is cancelled successfully

		request 4- Click Cancel	
TC-99	Attempt to manage booking requests by an unauthorized user	1- Log in as a buyer 2- Attempt to access booking requests management	The user is prevented from performing the operation

Table 62 Test cases for notifications system

Initial Test Cases – Notifications System

Test Case ID	Test Case Title	Test Steps	Expected Result
TC-100	Receive notification upon order acceptance	1- Seller accepts purchase order 2- Buyer navigates to notifications page	A new notification is displayed to the buyer indicating order acceptance
TC-101	Receive notification upon purchase order rejection	1- Purchase order rejected by the seller 2- Buyer navigates to the notifications page	A new notification is displayed to the buyer indicating the order rejection
TC-102	Receive notification upon order status update	1- Order status updated by the seller 2- Buyer navigates to the notifications page	A new notification is displayed showing the updated order status
TC-103	Receive notification upon bazaar reservation acceptance	1- Bazaar reservation request accepted by the organizer 2- Seller navigates to the notifications page	A new notification is displayed to the seller confirming the reservation acceptance

TC-104	Receive notification upon bazaar reservation rejection	1- Bazaar reservation request rejected by the organizer 2- Seller navigates to the notifications page	A new notification is displayed to the seller indicating the reservation rejection
TC-105	Display the count of unread notifications	1- Presence of new unread notifications 2- Open the control panel	The number of unread notifications is displayed next to the notifications icon
TC-106	Display notifications page	1- Log in as a user 2- Navigate to the notifications page	All notifications related to the user are displayed

10. Initial Requirement Trackability Matrix (RTM)

Req-ID	Requirement Title	Associated Dependency	SRS Section	High Level Design	Detailed Design	Coding	Component Test	Integration Test	System Test	Test Case ID(s)	Changed Requests No
REQ-1.1	The system must allow the user to create an account.	Account Management	§3 SRS – REQ-1.1 (p. 74)	-	-	-	-	-	-		
REQ-1.2	The system must allow the user to log in using a username and password.	Account Management	§3 SRS – REQ-1.2 (p. 75)	-	-	-	-	-	-		
REQ-1.3	The system must allow the user to log out of their account.	Account Management	§3 SRS – REQ-1.3 (p. 75)	-	-	-	-	-	-		
REQ-2.1	The system must allow the seller to add their shop information.	Shop Management	§3 SRS – REQ-2.1 (p. 75)	-	-	-	-	-	-		
REQ-2.2	The system must allow the seller to edit their shop information.	Shop Management	§3 SRS – REQ-2.2 (p. 75)	-	-	-	-	-	-		
REQ-2.3	The system must allow the seller to delete their shop.	Shop Management	§3 SRS – REQ-2.3 (p. 75)	-	-	-	-	-	-		
REQ-2.4	The system must allow the administrator to create a shop for any seller when needed.	Shop Management	§3 SRS – REQ-2.4 (p. 75)	-	-	-	-	-	-		
REQ-2.5	The system must allow the administrator to delete a shop.	Shop Management	§3 SRS – REQ-2.5 (p. 75)	-	-	-	-	-	-		
REQ-2.6	The system must allow the administrator to View registered shops.	Shop Management	§3 SRS – REQ-2.6 (p. 75)	-	-	-	-	-	-		
REQ-2.7	The system allows displaying all products belonging to a specific shop on a dedicated shop page.	Shop Management	§3 SRS – REQ-2.7 (p. 75)	-	-	-	-	-	-		
REQ-3.1	The system must allow the seller to add new products.	Product Management	§3 SRS – REQ-3.1 (p. 76)	-	-	-	-	-	-		
REQ-3.2	The system must allow the seller to edit product information.	Product Management	§3 SRS – REQ-3.2 (p. 76)	-	-	-	-	-	-		
REQ-3.3	The system must allow the seller to delete products.	Product Management	§3 SRS – REQ-3.3 (p. 76)	-	-	-	-	-	-		
REQ-3.4	The system must display the seller's products on their personal shop page.	Product Management	§3 SRS – REQ-3.4 (p. 76)	-	-	-	-	-	-		
REQ-3.5	The system must allow the seller to classify products into specific categories.	Product Management	§3 SRS – REQ-3.5 (p. 76)	-	-	-	-	-	-		
REQ-3.6	The system must allow the buyer to view products.	Product Management	§3 SRS – REQ-3.6 (p. 76)	-	-	-	-	-	-		
REQ-3.7	The system must allow the buyer to view product details.	Product Management	§3 SRS – REQ-3.7 (p. 76)	-	-	-	-	-	-		
REQ-4.1	The system must allow the buyer to add products to the basket.	Basket Management	§3 SRS – REQ-4.1 (p. 76)	-	-	-	-	-	-		
REQ-4.2	The system must allow the buyer to remove products from the basket.	Basket Management	§3 SRS – REQ-4.2 (p. 76)	-	-	-	-	-	-		

REQ-4.3	The system must allow the buyer to view the contents of the basket before confirming the order.	Basket Management	§3 SRS – REQ-4.3 (p. 76)	-	-	-	-	-	-	-	-
REQ-5.1	The system must allow the seller to view incoming purchase orders to their shop.	Order & Purchase Management	§3 SRS – REQ-5.1 (p. 77)	-	-	-	-	-	-	-	-
REQ-5.2	The system must allow the seller to accept purchase orders.	Order & Purchase Management	§3 SRS – REQ-5.2 (p. 77)	-	-	-	-	-	-	-	-
REQ-5.3	The system must allow the seller to reject purchase orders.	Order & Purchase Management	§3 SRS – REQ-5.3 (p. 77)	-	-	-	-	-	-	-	-
REQ-5.4	The system must allow the buyer to place a purchase order for products in the cart.	Order & Purchase Management	§3 SRS – REQ-5.4 (p. 77)	-	-	-	-	-	-	-	-
REQ-5.5	The system must allow the buyer to choose a communication method.	Order & Purchase Management	§3 SRS – REQ-5.5 (p. 77)	-	-	-	-	-	-	-	-
REQ-5.6	The system must allow the buyer to choose a delivery method.	Order & Purchase Management	§3 SRS – REQ-5.6 (p. 77)	-	-	-	-	-	-	-	-
REQ-5.7	The system must allow the buyer to view their order status (Under Review / Accepted / Rejected).	Order & Purchase Management	§3 SRS – REQ-5.7 (p. 77)	-	-	-	-	-	-	-	-
REQ-5.8	The system must allow the seller to update the order status according to execution stages.	Order & Purchase Management	§3 SRS – REQ-5.8 (p. 77)	-	-	-	-	-	-	-	-
REQ-5.9	The system must allow the buyer to view their order status at any time (Being Prepared / Prepared / Out for Delivery / Delivered).	Order & Purchase Management	§3 SRS – REQ-5.9 (p. 77)	-	-	-	-	-	-	-	-
REQ-5.10	The system must allow the buyer to view their previous order history.	Order & Purchase Management	§3 SRS – REQ-5.10 (p. 77)	-	-	-	-	-	-	-	-
REQ-6.1	The system must allow the buyer to add a comment on a product after a purchase.	Comment Management	§3 SRS – REQ-6.1 (p. 78)	-	-	-	-	-	-	-	-
REQ-6.2	The system must allow the seller and buyer to view comments on products.	Comment Management	§3 SRS – REQ-6.2 (p. 78)	-	-	-	-	-	-	-	-
REQ-6.3	The system must allow the seller to reply to comments on their products.	Comment Management	§3 SRS – REQ-6.3 (p. 78)	-	-	-	-	-	-	-	-
REQ-7.1	The system must allow the buyer to search for products by category.	Search & View	§3 SRS – REQ-7.1 (p. 78)	-	-	-	-	-	-	-	-
REQ-7.2	The system must allow the buyer to filter product results by price from lowest to highest and vice versa.	Search & View	§3 SRS – REQ-7.2 (p. 78)	-	-	-	-	-	-	-	-
REQ-8.1	The system must allow the seller to create a promotional advertisement about their products or shop.	Advertisement Management	§3 SRS – REQ-8.1 (p. 78)	-	-	-	-	-	-	-	-

REQ-8.2	The system must allow the seller to edit advertisement details.	Advertisement Management	§3 SRS – REQ-8.2 (p. 78)	-	-	-	-	-	-			
REQ-8.3	The system must allow the seller to delete unwanted advertisements.	Advertisement Management	§3 SRS – REQ-8.3 (p. 78)	-	-	-	-	-	-			
REQ-8.4	The system must allow the buyer to view all active advertisements.	Advertisement Management	§3 SRS – REQ-8.4 (p. 79)	-	-	-	-	-	-			
REQ-9.1	The system must allow the seller to create a Flash offer on a specific product.	Flash Offers Management	§3 SRS – REQ-9.1 (p. 79)	-	-	-	-	-	-			
REQ-9.2	The system must allow the seller to edit Flash offer details.	Flash Offers Management	§3 SRS – REQ-9.2 (p. 79)	-	-	-	-	-	-			
REQ-9.3	The system must allow the seller to delete a Flash offer.	Flash Offers Management	§3 SRS – REQ-9.3 (p. 79)	-	-	-	-	-	-			
REQ-9.4	The system must allow the buyer to view all Flash offers.	Flash Offers Management	§3 SRS – REQ-9.4 (p. 79)	-	-	-	-	-	-			
REQ-9.5	The system must automatically deactivate Flash offers after the specified period has ended.	Flash Offers Management	§3 SRS – REQ-9.5 (p. 79)	-	-	-	-	-	-			
REQ-10.1	The system must allow the buyer to add products to the favorites list.	WishList Management	§3 SRS – REQ-10.1 (p. 79)	-	-	-	-	-	-			
REQ-10.2	The system must allow the buyer to remove products from the favorites list.	WishList Management	§3 SRS – REQ-10.2 (p. 79)	-	-	-	-	-	-			
REQ-10.3	The system must allow the buyer to view all products added to the favorites list.	WishList Management	§3 SRS – REQ-10.3 (p. 80)	-	-	-	-	-	-			
REQ-10.4	The system must allow the buyer to navigate directly to the product page from the favorites list.	WishList Management	§3 SRS – REQ-10.4 (p. 80)	-	-	-	-	-	-			
REQ-10.5	The system must permanently save the WishList for each user to display it upon subsequent login.	WishList Management	§3 SRS – REQ-10.5 (p. 80)	-	-	-	-	-	-			
REQ-11.1	The system must display the number of received orders for the seller.	Reporting System	§3 SRS – REQ-11.1 (p. 80)	-	-	-	-	-	-			
REQ-11.2	The system must display the number of products listed in the shop for the seller.	Reporting System	§3 SRS – REQ-11.2 (p. 80)	-	-	-	-	-	-			
REQ-11.3	The system must display the number of shop visitors for the seller.	Reporting System	§3 SRS – REQ-11.3 (p. 80)	-	-	-	-	-	-			
REQ-11.4	The system must display general reports about shop activity for the seller.	Reporting System	§3 SRS – REQ-11.4 (p. 80)	-	-	-	-	-	-			
REQ-12.1	The system must allow the creation of a direct conversation between the buyer and seller.	Direct Messaging System	§3 SRS – REQ-12.1 (p. 80)	-	-	-	-	-	-			

REQ-12.2	The system must allow the seller and buyer to send and receive text messages.	Direct Messaging System	§3 SRS – REQ-12.2 (p. 81)	-	-	-	-	-	-	-	-
REQ-12.3	The system must allow the seller and buyer to view previous conversation history.	Direct Messaging System	§3 SRS – REQ-12.3 (p. 81)	-	-	-	-	-	-	-	-
REQ-12.4	The system must allow the seller and buyer to attach images or links within messages.	Direct Messaging System	§3 SRS – REQ-12.4 (p. 81)	-	-	-	-	-	-	-	-
REQ-12.5	The system must display notifications for the seller and buyer when a new message arrives.	Direct Messaging System	§3 SRS – REQ-12.5 (p. 81)	-	-	-	-	-	-	-	-
REQ-13.1	The system must allow the buyer to enter criteria such as budget or occasion type.	Smart Suggestions Feature	§3 SRS – REQ-13.1 (p. 81)	-	-	-	-	-	-	-	-
REQ-13.2	The system must analyze the inputs and suggest suitable products for the buyer as gifts.	Smart Suggestions Feature	§3 SRS – REQ-13.2 (p. 81)	-	-	-	-	-	-	-	-
REQ-13.3	The system must make suggestions using Groq API taking into account the specified criteria.	Smart Suggestions Feature	§3 SRS – REQ-13.3 (p. 81)	-	-	-	-	-	-	-	-
REQ-13.4	The system must allow the buyer to view the list of suggested products.	Smart Suggestions Feature	§3 SRS – REQ-13.4 (p. 81)	-	-	-	-	-	-	-	-
REQ-13.5	The system must allow the buyer to access the product details page by clicking on it.	Smart Suggestions Feature	§3 SRS – REQ-13.5 (p. 81)	-	-	-	-	-	-	-	-
REQ-14.1	The system must allow the bazaar organizer to create a new bazaar.	Bazaar Management	§3 SRS – REQ-14.1 (p. 82)	-	-	-	-	-	-	-	-
REQ-14.2	The system must allow the bazaar organizer to edit bazaar data.	Bazaar Management	§3 SRS – REQ-14.2 (p. 82)	-	-	-	-	-	-	-	-
REQ-14.3	The system must allow the bazaar organizer to delete an existing bazaar.	Bazaar Management	§3 SRS – REQ-14.3 (p. 82)	-	-	-	-	-	-	-	-
REQ-14.4	The system must allow the seller to view all available bazaars.	Bazaar Management	§3 SRS – REQ-14.4 (p. 82)	-	-	-	-	-	-	-	-
REQ-14.5	The system must allow the seller to view bazaar details.	Bazaar Management	§3 SRS – REQ-14.5 (p. 82)	-	-	-	-	-	-	-	-
REQ-15.1	The system must allow the seller to request a table reservation within the bazaar.	Bazaar Table Reservation Management	§3 SRS – REQ-15.1 (p. 82)	-	-	-	-	-	-	-	-
REQ-15.2	The system must allow the seller to cancel a reservation request before the bazaar start date.	Bazaar Table Reservation Management	§3 SRS – REQ-15.2 (p. 82)	-	-	-	-	-	-	-	-
REQ-15.3	The system must allow the bazaar organizer to view incoming reservation requests.	Bazaar Table Reservation Management	§3 SRS – REQ-15.3 (p. 82)	-	-	-	-	-	-	-	-

REQ-15.4	The system must allow the bazaar organizer to accept reservation requests.	Bazaar Table Reservation Management	§3 SRS – REQ-15.4 (p. 83)	-	-	-	-	-	-		
REQ-15.5	The system must allow the bazaar organizer to reject reservation requests.	Bazaar Table Reservation Management	§3 SRS – REQ-15.5 (p. 83)	-	-	-	-	-	-		
REQ-16.1	The system must allow the buyer to view a notification when a purchase order is accepted.	Notification System	§3 SRS – REQ-16.1 (p. 83)	-	-	-	-	-	-		
REQ-16.2	The system must allow the buyer to view a notification when a purchase order is rejected.	Notification System	§3 SRS – REQ-16.2 (p. 83)	-	-	-	-	-	-		
REQ-16.3	The system must allow the buyer to view a notification when the order status is updated.	Notification System	§3 SRS – REQ-16.3 (p. 83)	-	-	-	-	-	-		
REQ-16.4	The system must allow the buyer to view a notification when their comment is replied to.	Notification System	§3 SRS – REQ-16.4 (p. 83)	-	-	-	-	-	-		
REQ-16.5	The system must allow the seller to view a notification when a new order arrives.	Notification System	§3 SRS – REQ-16.5 (p. 83)	-	-	-	-	-	-		
REQ-16.6	The system must allow the seller to view a notification when a new comment is added to a product.	Notification System	§3 SRS – REQ-16.6 (p. 83)	-	-	-	-	-	-		
REQ-16.7	The system must allow the seller to view a notification when a bazaar reservation is accepted.	Notification System	§3 SRS – REQ-16.7 (p. 84)	-	-	-	-	-	-		
REQ-16.8	The system must allow the seller to view a notification when a bazaar reservation is rejected.	Notification System	§3 SRS – REQ-16.8 (p. 84)	-	-	-	-	-	-		
REQ-16.9	The system must allow the bazaar organizer to view a notification when a new reservation request arrives.	Notification System	§3 SRS – REQ-16.9 (p. 84)	-	-	-	-	-	-		
REQ-16.10	The system must allow the bazaar organizer to view a notification when a seller cancels their reservation.	Notification System	§3 SRS – REQ-16.10 (p. 84)	-	-	-	-	-	-		
NFR-1	The system must be scalable to support an increasing number of users without performance degradation.	Scalability	§3 SRS – NFR-1 (p. 84)	-	-	-	-	-	-		
NFR-2	The system must maintain operational readiness to ensure continuous availability of features.	Availability	§3 SRS – NFR-2 (p. 84)	-	-	-	-	-	-		
NFR-3	Basic authentication must be implemented and passwords must be stored in encrypted form.	Security	§3 SRS – NFR-3 (p. 84)	-	-	-	-	-	-		

NFR-4	The system must support switching the user interface language between Arabic and English easily.	Usability	§3 SRS – NFR-4 (p. 84)	-	-	-	-	-	-		
NFR-5	The code must be standardized and well-documented for ease of maintenance.	Maintainability	§3 SRS – NFR-5 (p. 84)	-	-	-	-	-	-		

Table 63 Initial Requirements Traceability Matrix (RTM)

Chapter5

System Design

1. Introduction

This chapter presents the design phase of the Syrian System for Marketing and Handicrafts. It covers the system architecture diagram, component diagrams, deployment diagram, along with the design patterns used illustrating the mechanism of integrating the user interface with the business logic and the database, and the organized and effective integration of external AI services, contributing to building a coherent and maintainable system.

2. System Architecture System architecture

The Syrian System for Marketing and Handicrafts relies on a hybrid architecture called Hybrid Modular Architecture, which combines several integrated architectural patterns including:

- Django Model–View–Template (MVT) pattern
- Principles of Layered Architecture
- Dividing the system into independent applications under the concept of Modular Django Apps
- Using independent service components following the Service-Oriented Components approach

This integration aims to achieve clear organization of the system structure, reduce coupling between components, improve maintainability, development, and reusability, while maintaining future scalability flexibility. Figure (90) illustrates the overall system architecture diagram, showing the main layers, request flow, division of the system into independent applications, as well as the integration

of the system with the external AI service Groq API used for generating gift suggestions.

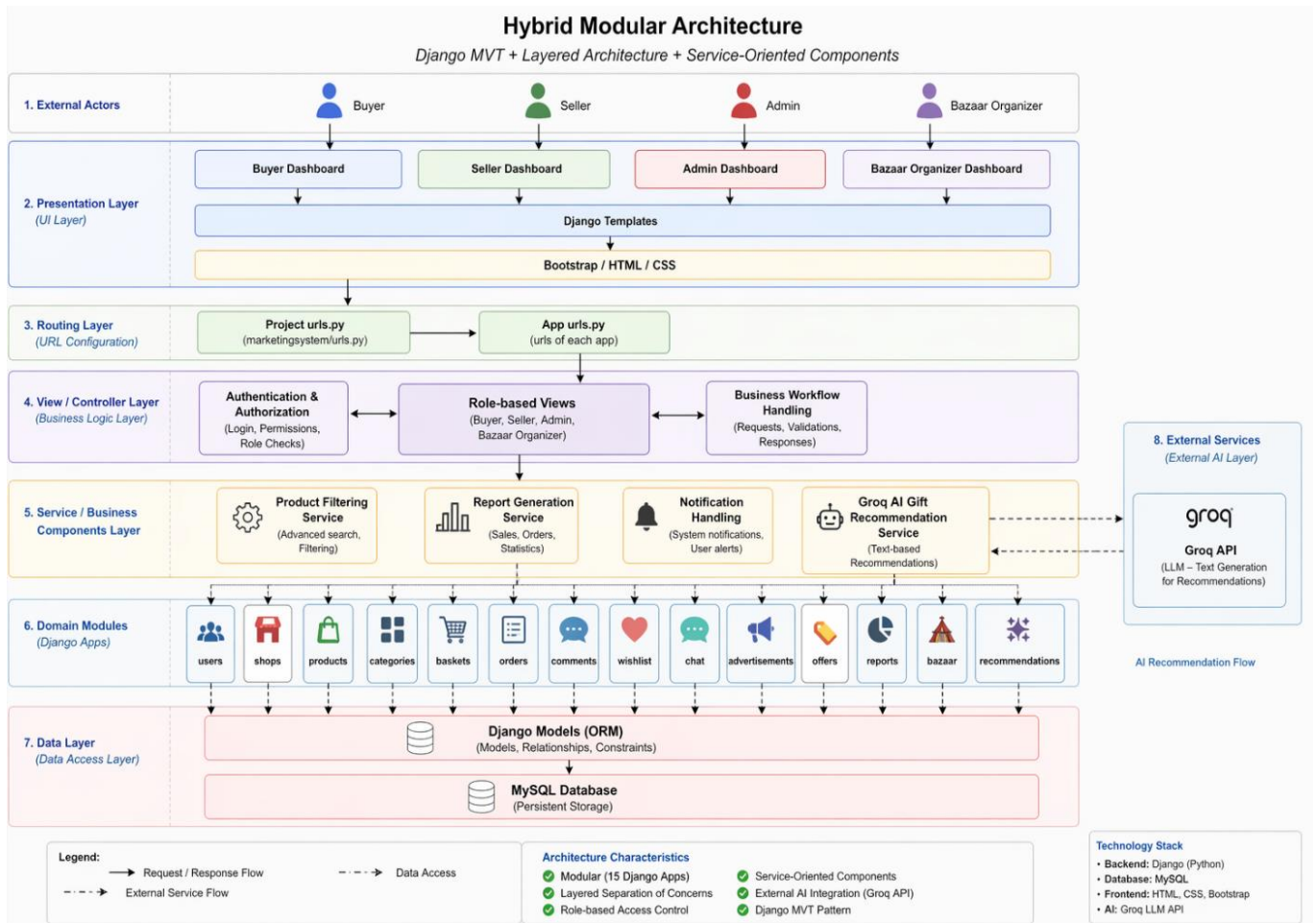


Figure 90 System Architecture

2.1 Hybrid Modular Architecture

This architecture relies on combining several architectural concepts within a single system, organizing the system in clear layers while simultaneously dividing it into independent modules that represent different Django applications.

In this system:

- Layered Architecture is used to organize workflow and responsibilities.

- Django MVT is used to organize the presentation layer and request processing mechanism.
- The system is divided into independent applications (Apps) such as Products, Orders, Bazaar, and Chats.
- Support services and complex processing are separated into independent service components (Services).

This approach contributes to:

- Improving project organization.
- Facilitating system maintenance and development.
- Reducing function overlap.
- Enabling separate component testing.
- Supporting future scalability and easy addition of new features.

The workflow inside the system starts from the users (Actors), passing through the presentation, routing, processing, and services layers, then reaches the data layer and the database.

2.1.1 Architectural Layers and Responsibilities

First, External Actors:

This layer represents the entities interacting with the system, including:

- **Buyer**

The buyer responsible for browsing products, placing orders, commenting, using favorites, messaging, and tracking their orders.

- **Seller**

The seller responsible for managing their store, products, offers, advertisements, orders, and bazaars.

- **Admin**

The administrator responsible for system monitoring, store management, and general supervision.

- **Bazaar Organizer**

The bazaar organizer responsible for managing bazaars and their reservation requests.

- Each user has different permissions and pages according to their role .within the system

Second: Presentation Layer (UI Layer)

This layer represents the system's visual user interface, responsible for .displaying pages and receiving user interactions

This layer includes:

- Buyer Dashboard
- Seller Dashboard
- Admin Dashboard
- Bazaar Organizer Dashboard
- Django Templates
- Bootstrap / HTML / CSS

The interfaces were developed using:

- HTML
- CSS

- Bootstrap
- Django Templates

The system also relies on the Server-Rendered UI approach, where HTML pages are generated on the server and sent directly to the browser without using SPA or React.

The interfaces rely on the Template Inheritance principle through the base file `base.html` to unify design and shared elements within the system.

Third: Routing Layer (URL Configuration):

This layer represents the request routing point within the system, and it is responsible for linking URLs to the appropriate handlers

It includes:

- `marketingsystem/urls.py`
- The `urls.py` files specific to each App.

This layer relies on hierarchical routing organization, where the main project file directs the request to the appropriate application, then the application file links the URL to the required View.

This organization contributes to:

- Clear system structure.
- Ease of request tracing.
- Separation of different applications' functionalities.

Fourth: View / Controller Layer

(Business Logic Coordination Layer)

This layer represents the part responsible for controlling the flow of operations within the system; it is the layer that receives requests and coordinates their execution

It includes:

• Authentication & Authorization

Responsible for:

- Login and logout.
- Authorization verification.
- Verification of roles (Buyer / Seller / Admin / Bazaar Organizer).

This is implemented using the Django Authentication System and decorators for role and permission verification.

• Role-based Views

These are the Views specific to each role within the system, where appropriate operations are executed based on the current user.

This layer includes operations such as:

- Displaying products.
- Creating orders.
- Shop management .
- Bazaar management.

- Booking processing.
- Notification management.

- **Business Workflow Handling**

Represents the part responsible for coordinating the workflow within the system, such as:

- Data validation.
- Enforcing business rules.
- Preparing responses.
- Managing the flow of requests between different layers.

This layer acts as an intermediary between the UI, services, and the internal logic of the system.

Fifth: Service / Business Components Layer

This layer represents independent service components used to support system functions and perform complex operations separately from the Views.

Among the key services used are:

- **Product Filtering Service**

A service responsible for performing advanced product search and filtering, such as filtering by price or category.

- **Report Generation Service**

A service responsible for generating reports and statistics related to sales, orders, and products.

- **Notification Handling**

A service responsible for managing system notifications and user alerts according to various events within the system.

- **Groq AI Gift Recommendation Service**

An independent service responsible for implementing the gift recommendation feature using artificial intelligence

This service performs:

1. Collecting user data and required inputs.
2. Gathering textual product descriptions from the database.
3. Sending data to Groq API.
4. Receiving the resulting suggestions.
5. Returning the results to the system for displaying to the user.

These services were separated into an independent layer to achieve:

- Reducing the complexity of the Views.
- Increasing reusability.
- Facilitating expansion and maintenance.
- Preventing direct coupling between the frontend and external services.

Sixth: Domain Modules Layer (Django Apps)

The system was divided into a set of independent applications (Modular Django Apps), where each application represents a specific functional domain within the system.

The core applications include:

- users
- shops
- products
- categories
- baskets
- orders
- comments
- wishlist
- chat
- advertisements
- offers
- reports
- bazaar
- recommendations

This division helps in:

- Organizing the project.
- Reducing component interdependency.
- Facilitating new feature development.
- Improving maintainability.

Seventh: Data Layer (Data Access Layer)

This layer represents the data access layer within the system, and includes:

• Django Models (ORM)

They represent the system entities and their various relationships, such as:

- users, shops, products, orders, comments, chats, bazaars, and notifications

They also define relationships and constraints using Django ORM.

• MySQL Database

Represents the primary database used for persistent data storage within the system.

MySQL was chosen due to:

- stability, good performance, and excellent compatibility with Django.

Eighth: External Services Layer (External AI Layer)

This layer represents the external services the system depends on.

In this project, the following are used:

- **Groq API**

A text-based artificial intelligence service (LLM-based API) used to generate product gift suggestions based on:

- budget, occasion type, age, gender, relationship, and product descriptions.

Groq API is accessed via a key stored inside a .env file and is used only when needed and within an independent service inside the system.

- The Groq API provides interaction capabilities with Large Language Models

(Large Language Models - LLMs) by sending text requests and receiving generated responses based on content and context analysis. In this system, the service was used to generate smart gift suggestions based on user input and linking them with the textual descriptions of the products stored in the database. Integration with the service was implemented using HTTP requests within an independent Service Component to separate external services from the rest of the system components and to achieve higher maintainability and extensibility, according to the official Groq API Documentation [5].

2.2 Django Model–View–Template (MVT) Pattern

–The system relies within the presentation layer on the Model–View Template (MVT) pattern provided by the Django framework, which aims to separate:

- Data representation, processing logic, presentation interface.

It is implemented within the project as follows:

First: Model

Models represent the data layer within the system.

Each app contains a models.py file to define its entities and their relationships, such as:

- Users, products, stores, orders, bazaars, bookings, comments, and chats.

This data is managed via Django ORM and connected to a MySQL database.

Second: View

Views represent the processing and control layer.

Views perform:

1. Receiving HTTP requests.
2. Verifying permissions and roles.
3. Executing the requested operations.
4. Handling services and Models.
5. Preparing data.
6. Returning HTML pages to the user.

They act as an intermediate layer coordinating work between the interface, system logic, and various services.

Third: Template

Templates represent the visual presentation interface within the system.

They are used to display data coming from Views using:

- HTML, CSS, Bootstrap and Django Templates.

They also rely on inheritance via base.html to unify design.

Templates contain no business logic; their function is limited strictly to data presentation.

2.3 Integration of MVT with Hybrid Architecture

MVT is not used in this system as an independent pattern, but as part of the multi-layer hybrid architecture, where:

- Templates, Views, and URLs represent the presentation and processing layer.
- Complex logic is implemented inside services and independent components.
- Database interaction is handled only through Models and Django ORM.
- External services like Groq API are separated into an independent layer.

This integration combines:

- Django MVT
- Layered Architecture
- Modular Architecture
- Service-Oriented Components

Clear system organization with high maintainability, extensibility, and future scalability.

3. Component Diagram

The component diagram represents the functional structure of the Syrian ,system for marketing and handicrafts built using the Django framework illustrating the core software components of the system and their relationships, focusing on the flow of requests and data between the user interface, processing components, and the database.

According to this diagram, the system consists of five main groups of components: Client Layer, Presentation Layer, Business Logic Layer, Data Access Layer, and External Services Components.



Figure 91 Component diagram

Component Functions

3.1 Client Layer

This layer represents the direct interaction point with users and consists of four main user interfaces, each with different privileges and functions:

Buyer UI: A dedicated interface for the buyer, allowing product browsing, cart management, placing orders, tracking their status, and interacting with comments and notifications.

Seller UI: A dedicated interface for the seller, enabling management of his store and products, order tracking, managing offers and advertisements, and handling bookings and notifications.

Admin UI: A dedicated interface for the system administrator, used for overseeing users and stores and managing general operations within the system.

Bazaar Organizer UI: A dedicated interface for the bazaar organizer, enabling creation of bazaars, managing their information, and following up on table booking requests with acceptance or rejection.

All these interfaces rely on the Templates component, responsible for rendering pages using HTML templates linked to the Django framework contributing to separating the presentation layer from the processing logic inside the system.

3.2 Presentation Layer

This layer forms the link between user interfaces and the rest of the system components, consisting of two main components:

Django URLs / API Gateway: Responsible for receiving incoming HTTP requests from user interfaces and routing them to the appropriate Views inside the system.

Django Views: Represent the first processing point for requests, where they receive requests, verify permissions, and execute the appropriate operations before passing the results to the templates responsible for rendering.

3.3 Business Logic Layer

This layer contains the core functional components of the system, where the essential operations related to the workflow inside the application are executed, comprising the following components:

User Management: Responsible for managing users, login, permissions, and various accounts within the system.

Shop Management: Handles managing stores, creating them, and linking them with sellers.

Product Catalog: Responsible for organizing products and categories and displaying them within the system.

Order & Basket System: Deals with basket operations, order creation, and tracking order status.

Communication (Chat / Comments): Provides a communication system through chats and comments between users.

Marketing Ads / Offers: Responsible for managing advertisements and marketing offers related to products.

Bazaar & Booking Management: Responsible for managing bazaars and sellers' table booking requests.

Notifications System: Handles creating notifications related to orders, comments, bookings, and various system states.

Reporting & Analytics: Provides reports and statistics related to the system and stores.

AI Recommendation Engine: An intelligent component responsible for generating product gift suggestions based on product data and descriptions within the system.

All these components communicate with the Data Access Layer to perform read and write operations, and the AI Recommendation Engine integrates .with an external AI service

3.4 Data Access Layer

This layer is represented by the Django ORM / Models component, which is responsible for direct interaction with the database using objects instead of traditional SQL queries.

This layer provides a unified interface for the system's various components to perform add, edit, delete, and query operations in an organized and secure manner.

3.5 Database Layer

The MySQL Database represents the main repository for all system data, such as users, stores, products, orders, messages, bookings, and notifications data.

These data are accessed via Django ORM, which contributes to ensuring data integrity and operation consistency.

3.6 Integration with External Services

,The diagram illustrates the system's integration with an external service Groq AI API, which is accessed via HTTPS protocol through the AI Recommendation Engine component.

This integration is used to perform intelligent processing and generate product suggestions, without the service being a direct part of the system's database.

4. Design Patterns in Use

4.1 Design Pattern: Decorator

Decorator" is a Structural Pattern used in our project to add additional" functionality to Views without directly modifying them. In Django framework, this pattern is commonly used via Decorators such as:

- `@login_required`
- `@seller_required`

- `@buyer_required`
- `@bazar_organizer_required`

These decorators are used to restrict access to certain system pages based on login status or user role. For example, unregistered users are prevented from accessing specific pages, or only sellers are allowed to access the product management page.

Purpose of this pattern in the project:

To achieve precise control over permissions, separating access verification logic from the core processing logic inside Views, making the code cleaner and easier to maintain.

4.2 Design Pattern: Template Method

Template Method" is a Behavioral Pattern used in the project by adopting a "base HTML template (base.html) that is extended in other project pages using `{% extends "base.html" %}`.

This base template contains the general structure of all pages such as the navigation bar and headers, while variable parts like the page title and content are specified inside blocks (`{% block content %}`).

Purpose of this pattern in the project:

To reuse the overall page structure, provide consistent and easily developed interfaces, while separating each page's content from the general design.

4.3 Design Pattern: Singleton

Singleton" is a Creational Pattern implicitly used in the Django project through the settings.py file, which represents a single reference point containing all project settings (such as database settings, paths, media files, etc).

Purpose of this pattern in the project:

To ensure there is only one instance of the settings object within the project, facilitating centralized configuration management without duplication or conflict.

5. Site maps

5.1 System Overview Site Map

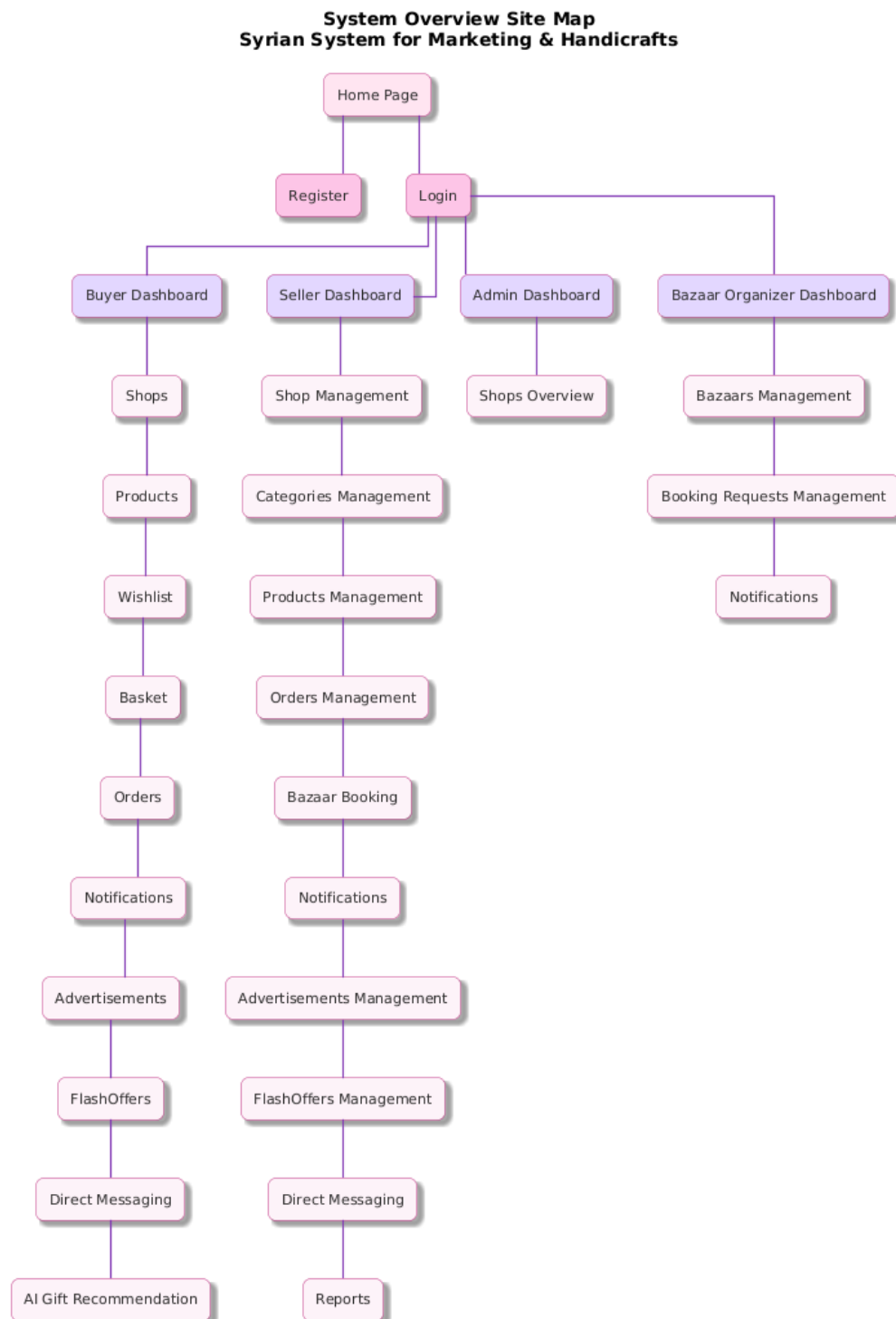


Figure 92 System Overview Site Map

5.2 Seller Site Map

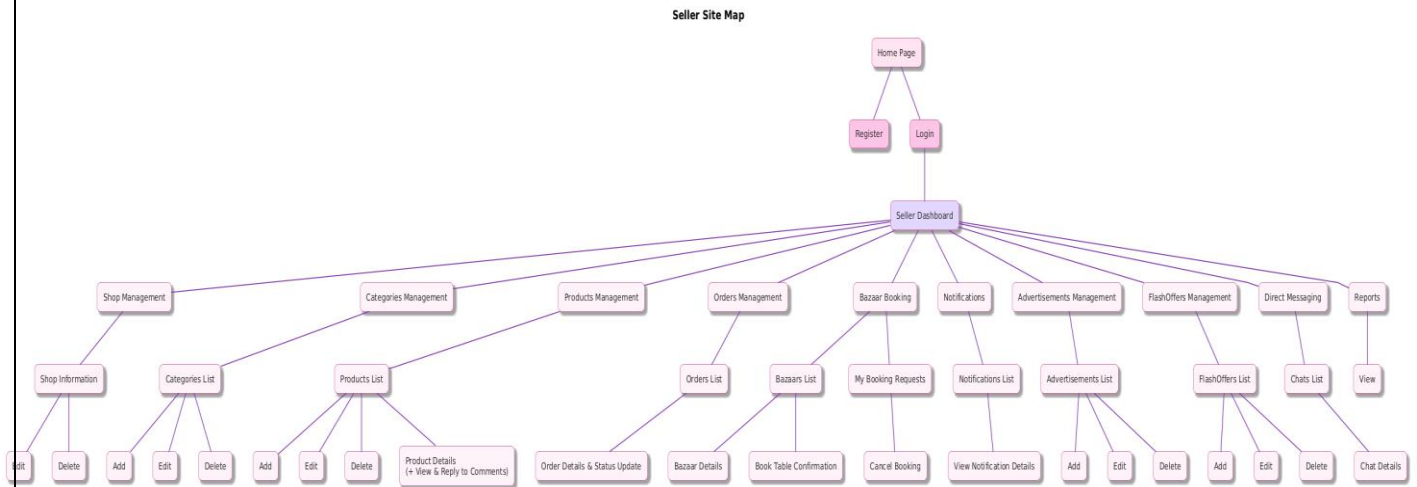


Figure 93 Seller Site Map

5.3 Buyer Site Map

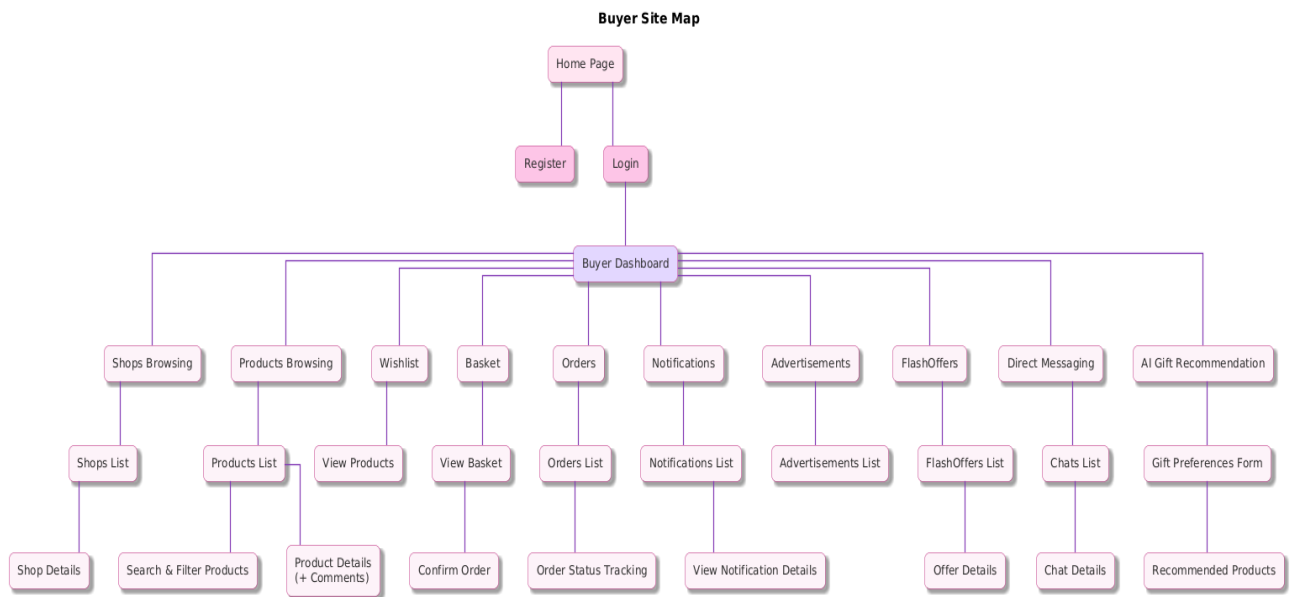


Figure 94 Buyer Site Map

5.4 Admin Site Map

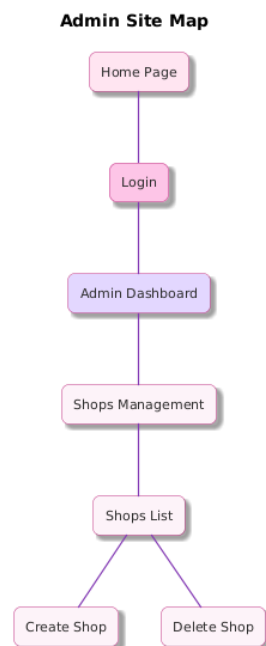


Figure 95 Admin Site Map

5.5 Bazar Organizer Site Map

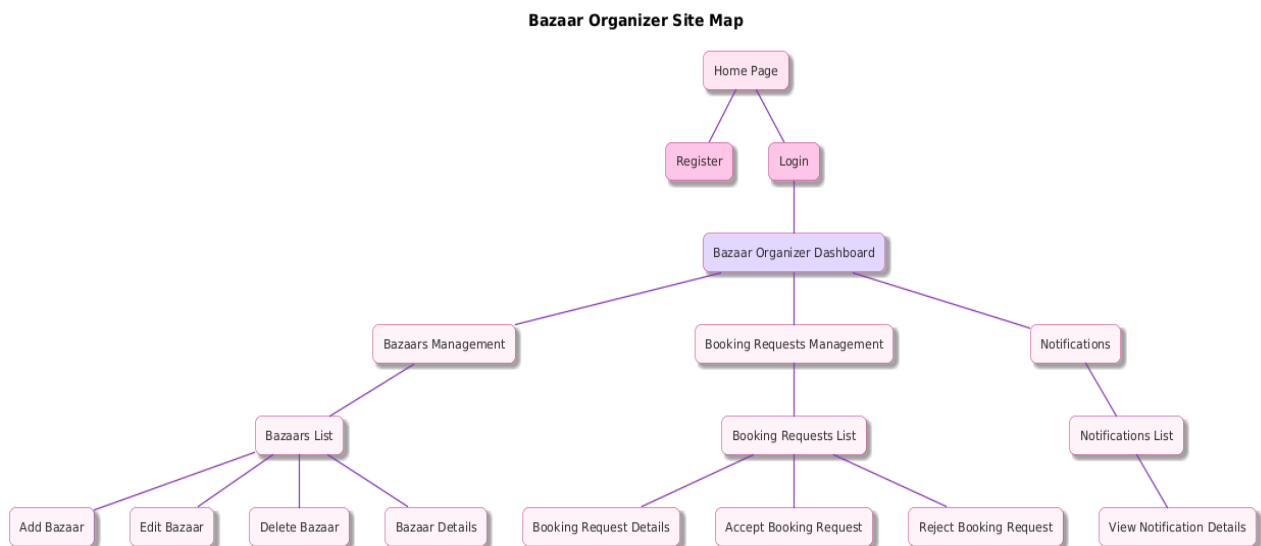


Figure 96 Bazar Organizer Site Map

Chapter6

Practical implementation

1. Introduction

In this chapter, we will present the practical aspect of system implementation, explaining the tools and techniques used during the development process. We will also review the system interfaces and conclude with the execution of some test cases to verify the system functions in multiple scenarios.

2. Tools Used

1. Python Django

Django is a high-level framework built with Python, aiming to accelerate the development process of web applications and provide a clean, practical design. Django provides ready-made components such as Models, Views, and Templates, reducing the need to write repetitive code. It also offers an extensible and easy-to-maintain environment and was relied upon to build the system's backend.

2. MySQL

MySQL is an open-source relational database management system, widely used for storing, managing, and retrieving data. It was used in this project as the main database to store user information, products, orders, stores, and other system data.

3. HTML, CSS, Bootstrap

HTML was used to create the general structure of user interfaces, CSS for styling and designing the overall appearance, with reliance on the Bootstrap

library to provide a responsive and modern design quickly and efficiently, Bootstrap helped build organized and consistent interfaces that support various devices.

4. Visual Studio Code (VS Code)

VS Code is one of the most popular development environments offering advanced features such as code completion (IntelliSense), debugging, and project management. It was used in the full development of the system, both front-end and back-end, due to its simplicity and support for a large number of useful extensions.

5. GitHub

GitHub is a code hosting platform using Git, used for tracking changes and sharing code among team members. It was relied upon in this project for version control, backups, and collaborative programming, which helped document changes and organize the development process.

6. Draw.io

Draw.io (currently known as diagrams.net) is a free web-based tool used for creating diagrams and illustrative charts. It allows users to draw flowcharts, network diagrams, database diagrams, UML diagrams, and many other types easily.

3. System Interfaces

- Home Page

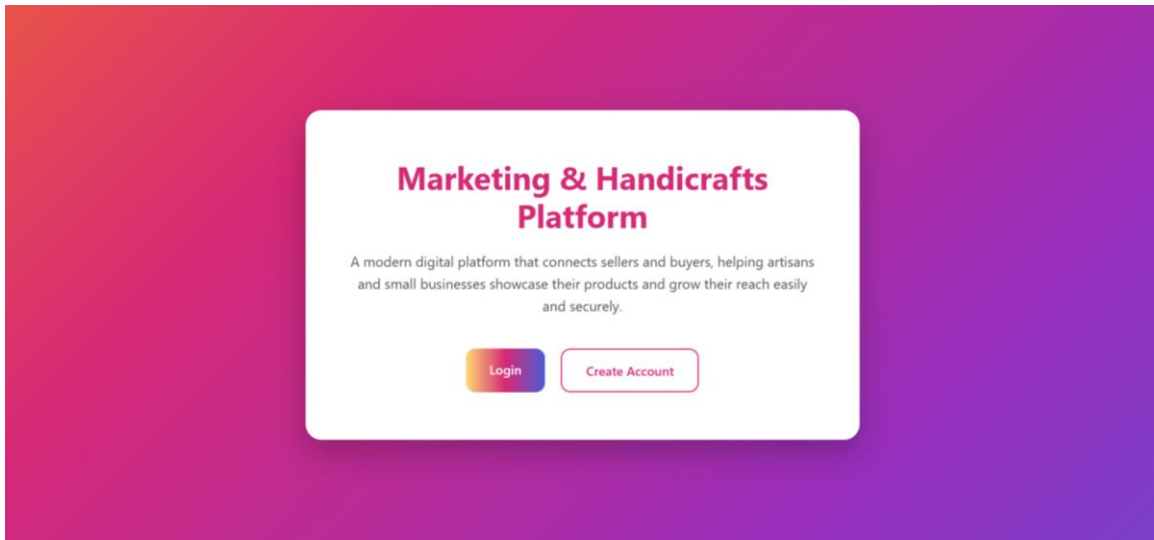


Figure (87): System Home Page Interface

- Create New Account

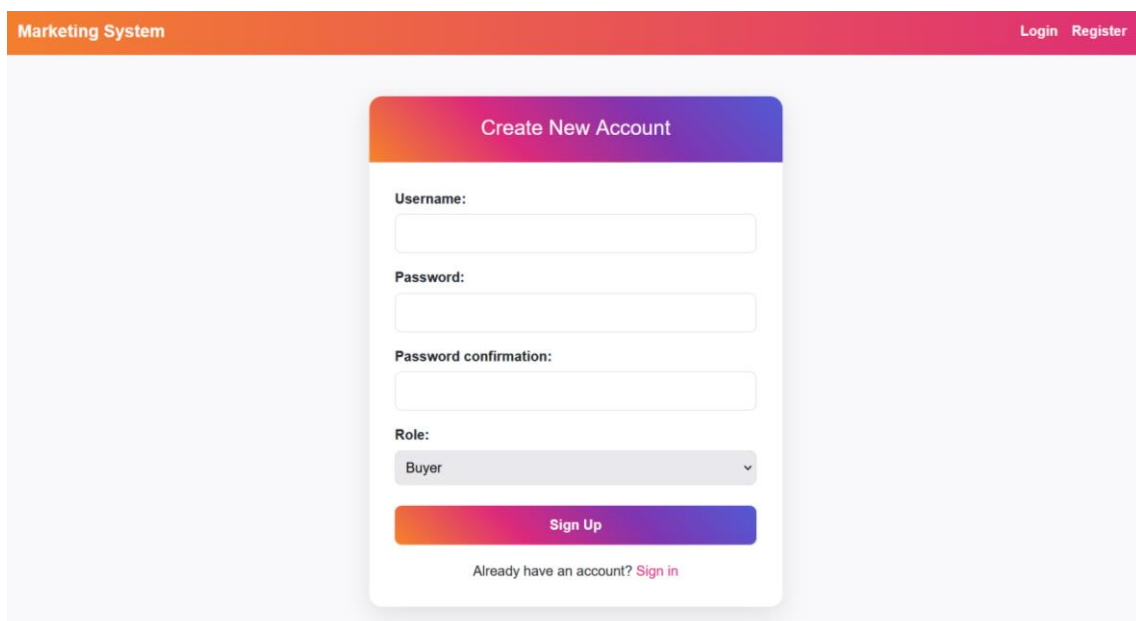
The image displays a 'Create New Account' form. At the top, there is a header bar with 'Marketing System' on the left and 'Login Register' on the right. The form itself is a white card with a pink-to-purple gradient header that says 'Create New Account'. It contains four input fields: 'Username:', 'Password:', 'Password confirmation:', and 'Role:'. The 'Role' field is a dropdown menu currently showing 'Buyer'. Below the inputs is a large 'Sign Up' button with a gradient. At the very bottom of the card, there is a link that says 'Already have an account? Sign in'.

Figure (88): New Account Creation Interface

- **Login**

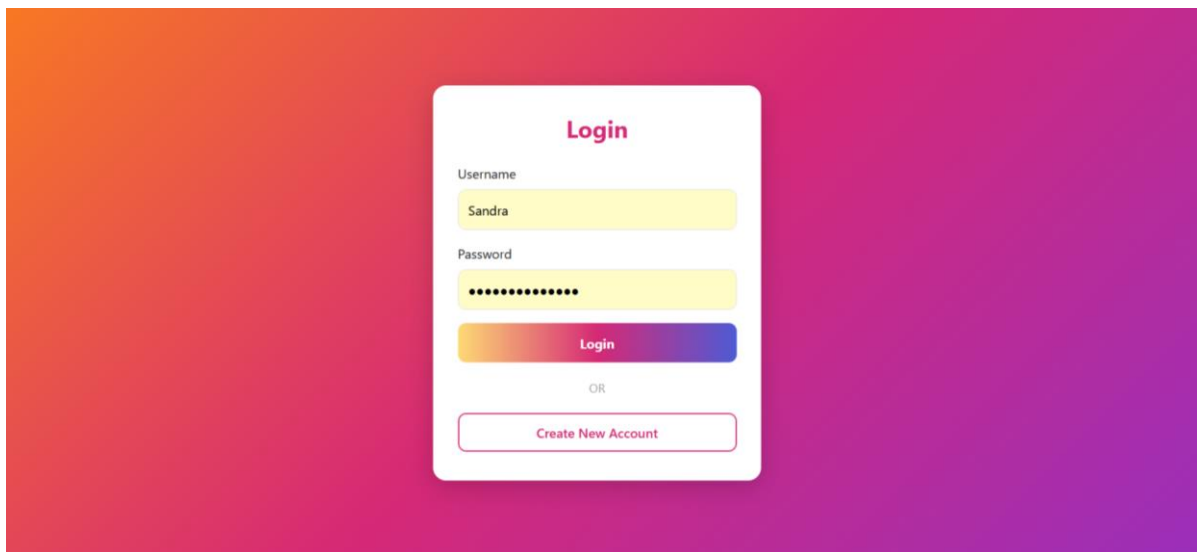


Figure (89): Login Interface

- **View Account**

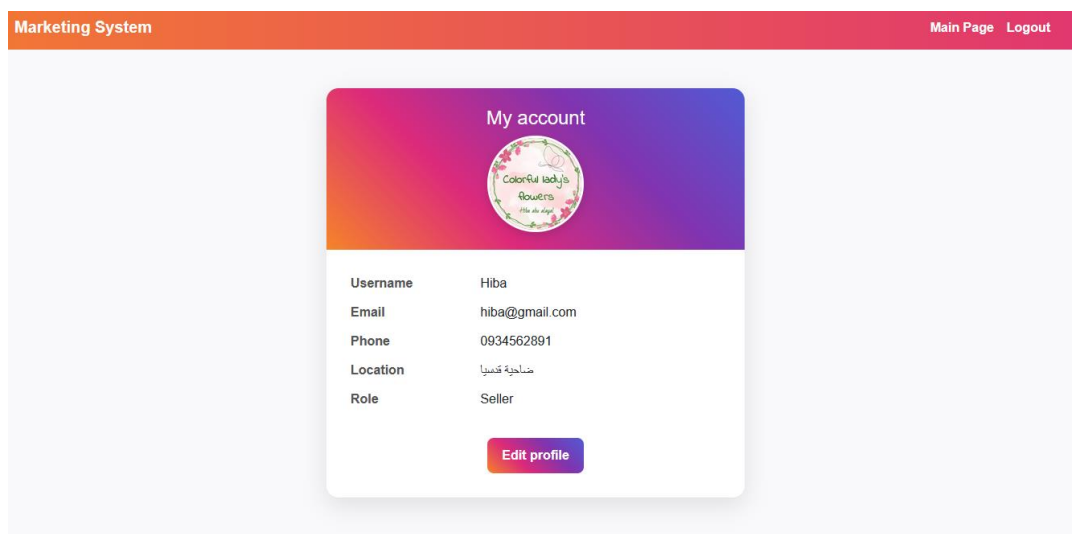


Figure (89): Account Viewing Interface

- **Edit Account Information**

Marketing System Main Page Logout

Edit account

Email
hiba@gmail.com

Phone number
0934562891

Location
مساحة قديم

Profile image
Currently: [profile_images/FB_IMG_1765980994545.jpg](#) ☐
Clear

Change:
Browse... No file selected.

Save changes

[Back to profile](#)

Figure (89): Account Information Editing Interface

1. Seller Interfaces

- **Seller Main Interface**

Marketing System Logout

Seller Panel

[Dashboard](#)

- [My Shop](#)
- [Categories](#)
- [Products](#)
- [Orders](#)
- [Advertisements](#)
- [Flash Offers](#)
- [Messages](#)
- [My Report](#)

[Logout](#)

Welcome, nour_nasan_nour
Manage your shop, products, and customer orders.

You don't have a shop yet
Create your shop to start selling products.

Create Shop

Figure (90): Seller Home Page Interface

- **Create New Shop**

Marketing System Main Page Logout

Create a New Shop

Name:

Description:

Address:

Phone number:

Email:

Figure (91): New Shop Creation Interface

- **View Shop Information**

Marketing System Main Page Logout

Your Shop

Manage your shop professionally

 **Sandra Saeed – Handmade Wood Art**
• Active Shop

Sandra Saeed is a handmade wood art brand specializing in carved wooden accessories, decorative pieces, and meaningful keepsakes. Each product is carefully crafted from natural wood, blending creativity, storytelling, and fine craftsmanship. The collection includes wooden pendants, keychains, sculptures, and artistic home décor—each piece carrying warmth, originality, and a handcrafted soul.

 ADDRESS
Syria

 PHONE NUMBER
0922218594

 EMAIL
sandra@gmail.com

Figure (92): Shop Information Viewing Interface

- **Edit Shop Information**

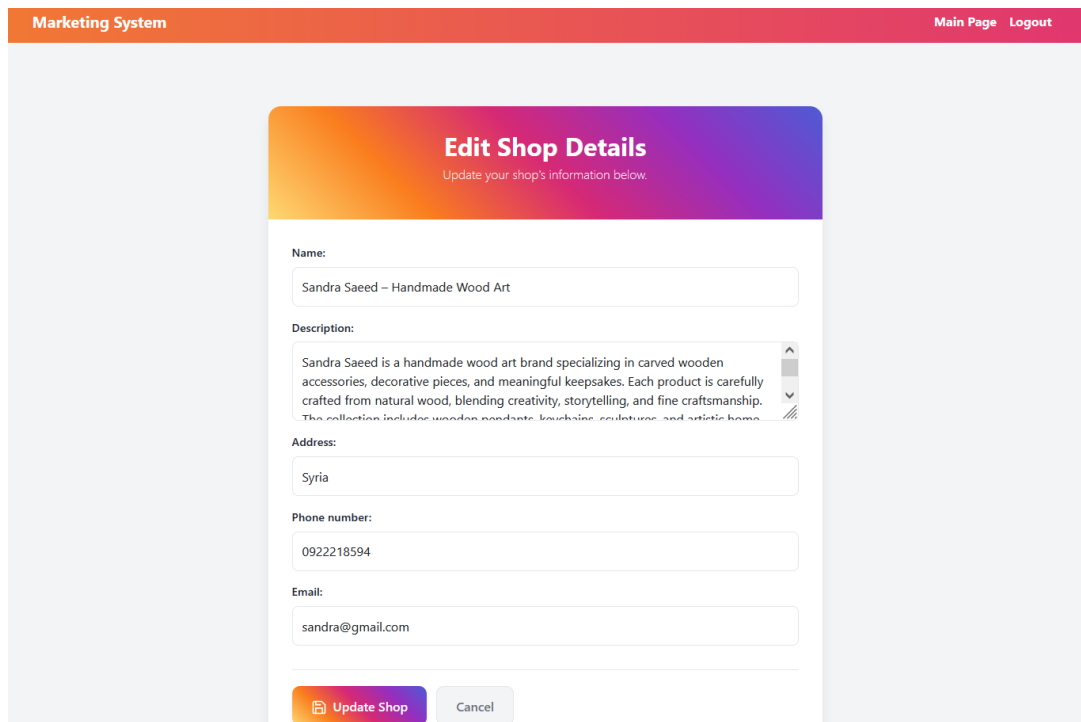


Figure (93): Shop Information Editing Interface

- **Confirm Shop Deletion**

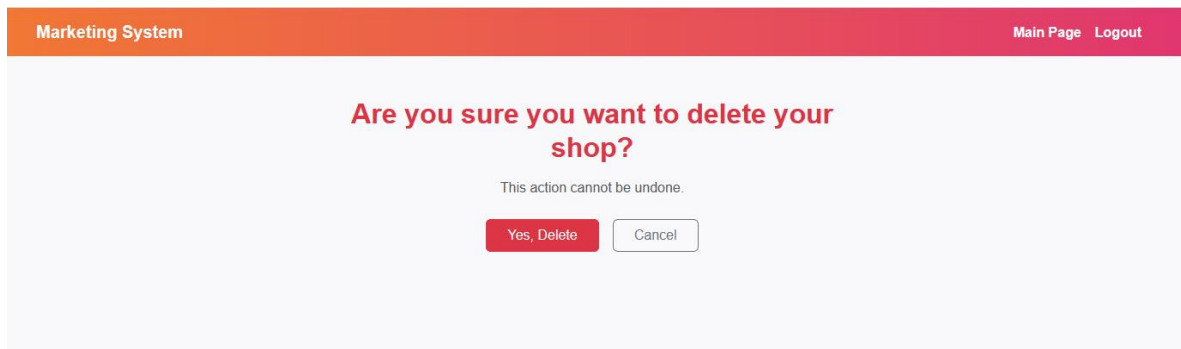


Figure (94): Shop Deletion Confirmation Interface

- **Seller dashboared**

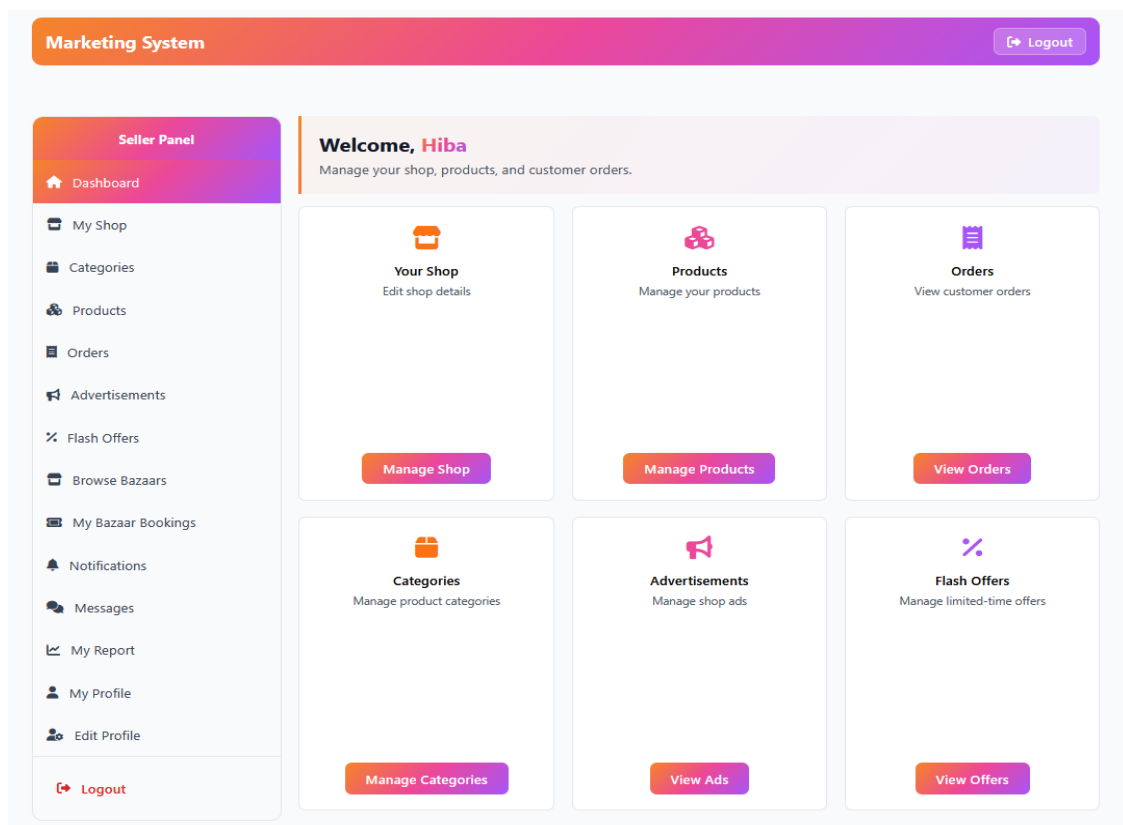


Figure (95): Seller dasheborad Interface

- **Category Management**

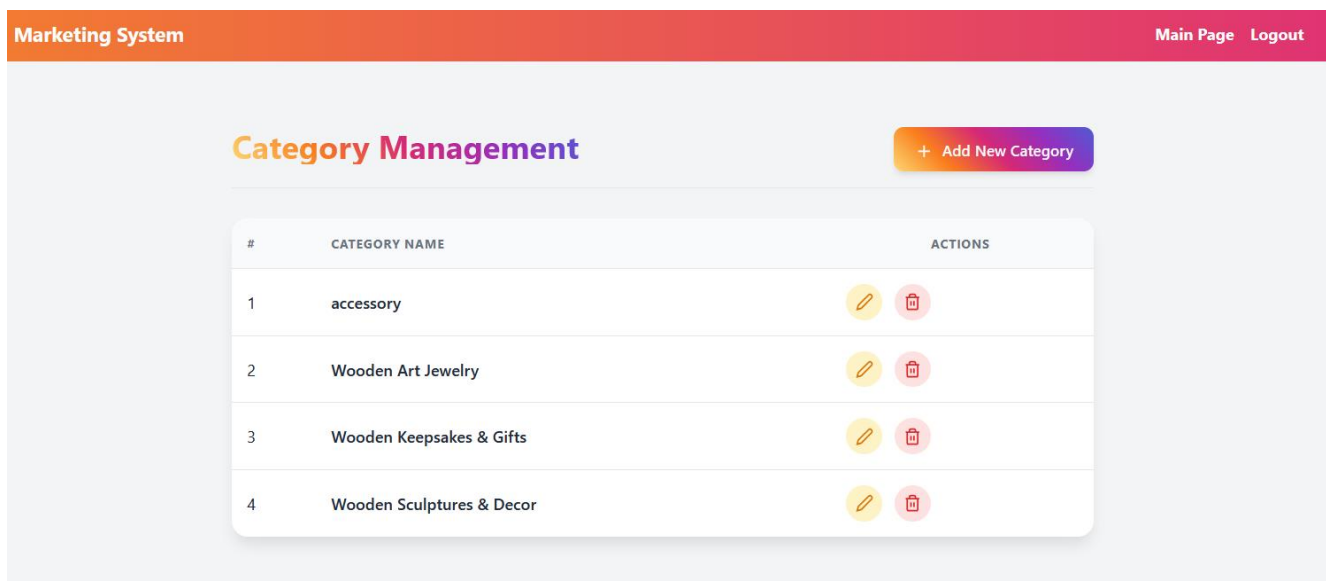


Figure (96): Category Management Interface

- **Add New Category**

Figure (97): Add New Category Interface

- **Edit Category**

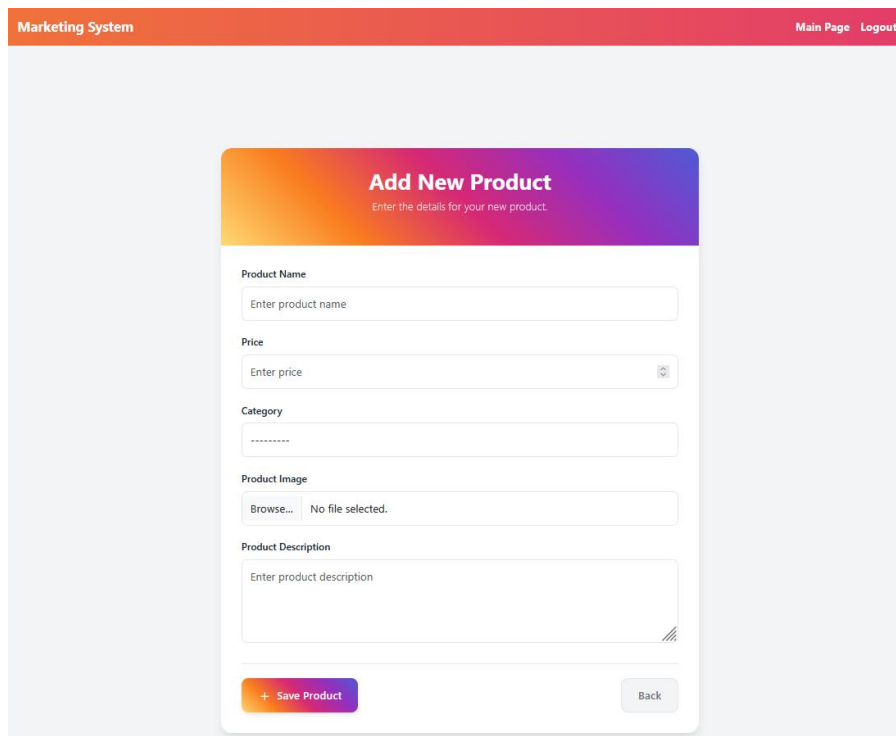
Figure (98): Edit Category Interface

- **Product Management**

#	NAME	PRICE	CATEGORY	IMAGE	DESCRIPTION	ACTIONS
1	Wizard Boy Wooden Keychain	50000.00 SYP	Wooden Keepsakes & Gifts		A charming handmade wooden keychain featuring a w...	  
2	Anime Couple Wooden Mini Art Frame	100000.00 SYP	Wooden Keepsakes & Gifts		A handcrafted wooden mini frame featuring an anim...	  
3	Sun & Moon Wooden Necklace and Bracelet Set	100000.00 SYP	Wooden Art Jewelry		A symbolic handmade jewelry set featuring a woode...	  
4	Wooden Balance Scale Sculpture	125000.00 SYP	Wooden Sculptures & Decor		An artistic handcrafted wooden sculpture inspired...	  
5	Mini Wooden Book Necklace	65000.00 SYP	Wooden Art Jewelry		A delicate handmade necklace featuring a miniatur...	  
6	Personalized Doctor Wooden Keepsake	75000.00 SYP	Wooden Keepsakes & Gifts		A handcrafted wooden keepsake featuring a stethos...	  
7	Lion & Sun Wooden Wallet Charm	60000.00 SYP	accessory		A bold handmade wooden wallet accessory featuring...	  

Figure (99): Product Management Interface

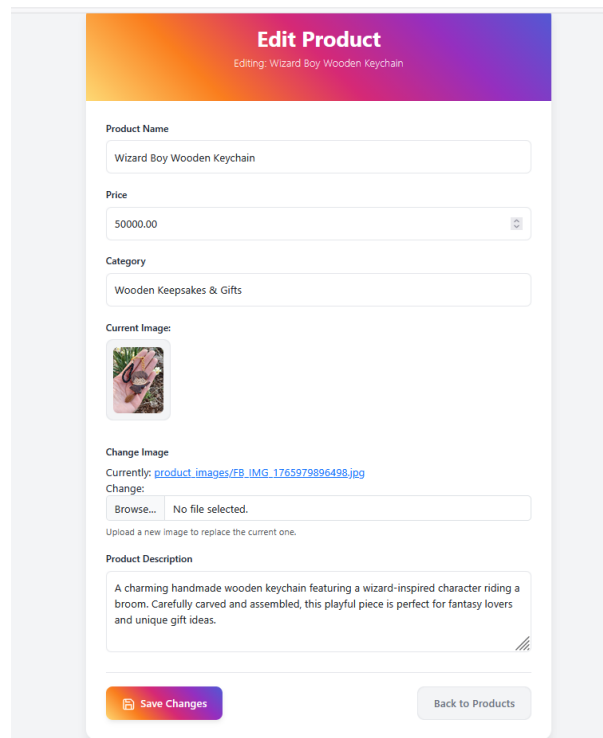
- **Add New Product**



The 'Add New Product' interface is a web form with a header bar containing 'Marketing System' and links to 'Main Page' and 'Logout'. The form itself has a colorful gradient header with the title 'Add New Product' and the instruction 'Enter the details for your new product.' Below the header, the form contains several input fields: 'Product Name' (text), 'Price' (number with a currency icon), 'Category' (text), 'Product Image' (file upload with 'Browse...' button and 'No file selected.' message), and 'Product Description' (text area). At the bottom, there are two buttons: '+ Save Product' and 'Back'.

Figure (100): Add New Product Interface

- **Edit Product**



The 'Edit Product' interface is a web form with a header bar containing the title 'Edit Product' and the instruction 'Editing: Wizard Boy Wooden Keychain'. Below the header, the form contains several input fields: 'Product Name' (text, pre-filled with 'Wizard Boy Wooden Keychain'), 'Price' (number, pre-filled with '50000.00'), 'Category' (text, pre-filled with 'Wooden Keepsakes & Gifts'), 'Current Image' (image preview showing a wooden keychain), 'Change Image' (file upload with 'Browse...' button and 'No file selected.' message), and 'Product Description' (text area, pre-filled with 'A charming handmade wooden keychain featuring a wizard-inspired character riding a broom. Carefully carved and assembled, this playful piece is perfect for fantasy lovers and unique gift ideas.'). At the bottom, there are two buttons: 'Save Changes' and 'Back to Products'.

Figure (101): Edit Product Interface

- **Confirm Delete Product**

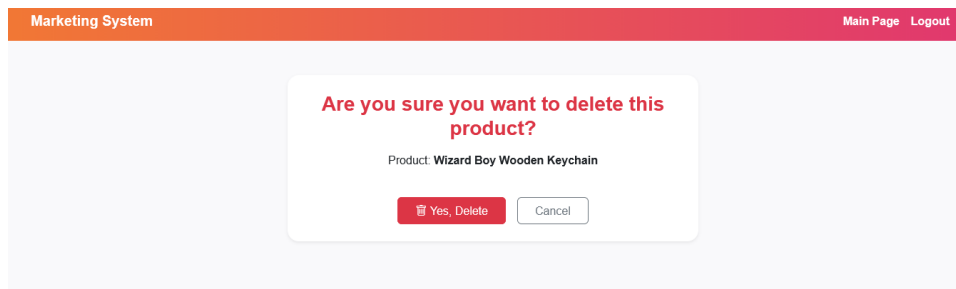


Figure (102): Confirm Delete Product Interface

- **View Product Details with Comments**

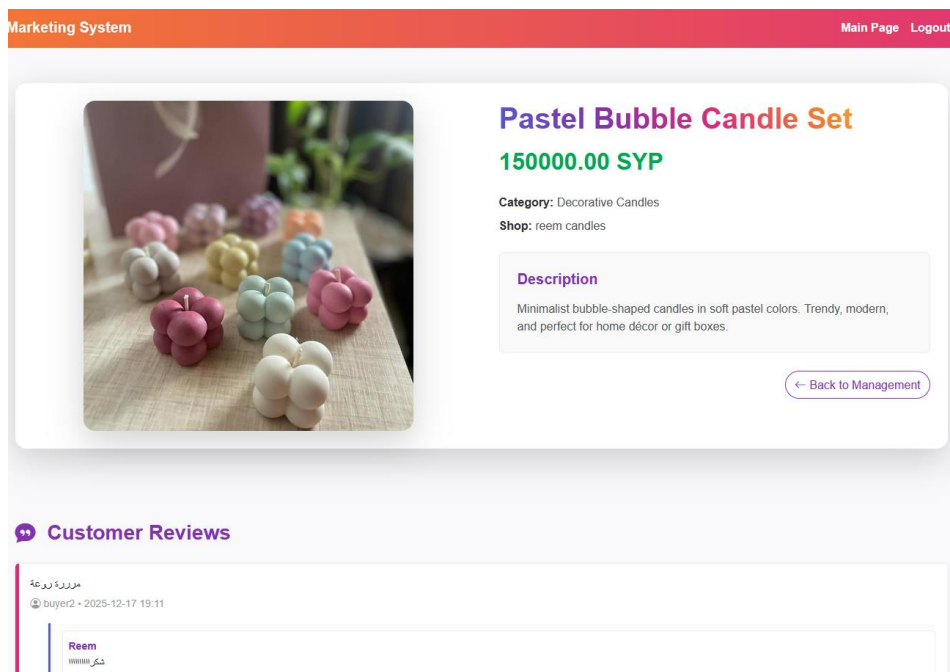


Figure (103): Product Details Interface Containing Comments

- **View Orders**

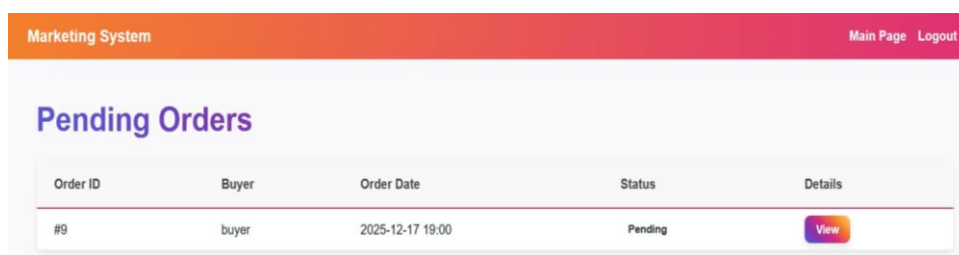


Figure (104): View Orders Interface

- **View Order Details**

Marketing System

Main PageLogout

Order Details #9

Buyer
buyer

Status
Pending

Basket ID
#1

Contact
phone

Products

Seven Chakra Healing Bracelet

Qty: 1

Approve Order

Reject Order

Figure (105): View Order Details Interface

- **Update Order Status**

Marketing System

Main PageLogout

Order Details #13

Buyer
buyer2

Status
Approved

Basket ID
#2

Contact
email

Products

Elegant Arabic Letter Necklace

Qty: 2

Custom Arabic Name Keychain

Qty: 1

Order Tracking

Preparing

2026-01-27 18:31

Prepared

2026-01-27 18:31

Update Order Status

Figure (106): Update Order Status Interface

- **Confirm Order Acceptance**

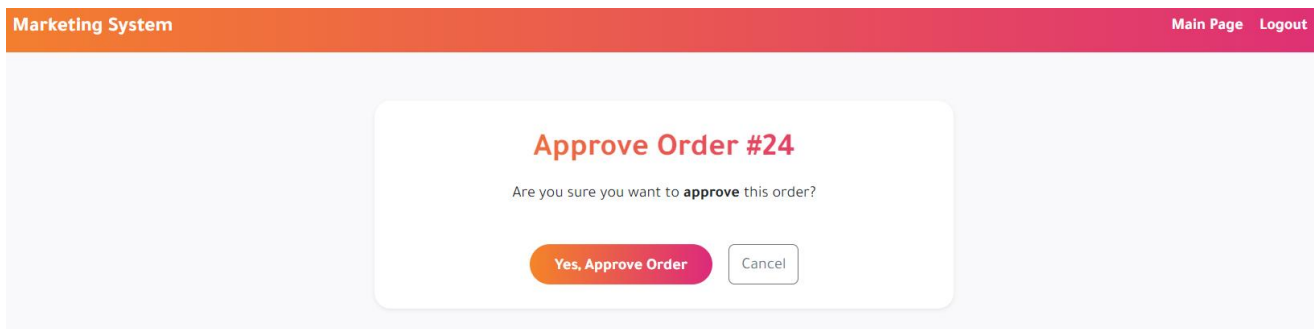


Figure (107): Confirm Order Acceptance Interface

- **Confirm Order Rejection**

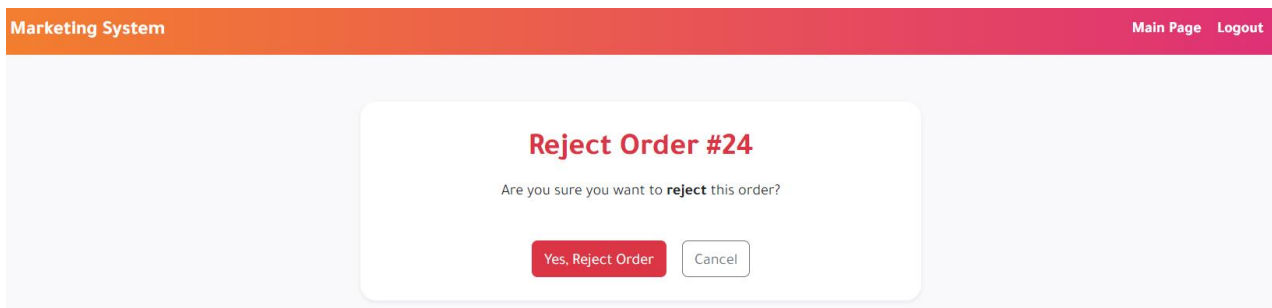


Figure (108): Confirm Order Rejection Interface

- **Manage Advertisements**

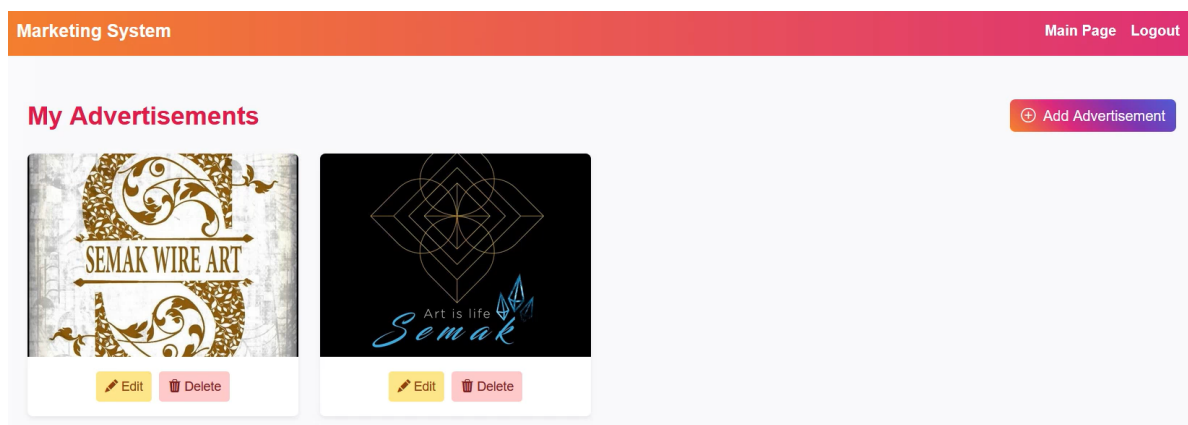


Figure (109): Manage Advertisements Interface

- **Add Advertisement**

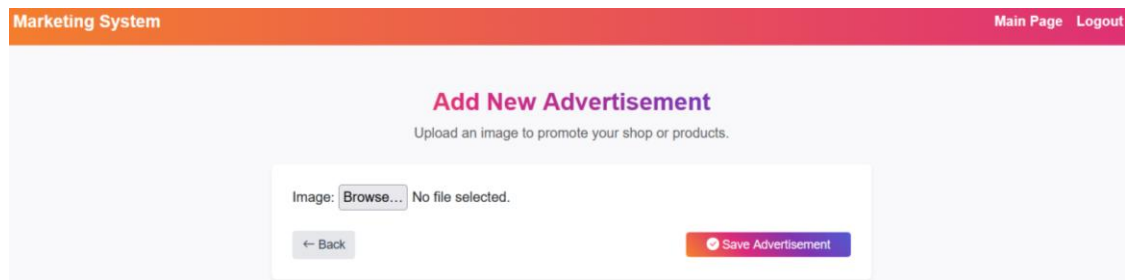


Figure (110): Add New Advertisement Interface

- **Edit Advertisement**

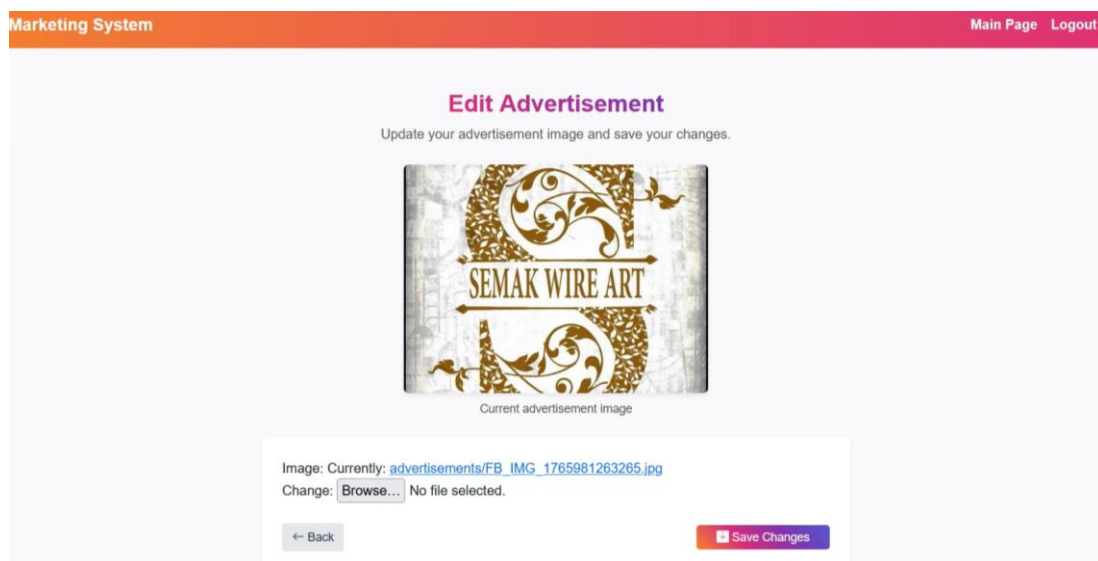


Figure (111): Edit Advertisement Interface

- **Confirm Delete Advertisement**

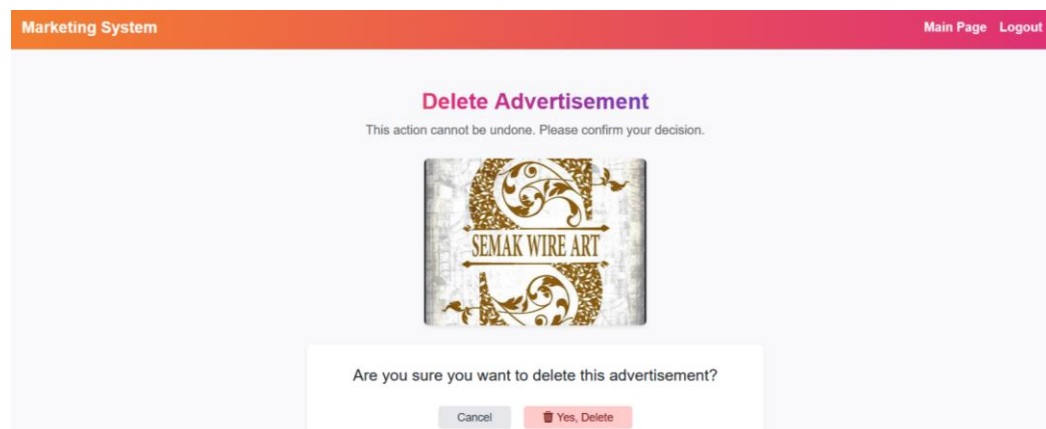


Figure (112): Confirm Delete Advertisement Interface

- **Manage Flash Offers**

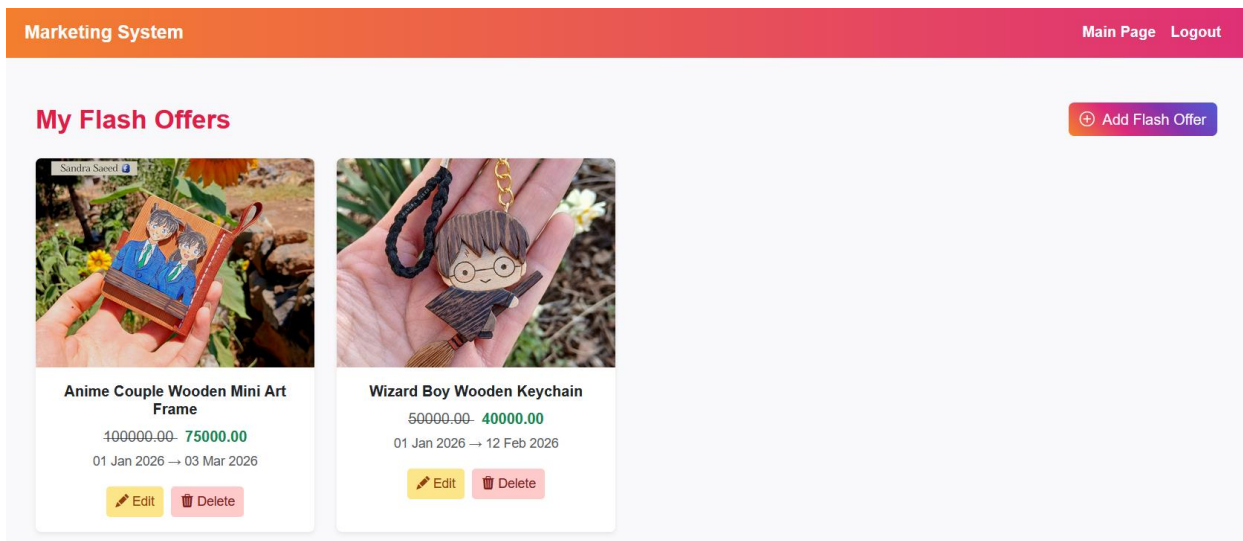


Figure (113): Manage Flash Offers Interface

- **Add Flash Offer**

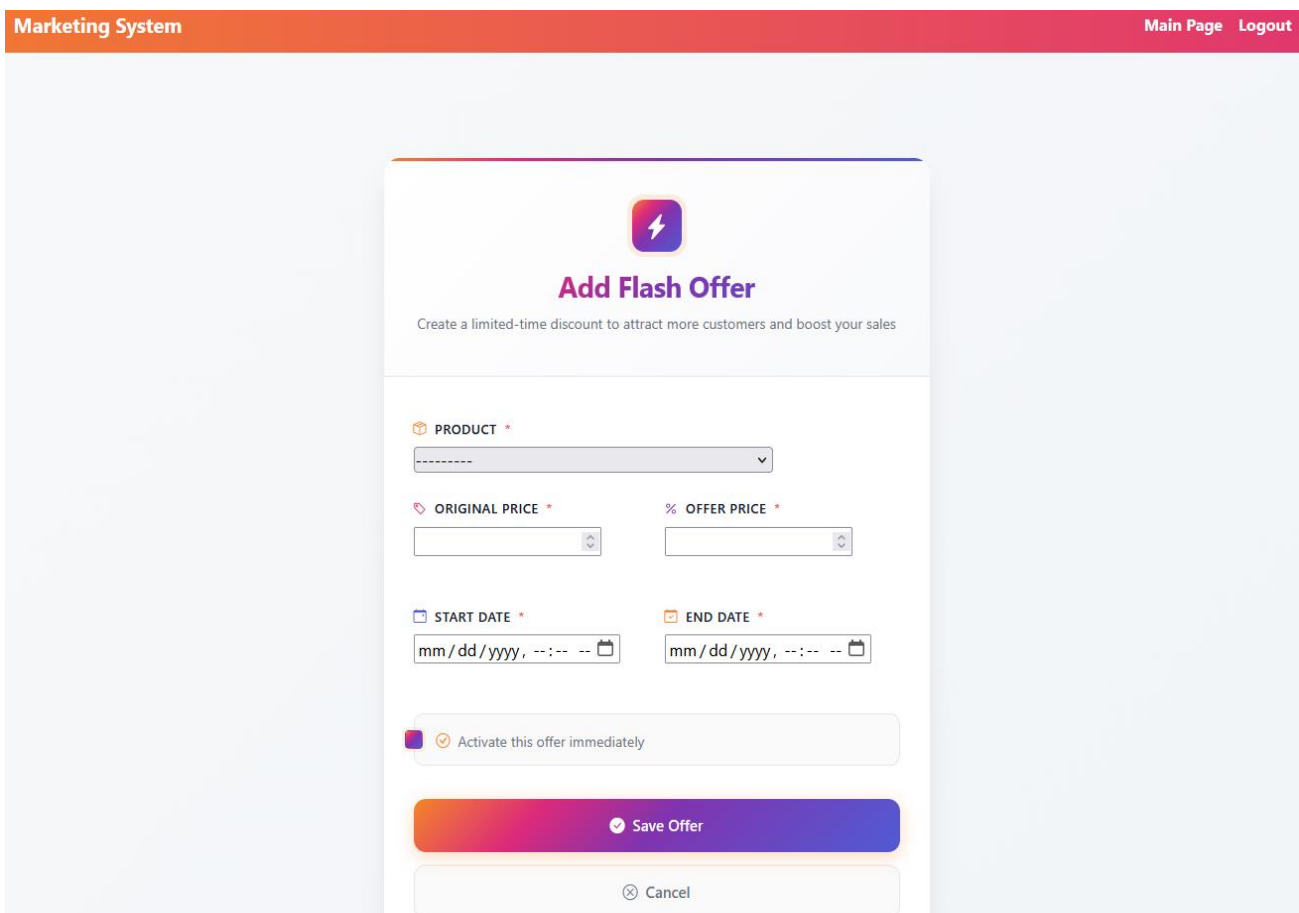
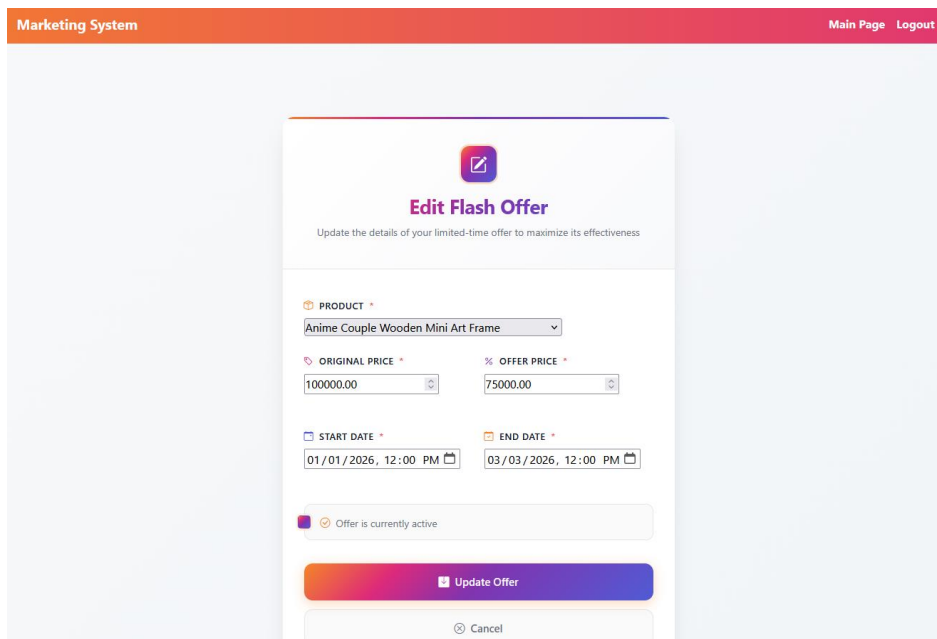


Figure (114): Add Flash Offer Interface

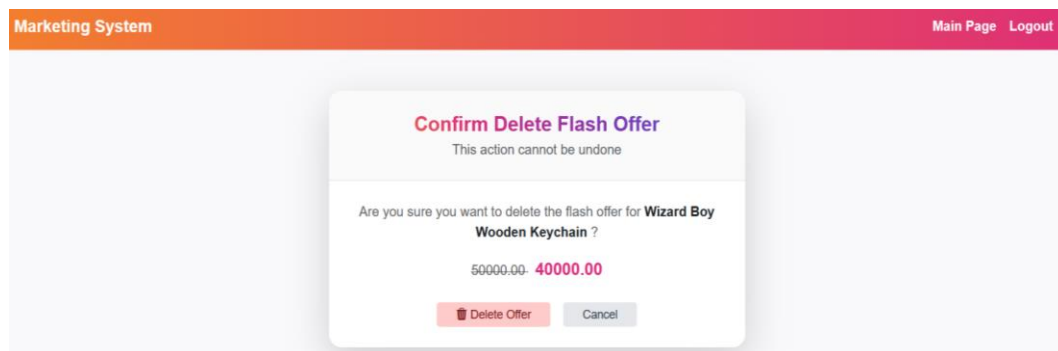
- **Edit Flash Offer**



The interface for editing a flash offer is displayed within a modal window. At the top, it says "Edit Flash Offer" with a subtext "Update the details of your limited-time offer to maximize its effectiveness". Below this, there are several input fields: a "PRODUCT" dropdown menu showing "Anime Couple Wooden Mini Art Frame", an "ORIGINAL PRICE" field with "100000.00", an "OFFER PRICE" field with "75000.00", a "START DATE" field with "01/01/2026, 12:00 PM", and an "END DATE" field with "03/03/2026, 12:00 PM". A status bar indicates "Offer is currently active". At the bottom, there are two buttons: "Update Offer" and "Cancel".

Figure (115): Edit Flash Offer Interface

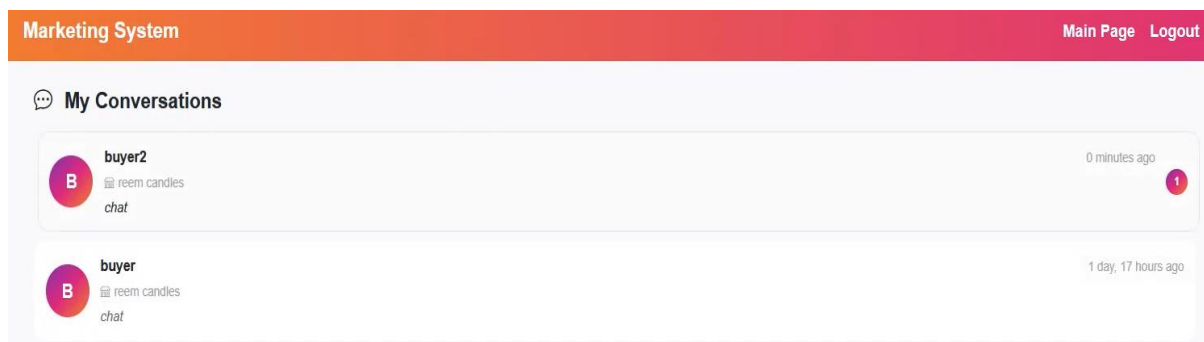
- **Confirm Delete Flash Offer**



The interface for confirming the deletion of a flash offer is shown in a modal window. It starts with the title "Confirm Delete Flash Offer" and the warning "This action cannot be undone". The main text asks, "Are you sure you want to delete the flash offer for Wizard Boy Wooden Keychain?". Below this, the original price "50000.00" and the offer price "40000.00" are displayed. At the bottom, there are two buttons: "Delete Offer" and "Cancel".

Figure (116): Confirm Delete Flash Offer Interface

- **View Conversations**



The interface for viewing seller conversations is shown. It has a header "Marketing System" and a sub-header "My Conversations". Below this, there are two conversation entries. The first entry is from "buyer2" with a status of "reem candles chat" and a timestamp of "0 minutes ago". The second entry is from "buyer" with a status of "reem candles chat" and a timestamp of "1 day, 17 hours ago".

Figure (117): Seller Conversations Interface

- **Start Conversation**

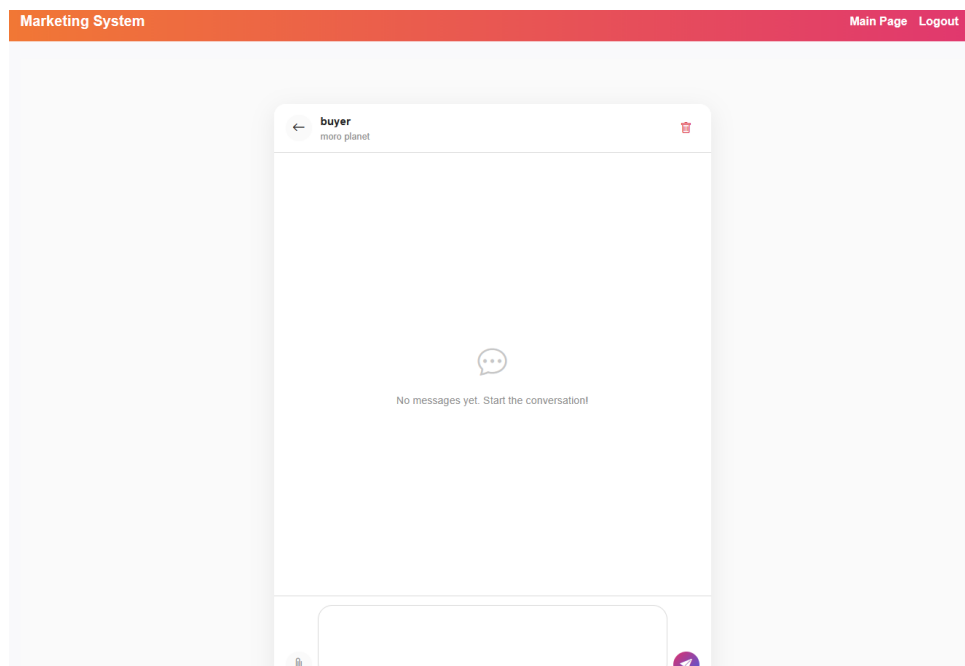


Figure (118): Start New Conversation Interface for Seller

- **Seller Conversation**

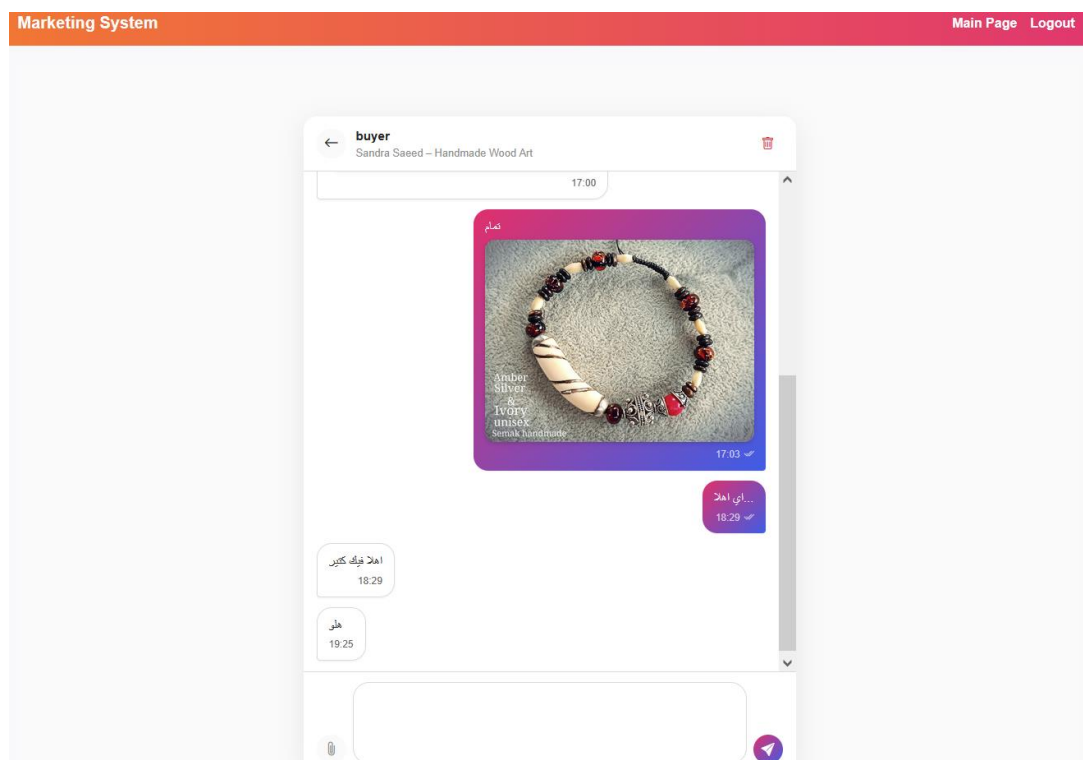


Figure (119): Seller Conversation Interface

- **View Reports**

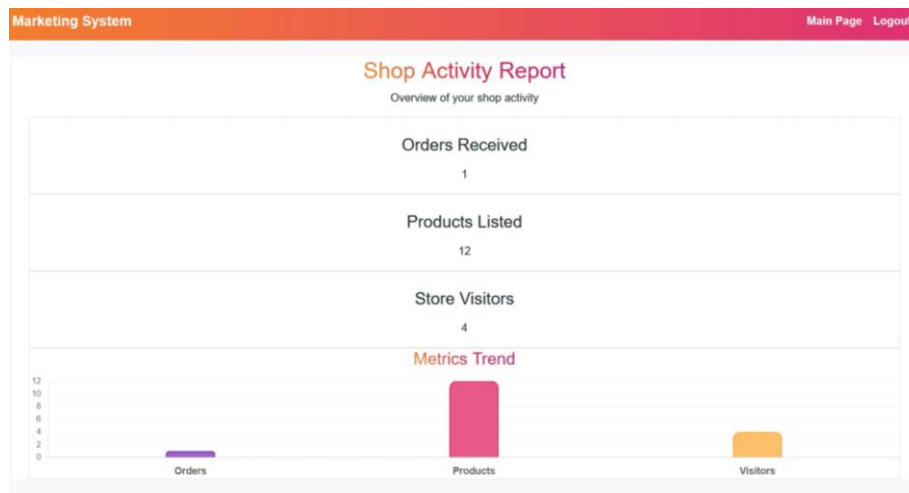


Figure (120): View Reports Interface

- **View Bazaars**

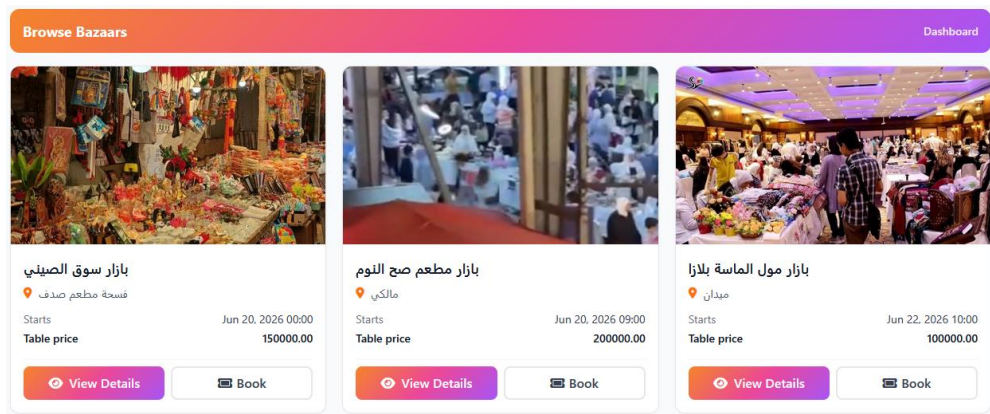


Figure (121): View Bazaars Interface

- **View Bazaar Booking Requests Status**

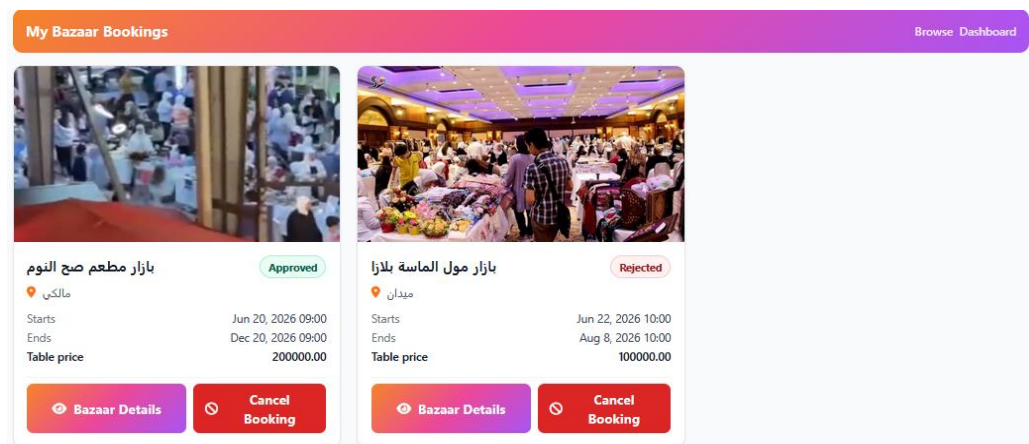


Figure (122): Bazaar Booking Requests Status Interface

- **View Notifications**

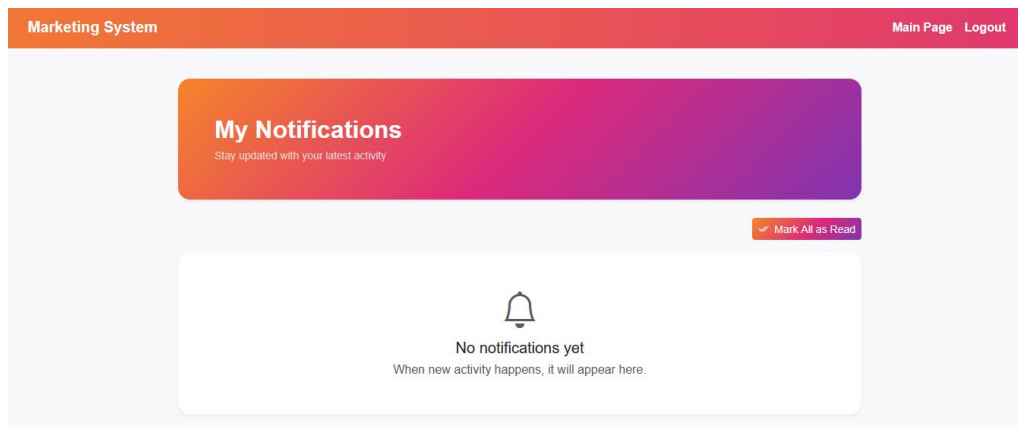


Figure (123): View Notifications Interface

2. Buyer Interfaces

- **Buyer Dashboard**

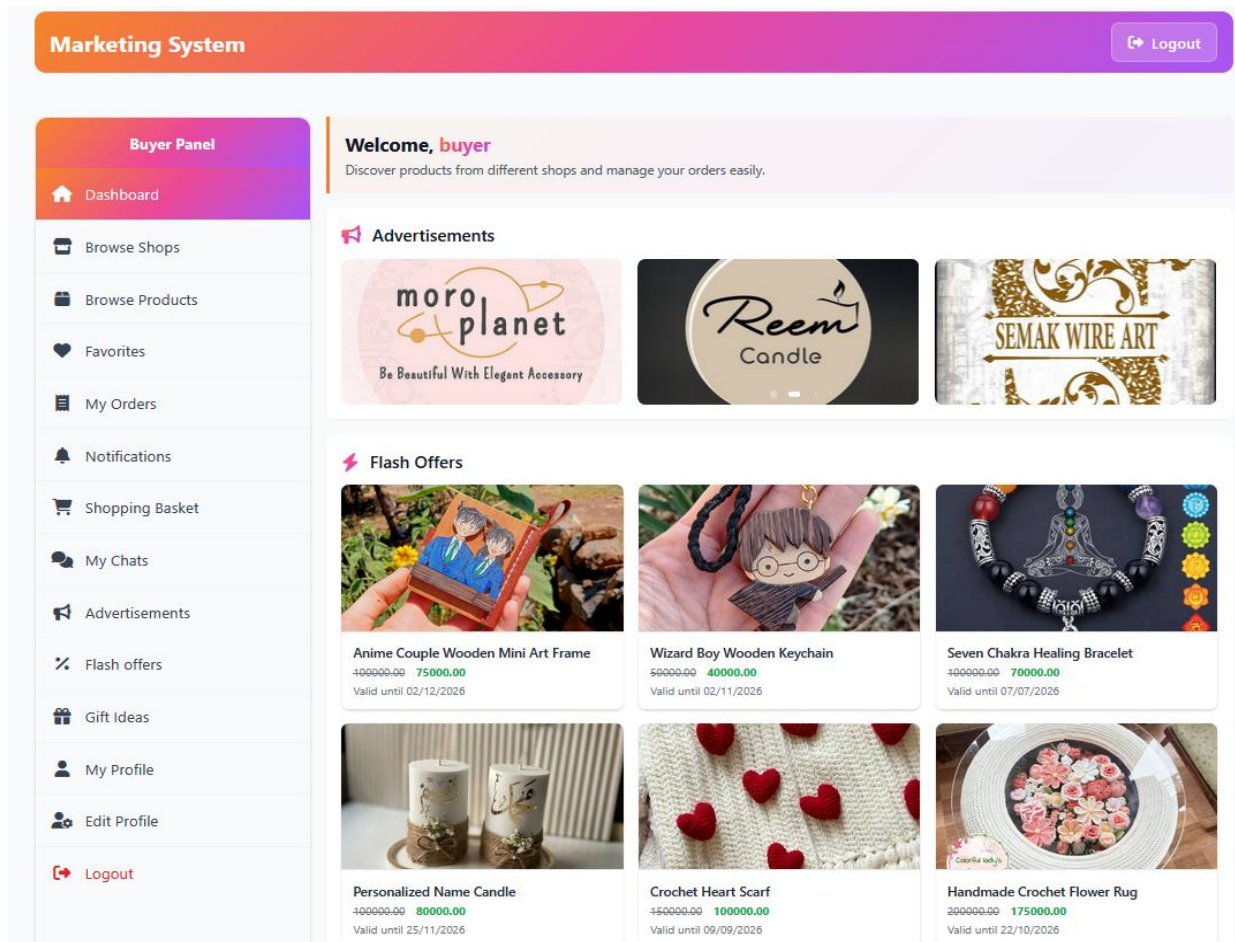


Figure (124): Buyer Dashboard Interface

- **View Shops**

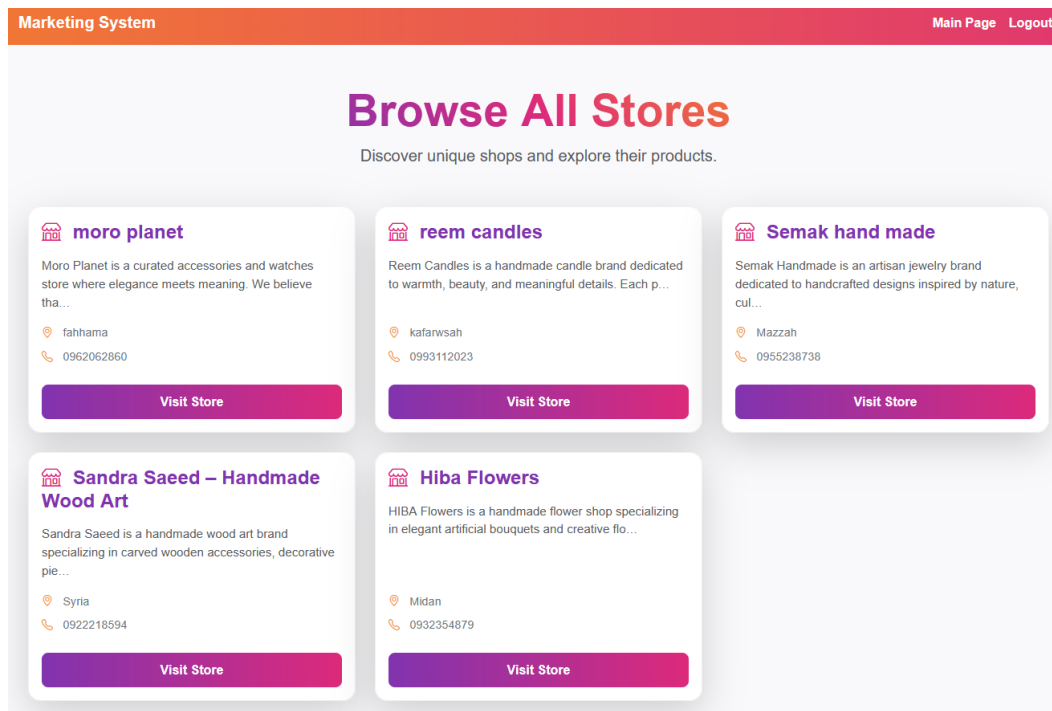


Figure (125): Shops Display Interface

- **Display Shop Details Page**

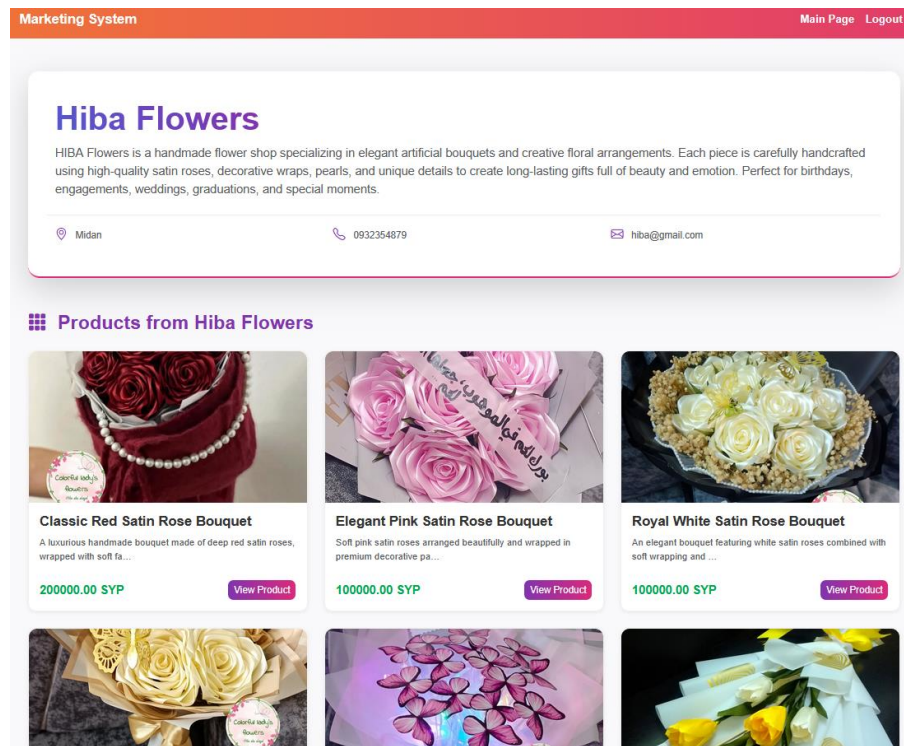


Figure (126): Store Details Display Interface

- Display Products

Marketing System
Main Page
Logout

Browse Products
View Basket

Category
Price Sort
All Categories
↑
↓



Elegant Arabic Letter Necklace
100000.00 SYP
Category: Necklaces & Pendants
Shop: more planet

View Details



Classic Luxury Bracelet – Cartier Style
60000.00 SYP
Category: Accessory
Shop: more planet

View Details



Feminine Crystal Bracelet
50000.00 SYP
Category: Accessory
Shop: more planet

View Details



Elegant Dress Sculpted Candle
50000.00 SYP
Category: Decorative Candles
Shop: reem candles

View Details



Handcrafted Floral Candle Plate
50000.00 SYP
Category: Floral Candles
Shop: reem candles

View Details



Flower Garden Candle
60000.00 SYP
Category: Floral Candles
Shop: reem candles

View Details



Personalized Name Candle
100000.00 SYP



Pastel Bubble Candle Set
150000.00 SYP



Sunflower Inspired Candle
40000.00 SYP

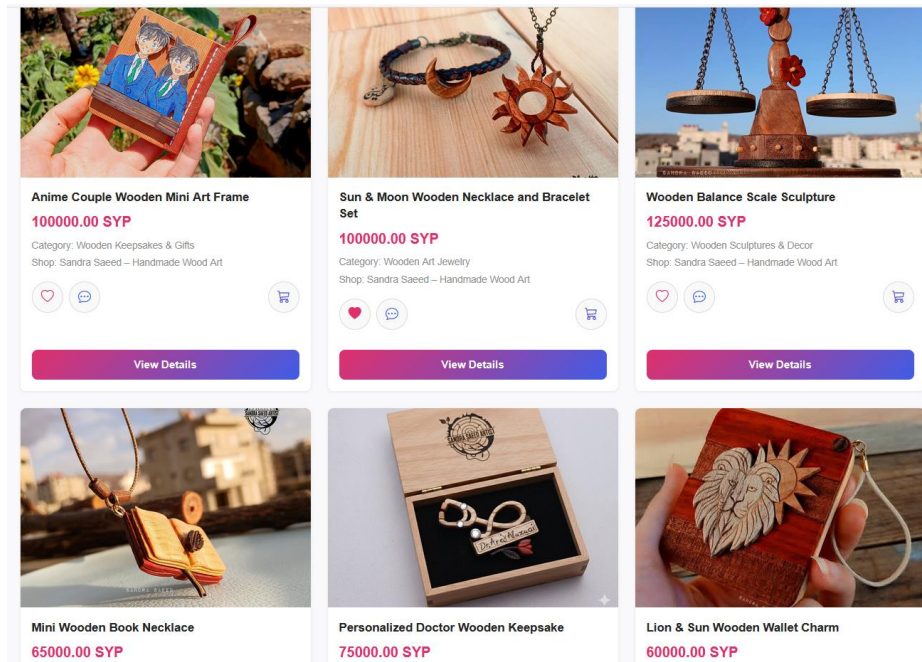


Figure (127): Products Display Interface

- **Display Product Details**

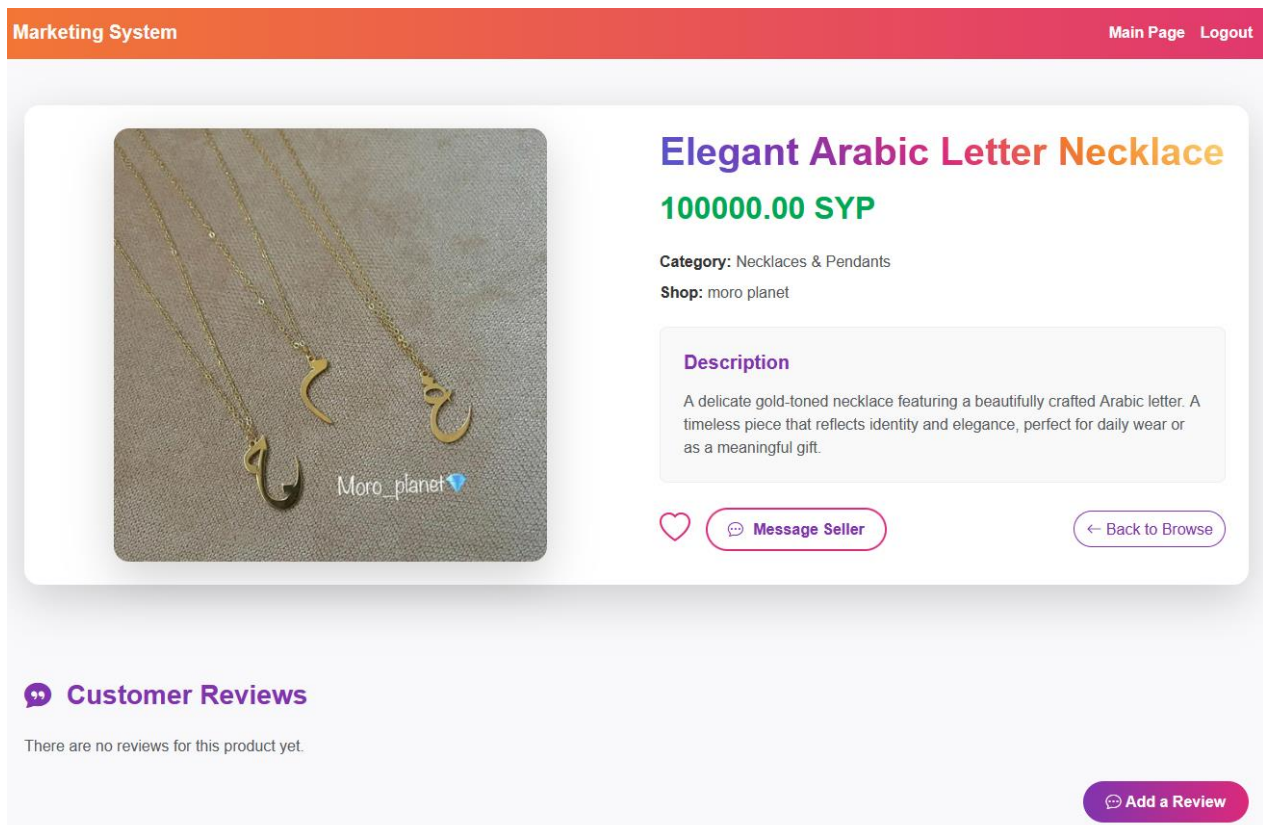


Figure (128): Product Details Display Interface

- **Display Empty Favorites List**

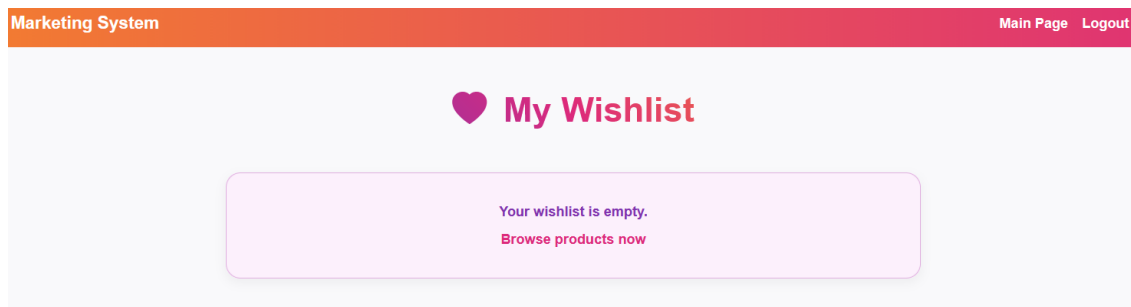


Figure (129): Empty Favorites List Display Interface

- **Display Favorites List**

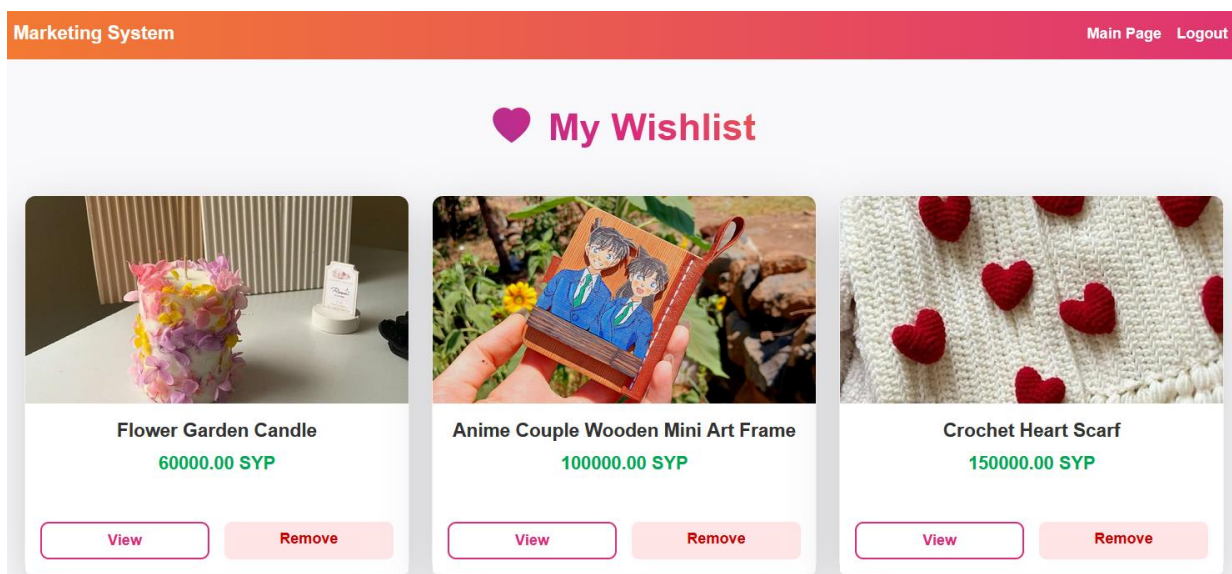


Figure (130): Favorites List Display Interface

- **Display Empty Basket**

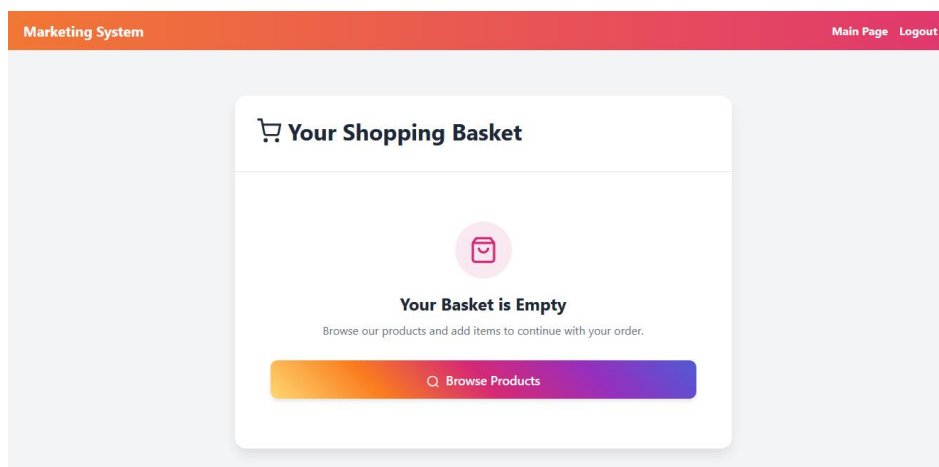


Figure (130): Empty Basket Display Interface

- **Display Basket**

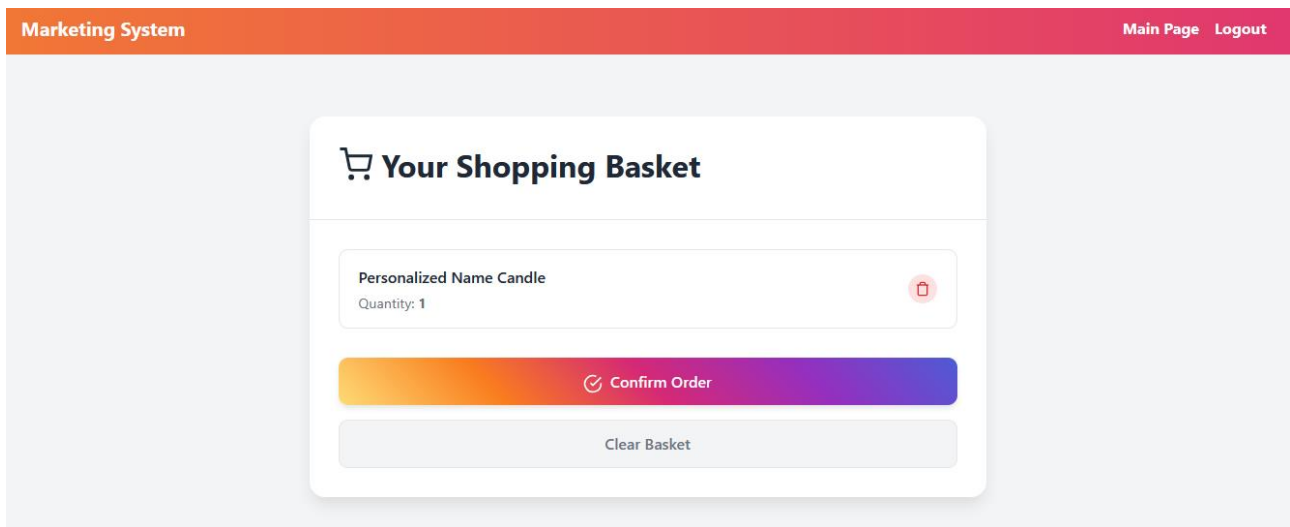


Figure (131) :Basket Display Interface

- **Empty the Basket**

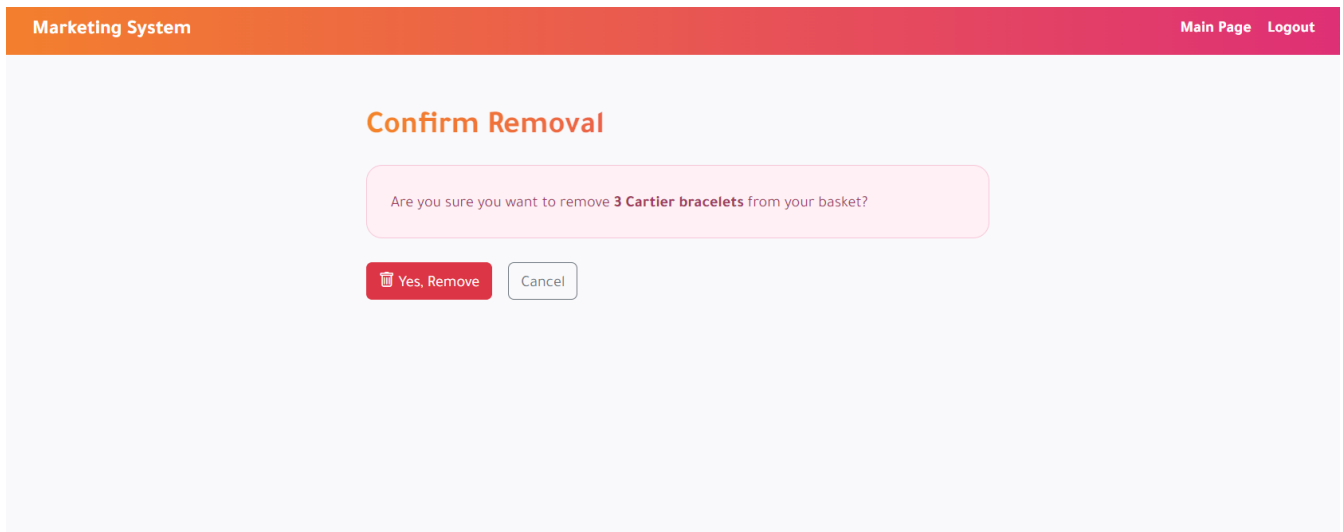


Figure (132): Empty Basket Interface

- **Purchase Order Confirmation**

Marketing System

Main PageLogout

Confirm Your Order

Please review your order and confirm the details.

Basket Contents:

Personalized Name Candle

Qty: 1

Preferred Contact Method:

☐ Phone

☐ Email

Delivery Method:

☐ Internal

☐ Manual

☐ Shipping

✔ Confirm Order

Cancel

Figure (133): Purchase Order Confirmation Interface

- **Display Order Status**

Marketing System

Main PageLogout

My Orders

Order ID	Status	Tracking
#14	Pending	Not available
#13	Approved	View Tracking
#12	Approved	View Tracking
#11	Delivered	Not available

Figure (134): Order Status Display Interface

- **Track Order Status**

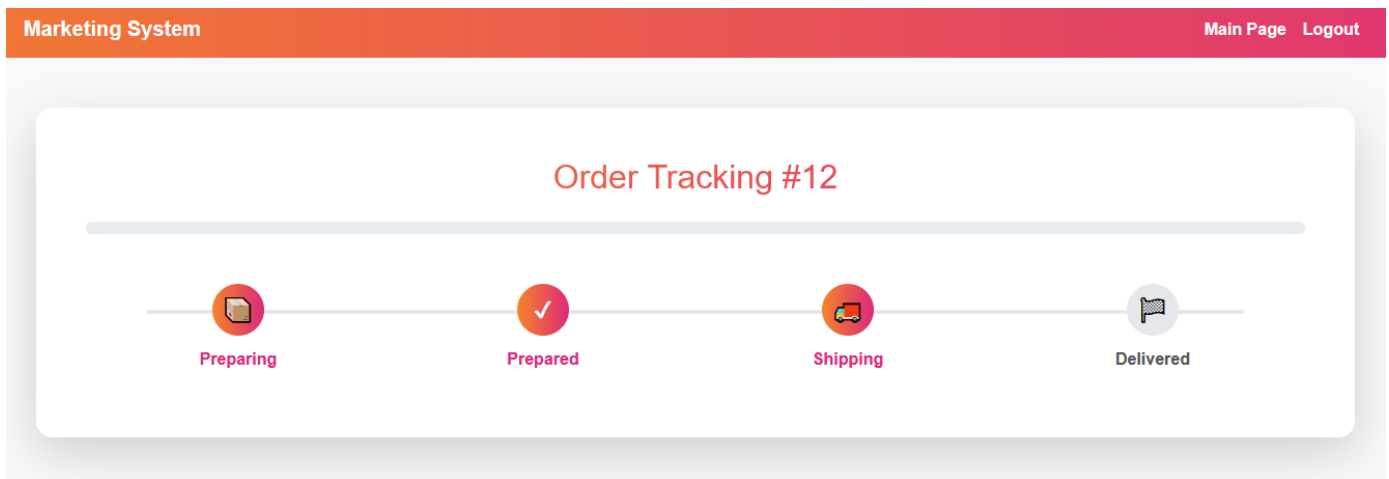


Figure (135): Order Status Tracking Interface

- **Add Comment on Product**

The interface for adding a comment is displayed within a header bar. The header bar has an orange section on the left labeled "Marketing System" and a pink section on the right with links for "Main Page" and "Logout". The main content area is a white card with a light gray shadow. At the top of the card, the text "Add Comment for: Necklace" is centered in a pink font. Below this, the text "Your comment:" is followed by a large text input field with the placeholder text "Write your comment here...". At the bottom left of the input field, there is an orange button with a speech bubble icon and the text "Submit Comment". At the bottom right of the input field, there is a gray button with a left arrow icon and the text "Back".

Figure (136): Add Comment on Product Interface

- **Display Comments**

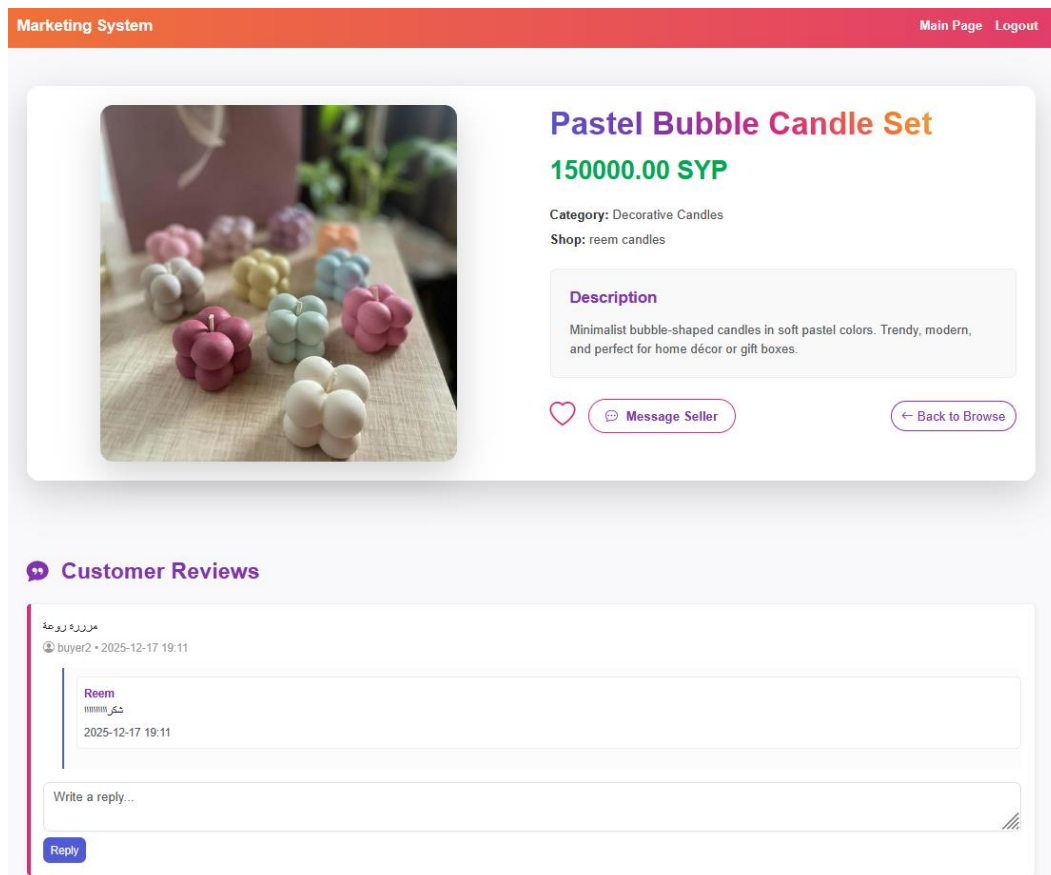


Figure (137): Comments Display Interface

- **View Conversations**

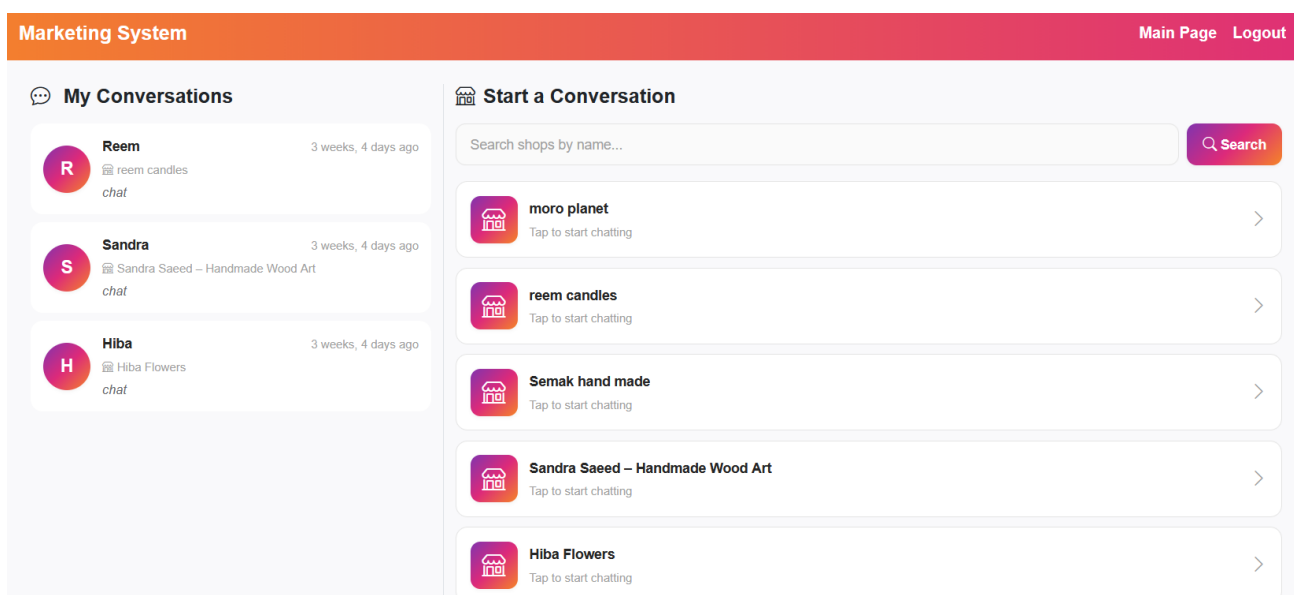


Figure (138): Buyer Conversations Display Interface

- **Start Conversation**

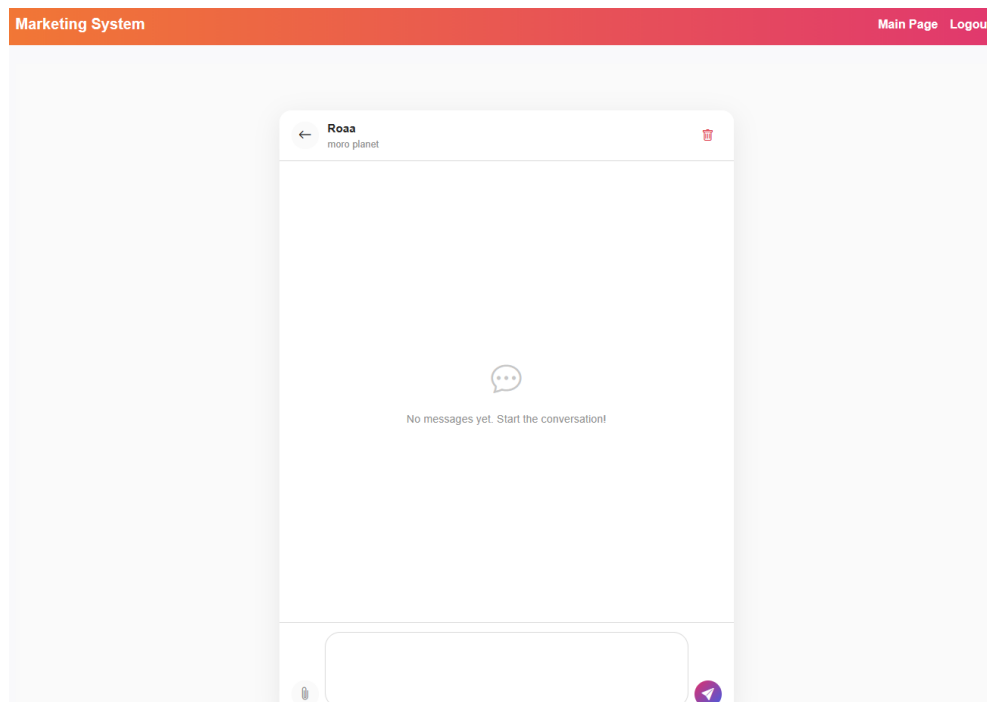


Figure (139): Buyer New Conversation Start Interface

- **Buyer Conversation**

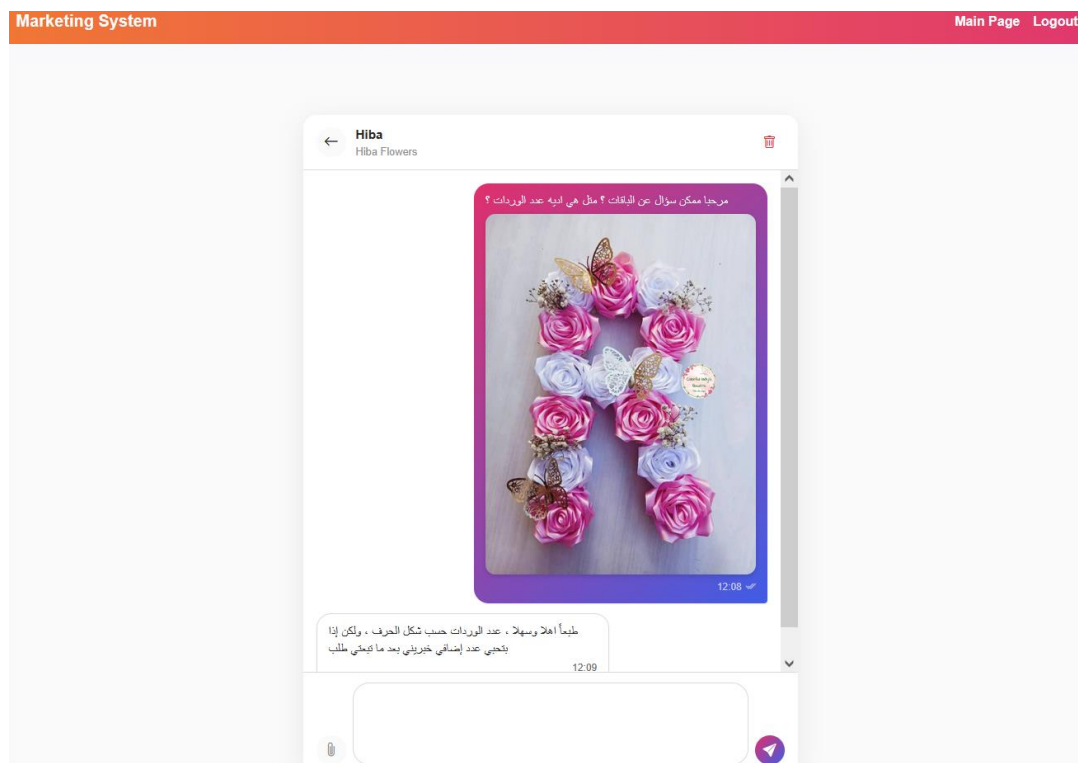


Figure (140): Buyer Conversation Interface

- Display Advertisements

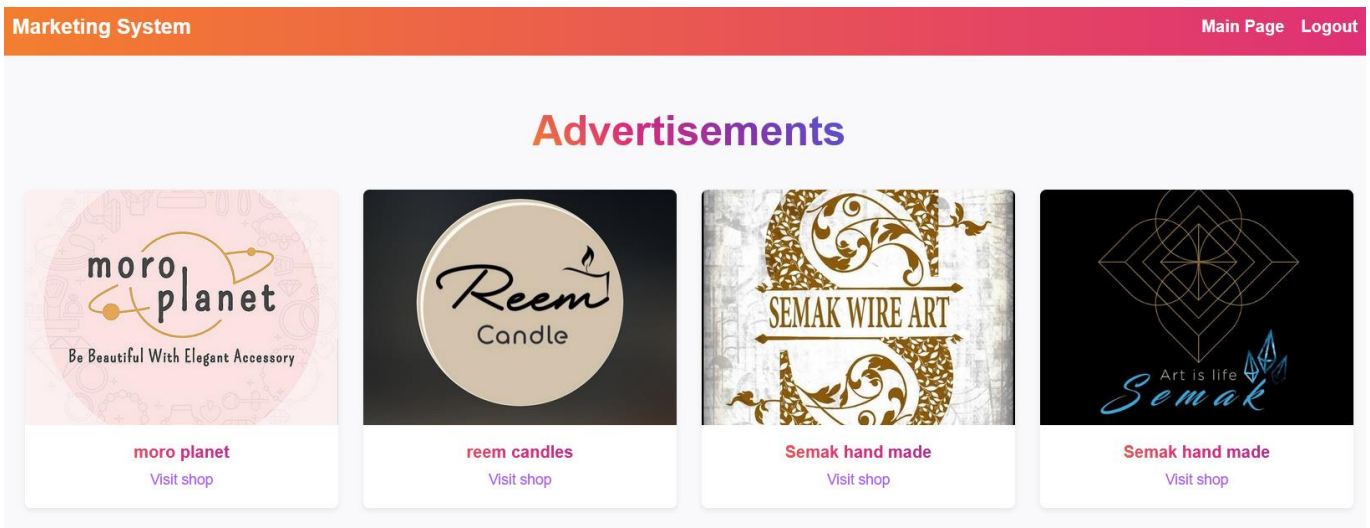


Figure (141): Advertisements Display Interface

- Display Flash Offers

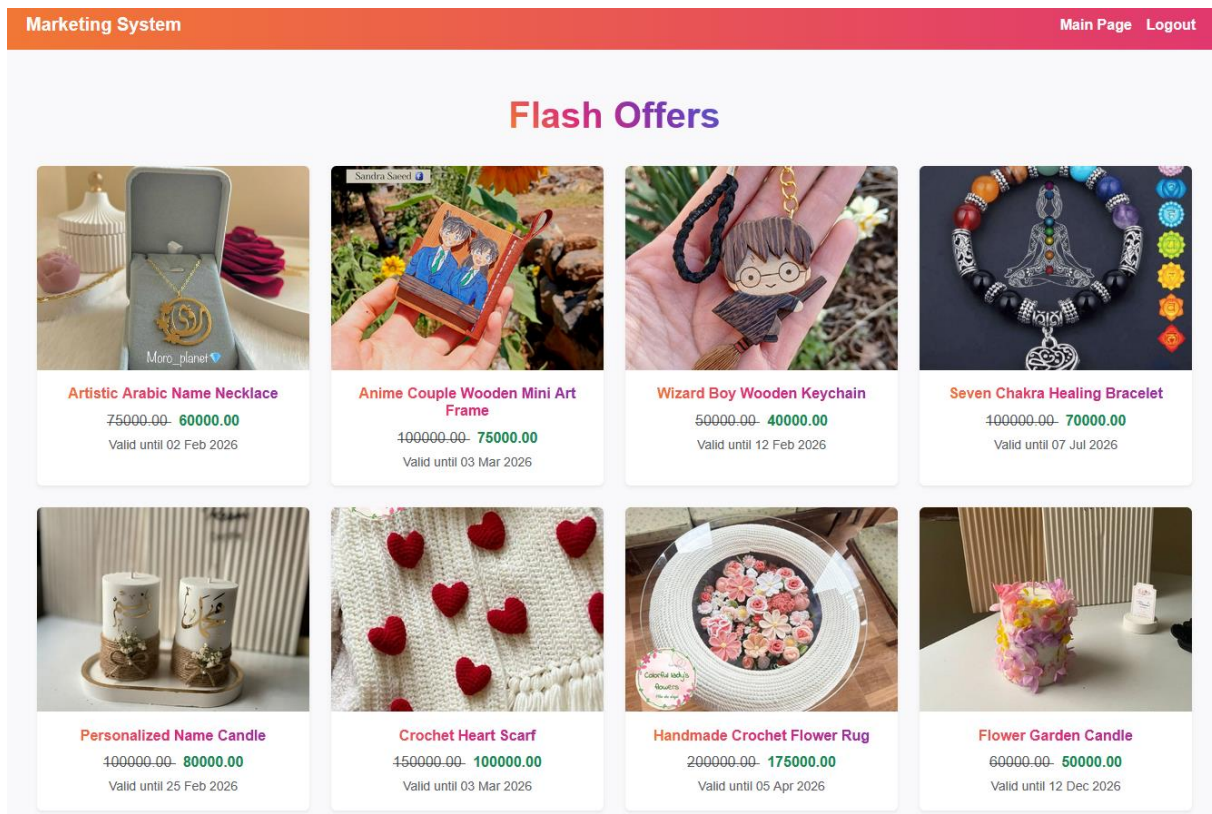


Figure (142): Flash Offers Display Interface

- **Gift Suggestion Interface Using Artificial Intelligence**

Marketing System

Main PageLogout

AI Gift Recommendation

Let our smart system suggest the perfect gift for every occasion

Budget

Occasion

Birthday

Recipient Gender

Any

Age Range

Any

Relationship

Any

Recommend Gift

Figure (143): Gift Suggestion Interface Using Artificial Intelligence

- **Display Products Suggested as Gifts**

Marketing System

Main PageLogout

AI Gift Recommendation

Let our smart system suggest the perfect gift for every occasion

Budget

100000

Occasion

Graduation

Recipient Gender

Female

Age Range

Adult

Relationship

Friend

Recommend Gift

Recommended Products

Personalized Name Candle

100000.00

A custom candle featuring elegant Arabic calligraphy, designed especially for meaningful gifts and special memories.

Sunflower Inspired Candle

40000.00

A cheerful candle inspired by sunflower blooms, symbolizing warmth and happiness. Ideal for uplifting home décor.

Figure (144): Suggested Gifts Display Interface

- **View Notifications**

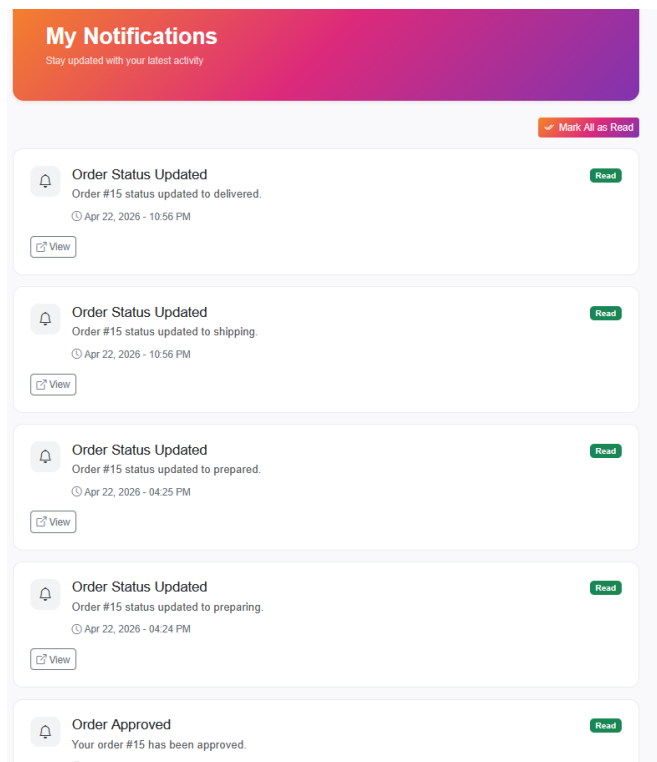


Figure (145): Notifications Display Interface

3. Administrator Interfaces

- **Administrator Dashboard**

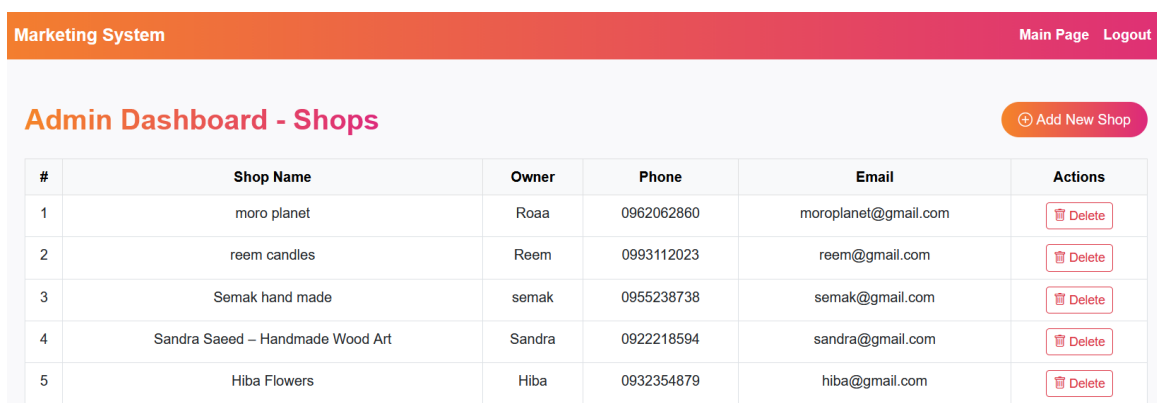


Figure (146): Administrator Dashboard Interface

- **Create Shop**

The screenshot shows a web interface for creating a new shop. At the top, there's a header bar with 'Marketing System' on the left and 'Main Page' and 'Logout' on the right. The main content area is a light gray box containing a white form titled 'Create New Shop'. The form has the following elements: a dropdown menu for 'Select Shop Owner', text input fields for 'Shop Name', 'Store Description', 'Shop Address', 'Phone Number', and 'Email Address'. At the bottom of the form, there is an orange 'Save Shop' button with a plus icon and a white 'Back' button.

Figure (147): Shop Creation Interface

4. Bazaar Organizer Interfaces

- **Bazaar Organizer Dashboard**

The screenshot displays the 'Bazaar Organizer' dashboard. The top header bar is orange and pink, with 'Marketing System' on the left and a 'Logout' button on the right. A sidebar on the left, titled 'Bazar Organizer', contains a list of navigation items: 'Dashboard' (active), 'Add Bazaar', 'My Bazaars', 'Booking Requests', 'Notifications', 'My Profile', 'Edit Profile', and a 'Logout' button at the bottom. The main content area has a light purple background. It starts with a 'Welcome, mall' message and a sub-header 'Create and manage your bazaars from the sidebar, or use the quick actions below.' Below this, there are three white cards with orange accents: 'Add Bazaar' (with a plus icon and 'Create a new bazaar listing'), 'My Bazaars' (with a storefront icon and 'View and edit your bazaars'), and 'Booking Requests' (with a calendar icon and 'View and manage booking requests'). Each card has an orange 'Open' button at the bottom.

Figure (148): Bazaar Organizer Dashboard Interface

- Add Bazaar

New bazaar

My Bazaars

Title

بازار سوق الصيني

Description

بازار يشمل منتجات يدوية تشبه المنتجات الصينية ولكنها صنع يدوي سوري

Main image

Browse...

photo_2026-05-08_16-20-03.jpg

Location

فسحة مطعم صدف

Start datetime

06/20/2026, 12:00 AM

End datetime

12/02/2026, 08:00 PM

Table price

150000

Create bazaar

Cancel

Figure (149): Add Bazaar Interface

- Bazaar Management

My Bazaars

Dashboard + Add

بازار مول الماسة بلازا

ميدان

Starts Jun 22, 2026 10:00

Ends Aug 8, 2026 10:00

Table price 100000.00

Edit Delete

بازار مطعم صبح النوم

مالكي

Starts Jun 20, 2026 09:00

Ends Dec 20, 2026 09:00

Table price 200000.00

Edit Delete

بازار سوق الصيني

فسحة مطعم صدف

Starts Jun 20, 2026 00:00

Ends Dec 2, 2026 20:00

Table price 150000.00

Edit Delete

Figure (150): Bazaar Management Interface

• Bazaar Table Booking Management

Booking Requests				Dashboard	My Bazaars
Bazaar	Seller	Status	Actions		
بازار مول الماسة بلارا Jun 22, 2026 10:00	Roa (Seller)	Cancelled	No actions		
بازار مطعم صبح التوم Jun 20, 2026 09:00	Hiba (Seller)	Approved	No actions		
بازار مول الماسة بلارا Jun 22, 2026 10:00	Hiba (Seller)	Rejected	No actions		
بازار مول الماسة بلارا Jun 22, 2026 10:00	Reem (Seller)	Cancelled	No actions		

Figure (150): Bazaar Table Booking Management Interface

• View Notifications

Marketing System
Main Page
Logout

My Notifications

Stay updated with your latest activity

Mark All as Read

Bazaar Booking Cancelled

Roa cancelled booking for bazaar "بازار مول الماسة بلارا".

Apr 22, 2026 - 04:01 PM

View

Read

New Bazaar Booking Request

Roa requested booking for bazaar "بازار مول الماسة بلارا".

Apr 22, 2026 - 03:59 PM

View

Read

Figure (151): Notifications Display Interface

4. Test Cases Execution

Table 64 Test cases execution for account management

Test case id	Test case title	Tested data	Expected result	Actual result	Pass/Fail
TC-01	Login with valid credentials	Username: valid_user / Password: valid_password	Successful login and navigation to the appropriate interface	Successfully logged in and navigated to the appropriate interface	Pass
TC-02	Login with incorrect password	Username: valid_user / Password: wrong_password	An error message indicating invalid login credentials appears	An error message indicating invalid login credentials appeared	Pass
TC-03	Login with incorrect username	Username: unknown_user / Password: any	An error message indicating invalid login credentials appears	An error message indicating invalid login credentials appeared	Pass
TC-04	Login with fields left empty	Username: (empty) / Password: (empty)	A message requesting entry of all required fields appears	A message requesting entry of all required fields appeared	Pass
TC-05	Login with three consecutive	attempts with 3 incorrect password	Temporary lockout with a retry later message	Temporarily locked out with a retry	Pass

	failed attempts			later message appeared	
TC-06	Logout from the account	Session: logged-in user	Logout and redirect to login page	Logged out and redirected to login page	Pass

Table 65 Test cases execution for shop management

Test case id	Test case title	Tested data	Expected result	Actual result	Pass/Fail
TC-07	Create a new shop by the seller	Seller account + Shop data (valid)	The shop is successfully created and displayed on the seller's page	The shop was successfully created and displayed on the seller's page	Pass
TC-08	Create a shop with a missing required field	Seller account + Missing required field	A message requesting all required information appears	A message requesting all required information appeared	Pass
TC-09	Edit shop information by the seller	Seller account + Update shop fields	Shop information is successfully updated	Shop information was successfully updated	Pass
TC-10	Edit shop information with a field left empty	Seller account + Clear required field	A message requesting all required information appears	A message requesting all required information appeared	Pass

TC-11	Delete shop by seller	Seller account + Confirm delete	The shop is successfully deleted from the system	The shop was successfully deleted from the system	Pass
TC-12	Create new shop by admin	Admin account + Select seller + Shop data (valid)	A new shop is created and linked to the selected seller	A new shop was created and linked to the selected seller	Pass
TC-13	Delete shop by admin	Admin account + Select shop + Confirm delete	The shop is successfully deleted from the system	The shop was successfully deleted from the system	Pass

Table 66 Test cases execution for product management

Test case id	Test case title	Tested data	Expected result	Actual result	Pass/Fail
TC-14	Add new product by seller	Seller account + Product data (valid)	The product is successfully added and appears among the shop products	The product was added and appeared among the shop products	Pass
TC-15	Add product with a field left empty	Seller account + Missing required field	A message requesting all required information appears	A message requesting all required information appeared	Pass

TC-16	Add an existing product	Seller account + Duplicate product (same key data)	A message appears indicating that the product already exists	A message appeared indicating that the product already exists	Pass
TC-17	Edit product information by seller	Seller account + Update product fields	Product information is updated successfully	Product information was updated successfully	Pass
TC-18	Edit product with a field left empty	Seller account + Clear required field	A message requesting all required information appears	A message requesting all required information appeared	Pass
TC-19	Delete product by seller	Seller account + Confirm delete	The product is successfully deleted from the shop	The product was successfully deleted from the shop	Pass
TC-20	Attempt to edit product by unauthorized user	Buyer account attempts seller action	User is prevented from performing the operation and access is denied	User was prevented and access denied	Pass

Table 67 Test cases execution for category management

Test case id	Test case title	Tested data	Expected result	Actual result	Pass/Fail
TC-21	Add new category by seller	Seller account + Category name (valid)	The category is successfully added and appears in the category list	The category was added and appeared in the category list	Pass
TC-22	Add category with category name left empty	Seller account + Category name: (empty)	A message requesting the category name to be entered appears	A message requesting the category name to be entered appeared	Pass
TC-23	Adding an already existing category	Seller account + Existing category name	A message indicating that the category already exists appears	A message indicating that the category already exists appeared	Pass
TC-24	Editing an existing category name	Seller account + Rename category	The category name is successfully updated	The category name was successfully updated	Pass
TC-25	Editing a category leaving the name empty	Seller account + New name: (empty)	A message requesting the category name to be entered appears	A message requesting the category name to be entered appeared	Pass

TC-26	Deleting a category by the seller	Seller account + Confirm delete	The category is successfully deleted from the system	The category was successfully deleted from the system	Pass
TC-27	Attempt to manage categories by an unauthorized user	Buyer account attempts category management	The user is prevented from accessing category management	The user was prevented from accessing category management	Pass

Table 68 Test cases execution for basket management

Test case id	Test case title	Tested data	Expected result	Actual result	Pass/Fail
TC-28	Adding a product to the basket from the same store	Buyer account + Product from Shop A	The product is successfully added to the basket	The product was successfully added to the basket	Pass
TC-29	Adding another product from the same store to the basket	Buyer account + Another product from Shop A	The second product is successfully added to the basket	The second product was successfully added to the basket	Pass
TC-30	Adding a product from a different store to the basket	Buyer account + Product from Shop B while basket has Shop A	An alert message appears indicating that purchase must be from	An alert message indicating that purchase must be from one store	Pass

			one store only	only appeared	
TC-31	Removing a product from the basket	Buyer account + Remove item	The product is successfully removed from the basket	The product was successfully removed from the basket	Pass
TC-32	Displaying basket contents	Buyer account + Open basket page	All products in the basket are displayed	All products in the basket were displayed	Pass
TC-33	Accessing basket without login	Guest user opens basket URL	Redirect to login page	Redirected to login page	Pass
TC-34	Attempt to manage basket by unauthorized user	Seller account attempts basket action	User is prevented from performing action	User was prevented from performing action	Pass

Table 69 Test cases execution for orders management

Test case id	Test case title	Tested data	Expected result	Actual result	Pass/Fail
TC-35	Creating a purchase order by buyer	Buyer account + Basket (not empty) + Contact method + Delivery method	Order is successfully created and saved with status Pending Review	Order was created and saved with status Pending Review	Pass

TC-36	Attempt to create order with empty basket	Buyer account + Empty basket	A message appears indicating no products in basket	A message appeared indicating no products in basket	Pass
TC-37	Purchase order acceptance by seller	Seller account (shop owner) + Pending order	Order is accepted and its status updated to Accepted	Order was accepted and its status updated to Accepted	Pass
TC-38	Purchase order rejection by seller	Seller account (shop owner) + Pending order	Order is rejected and its status updated to Rejected	Order was rejected and its status updated to Rejected	Pass
TC-39	Displaying order status by buyer	Buyer account + Existing order	Order is displayed with current status	Order was displayed with current status	Pass
TC-40	Updating order status to Preparing	Seller account + Approved order	Order status is successfully updated to Preparing	Order status was updated to Preparing	Pass
TC-41	Updating order status to Prepared	Seller account + Order in (Preparing)	Order status is updated to Prepared	Order status was updated to Prepared	Pass
TC-42	Updating order status to Out for Delivery	Seller account + Order in (Prepared)	Order status is updated to Out for Delivery	Order status was updated to Out for Delivery	Pass

TC-43	Update order status to Delivered	Seller account + Order in (In Delivery)	The order status is updated to Delivered and the order is closed	The order status has been updated to Delivered and the order closed	Pass
TC-44	Attempt to update status of an unacceptable order	Seller account + Order status: rejected/pending	The action is blocked with an explanatory message	The action was blocked and an explanatory message appeared	Pass
TC-45	Attempt to track order by unauthorized user	Seller account tries to access order not in his shop	The user is prevented from accessing the order	The user was prevented from accessing the order	Pass

Table 70 Test cases execution for search and filtering

Test case id	Test case title	Tested data	Expected result	Actual result	Pass/Fail
TC-46	Display products by selected category	Select category with products	Only products belonging to the selected category are displayed	Only products of the selected category were displayed	Pass
TC-47	Display products within an empty category	Select empty category	A message appears indicating no products exist in this category	A message appeared indicating no products exist in this category	Pass

TC-48	Filter products by price from low to high	Apply price sort: low to high	Products are sorted ascending by price	Products were sorted ascending by price	Pass
TC-49	Filter products by price from high to low	Apply price sort: high to low	Products are sorted descending by price	Products were sorted descending by price	Pass
TC-50	Apply price filtering within a selected category	Select category + Apply price sort	Display selected category products sorted by price	Displayed selected category products sorted by price	Pass

Table 71 Test cases execution for comments management

Test case id	Test case title	Tested data	Expected result	Actual result	Pass/Fail
TC-51	Add comment after purchasing and receiving the product	Buyer account + Delivered order for product	The comment is successfully added and displayed below the product	The comment was added and displayed below the product	Pass
TC-52	Attempt to add comment without purchasing the product	Buyer account + Product not purchased	Adding comment is blocked with an explanatory message	Adding comment was blocked and an explanatory message appeared	Pass

TC-53	Attempt to add an empty comment	Buyer account + Comment: (empty)	Display a message requesting comment text input	A message requesting comment text input appeared	Pass
TC-54	Display comments related to a product	Open product details with comments	Show all comments associated with the product	All comments associated with the product were displayed	Pass
TC-55	Seller's reply to a comment	Seller account + Reply text (valid)	Successfully add the seller's reply below the comment	The seller's reply was added below the comment	Pass
TC-56	Attempt by seller to reply with an empty comment	Seller account + Reply: (empty)	Display a message requesting reply text input	A message requesting reply text input appeared	Pass
TC-57	Attempt to add a reply by an unauthorized user	Buyer account tries to reply	Prevent the user from performing the operation	User was prevented from performing action	Pass

Table 72 Test cases execution for advertisements management

Test case id	Test case title	Tested data	Expected result	Actual result	Pass/Fail
TC-58	Add a new advertisement by the seller	Seller account + Ad data + Image	Successfully add the advertisement and show it within the ads list	The advertisement was added and displayed within the ads list	Pass
TC-59	Add an advertisement without attaching an image	Seller account + Ad data + No image	Display a message requesting to attach an image to the advertisement	A message requesting to attach an image to the advertisement appeared	Pass
TC-60	Modify an existing advertisement by the seller	Seller account + Update ad fields	Successfully update the advertisement data	Advertisement data was successfully updated	Pass
TC-61	Modify an advertisement leaving a mandatory field empty	Seller account + Clear required field	A message requesting entry of all required fields appears	A message requesting entry of all required fields appeared	Pass
TC-62	Delete an advertisement by the seller	Seller account + Confirm delete	Successfully delete the advertisement from the system	The advertisement was successfully deleted from the system	Pass
TC-63	Attempt to manage advertisements	Buyer account	Prevent the user from	The user was prevented from	Pass

	by an unauthorized user	attempts ads management	accessing ad management	accessing ad management	
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Table 73 Test cases execution for flash offers management

Test case id	Test case title	Tested data	Expected result	Actual result	Pass/Fail
TC-66	Add new temporary offer by seller	Seller account + Product + New price + Duration	Temporary offer created and activated successfully	The temporary offer was created and activated	Pass
TC-67	Add temporary offer with a mandatory field left empty	Seller account + Missing required field	A message requesting all required information appears	A message requesting all required information appeared	Pass
TC-68	Add temporary offer with invalid price	Seller account + Invalid price	Offer creation rejected with explanatory message	Offer creation was rejected and an explanatory message appeared	Pass
TC-69	Modify existing temporary offer	Seller account + Update offer fields	Temporary offer updated successfully	The temporary offer was updated successfully	Pass
TC-70	Delete temporary	Seller account +	Temporary offer deleted successfully	The temporary offer was	Pass

	offer by seller	Confirm delete	from the system	successfully deleted from the system	
TC-71	Temporary offer duration expires automatically	Active offer reaches end time	Offer stops automatically and original product price is restored	The offer stopped automatically and the original price was restored	Pass
TC-72	Unauthorized user attempts temporary offer management	Buyer account attempts offer management	Prevent the user from performing the operation	User was prevented from performing action	Pass

Table 74 Test cases execution for wishlist management

Test case id	Test case title	Tested data	Expected result	Actual result	Pass/Fail
TC-73	Add product to wishlist	Buyer account + Click add to wishlist	Product added to wishlist successfully	The product was added to the wishlist	Pass
TC-74	Remove product from wishlist	Buyer account + Remove wishlist item	Product removed from wishlist successfully	The product was removed from the wishlist	Pass
TC-75	Display Favorites List	Buyer account + Open wishlist page	Display all products added to wishlist	All products added to the wishlist were displayed	Pass

TC-76	View wishlist while it is empty	Buyer account + Empty wishlist	Message appears indicating the wishlist is empty	A message appeared indicating the wishlist is empty	Pass
TC-77	Attempt to use wishlist without login	Guest user tries add to wishlist	Redirect to login page	Redirected to login page	Pass
TC-78	Attempt to manage wishlist by unauthorized user	Seller account attempts wishlist	Prevent the user from performing the operation	User was prevented from performing action	Pass

Table 75 Test cases execution for reports system

Test case id	Test case title	Tested data	Expected result	Actual result	Pass/Fail
TC-79	Display orders report by seller	Seller account + Existing orders	Display orders report of seller's store	Orders report of seller's store displayed	Pass
TC-80	Display orders report when no orders exist	Seller account + No orders	Display message indicating no data to show	Message indicating no data to show appeared	Pass
TC-81	Attempt to access reports system by unauthorized user	Buyer account attempts reports page	Prevent user from accessing reports system	User was prevented from accessing reports system	Pass

Table 76 Test cases execution for messaging system

Test case id	Test case title	Tested data	Expected result	Actual result	Pass/Fail
TC-82	Send valid text message	Buyer/Seller account + Message text	Send the message and display it in the conversation	Message was sent and appeared in the conversation	Pass
TC-83	Attempt to send empty message	Message: (empty)	Display alert message requesting message text input	Alert message requesting message text input appeared	Pass
TC-84	Display conversation history	Open messages page	Display all user's conversations	All user's conversations displayed	Pass
TC-85	Attempt to access conversation not owned by user	Open conversation not owned by user	Prevent user from accessing the conversation	User was prevented from accessing the conversation	Pass

Table 77 Test cases execution for AI gift recommendation feature

Test case id	Test case title	Tested data	Expected result	Actual result	Pass/Fail
TC-86	Get gift suggestion using AI	Buyer input: budget + gender + age + relation + occasion (valid)	Display list of suggested products	List of suggested products displayed	Pass

TC-87	Attempt to use feature with incomplete data	Buyer input: missing required field(s)	Display message requesting all required data to be entered	Message requesting all required data to be entered appeared	Pass
TC-88	Failed to connect to AI service	Simulated API failure / timeout	A message appears indicating the service is currently unavailable	A message appeared indicating the service is currently unavailable	Pass

Table 78 Test cases execution for bazaar management

Test case id	Test case title	Tested data	Expected result	Actual result	Pass/Fail
TC-89	Creating a new bazaar by the bazaar organizer	Bazaar organizer + Valid bazaar data	The bazaar is successfully created and displayed within the list of bazaars	The bazaar was successfully created and displayed within the list of bazaars	Pass
TC-90	Creating a bazaar while leaving a required field empty	Bazaar organizer + Missing required field	A message requesting all required information appears	A message requesting all required information appeared	Pass
TC-91	Modifying data of an existing bazaar	Bazaar organizer + Update bazaar data	The bazaar data is successfully updated	The bazaar data was successfully updated	Pass

TC-92	Deleting a bazaar by the bazaar organizer	Bazaar organizer + Confirm delete	The bazaar is successfully deleted from the system	The bazaar was successfully deleted from the system	Pass
TC-93	Attempt to manage bazaars by an unauthorized user	Buyer/Seller account attempts bazaar management	The user is prevented from accessing bazaar management	The user was prevented from accessing bazaar management	Pass

Table 79 Test cases execution for bazaar table booking management

Test case id	Test case title	Tested data	Expected result	Actual result	Pass/Fail
TC-94	Sending a table booking request in a bazaar	Seller account + Bazaar booking request	The booking request is successfully sent with a status of pending review	The booking request was successfully sent with a status of pending review	Pass
TC-95	Attempt to book more than one table in the same bazaar	Seller account + Duplicate booking request	A message appears indicating that booking more than one table in the same bazaar is not allowed	A message appeared indicating that booking more than one table in the same bazaar is not allowed	Pass
TC-96	Accepting a booking	Bazaar organizer +	The booking request is	The booking request was	Pass

	request by the bazaar organizer	Pending booking request	accepted and its status is updated to accepted	accepted and its status was updated to accepted	
TC-97	Rejecting a booking request by the bazaar organizer	Bazaar organizer + Pending booking request	The booking request is rejected and its status is updated to rejected	The booking request was rejected and its status was updated to rejected	Pass
TC-98	Cancelling a booking request before the bazaar date	Seller account + Existing booking request	The booking request is successfully cancelled	The booking request was successfully cancelled	Pass
TC-99	Attempt to manage booking requests by an unauthorized user	Buyer account attempts booking management	User is prevented from performing action	User was prevented from performing action	Pass

Table 80 Test cases execution for notifications system

Test case id	Test case title	Tested data	Expected result	Actual result	Pass/Fail
TC-100	Receiving notification upon purchase order approval	Approved order	A new notification is displayed to the buyer upon order approval	A new notification was displayed to the buyer	Pass

				upon order approval	
TC-101	Receiving notification upon purchase order rejection	Rejected order	A new notification is displayed to the buyer upon order rejection	A new notification was displayed to the buyer upon order rejection	Pass
TC-102	Receiving notification upon order status update	Order status update	A new notification is displayed indicating the order status update	A new notification was displayed indicating the order status update	Pass
TC-103	Receiving notification upon bazaar booking acceptance	Accepted bazaar booking	A new notification is displayed to the seller upon acceptance of the booking request	A new notification was displayed to the seller upon acceptance of the booking request	Pass
TC-104	Receiving notification upon bazaar booking rejection	Rejected bazaar booking	A new notification is displayed to the seller upon rejection of the booking request	A new notification was displayed to the seller upon rejection of the booking request	Pass

TC-105	Display number of unread notifications	Unread notifications exist	The number of unread notifications is displayed next to the notifications icon	The number of unread notifications was displayed next to the notifications icon	Pass
TC-106	Show notifications page	Open notifications page	All notifications related to the user are displayed	All notifications related to the user were displayed	Pass

5. Final Requirements Traceability Matrix (RTM)

Req-ID	Requirement Title	Associated Dependency	SRS Section	High Level Design	Detailed Design	Coding	Component Test	Integration Test	System Test	Test Case ID(s)	Changed Requests No
REQ-1.1	The system must allow the user to create an account.	Account Management	§3 SRS – REQ-1.1 (p. 74)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Account Management (p. 162)	users app – CustomUser	Manual functional test (TC-REQ-1.1) – p. 183	End-to-end scenario test (TC-REQ-1.1-INT) – p. 267	System workflow test (ST-REQ-1.1) – p. 267	TC-REQ-1.1-01, TC-REQ-1.1-02	
REQ-1.2	The system must allow the user to log in using a username and password.	Account Management	§3 SRS – REQ-1.2 (p. 75)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Account Management (p. 162)	users app – CustomUser	Manual functional test (TC-REQ-1.2) – p. 183	End-to-end scenario test (TC-REQ-1.2-INT) – p. 267	System workflow test (ST-REQ-1.2) – p. 267	TC-REQ-1.2-01, TC-REQ-1.2-02	
REQ-1.3	The system must allow the user to log out of their account.	Account Management	§3 SRS – REQ-1.3 (p. 75)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Account Management (p. 162)	users app – CustomUser	Manual functional test (TC-REQ-1.3) – p. 183	End-to-end scenario test (TC-REQ-1.3-INT) – p. 267	System workflow test (ST-REQ-1.3) – p. 267	TC-REQ-1.3-01, TC-REQ-1.3-02	
REQ-2.1	The system must allow the seller to add their shop information.	Shop Management	§3 SRS – REQ-2.1 (p. 75)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Shop Management (p. 162)	shops app – Shop	Manual functional test (TC-REQ-2.1) – p. 184	End-to-end scenario test (TC-REQ-2.1-INT) – p. 268	System workflow test (ST-REQ-2.1) – p. 268	TC-REQ-2.1-01, TC-REQ-2.1-02	
REQ-2.2	The system must allow the seller to edit their shop information.	Shop Management	§3 SRS – REQ-2.2 (p. 75)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Shop Management (p. 162)	shops app – Shop	Manual functional test (TC-REQ-2.2) – p. 184	End-to-end scenario test (TC-REQ-2.2-INT) – p. 268	System workflow test (ST-REQ-2.2) – p. 268	TC-REQ-2.2-01, TC-REQ-2.2-02	
REQ-2.3	The system must allow the seller to delete their shop.	Shop Management	§3 SRS – REQ-2.3 (p. 75)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Shop Management (p. 162)	shops app – Shop	Manual functional test (TC-REQ-2.3) – p. 184	End-to-end scenario test (TC-REQ-2.3-INT) – p. 268	System workflow test (ST-REQ-2.3) – p. 268	TC-REQ-2.3-01, TC-REQ-2.3-02	
REQ-2.4	The system must allow the administrator to create a shop for any seller when needed.	Shop Management	§3 SRS – REQ-2.4 (p. 75)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Shop Management (p. 162)	shops app – Shop	Manual functional test (TC-REQ-2.4) – p. 184	End-to-end scenario test (TC-REQ-2.4-INT) – p. 268	System workflow test (ST-REQ-2.4) – p. 268	TC-REQ-2.4-01, TC-REQ-2.4-02	
REQ-2.5	The system must allow the administrator to delete a shop.	Shop Management	§3 SRS – REQ-2.5 (p. 75)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Shop Management (p. 162)	shops app – Shop	Manual functional test (TC-REQ-2.5) – p. 184	End-to-end scenario test (TC-REQ-2.5-INT) – p. 268	System workflow test (ST-REQ-2.5) – p. 268	TC-REQ-2.5-01, TC-REQ-2.5-02	
REQ-2.6	The system must allow the administrator to View registered shops.	Shop Management	§3 SRS – REQ-2.6 (p. 75)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Shop Management (p. 162)	shops app – Shop	Manual functional test (TC-REQ-2.6) – p. 184	End-to-end scenario test (TC-REQ-2.6-INT) – p. 268	System workflow test (ST-REQ-2.6) – p. 268	TC-REQ-2.6-01, TC-REQ-2.6-02	
REQ-2.7	The system allows displaying all products belonging to a specific shop on a dedicated shop page.	Shop Management	§3 SRS – REQ-2.7 (p. 75)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Shop Management (p. 162)	shops app – Shop	Manual functional test (TC-REQ-2.7) – p. 184	End-to-end scenario test (TC-REQ-2.7-INT) – p. 268	System workflow test (ST-REQ-2.7) – p. 268	TC-REQ-2.7-01, TC-REQ-2.7-02	
REQ-3.1	The system must allow the seller to add new products.	Product Management	§3 SRS – REQ-3.1 (p. 76)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Product Management (p. 163)	products app – Product	Manual functional test (TC-REQ-3.1) – p. 185	End-to-end scenario test (TC-REQ-3.1-INT) – p. 269	System workflow test (ST-REQ-3.1) – p. 269	TC-REQ-3.1-01, TC-REQ-3.1-02	
REQ-3.2	The system must allow the seller to edit product information.	Product Management	§3 SRS – REQ-3.2 (p. 76)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Product Management (p. 163)	products app – Product	Manual functional test (TC-REQ-3.2) – p. 185	End-to-end scenario test (TC-REQ-3.2-INT) – p. 269	System workflow test (ST-REQ-3.2) – p. 269	TC-REQ-3.2-01, TC-REQ-3.2-02	
REQ-3.3	The system must allow the seller to delete products.	Product Management	§3 SRS – REQ-3.3 (p. 76)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Product Management (p. 163)	products app – Product	Manual functional test (TC-REQ-3.3) – p. 185	End-to-end scenario test (TC-REQ-3.3-INT) – p. 269	System workflow test (ST-REQ-3.3) – p. 269	TC-REQ-3.3-01, TC-REQ-3.3-02	
REQ-3.4	The system must display the seller's products on their personal shop page.	Product Management	§3 SRS – REQ-3.4 (p. 76)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Product Management (p. 163)	products app – Product	Manual functional test (TC-REQ-3.4) – p. 185	End-to-end scenario test (TC-REQ-3.4-INT) – p. 269	System workflow test (ST-REQ-3.4) – p. 269	TC-REQ-3.4-01, TC-REQ-3.4-02	
REQ-3.5	The system must allow the seller to classify products into specific categories.	Product Management	§3 SRS – REQ-3.5 (p. 76)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Product Management (p. 163)	products app – Product	Manual functional test (TC-REQ-3.5) – p. 185	End-to-end scenario test (TC-REQ-3.5-INT) – p. 269	System workflow test (ST-REQ-3.5) – p. 269	TC-REQ-3.5-01, TC-REQ-3.5-02	

REQ-3.6	The system must allow the buyer to view products.	Product Management	§3 SRS – REQ-3.6 (p. 76)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Product Management (p. 163)	products app – Product	Manual functional test (TC-REQ-3.6) – p. 185	End-to-end scenario test (TC-REQ-3.6-INT) – p. 269	System workflow test (ST-REQ-3.6) – p. 269	TC-REQ-3.6-01, TC-REQ-3.6-02	
REQ-3.7	The system must allow the buyer to view product details.	Product Management	§3 SRS – REQ-3.7 (p. 76)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Product Management (p. 163)	products app – Product	Manual functional test (TC-REQ-3.7) – p. 185	End-to-end scenario test (TC-REQ-3.7-INT) – p. 269	System workflow test (ST-REQ-3.7) – p. 269	TC-REQ-3.7-01, TC-REQ-3.7-02	
REQ-4.1	The system must allow the buyer to add products to the basket.	Basket Management	§3 SRS – REQ-4.1 (p. 76)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Basket Management (p. 163)	baskets app – Basket / BasketItem	Manual functional test (TC-REQ-4.1) – p. 189	End-to-end scenario test (TC-REQ-4.1-INT) – p. 272	System workflow test (ST-REQ-4.1) – p. 272	TC-REQ-4.1-01, TC-REQ-4.1-02	
REQ-4.2	The system must allow the buyer to remove products from the basket.	Basket Management	§3 SRS – REQ-4.2 (p. 76)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Basket Management (p. 163)	baskets app – Basket / BasketItem	Manual functional test (TC-REQ-4.2) – p. 189	End-to-end scenario test (TC-REQ-4.2-INT) – p. 272	System workflow test (ST-REQ-4.2) – p. 272	TC-REQ-4.2-01, TC-REQ-4.2-02	
REQ-4.3	The system must allow the buyer to view the contents of the basket before confirming the order.	Basket Management	§3 SRS – REQ-4.3 (p. 76)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Basket Management (p. 163)	baskets app – Basket / BasketItem	Manual functional test (TC-REQ-4.3) – p. 189	End-to-end scenario test (TC-REQ-4.3-INT) – p. 272	System workflow test (ST-REQ-4.3) – p. 272	TC-REQ-4.3-01, TC-REQ-4.3-02	
REQ-5.1	The system must allow the seller to view incoming purchase orders to their shop.	Order & Purchase Management	§3 SRS – REQ-5.1 (p. 77)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Order & Purchase Management (p. 164)	orders app – Order / OrderTracking	Manual functional test (TC-REQ-5.1) – p. 190	End-to-end scenario test (TC-REQ-5.1-INT) – p. 273	System workflow test (ST-REQ-5.1) – p. 273	TC-REQ-5.1-01, TC-REQ-5.1-02	
REQ-5.2	The system must allow the seller to accept purchase orders.	Order & Purchase Management	§3 SRS – REQ-5.2 (p. 77)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Order & Purchase Management (p. 164)	orders app – Order / OrderTracking	Manual functional test (TC-REQ-5.2) – p. 190	End-to-end scenario test (TC-REQ-5.2-INT) – p. 273	System workflow test (ST-REQ-5.2) – p. 273	TC-REQ-5.2-01, TC-REQ-5.2-02	
REQ-5.3	The system must allow the seller to reject purchase orders.	Order & Purchase Management	§3 SRS – REQ-5.3 (p. 77)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Order & Purchase Management (p. 164)	orders app – Order / OrderTracking	Manual functional test (TC-REQ-5.3) – p. 190	End-to-end scenario test (TC-REQ-5.3-INT) – p. 273	System workflow test (ST-REQ-5.3) – p. 273	TC-REQ-5.3-01, TC-REQ-5.3-02	
REQ-5.4	The system must allow the buyer to place a purchase order for products in the cart.	Order & Purchase Management	§3 SRS – REQ-5.4 (p. 77)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Order & Purchase Management (p. 164)	orders app – Order / OrderTracking	Manual functional test (TC-REQ-5.4) – p. 190	End-to-end scenario test (TC-REQ-5.4-INT) – p. 273	System workflow test (ST-REQ-5.4) – p. 273	TC-REQ-5.4-01, TC-REQ-5.4-02	
REQ-5.5	The system must allow the buyer to choose a communication method.	Order & Purchase Management	§3 SRS – REQ-5.5 (p. 77)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Order & Purchase Management (p. 164)	orders app – Order / OrderTracking	Manual functional test (TC-REQ-5.5) – p. 190	End-to-end scenario test (TC-REQ-5.5-INT) – p. 273	System workflow test (ST-REQ-5.5) – p. 273	TC-REQ-5.5-01, TC-REQ-5.5-02	
REQ-5.6	The system must allow the buyer to choose a delivery method.	Order & Purchase Management	§3 SRS – REQ-5.6 (p. 77)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Order & Purchase Management (p. 164)	orders app – Order / OrderTracking	Manual functional test (TC-REQ-5.6) – p. 190	End-to-end scenario test (TC-REQ-5.6-INT) – p. 273	System workflow test (ST-REQ-5.6) – p. 273	TC-REQ-5.6-01, TC-REQ-5.6-02	
REQ-5.7	The system must allow the buyer to view their order status (Under Review / Accepted / Rejected).	Order & Purchase Management	§3 SRS – REQ-5.7 (p. 77)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Order & Purchase Management (p. 164)	orders app – Order / OrderTracking	Manual functional test (TC-REQ-5.7) – p. 190	End-to-end scenario test (TC-REQ-5.7-INT) – p. 273	System workflow test (ST-REQ-5.7) – p. 273	TC-REQ-5.7-01, TC-REQ-5.7-02	
REQ-5.8	The system must allow the seller to update the order status according to execution stages.	Order & Purchase Management	§3 SRS – REQ-5.8 (p. 77)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Order & Purchase Management (p. 164)	orders app – Order / OrderTracking	Manual functional test (TC-REQ-5.8) – p. 190	End-to-end scenario test (TC-REQ-5.8-INT) – p. 273	System workflow test (ST-REQ-5.8) – p. 273	TC-REQ-5.8-01, TC-REQ-5.8-02	
REQ-5.9	The system must allow the buyer to view their order status at any time (Being Prepared / Prepared / Out for Delivery / Delivered).	Order & Purchase Management	§3 SRS – REQ-5.9 (p. 77)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Order & Purchase Management (p. 164)	orders app – Order / OrderTracking	Manual functional test (TC-REQ-5.9) – p. 190	End-to-end scenario test (TC-REQ-5.9-INT) – p. 273	System workflow test (ST-REQ-5.9) – p. 273	TC-REQ-5.9-01, TC-REQ-5.9-02	
REQ-5.10	The system must allow the buyer to view their previous order history.	Order & Purchase Management	§3 SRS – REQ-5.10 (p. 77)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Order & Purchase Management (p. 164)	orders app – Order / OrderTracking	Manual functional test (TC-REQ-5.10) – p. 190	End-to-end scenario test (TC-REQ-5.10-INT) – p. 273	System workflow test (ST-REQ-5.10) – p. 273	TC-REQ-5.10-01, TC-REQ-5.10-02	

REQ-6.1	The system must allow the buyer to add a comment on a product after a purchase.	Comment Management	§3 SRS – REQ-6.1 (p. 78)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Comment Management (p. 164)	comments app – Comment	Manual functional test (TC-REQ-6.1) – p. 193	End-to-end scenario test (TC-REQ-6.1-INT) – p. 276	System workflow test (ST-REQ-6.1) – p. 276	TC-REQ-6.1-01, TC-REQ-6.1-02	
REQ-6.2	The system must allow the seller and buyer to view comments on products.	Comment Management	§3 SRS – REQ-6.2 (p. 78)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Comment Management (p. 164)	comments app – Comment	Manual functional test (TC-REQ-6.2) – p. 193	End-to-end scenario test (TC-REQ-6.2-INT) – p. 276	System workflow test (ST-REQ-6.2) – p. 276	TC-REQ-6.2-01, TC-REQ-6.2-02	
REQ-6.3	The system must allow the seller to reply to comments on their products.	Comment Management	§3 SRS – REQ-6.3 (p. 78)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Comment Management (p. 164)	comments app – Comment	Manual functional test (TC-REQ-6.3) – p. 193	End-to-end scenario test (TC-REQ-6.3-INT) – p. 276	System workflow test (ST-REQ-6.3) – p. 276	TC-REQ-6.3-01, TC-REQ-6.3-02	
REQ-7.1	The system must allow the buyer to search for products by category.	Search & View	§3 SRS – REQ-7.1 (p. 78)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Search & View (p. 163)	products app – Product (search/filter)	Manual functional test (TC-REQ-7.1) – p. 192	End-to-end scenario test (TC-REQ-7.1-INT) – p. 275	System workflow test (ST-REQ-7.1) – p. 275	TC-REQ-7.1-01, TC-REQ-7.1-02	
REQ-7.2	The system must allow the buyer to filter product results by price from lowest to highest and vice versa.	Search & View	§3 SRS – REQ-7.2 (p. 78)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Search & View (p. 163)	products app – Product (search/filter)	Manual functional test (TC-REQ-7.2) – p. 192	End-to-end scenario test (TC-REQ-7.2-INT) – p. 275	System workflow test (ST-REQ-7.2) – p. 275	TC-REQ-7.2-01, TC-REQ-7.2-02	
REQ-8.1	The system must allow the seller to create a promotional advertisement about their products or shop.	Advertisement Management	§3 SRS – REQ-8.1 (p. 78)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Advertisement Management (p. 165)	advertisement app – Advertisement	Manual functional test (TC-REQ-8.1) – p. 194	End-to-end scenario test (TC-REQ-8.1-INT) – p. 278	System workflow test (ST-REQ-8.1) – p. 278	TC-REQ-8.1-01, TC-REQ-8.1-02	
REQ-8.2	The system must allow the seller to edit advertisement details.	Advertisement Management	§3 SRS – REQ-8.2 (p. 78)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Advertisement Management (p. 165)	advertisement app – Advertisement	Manual functional test (TC-REQ-8.2) – p. 194	End-to-end scenario test (TC-REQ-8.2-INT) – p. 278	System workflow test (ST-REQ-8.2) – p. 278	TC-REQ-8.2-01, TC-REQ-8.2-02	
REQ-8.3	The system must allow the seller to delete unwanted advertisements.	Advertisement Management	§3 SRS – REQ-8.3 (p. 78)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Advertisement Management (p. 165)	advertisement app – Advertisement	Manual functional test (TC-REQ-8.3) – p. 194	End-to-end scenario test (TC-REQ-8.3-INT) – p. 278	System workflow test (ST-REQ-8.3) – p. 278	TC-REQ-8.3-01, TC-REQ-8.3-02	
REQ-8.4	The system must allow the buyer to view all active advertisements.	Advertisement Management	§3 SRS – REQ-8.4 (p. 79)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Advertisement Management (p. 165)	advertisement app – Advertisement	Manual functional test (TC-REQ-8.4) – p. 194	End-to-end scenario test (TC-REQ-8.4-INT) – p. 278	System workflow test (ST-REQ-8.4) – p. 278	TC-REQ-8.4-01, TC-REQ-8.4-02	
REQ-9.1	The system must allow the seller to create a Flash offer on a specific product.	Flash Offers Management	§3 SRS – REQ-9.1 (p. 79)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Flash Offers Management (p. 165)	offers app – OfferFlash	Manual functional test (TC-REQ-9.1) – p. 196	End-to-end scenario test (TC-REQ-9.1-INT) – p. 279	System workflow test (ST-REQ-9.1) – p. 279	TC-REQ-9.1-01, TC-REQ-9.1-02	
REQ-9.2	The system must allow the seller to edit Flash offer details.	Flash Offers Management	§3 SRS – REQ-9.2 (p. 79)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Flash Offers Management (p. 165)	offers app – OfferFlash	Manual functional test (TC-REQ-9.2) – p. 196	End-to-end scenario test (TC-REQ-9.2-INT) – p. 279	System workflow test (ST-REQ-9.2) – p. 279	TC-REQ-9.2-01, TC-REQ-9.2-02	
REQ-9.3	The system must allow the seller to delete a Flash offer.	Flash Offers Management	§3 SRS – REQ-9.3 (p. 79)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Flash Offers Management (p. 165)	offers app – OfferFlash	Manual functional test (TC-REQ-9.3) – p. 196	End-to-end scenario test (TC-REQ-9.3-INT) – p. 279	System workflow test (ST-REQ-9.3) – p. 279	TC-REQ-9.3-01, TC-REQ-9.3-02	
REQ-9.4	The system must allow the buyer to view all Flash offers.	Flash Offers Management	§3 SRS – REQ-9.4 (p. 79)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Flash Offers Management (p. 165)	offers app – OfferFlash	Manual functional test (TC-REQ-9.4) – p. 196	End-to-end scenario test (TC-REQ-9.4-INT) – p. 279	System workflow test (ST-REQ-9.4) – p. 279	TC-REQ-9.4-01, TC-REQ-9.4-02	
REQ-9.5	The system must automatically deactivate Flash offers after the specified period has ended.	Flash Offers Management	§3 SRS – REQ-9.5 (p. 79)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Flash Offers Management (p. 165)	offers app – OfferFlash	Manual functional test (TC-REQ-9.5) – p. 196	End-to-end scenario test (TC-REQ-9.5-INT) – p. 279	System workflow test (ST-REQ-9.5) – p. 279	TC-REQ-9.5-01, TC-REQ-9.5-02	
REQ-10.1	The system must allow the buyer to add products to the favorites list.	Wishlist Management	§3 SRS – REQ-10.1 (p. 79)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Wishlist Management (p. 165)	wishlist app – Wishlist	Manual functional test (TC-REQ-10.1) – p. 197	End-to-end scenario test (TC-REQ-10.1-INT) – p. 280	System workflow test (ST-REQ-10.1) – p. 280	TC-REQ-10.1-01, TC-REQ-10.1-02	
REQ-10.2	The system must allow the buyer to remove products from the favorites list.	Wishlist Management	§3 SRS – REQ-10.2 (p. 79)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Wishlist Management (p. 165)	wishlist app – Wishlist	Manual functional test (TC-REQ-10.2) – p. 197	End-to-end scenario test (TC-REQ-10.2-INT) – p. 280	System workflow test (ST-REQ-10.2) – p. 280	TC-REQ-10.2-01, TC-REQ-10.2-02	

REQ-10.3	The system must allow the buyer to view all products added to the favorites list.	WishList Management	§3 SRS – REQ-10.3 (p. 80)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – WishList Management (p. 165)	wishlist app – Wishlist	Manual functional test (TC-REQ-10.3) – p. 197	End-to-end scenario test (TC-REQ-10.3-INT) – p. 280	System workflow test (ST-REQ-10.3) – p. 280	TC-REQ-10.3-01, TC-REQ-10.3-02	
REQ-10.4	The system must allow the buyer to navigate directly to the product page from the favorites list.	WishList Management	§3 SRS – REQ-10.4 (p. 80)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – WishList Management (p. 165)	wishlist app – Wishlist	Manual functional test (TC-REQ-10.4) – p. 197	End-to-end scenario test (TC-REQ-10.4-INT) – p. 280	System workflow test (ST-REQ-10.4) – p. 280	TC-REQ-10.4-01, TC-REQ-10.4-02	
REQ-10.5	The system must permanently save the Wishlist for each user to display it upon subsequent login.	WishList Management	§3 SRS – REQ-10.5 (p. 80)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – WishList Management (p. 165)	wishlist app – Wishlist	Manual functional test (TC-REQ-10.5) – p. 197	End-to-end scenario test (TC-REQ-10.5-INT) – p. 280	System workflow test (ST-REQ-10.5) – p. 280	TC-REQ-10.5-01, TC-REQ-10.5-02	
REQ-11.1	The system must display the number of received orders for the seller.	Reporting System	§3 SRS – REQ-11.1 (p. 80)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Reporting System (p. 165)	reports app – Aggregations across apps	Manual functional test (TC-REQ-11.1) – p. 198	End-to-end scenario test (TC-REQ-11.1-INT) – p. 281	System workflow test (ST-REQ-11.1) – p. 281	TC-REQ-11.1-01, TC-REQ-11.1-02	
REQ-11.2	The system must display the number of products listed in the shop for the seller.	Reporting System	§3 SRS – REQ-11.2 (p. 80)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Reporting System (p. 165)	reports app – Aggregations across apps	Manual functional test (TC-REQ-11.2) – p. 198	End-to-end scenario test (TC-REQ-11.2-INT) – p. 281	System workflow test (ST-REQ-11.2) – p. 281	TC-REQ-11.2-01, TC-REQ-11.2-02	
REQ-11.3	The system must display the number of shop visitors for the seller.	Reporting System	§3 SRS – REQ-11.3 (p. 80)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Reporting System (p. 165)	reports app – Aggregations across apps	Manual functional test (TC-REQ-11.3) – p. 198	End-to-end scenario test (TC-REQ-11.3-INT) – p. 281	System workflow test (ST-REQ-11.3) – p. 281	TC-REQ-11.3-01, TC-REQ-11.3-02	
REQ-11.4	The system must display general reports about shop activity for the seller.	Reporting System	§3 SRS – REQ-11.4 (p. 80)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Reporting System (p. 165)	reports app – Aggregations across apps	Manual functional test (TC-REQ-11.4) – p. 198	End-to-end scenario test (TC-REQ-11.4-INT) – p. 281	System workflow test (ST-REQ-11.4) – p. 281	TC-REQ-11.4-01, TC-REQ-11.4-02	
REQ-12.1	The system must allow the creation of a direct conversation between the buyer and seller.	Direct Messaging System	§3 SRS – REQ-12.1 (p. 80)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Direct Messaging System (p. 166)	chat app – Conversation / Message	Manual functional test (TC-REQ-12.1) – p. 199	End-to-end scenario test (TC-REQ-12.1-INT) – p. 282	System workflow test (ST-REQ-12.1) – p. 282	TC-REQ-12.1-01, TC-REQ-12.1-02	
REQ-12.2	The system must allow the seller and buyer to send and receive text messages.	Direct Messaging System	§3 SRS – REQ-12.2 (p. 81)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Direct Messaging System (p. 166)	chat app – Conversation / Message	Manual functional test (TC-REQ-12.2) – p. 199	End-to-end scenario test (TC-REQ-12.2-INT) – p. 282	System workflow test (ST-REQ-12.2) – p. 282	TC-REQ-12.2-01, TC-REQ-12.2-02	
REQ-12.3	The system must allow the seller and buyer to view previous conversation history.	Direct Messaging System	§3 SRS – REQ-12.3 (p. 81)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Direct Messaging System (p. 166)	chat app – Conversation / Message	Manual functional test (TC-REQ-12.3) – p. 199	End-to-end scenario test (TC-REQ-12.3-INT) – p. 282	System workflow test (ST-REQ-12.3) – p. 282	TC-REQ-12.3-01, TC-REQ-12.3-02	
REQ-12.4	The system must allow the seller and buyer to attach images or links within messages.	Direct Messaging System	§3 SRS – REQ-12.4 (p. 81)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Direct Messaging System (p. 166)	chat app – Conversation / Message	Manual functional test (TC-REQ-12.4) – p. 199	End-to-end scenario test (TC-REQ-12.4-INT) – p. 282	System workflow test (ST-REQ-12.4) – p. 282	TC-REQ-12.4-01, TC-REQ-12.4-02	
REQ-12.5	The system must display notifications for the seller and buyer when a new message arrives.	Direct Messaging System	§3 SRS – REQ-12.5 (p. 81)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Direct Messaging System (p. 166)	chat app – Conversation / Message	Manual functional test (TC-REQ-12.5) – p. 199	End-to-end scenario test (TC-REQ-12.5-INT) – p. 282	System workflow test (ST-REQ-12.5) – p. 282	TC-REQ-12.5-01, TC-REQ-12.5-02	
REQ-13.1	The system must allow the buyer to enter criteria such as budget or occasion type.	Smart Suggestions Feature	§3 SRS – REQ-13.1 (p. 81)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Smart Suggestions Feature (p. 167)	recommendations app – Recommendation (Groq API)	Manual functional test (TC-REQ-13.1) – p. 200	End-to-end scenario test (TC-REQ-13.1-INT) – p. 282	System workflow test (ST-REQ-13.1) – p. 282	TC-REQ-13.1-01, TC-REQ-13.1-02	
REQ-13.2	The system must analyze the inputs and suggest suitable products for the buyer as gifts.	Smart Suggestions Feature	§3 SRS – REQ-13.2 (p. 81)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Smart Suggestions Feature (p. 167)	recommendations app – Recommendation (Groq API)	Manual functional test (TC-REQ-13.2) – p. 200	End-to-end scenario test (TC-REQ-13.2-INT) – p. 282	System workflow test (ST-REQ-13.2) – p. 282	TC-REQ-13.2-01, TC-REQ-13.2-02	
REQ-13.3	The system must make suggestions using Groq API taking into account the specified criteria.	Smart Suggestions Feature	§3 SRS – REQ-13.3 (p. 81)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Smart Suggestions Feature (p. 167)	recommendations app – Recommendation (Groq API)	Manual functional test (TC-REQ-13.3) – p. 200	End-to-end scenario test (TC-REQ-13.3-INT) – p. 282	System workflow test (ST-REQ-13.3) – p. 282	TC-REQ-13.3-01, TC-REQ-13.3-02	
REQ-13.4	The system must allow the buyer to view the list of suggested products.	Smart Suggestions Feature	§3 SRS – REQ-13.4 (p. 81)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Smart Suggestions	recommendations app – Recommendation (Groq API)	Manual functional test (TC-REQ-13.4) – p. 200	End-to-end scenario test (TC-REQ-13.4-INT) – p. 282	System workflow test (ST-REQ-13.4) – p. 282	TC-REQ-13.4-01, TC-REQ-13.4-02	

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REQ-13.5	The system must allow the buyer to access the product details page by clicking on it.	Smart Suggestions Feature	§3 SRS – REQ-13.5 (p. 81)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Smart Suggestions Feature (p. 167)	recommendations app – Recommendation (Groq API)	Manual functional test (TC-REQ-13.5) – p. 200	End-to-end scenario test (TC-REQ-13.5-INT) – p. 282	System workflow test (ST-REQ-13.5) – p. 282	TC-REQ-13.5-01, TC-REQ-13.5-02	
REQ-14.1	The system must allow the bazaar organizer to create a new bazaar.	Bazaar Management	§3 SRS – REQ-14.1 (p. 82)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Bazaar Management (p. 167)	bazaar app – Bazaar	Manual functional test (TC-REQ-14.1) – p. 200	End-to-end scenario test (TC-REQ-14.1-INT) – p. 283	System workflow test (ST-REQ-14.1) – p. 283	TC-REQ-14.1-01, TC-REQ-14.1-02	
REQ-14.2	The system must allow the bazaar organizer to edit bazaar data.	Bazaar Management	§3 SRS – REQ-14.2 (p. 82)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Bazaar Management (p. 167)	bazaar app – Bazaar	Manual functional test (TC-REQ-14.2) – p. 200	End-to-end scenario test (TC-REQ-14.2-INT) – p. 283	System workflow test (ST-REQ-14.2) – p. 283	TC-REQ-14.2-01, TC-REQ-14.2-02	
REQ-14.3	The system must allow the bazaar organizer to delete an existing bazaar.	Bazaar Management	§3 SRS – REQ-14.3 (p. 82)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Bazaar Management (p. 167)	bazaar app – Bazaar	Manual functional test (TC-REQ-14.3) – p. 200	End-to-end scenario test (TC-REQ-14.3-INT) – p. 283	System workflow test (ST-REQ-14.3) – p. 283	TC-REQ-14.3-02	
REQ-14.4	The system must allow the seller to view all available bazaars.	Bazaar Management	§3 SRS – REQ-14.4 (p. 82)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Bazaar Management (p. 167)	bazaar app – Bazaar	Manual functional test (TC-REQ-14.4) – p. 200	End-to-end scenario test (TC-REQ-14.4-INT) – p. 283	System workflow test (ST-REQ-14.4) – p. 283	TC-REQ-14.4-01, TC-REQ-14.4-02	
REQ-14.5	The system must allow the seller to view bazaar details.	Bazaar Management	§3 SRS – REQ-14.5 (p. 82)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Bazaar Management (p. 167)	bazaar app – Bazaar	Manual functional test (TC-REQ-14.5) – p. 200	End-to-end scenario test (TC-REQ-14.5-INT) – p. 283	System workflow test (ST-REQ-14.5) – p. 283	TC-REQ-14.5-01, TC-REQ-14.5-02	
REQ-15.1	The system must allow the seller to request a table reservation within the bazaar.	Bazaar Table Reservation Management	§3 SRS – REQ-15.1 (p. 82)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Bazaar Table Reservation Management (p. 167)	bazaar app – BazaarTable / Reservation	Manual functional test (TC-REQ-15.1) – p. 202	End-to-end scenario test (TC-REQ-15.1-INT) – p. 284	System workflow test (ST-REQ-15.1) – p. 284	TC-REQ-15.1-01, TC-REQ-15.1-02	
REQ-15.2	The system must allow the seller to cancel a reservation request before the bazaar start date.	Bazaar Table Reservation Management	§3 SRS – REQ-15.2 (p. 82)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Bazaar Table Reservation Management (p. 167)	bazaar app – BazaarTable / Reservation	Manual functional test (TC-REQ-15.2) – p. 202	End-to-end scenario test (TC-REQ-15.2-INT) – p. 284	System workflow test (ST-REQ-15.2) – p. 284	TC-REQ-15.2-01, TC-REQ-15.2-02	
REQ-15.3	The system must allow the bazaar organizer to view incoming reservation requests.	Bazaar Table Reservation Management	§3 SRS – REQ-15.3 (p. 82)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Bazaar Table Reservation Management (p. 167)	bazaar app – BazaarTable / Reservation	Manual functional test (TC-REQ-15.3) – p. 202	End-to-end scenario test (TC-REQ-15.3-INT) – p. 284	System workflow test (ST-REQ-15.3) – p. 284	TC-REQ-15.3-01, TC-REQ-15.3-02	
REQ-15.4	The system must allow the bazaar organizer to accept reservation requests.	Bazaar Table Reservation Management	§3 SRS – REQ-15.4 (p. 83)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Bazaar Table Reservation Management (p. 167)	bazaar app – BazaarTable / Reservation	Manual functional test (TC-REQ-15.4) – p. 202	End-to-end scenario test (TC-REQ-15.4-INT) – p. 284	System workflow test (ST-REQ-15.4) – p. 284	TC-REQ-15.4-01, TC-REQ-15.4-02	
REQ-15.5	The system must allow the bazaar organizer to reject reservation requests.	Bazaar Table Reservation Management	§3 SRS – REQ-15.5 (p. 83)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Bazaar Table Reservation Management (p. 167)	bazaar app – BazaarTable / Reservation	Manual functional test (TC-REQ-15.5) – p. 202	End-to-end scenario test (TC-REQ-15.5-INT) – p. 284	System workflow test (ST-REQ-15.5) – p. 284	TC-REQ-15.5-01, TC-REQ-15.5-02	
REQ-16.1	The system must allow the buyer to view a notification when a purchase order is accepted.	Notification System	§3 SRS – REQ-16.1 (p. 83)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Notification System (p. 168)	notifications app – Notification	Manual functional test (TC-REQ-16.1) – p. 203	End-to-end scenario test (TC-REQ-16.1-INT) – p. 285	System workflow test (ST-REQ-16.1) – p. 285	TC-REQ-16.1-01, TC-REQ-16.1-02	
REQ-16.2	The system must allow the buyer to view a notification when a purchase order is rejected.	Notification System	§3 SRS – REQ-16.2 (p. 83)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Notification System (p. 168)	notifications app – Notification	Manual functional test (TC-REQ-16.2) – p. 203	End-to-end scenario test (TC-REQ-16.2-INT) – p. 285	System workflow test (ST-REQ-16.2) – p. 285	TC-REQ-16.2-01, TC-REQ-16.2-02	
REQ-16.3	The system must allow the buyer to view a notification when the order status is updated.	Notification System	§3 SRS – REQ-16.3 (p. 83)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Notification System (p. 168)	notifications app – Notification	Manual functional test (TC-REQ-16.3) – p. 203	End-to-end scenario test (TC-REQ-16.3-INT) – p. 285	System workflow test (ST-REQ-16.3) – p. 285	TC-REQ-16.3-01, TC-REQ-16.3-02	
REQ-16.4	The system must allow the buyer to view a notification when their comment is replied to.	Notification System	§3 SRS – REQ-16.4 (p. 83)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Notification System (p. 168)	notifications app – Notification	Manual functional test (TC-REQ-16.4) – p. 203	End-to-end scenario test (TC-REQ-16.4-INT) – p. 285	System workflow test (ST-REQ-16.4) – p. 285	TC-REQ-16.4-01, TC-REQ-16.4-02	

REQ-16.5	The system must allow the seller to view a notification when a new order arrives.	Notification System	§3 SRS – REQ-16.5 (p. 83)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Notification System (p. 168)	notifications app – Notification	Manual functional test (TC-REQ-16.5) – p. 203	End-to-end scenario test (TC-REQ-16.5-INT) – p. 285	System workflow test (ST-REQ-16.5) – p. 285	TC-REQ-16.5-01, TC-REQ-16.5-02	
REQ-16.6	The system must allow the seller to view a notification when a new comment is added to a product.	Notification System	§3 SRS – REQ-16.6 (p. 83)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Notification System (p. 168)	notifications app – Notification	Manual functional test (TC-REQ-16.6) – p. 203	End-to-end scenario test (TC-REQ-16.6-INT) – p. 285	System workflow test (ST-REQ-16.6) – p. 285	TC-REQ-16.6-01, TC-REQ-16.6-02	
REQ-16.7	The system must allow the seller to view a notification when a bazaar reservation is accepted.	Notification System	§3 SRS – REQ-16.7 (p. 84)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Notification System (p. 168)	notifications app – Notification	Manual functional test (TC-REQ-16.7) – p. 203	End-to-end scenario test (TC-REQ-16.7-INT) – p. 285	System workflow test (ST-REQ-16.7) – p. 285	TC-REQ-16.7-01, TC-REQ-16.7-02	
REQ-16.8	The system must allow the seller to view a notification when a bazaar reservation is rejected.	Notification System	§3 SRS – REQ-16.8 (p. 84)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Notification System (p. 168)	notifications app – Notification	Manual functional test (TC-REQ-16.8) – p. 203	End-to-end scenario test (TC-REQ-16.8-INT) – p. 285	System workflow test (ST-REQ-16.8) – p. 285	TC-REQ-16.8-01, TC-REQ-16.8-02	
REQ-16.9	The system must allow the bazaar organizer to view a notification when a new reservation request arrives.	Notification System	§3 SRS – REQ-16.9 (p. 84)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Notification System (p. 168)	notifications app – Notification	Manual functional test (TC-REQ-16.9) – p. 203	End-to-end scenario test (TC-REQ-16.9-INT) – p. 285	System workflow test (ST-REQ-16.9) – p. 285	TC-REQ-16.9-01, TC-REQ-16.9-02	
REQ-16.10	The system must allow the bazaar organizer to view a notification when a seller cancels their reservation.	Notification System	§3 SRS – REQ-16.10 (p. 84)	§Ch.5 §3 Component Diagram (p. 222)	§5.4 Class Diagram – Notification System (p. 168)	notifications app – Notification	Manual functional test (TC-REQ-16.10) – p. 203	End-to-end scenario test (TC-REQ-16.10-INT) – p. 285	System workflow test (ST-REQ-16.10) – p. 285	TC-REQ-16.10-01, TC-REQ-16.10-02	
NFR-1	The system must be scalable to support an increasing number of users without performance degradation.	Scalability	§3 SRS – NFR-1 (p. 84)	§Ch.5 §2 System Architecture (p. 209)	§Ch.5 §4 Design Patterns (p. 224)	Implemented (Django config + architecture)	Manual functional test (TC-NFR-1) – p. 203	End-to-end scenario test (TC-NFR-1-INT) – p. 285	System workflow test (ST-NFR-1) – p. 285	TC-NFR-1-CHK	
NFR-2	The system must maintain operational readiness to ensure continuous availability of features.	Availability	§3 SRS – NFR-2 (p. 84)	§Ch.5 §2 System Architecture (p. 209)	§Ch.5 §4 Design Patterns (p. 224)	Implemented (Django config + architecture)	Manual functional test (TC-NFR-2) – p. 203	End-to-end scenario test (TC-NFR-2-INT) – p. 285	System workflow test (ST-NFR-2) – p. 285	TC-NFR-2-CHK	
NFR-3	Basic authentication must be implemented and passwords must be stored in encrypted form.	Security	§3 SRS – NFR-3 (p. 84)	§Ch.5 §2 System Architecture (p. 209)	§Ch.5 §4 Design Patterns (p. 224)	Implemented (Django config + architecture)	Manual functional test (TC-NFR-3) – p. 203	End-to-end scenario test (TC-NFR-3-INT) – p. 285	System workflow test (ST-NFR-3) – p. 285	TC-NFR-3-CHK	
NFR-4	The system must support switching the user interface language between Arabic and English easily.	Usability	§3 SRS – NFR-4 (p. 84)	§Ch.5 §2 System Architecture (p. 209)	§Ch.5 §4 Design Patterns (p. 224)	Implemented (Django config + architecture)	Manual functional test (TC-NFR-4) – p. 203	End-to-end scenario test (TC-NFR-4-INT) – p. 285	System workflow test (ST-NFR-4) – p. 285	TC-NFR-4-CHK	
NFR-5	The code must be standardized and well-documented for ease of maintenance.	Maintainability	§3 SRS – NFR-5 (p. 84)	§Ch.5 §2 System Architecture (p. 209)	§Ch.5 §4 Design Patterns (p. 224)	Implemented (Django config + architecture)	Manual functional test (TC-NFR-5) – p. 203	End-to-end scenario test (TC-NFR-5-INT) – p. 285	System workflow test (ST-NFR-5) – p. 285	TC-NFR-5-CHK	

Table 81 Final Requirements Traceability Matrix (RTM)

Chapter 7

Report Conclusion, Recommendations, and Future Work

1. Introduction

This chapter reviews the final results achieved within the Syrian System for Marketing and Handicrafts project, detailing the main accomplishments of the system, analyzing strengths and challenges faced during development, and presenting proposals and future recommendations that could contribute to the system's ongoing development and improvement.

2. Project Summary and Key Achievements

The Syrian System for Marketing and Handicrafts project is an integrated electronic platform developed to support artisans, productive families, and small business owners by providing an online environment that helps them create their own shops, display, market, and manage their handmade products easily and securely.

The system provides a comprehensive set of functionalities covering various core operations within the platform, including account, shop, product, category, basket, and order management, as well as facilitating communication between sellers and buyers through chat and comments systems.

Additional features were developed to enhance system quality and practical value, such as the notifications system, flash offers, wishlist, along with bazaar management and table booking features, which made the system more comprehensive and realistic compared to traditional systems.

The project is also distinguished by integrating artificial intelligence technologies by relying on Groq API to provide intelligent gift recommendations based on user-provided information such as occasion, age, gender, and budget, thereby enhancing the user experience and adding an intelligent aspect to the system.

3. Difficulties and Challenges

Designing a multi-role system supporting buyer, seller, admin, and bazaar organizer.

Organizing relationships between different entities within the database .

Developing a notifications system integrated with multiple events inside the system.

Implementing bazaar booking system and managing acceptance/rejection cases.

Maintaining consistency in interface design and user experience .

Integrating an external artificial intelligence service with the system .

Managing a large number of software modules and ensuring their integration .

4. Strengths and Weaknesses Analysis

First: Strengths

- Adoption of a clear and scalable architecture using Django MVT and Layered Architecture.
- Support for multiple user types with independent permissions for each role.
- Provision of advanced features such as notifications system, bazaars, and smart recommendations.
- Modern and user-friendly interface designs.
- Ability to easily develop the system and add future features.

Second: Weaknesses

- Lack of real electronic payment support in the current version.
- Dependence on internet connectivity to use AI features.
- Absence of a dedicated mobile application for the system.
- Need for additional performance improvements when the number of users grows.

5. Future Developments

Add full support for electronic payment gateways.

Develop a mobile application for Android.

Add an integrated shipping and delivery system.

Improve the AI system to provide more accurate recommendations.

Add advanced reports and analytics for sellers.

Enhance scalability and performance to support a larger number of users.

Add features based on user behavior analysis and product recommendations.

Add an advanced ratings and analytics system.

6. Recommendations

It is recommended to continue developing and expanding the system to become a fully integrated electronic platform supporting the handicraft sector more comprehensively, with a focus on improving user experience, enhancing security and performance, as well as supporting additional intelligent services based on AI and data analysis technologies.

It is also recommended to conduct advanced performance and security testing in future phases, and to deploy the system in a real operational environment to enable broader usage.

7. Conclusion

The Syrian System for Marketing and Handicrafts project presented a practical model for implementing modern software engineering concepts within an integrated electronic system aimed at supporting artisans and small business owners and enabling them digitally.

The system also demonstrated the possibility of building a flexible and scalable platform combining traditional e-commerce functionalities with some modern intelligent features, making it a suitable foundation for developing more advanced systems in the future.

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